

# Site Construction Manual

Clayton Built®

# USING THE MANUAL

This manual is organized into a series of steps that will take you through the entire installation process using only those pages required for the specific home being installed. First, review the entire manual, including the Introduction chapter. As you read it, identify sections of the manual that you will need; identify other documents or information you will need; construct lists of tools and materials required for your particular installation; and make sure you have everything you need before starting work.

After reviewing the entire manual, refer to the sequence of installation steps in the table below. Identify the pathway for your installation and follow the arrows downward. Select either **Single Section Home** or a **Multi-Section Home** and choose the column corresponding to the home's foundation type, either **Pier and Ground Anchor** or **Load-Bearing Perimeter Wall** (see **Definitions**, p. 11). Then complete the work in each of the sections starting with **Getting Started**.

If using an alternative (proprietary) foundation system, the installation process will change from that described in this manual. Consult the system manufacturer's directions for instructions. See page 15 for alternative foundation system criteria.

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|---|---|---|---|
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# APPENDIX AND QR CODE LIST

Please refer to the ***Appendix QR code*** for any appendix referenced in this document. Any DAPIA approved, plant specific details that may supplement or supersede information provided in the following installation manual will be located at the end.



*Appendix QR Code*

## Appendix List


Appendix A. Tag Units

Appendix B. Triple Section Homes

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## QR Code List


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# INTRODUCTION

This installation manual contains instructions that must be followed for the proper installation of the home. It complies with the HUD Installation Standards. Please read all instructions and any other documents (including addendum pages and supplements) that may apply to the specific home prior to commencing site work or installation.

This installation manual covers permits and site work through final inspection of the installation. It covers both single and multi-section homes installed over pier and anchor and load bearing crawl space walls. It contains instructions, including specifications and procedures, for the set and hookup of homes to be used as single-family and multi-family dwellings.

No manual can cover all circumstances that may exist for certain home designs or building sites. For questions, further clarification or if you encounter conditions at the site or in the design of the home or its foundation not covered by this manual, please contact the manufacturer, a registered engineer or registered architect.

Supplemental addendum pages may be included with this manual. Supplements include requirements not covered in this manual or that supersede the manual instructions.

**Once the home installation is complete, leave this manual with the home.**

## Important Notices

- The home manufacturer is not responsible for installation or for the materials supplied by the set-up crew at the time of installation. The installer is responsible for any deviations from the installation instructions of this manual.
- To keep the home in compliance with its warranty, the home installation must follow the procedures described in this manual or other procedures approved by the manufacturer. Deviation from the instructions in this manual may void the home's warranty. Any alterations or changes to the home shall be approved by a registered engineer or registered architect and may still be subject to warranty violations.
- When an installer does not provide support and anchorage in accordance with the approved manufacturer's installation instructions, or encounters site conditions (such as areas that are subject to flood damage or high seismic risk) or other conditions that prevent the use of the instructions provided in this manual, the installer must obtain special site-specific instructions from the manufacturer or use a design approved by a registered engineer or registered architect. Designs provided by registered professional engineers or registered architects may also be approved by the manufacturer and DAPIA. Alternative support and anchorage designs may be used which are state approved and acceptable to the local authority having jurisdiction.

## Safety

There are potential hazards associated with the installation of a manufactured home. Home installers are licensed and, as experienced professionals, should recognize these hazards, be qualified to work with them, and be capable of providing safe work practices and equipment that minimize the risks of injury.

Only qualified persons should install a manufactured home. The installer must possess a valid installation license as a manufactured home installer. As qualified professionals in the field of home installation, installers are the experts and must be aware of the hazards and conditions faced. Warnings are published throughout this manual as reminders. These reminders may not cover all hazards, all potential hazards, or all possible consequences of improper or unsafe installation practices.

Construction crews should be trained in the skills required and be supervised by experienced personnel. Installers should regularly inspect work performed by crews and subcontractors.

Obey OSHA regulations, particularly those related to home construction, such as Title 29 Code of Regulations Part 1926. For copies of OSHA regulations, call (202) 512-1800 or visit **www.osha.gov** on the web.

## Resources

### Office of Regulatory Affairs and Manufactured Housing

US Department of Housing and Urban Development
451 Seventh Street, SW, Room 9164
Washington, DC 20410-8000
Telephone: (202) 708-6423 or (800) 927-2891
FAX: (202) 708-4213

### State Administrative Agencies

A list of SAAs may be found on the web at **www.hud.gov/offices/hsg/sfh/mhs/mhssaa.cfm** or by contacting the Office of Regulatory Affairs and Manufactured Housing or in the Homeowners Manual.

## Federal Pre-emption

This home was engineered, constructed and inspected in conformance with the Federal Manufactured Home Construction and Safety Standards of the US Department of Housing and Urban Development (24 CFR Part 3280, commonly referred to as the “HUD Code”) in effect on the date of construction. These Standards set forth minimum requirements for the design and construction of manufactured homes designed to be used as dwellings.

Individual states, counties and cities shall have no authority to establish standards regarding the construction or safety of a manufactured home. A metal certification label is affixed to each section of the home to certify that it has been constructed and inspected to comply with these Standards. The design plans and in-plant construction of all homes are inspected by independent third-party agencies to assure compliance with the Standards.

The installation of the home and any alterations made to the home shall conform to the requirements of the Federal Manufactured Home Construction and Safety Standards and the HUD Model Manufactured Home Installation Standards. These installation instructions are minimum requirements. Applicable local or state laws may have more stringent installation requirements than outlined in this manual and must be followed. Consult with the local authority having jurisdiction (LAHJ) for regulations that may require licenses and/or permits, or which may affect procedures described in this manual.

## Definitions

**Anchor assembly.** Any device or other means designed to transfer home anchoring loads to the ground.

**Anchoring equipment.** Ties, straps, cables, turnbuckles, chains, and other approved components, including tensioning devices that are used to secure a home to anchor assemblies.

**Anchoring system.** A combination of anchoring equipment and anchor assemblies that will, when properly designed and installed, resist the uplift, overturning, and lateral forces on the home.

**Basement.** A load bearing perimeter wall foundation that includes habitable space (finished or unfinished, heated or unheated) completely or partially below grade.

**Concrete anchor.** A specific anchoring assembly device designed to transfer home anchoring tension loads from straps to concrete foundation or slab system.

**Crawlspace.** The space underneath the home’s floor system, enclosed with either load- or non-load bearing perimeter walls. The ground may be covered with a concrete slab or by a plastic ground cover. Crawlspace walls must be vented and an access must be provided.

**Crossovers.** Utility interconnections between sections of multi-section homes, including heating and cooling ducts, electrical circuits, and water pipes, drain plumbing, and gas lines.

**Data plate.** An information sheet located on a cabinet door under the kitchen sink or on a wall panel or door face near the electrical panel, utility room or in the master bedroom closet. It contains a unique identification number and identifies the wind zone, roof load zone, and climatic zone for which the home was constructed.

**Diagonal tie.** A tie intended to resist horizontal or shear forces, but which may resist vertical, uplift, and overturning forces.

**Footing.** That portion of the support system that transmits loads directly to the soil.

**Ground anchor.** A specific anchoring assembly device designed to transfer home loads to the ground.

**H-Beam.** Steel H-beams, also called cross beams, are often used to support a home over a basement or crawlspace. They span across the foundation from sidewall to sidewall, typically with an intermediate support pier and footing (typically in the center point resulting in a line of piers under the centerline of a double section home).

**Information Packet.** A set of important documents provided with the home including warranties, information on high wind coverage, and other features of the specific home.

**Installation license.** The proof that an installer meets the requirements for installing manufactured homes under the HUD-administered installation program.

**Labelled.** Equipment or materials to which has been attached a label, symbol, or other identifying mark of a certified testing laboratory, inspection agency, or other organization concerned with product evaluation. The label indicates compliance with nationally recognized standards or tests to determine suitable usage in a specified manner.

**Listed or certified.** Included in a list published by a nationally recognized testing laboratory, inspection agency, or other organization concerned with product evaluation that maintains periodic inspection of production of listed equipment or materials, and whose listing states either that the equipment or material meets nationally recognized standards or has been tested and found suitable for use in a specified manner.

**Load-bearing perimeter wall foundation**. A support system for the home whereby the home is mechanically fastened to a structural wall(s) that transfers gravity, lateral and uplift loads to the ground.

**Local authority having jurisdiction (LAHJ)**. The state, city, county, municipality, utility, or organization that has local responsibilities that must be complied with during the installation of a manufactured home.

**Must.** Indicates a mandatory requirement.

**N/A.** Indicates not applicable.

**Pier.** That portion of the support system between the footing and the home, exclusive of shims. Types of piers include but are not limited to: (1) manufactured steel stands; (2) pressure-treated wood; (3) manufactured concrete stands; (4) concrete blocks; and (5) portions of foundation walls.

**Pier and ground anchor foundation.** A support system for the home that employs piers under the chassis and other locations to support gravity loads and employs ground anchors and tie downs (the stabilizing system) to resist lateral and uplift loads.

**Perimeter blocking.** Regularly spaced piers supporting the sidewalls and marriage line of the home. Some homes require perimeter blocking in addition to supports under the home's frame.

**Qualified (or licensed).** Has the necessary knowledge and skills gained from experience and training that will allow performance of the job safely, competently, and in accordance with all applicable codes, standards, rules and regulations. Meets all necessary qualification tests

including any license and certification requirements that may be in effect in the area where the home will be installed, including the requirements for installing manufactured homes under the HUD-administered installation program. The term does not incorporate a state-issued installation license or certification, except to the extent provided in this part. The term does not imply that HUD approves or recommends an installer or warrants the work of an installer and should not be used in any way that indicates HUD approval in violation of 18 U.S.C. 709.

**Ramada.** Any freestanding roof or shade structure installed or erected over a home or any portion of the home.

**Should.** Indicates a recommendation that is strongly advised but not mandatory.

**Shall.** Indicates a mandatory requirement.

**Site for a home.** A designated parcel of land designed for the accommodation of one home, its accessory buildings or structures, and accessory equipment, for the exclusive use of the occupants of the home.

**Skirting.** A weather-resistant material used to enclose the perimeter, under the living area of the home, from the bottom of the home to grade.

**Stabilizing system.** All components of the anchoring and support systems, such as piers, footings, ties, anchoring equipment, ground anchors, or any other materials and methods of construction that support and secure the home to the ground.

**Support system.** Pilings, columns, a combination of footings, piers, foundation walls, caps, and shims and any combination thereof that will, when properly installed, support and secure the home to the ground.

**Tie.** Straps, cable, or securing devices used to connect the home to anchoring assemblies.

**Utility connection.** The connection of the home to utilities that include, but are not limited to, electricity, water, sewer, gas, or fuel oil.

**Vertical tie.** A tie intended to resist uplifting and overturning forces.

**Wind Zone.** The areas designated on the Basic Wind Zone Map, as further defined in by the Manufactured Home Construction and Safety Standards

## Engineer's Stamp

Certain pages of this manual display the seal of a registered engineer. Federal guidelines only require the seal from one state to be displayed, but the details herein apply to all states.

## Symbols Used in the Manual



This icon indicates an important warning. It is critical to heed these warnings.

## Abbreviations

| | |
|---|---|
| **ABS** | Acrylonitrile Butadiene Styrene |
| **ANSI** | American National Standards Institute |
| **APA** | American Plywood Association |
| **ASTM** | American Society for Testing and Materials |
| **AWPA** | American Wood Preservers Association |
| **CFM** | Cubic feet per minute |
| **CFR** | Code of Federal Regulations |
| **DWV** | Drain, Waste, Vent |
| **EMT** | Electrical metallic tubing |
| **FEMA** | Federal Emergency Management Agency |
| **ft** | Foot/feet |
| **ga** | Gauge |
| **HUD** | US Department of Housing and Urban Development |
| **in** | Inch(es) |
| **LAHJ** | Local Authority Having Jurisdiction |
| **lb(s)** | Pound(s) |
| **max.** | Maximum |
| **MHCSS** | Manufactured Home Construction and Safety Standards |
| **min.** | Minimum |
| **mph** | Mile(s) per hour |
| **NEC** | National Electric Code |
| **NFIP** | National Flood Insurance Program |
| **NFPA** | National Fire Protection Association |
| **o.c.** | On center |
| **OSHA** | Occupational Safety and Health Administration |
| **oz** | Ounce(s) |
| **p.** | Page |
| **psf** | Pounds per square foot |
| **psi** | Pounds per square inch |
| **SAA** | State Administrative Agency |
| **sq ft** | Square foot/feet |

## Alternative Foundation Systems

Alternative foundation systems or designs are permitted in accordance with the following:

- System designs are prepared by a registered engineer or a registered architect or tested and certified by a registered engineer or registered architect in accordance with acceptable engineering practice and are manufactured and installed so as not to take the home out of compliance with the Manufactured Home Construction and Safety Standards.
- An Alternative Foundation and/or Tiedown system design must be submitted to the manufacturer if it is not listed on the following website: **www.goldenwestalbany.com**. Click the **Builders Resource** tab at the bottom of the page to view all DAPIA approved Alternative Tiedown systems. Prior to obtaining an alternative design contact the home building facility for available approved alternative designs or instructions for submitting an alternative design.

## Display and Storage of the Home

### Weather Protection

If the installation is not started immediately upon delivery of the home, the retailer and/or installer has the responsibility to ensure the exterior weather protection covering of marriage walls and the roof of homes has not been damaged during shipment. Inspect the home immediately upon the delivery and frequently during storage. Promptly repair tears in the home closure materials to prevent damage from the elements. Inspect and repair nail holes in roof shingles with asphalt cement or replace damaged shingles. Inspect and repair siding as needed.

### Supporting a Home for Display

When a new home is to be displayed at a retail location, temporarily block and support the home. Set up homes with single block piers (maximum height per **Table 14**, p. 57), metal piers or jack stands spaced no further apart than 12 feet o.c. beneath each I-beam. The tire and axle system may be used as one of these required supports, and the hitch jack may be used as another. Locate the first support no further than two feet from the rear end of the home (**Figure 1**). Place additional supports along the perimeter on either side of openings greater than four feet (i.e. sliding glass doors, bay windows, etc.).

For multi section homes, locate additional supports along the marriage line under support columns. These locations will be marked by the manufacturer.

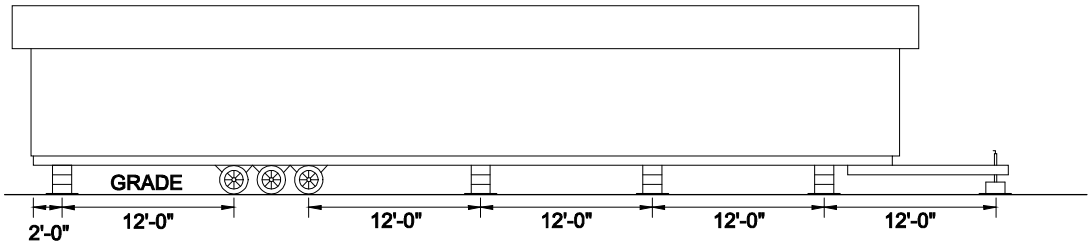



*Figure 1. Supporting a home for display*

For all homes, place footings below each support. Footings may be placed directly on the surface grade without excavation and may be ABS pads, 2 x 10 by 16 inch long pressure treated lumber or 16” x 16” by 4 inch thick concrete pads.

### Supporting a Home for Storage

To prevent damage to homes being stored at the manufacturer’s facility, model home center, or the home site, but not on display (i.e. people shall not be permitted inside the home) for a period exceeding 30 days, locate supports below each I-beam no further than two feet from each end of the home and at the approximate center of the home length.

Whether the home is being supported for display or storage the height of the home should be no higher than 48 inches as measured from the top of the ground to the bottom of the I-beam. In addition, it is extremely important that the roof/ridge vents are installed while the home is on display. Failure to install the roof/ridge vents may lead to significant damage to the roof and home.

# GETTING STARTED

This chapter covers a few steps that, taken now, will avoid problems later in the installation process.

**Follow the Steps below:**


▼ **Step 1. Locate The Data Plate** (p. 17)
▼ **Step 2. Confirm Wind Zone** (p. 17)
▼ **Step 3. Confirm Thermal Zone** (p. 19)
▼ **Step 4. Confirm Roof-Load Zone** (p. 22)
▼ **Step 5. Check Local Codes And Secure Permits** (p. 24)


## Step 1. Locate the Data Plate

Locate the data plate inside the home (**Figure 2**), typically inside a kitchen cabinet door or on a wall panel or door face near the electrical panel, utility room or bedroom closet.



*Figure 2. Sample data plate – shown as reference only, actual data plate may vary.*

The information on the data plate will be used to verify that the home was designed for the proper location.

## Step 2. Confirm Wind Zone

From **Table 1**, identify the wind zone for the home. Verify that the home conforms to the following rules and any special requirements determined by the LAHJ.

- No home may be located in a higher wind zone than that indicated on the data plate. (Example: a home designed for Wind Zone II cannot be placed in Wind Zone III.)
- A home may be located in a lower wind zone than that indicated on the data plate. (Example: a home designed for Wind Zone II can be placed in either Wind Zone II or I.) When a home is located in a lower wind zone than indicated on the data plate it may be installed per the requirements of the lower wind zone.
- Homes located within 1,500 feet of the coastline in Wind Zones II and III must be designed to withstand exposure 'D' conditions. This will be indicated on the data plate.

If the home does not conform to these rules, contact the manufacturer immediately.

*Table 1. Wind Zone by locality*

| **Wind Zone I** | |
|---|---|
| All areas except those areas listed below as being within Wind Zone II or III | |
| **Wind Zone II** | |
| **Alabama** | Counties of Baldwin and Mobile |
| **Florida** | All counties except those listed below as within Wind Zone III |
| **Georgia** | Counties of Bryan, Camden, Chatham, Glynn, Liberty, McIntosh |
| **Louisiana** | Parishes of Acadia, Allen, Ascension, Assumption, Calcasieu, Cameron, East Baton Rouge, East Feliciana, Evangeline, Iberia, Iberville, Jefferson Davis, Lafayette, Livingston, Pointe Coupee, St. Helena, St. James, St. John the Baptist, St. Landry, St. Martin, St. Tammany, Tangipahoa, Vermillion, Washington, West Baton Rouge, and West Feliciana |
| **Maine** | Counties of Hancock and Washington |
| **Massachusetts** | Counties of Barnstable, Bristol, Dukes, Nantucket, and Plymouth |
| **Mississippi** | Counties of George, Hancock, Harrison, Jackson, Pearl River, and Stone |
| **North Carolina** | Counties of Beaufort, Brunswick, Camden, Chowan, Columbus, Craven, Currituck, Jones, New Hanover, Onslow, Pamlico, Pasquotank, Pender, Perquimans, Tyrrell, and Washington |
| **South Carolina** | Counties of Beaufort, Berkeley, Charleston, Colleton, Dorchester, Georgetown, Horry, Jasper, and Williamsburg |
| **Texas** | Counties of Aransas, Brazoria, Calhoun, Cameron, Chambers, Galveston, Jefferson, Kenedy, Kleberg, Matagorda, Nueces, Orange, Refugio, San Patricio, and Willacy |
| **Virginia** | Cities of Chesapeake, Norfolk, Portsmouth, Princess Anne, and Virginia Beach |
| **Wind Zone III** | |
| **Hawaii** | Entire state |
| **Alaska** | Coastal regions (as determined by the 90 mph isotach on the ANSI/ASCE 7-88 map) |
| **Florida** | Counties of Broward, Charlotte, Collier, Dade, Franklin, Gulf, Hendry, Lee, Martin, Manatee, Monroe, Palm Beach, Pinellas, and Sarasota |
| **Louisiana** | Parishes of Jefferson, La Fourche, Orleans, Plaquemines, St. Bernard, St. Charles, St. Mary, and Terrebonne |
| **North Carolina** | Counties of Carteret, Dare, and Hyde |
| **Other** | All regions of the U.S. Territories of American Samoa, Guam, Northern Mariana Islands, Puerto Rico, Trust Territory of the Pacific Islands, and the United States Virgin Islands |

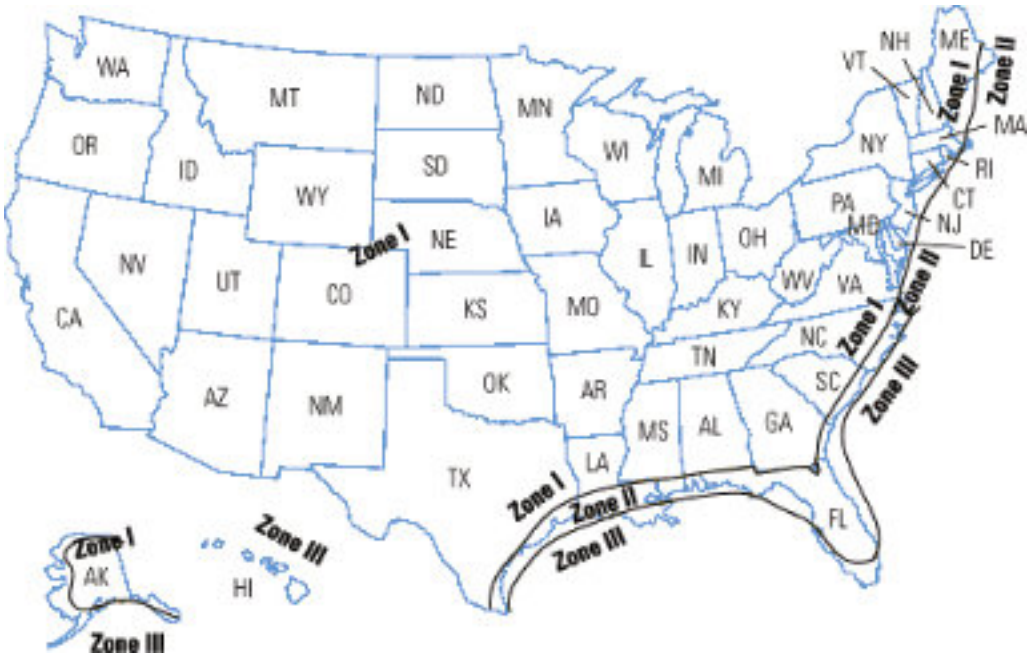



*Figure 3. Wind zone map*

## Step 3. Confirm Thermal Zone

From **Table 2**, identify the thermal (UO) zone for the home. Verify that the home conforms to the following rules.

- No home may be located in an area with a higher thermal zone number than that indicated on the data plate. (Example: a home designed for Thermal Zone 2 cannot be placed in Thermal Zone 3.)
- A home may be located in a lower thermal zone than that indicated on the data plate. (Example: a home designed for Thermal Zone 2 may be placed in either Thermal Zone 2 or 1.).
- In no case may a home designated for installation in the “Humid & Fringe Climate,” as identified on the data plate, be located outside of this region (**Table 2**).

If the home does not conform to these rules, contact the manufacturer immediately.

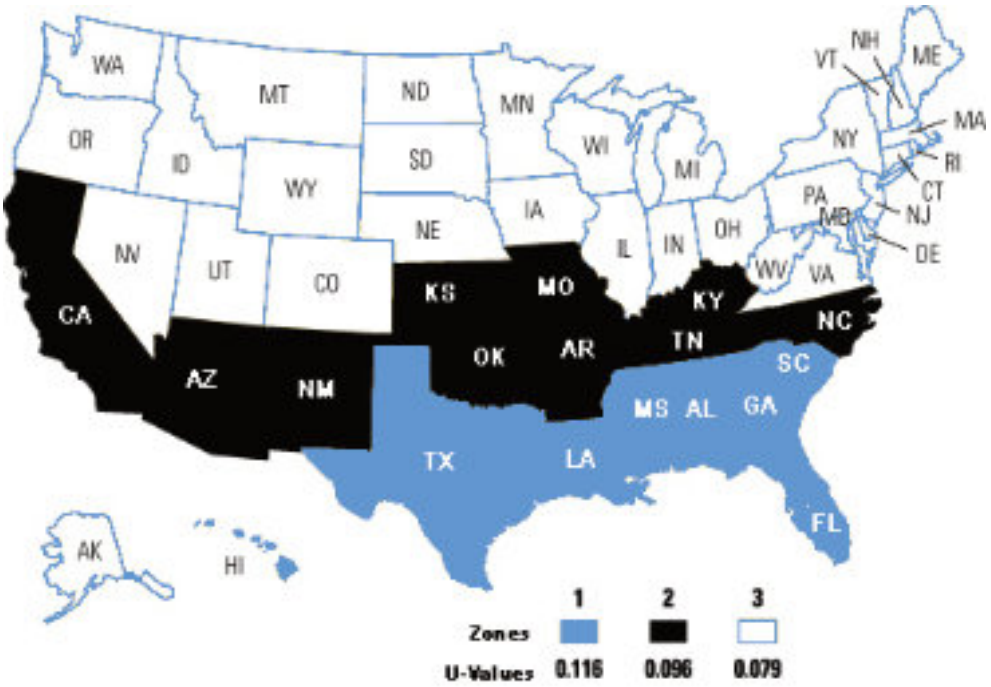



*Figure 4. Thermal (UO) zone map*

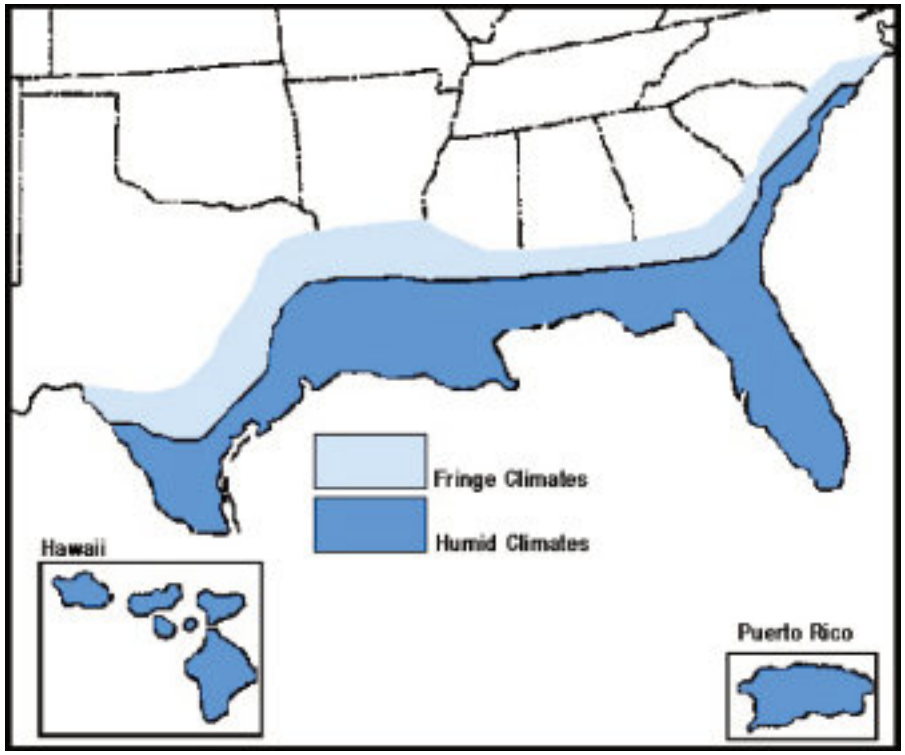



*Figure 5. Humid & fringe climate map*

*Table 2. Humid and fringe climate zones*

| Humid and Fringe Climate Zone | |
|---|---|
| **Alabama** | Counties of Baldwin, Barbour, Bullock, Butler, Choctaw, Clarke, Coffee, Conecuh, Covington, Cerenshaw, Dale, Escambia, Geneva, Henry, Houston, Lowndes, Marengo, Mobile, Monroe, Montgomery, Pike, Washington, and Wilcox |
| **Florida** | All counties and locations |
| **Georgia** | Counties of Appling, Atkinson, Bacon, Baker, Ben Hill, Berrien, Brantley, Brooks, Bryan, Calhoun, Camden, Charleton, Chatham, Clay, Clinch, Coffee, Colquitt, Cook, Crisp, Decatur, Dougherty, Early, Echols, Effingham, Evans, Glynn, Grady, Irwin, Jeff Davis, Lanier, Lee, Liberty, Long, Lowndes, McIntosh, Miller, Mitchell, Pierce, Quitman, Randolph, Seminole, Tattnall, Terrell, Thomas, Tift, Turner, Ware, Wayne, and Worth |
| **Hawaii** | All counties and locations |
| **Louisiana** | All counties and locations |
| **Mississippi** | Counties of Adams, Amite, Claiborne, Clarke, Copiah, Covington, Forrest, Franklin, George, Greene, Hancock, Harrison, Hinds, Issaquena, Jackson, Jasper, Jefferson, Jefferson Davis, Jones, Lamar, Lawrence, Lincoln, Marion, Pearl River, Perry, Pike, Rankin, Simpson, Smith, Stone, Walthall, Warren, Wayne, and Wilkinson |
| **North Carolina** | Counties of Brunswick, Carteret, Columbus, New Hanover, Onslow, and Pender |
| **South Carolina** | Counties of Beaufort, Berkeley, Charleston, Colleton, Dorchester, Georgetown, Horry and Jasper |
| **Texas** | Counties of Anderson, Angelina, Aransas, Atascosa, Austin, Bastrop, Bee, Bexar, Brazoria, Brazos, Brooks, Burleson, Caldwell, Calhoun, Cameron, Camp, Cass, Chambers, Cherokee, Colorado, Comal, De Witt, Dimmit, Duval, Falls, Fayette, Fort Bend, Franklin, Freestone, Frio, Galveston, Goliad, Gonzales, Gregg, Grimes, Guadalupe, Hardin, Harris, Harrison, Hays, Henderson, Hidalgo, Hopkins, Houston, Jackson, Jasper, Jefferson, Jim Hogg, Jim Wells, Karnes, Kaufman, Kenedy, Kinney, Kleberg, La Salle, Lavaca, Lee, Leon, Liberty, Limestone, Live Oak, Madison, Marion, Matagorda, Maverick, McMullen, Medina, Milam, Montgomery, Morris, Nacogdoches, Navarro, Newton, Nueces, Orange, Panola, Polk, Rains, Refugio, Robertson, Rusk, Sabine, San Augustine, San Jacinto, San Patricio, Shelby, Smith, Starr, Titus, Travis, Trinity, Tyler, Upshur, Uvalde, Val Verde, Van Zandt, Victoria, Walker, Waller, Washington, Webb, Wharton, Willacy, Williamson, Wilson, Wood, Zapata, and Zavala |

## Step 4. Confirm Roof Load Zone

From **Table 3**, identify the Roof Load Zone for the home. Verify that the home conforms to the following rules.

- No home may be placed in an area with a higher roof load than that indicated on the data plate. (Example: a home designed for the South (20 psf) Roof Load Zone cannot be placed in the Middle (30 psf) Roof Load Zone).
- A home may be located in an area with a lower roof load than that indicated on the data plate. (Example: a home designed for the Middle (30 psf) Roof Load Zone may be placed in the South (20 psf) Roof Load Zone). When a home is located in an area with a lower roof load than indicated on the data plate it may be installed per the requirements of the lower roof load area.
- There are special high roof load areas (primarily in mountains) not shown on the map. Contact the LAHJ or SAA for information about these areas. The home's data plate will indicate if the home has been designed for one of these high roof load areas.
- Ramadas may be used in areas with roof live loads greater than 40 psf. Ramadas are to be self-supporting, except that any connection to the home must be for weatherproofing only.

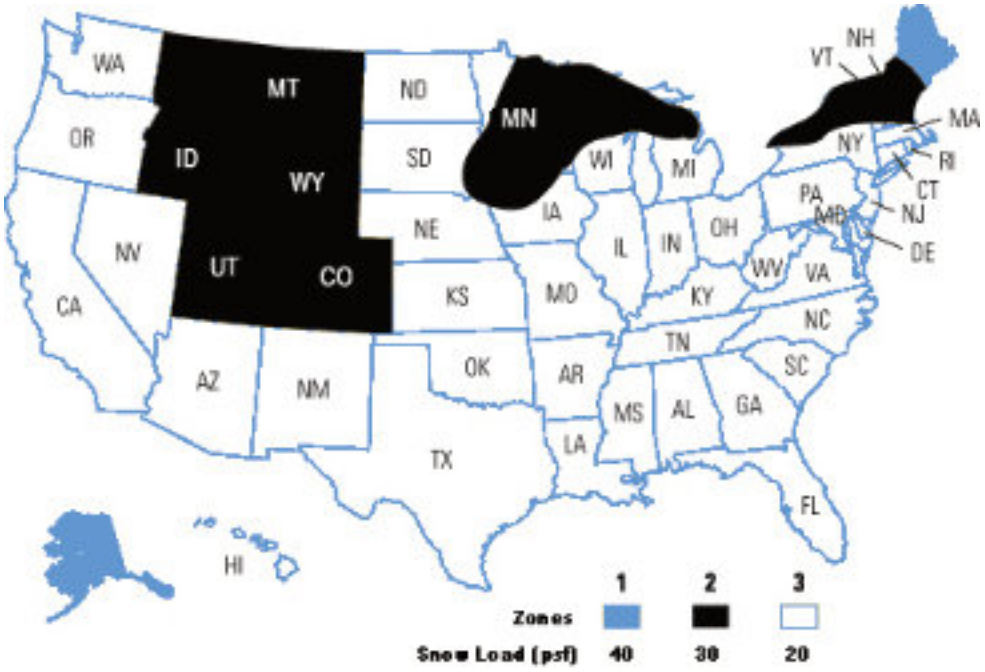



*Figure 6. Roof (snow) load map*

*Table 3. Roof loads by locality*

| | |
|---|---|
| **North (40 psf roof load)** | |
| **Alaska** | All counties |
| **Maine** | Counties of Aroostook, Piscataquis, Somerset, Penobscot, Waldo, Knox, Hancock, Washington |
| **Middle (30 psf roof load)** | |
| **Colorado** | All counties |
| **Idaho** | All counties |
| **Iowa** | Counties of: Buena Vista, Butler, Calhoun, Cerro Gordo, Cherokee, Chickasaw, Clay, Dickinson, Emmet, Floyd, Franklin, Hamilton, Hancock, Hardin, Howard, Humboldt, Ida, Kossuth, Lyon, Mitchell, O'Brien, Osceola, Palo Alto, Plymouth, Pocahontas, Sac, Sioux, Webster, Winnebago, Worth, Wright |
| **Maine** | Counties of Androscoggin, Cumberland, Franklin, Kanabec, Lincoln, Oxford, Sagadahoc, York |
| **Massachusetts** | County of Essex |
| **Michigan** | Counties of Alger, Alcona, Alpena, Antrim, Baraga, Benzie, Charlevoix, Cheboygan, Chippewa, Crawford, Delta, Dickson, Emmet, Gogebic, Grand Traverse, Houghton, Iron, Kalkaska, Keweenaw, Leelanau, Luce, Mackinac, Marquette, Menominee, Missaukee, Montmorency, Ogemaw, Ontonagon, Oscoda, Otsego, Presque Isle, Roscommon, Schoolcraft, Wexford |
| **Minnesota** | Counties of Aitkin, Anoka, Benton, Blue Earth, Brown, Cass, Carlton, Carver, Chippewa, Chisago, Cook, Cottonwood, Crow Wing, Dakota, Dodge, Douglas, Faribault, Fillmore, Freeborn, Goodhue, Grant, Hennepin, Hubbard, Itasca, Isanti, Jackson, Kandiyohi, Kanabec, Koochiching, Lac qui Parle, Lake, Le Sueur, Lincoln, Lyon, McLeod, Meeker, Morrison, Millie Lacs, Mower, Martin, Murray, Nicollet, Nobles, Olmsted, Pipestone, Pine, Pope, Ramsey, Redwood, Renville, Rice, Rock, St. Louis, Sibley, Scott, Steele, Sherburne, Swift, Stearns, Stevens, Todd, Wadena, Wright, Washington, Wabasha, Winona, Waseca, Watonwan, Yellow Medicine |
| **Montana** | All Counties |
| **New Hampshire** | All Counties |
| **New York** | Counties of Cayuga, Clinton, Essex, Erie, Franklin, Fulton, Genesee, Hamilton, Herkimer, Jefferson, Lewis, Livingston, Madison, Monroe, Montgomery, Niagara, Oneida, Onondaga Ontario, Orleans, Oswego, St. Lawrence, Saratoga, Schenectady, Seneca, Warren, Washington, Wayne, Wyoming, Yates |
| **South Dakota** | Counties of Brookings, Clay, Codington, Deuel, Grant, Hamlin, Hanson, Hutchinson, Kingsbury, Lake, Lincoln, McCook, Miner, Minnehaha, Moody, Turner, Union, Yankton |
| **Utah** | All Counties |
| **Vermont** | Counties of Addison, Caledonia, Chittenden, Essex, Franklin, Grand Isle, Lamoille, Orange, Orleans, Rutland, Washington, Windsor |
| **Wisconsin** | Counties of Ashland, Bayfield, Barron, Buffalo, Burnett, Clark, Chippewa, Door, Douglas, Dunn, Eau Claire, Florence, Forest, Iron, Jackson, Langlade, Lincoln, Marathon, Marinette, Menominee, Oconto, Oneida, Pepin, Pierce, Polk, Price, Rusk, St. Croix, Sawyer, Taylor, Trempealeau, Vilas, Washburn |
| **Wyoming** | All Counties |
| **South (20 psf roof load)** | |
| **Other** | The states and counties not listed for the Middle or North roof load zone above are deemed to be within the South roof load zone. |

Is the data plate present and the home placed in the appropriate wind, thermal and roof load zones?

► **YES**, go to **Step 5, Check local codes and secure permits,** (p.24).

► **NO,** Stop installation activities and notify the home retailer.

## Step 5. Check Local Codes and Secure Permits

Local regulations may set conditions for the siting and installation of a manufactured home. Consult the LAHJ, state manufactured housing association and the state SAA (See **Resources, p. 10**) for the specific local requirements, including:

- Building codes that may affect the construction of site built structures and infrastructure.
- Local requirements regulating the installation of manufactured homes.
- Setback requirements for property lines, streets, yards, and courts.
- Fire separation distances.
- Development covenants for the specific property.
- The locations of flood hazard areas and any special foundation requirements for homes installed in those areas.
- In some areas, building permits are required to install manufactured homes. Prior to making any alteration to the site and the home, contact the LAHJ to determine if plan approval and permits are required.
- The foundation specifications contained in this manual are NOT intended to address flood loads. If the home is located in the flood plain, consult a registered engineer.

► go to Prepare the Site (p. 25)

# PREPARE THE SITE

This chapter explains the process of planning the site, evaluating the soil, and preparing the site for construction of the home's support system.

**Follow the Steps below:**


▼ **Step 1. Plan Site Access** (p. 25)
▼ **Step 2. Determine Home Location And Layout** (p. 26)
▼ **Step 3. Clear And Grade The Site** (p. 26)
▼ **Step 4. Determine Soil Conditions** (p. 27)
▼ **Step 5. Determine Soil Bearing Capacity And Frost Line** (p. 29)
▼ **Step 6. Determine Ground Anchor Holding Capacity** (p. 32)


## Step 1. Plan Site Access

Planning the route to the site is typically the responsibility of the retailer or transportation company. Whoever is responsible must secure state permits from the states through which the home will pass.

In planning the route, avoid obstructions that might interfere with the passage of the home, such as low hanging wires and trees, low overpasses and bridges not suitable for the load. Contact the utility company if wires need to be moved. Do not allow branches, bushes or other foliage to scrape against the home as the home is moved to the site. Avoid ditches, berms, steep slopes and soft ground. Identify and fill any holes and soft spots into which the transporter's wheels may sink. Avoid moving over steep changes in grade (20 degrees or more).

If required, provide for home storage and staging areas on the site. Plan the delivery and staging of home sections and materials so that after all deliveries are complete, home sections and materials can be accessed for use and installed in the appropriate sequence. Orient home sections so they do not have to be rotated or excessively maneuvered during the installation process. Plan for temporary needs, such as dumpsters, portable toilets, crew parking, delivery vehicle drop-offs and concrete mixer deliveries.

Before moving the home to the site, inform the LAHJ and make sure the site is prepared and utilities are available. Final responsibility for site preparation, including soil stability and frost heave control, lies with the installer. An improperly prepared site may result in the denial of a foundation-related warranty claim.

## Step 2. Determine Home Location and Layout

The home location may have already been determined by others. If not, plan the home location and layout in compliance with the regulations researched in **Getting Started**, **STEP 5. CHECK LOCAL CODES AND SECURE PERMITS** (p. 24). Contact utilities for locations of existing infrastructure, such as underground cables, pipes and electrical lines. Comply with any LAHJ fire separation requirements or the requirements NFPA 501A, 2003 edition (Chapter 6). When planning the site improvements, consider the following:

- The home location should be level.
- Avoid contact with large trees, steep slopes, poorly drained areas and potential flood zones.
- Preserve trees and shrubs for shade, visual screens and windbreaks.
- Plan the driveway, parking areas, septic, well, other structures and utility lines.
- Consider future additions, such as screen rooms, porches and awnings.
- Select a home site away from areas of standing water and natural water paths.

## Step 3. Clear and Grade the Site

Trim overhanging foliage (tree limbs, etc. that could cause damage to the home) considering future growth, potential storms, swaying in wind and snow/ice-weighted branches. All organic material such as vegetation, wood, roots, twigs, dead branches, grass, brush and wood scraps must be removed in areas where footings are to be placed. Remove any debris that could become termite infested from the site and surrounding area. Properly dispose of all items.

Home site must be graded to drain surface water away from foundation walls and to prevent water build-up under the home. (**Figure 7**) The grade shall not be less than 5 inches within the first 10 feet around the perimeter of the home. Where proper slope away from home cannot be achieved, the site must be provided with drains, ditches, berms, swales or otherwise graded to drain water away from the structure, as shown in **Figure 8**. If home will have skirting, grading shall facilitate the drainage of surface water away from the home. Homes on concrete slabs or with perimeter foundation walls shall have drainage tiles, gravel, crushed stone drains, or other approved drainage systems. These must be installed at or below the lowest grade under the home or below the bottom of the slab and shall discharge by gravity or mechanical means to daylight.

Moisture under the home can result in structural damage to the floor system, foundation, anchoring system and other parts of the home. Failure to provide adequate slope/drainage can result in moisture-related problems such as mold, mildew and erosion. The home site must be prepared so that there will be no depressions in which surface water may accumulate beneath the home.

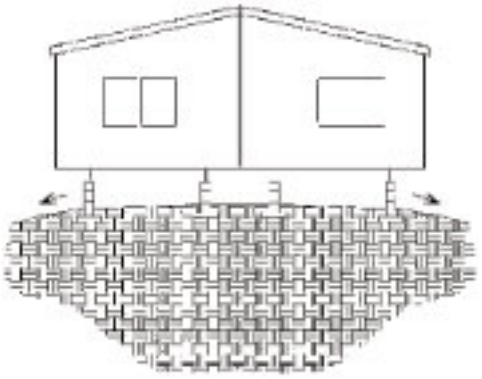


*Figure 7. Grade the soil away from the home*

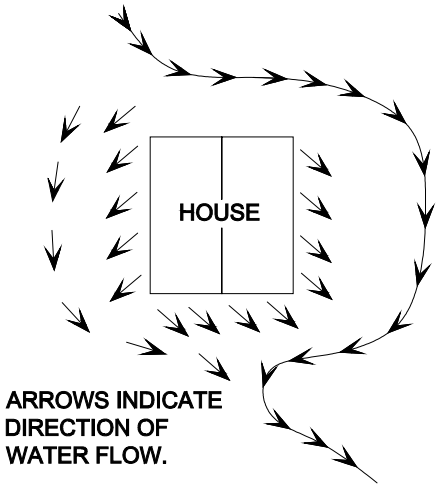



*Figure 8. Direct runoff away from the home*

Grade the ground so that water under porches, decks and recessed entries flows away from the home. If proper grading is not possible, use other methods such as a drain tile and automatic sump pump system to remove any water that may collect under the home.

The home is suitable for the installation of gutters and downspouts. Gutters and downspouts must be installed per the gutter and downspout manufacturer's installation instructions. When gutters and downspouts are installed, direct runoff away from the home.

## Step 4. Determine Soil Conditions

Examine the soil type under the proposed home location to make sure it is suitable for placement of a home. The design of the home's foundation, including footing/pier spacing and size, will in part be determined by the bearing capacity of the soil and, if ground anchors are used, by the soil's withdrawal strength. The soil under every portion of the foundation must meet the following criteria. Soil and fill under the home, but not directly supporting footings, are exempt from these criteria.

- The soil must be firm and undisturbed (not previously excavated) or fill compacted to at least 90% of its maximum relative density). Uncompacted fill will settle over time, causing the home to shift and become unlevel.
- Fill must not contain large debris.
- The soil must not be comprised of organic clays or peat. Organic material can decay, causing settlement and also may harbor pests that can infest the home.
- Consult a geologist or the LAHJ if you are unsure of the water table level.
- The soil must not be a highly expansive type.

Does the soil meet these criteria?

► **YES**, go to **Step 5, Determine Soil bearing capacity and frost line,** (p. 29).

► **NO**, Consult a registered engineer, registered architect, or geologist to determine a suitable soil bearing capacity.

## Step 5. Determine Soil Bearing Capacity and Frost Line

The soil under a home must be capable of withstanding the loads imposed by the weight of the home, its foundation and furnishings as well as any loads imposed by wind, snow or other climactic conditions.

### Soil Bearing Capacity

Determine the soil bearing capacity in pounds per square foot (psf) before designing a foundation. The higher the capacity (psf), the more weight the soil can hold without unduly compressing. As the soil bearing capacity increases, footings can be reduced in size or spaced farther apart.

Use one or more of the following methods to determine the site's soil bearing capacity:

- Test the soil. Hire a registered geologist, registered engineer, or registered architect to determine the soil classification and maximum allowable soil bearing capacity by testing the soil in accordance with generally accepted engineering practice. In addition, the registered professional may designate a footing capacity (lbs) based on a site-specific soils evaluation and footing design.
- Obtain soil records. The local office of the U.S. Department of Agriculture's Natural Resources Conservation Service (www.soils.usda.gov) and/or the LAHJ may have test results and/or soil analyses on file for the area.
- Conduct a pocket penetrometer test.
- Consult LAHJ for locally accepted soil bearing capacity.
- Determine soil bearing value by visual examination or by using torque probe test. If one of the options above is not available, the values in Table 4 can be used to establish soil bearing capacity by visual examination or use of torque probe. This method provides lower capacity values than the options above. Accurate soil identification typically requires special training or expertise. An engineer or building code official may be able to assist in classifying the soil found on the site.



Soil bearing capacity. Foundations on soils with bearing capacities less than 1,000 psf must be designed by a registered engineer or registered architect and approved by the LAHJ.



Table 4. Soil bearing capacity By soil type

| Soil Classification | | | | | |
|---|---|---|---|---|---|
| Classification Number | ASTM D 2487-00 or D 2488-00 (incorporated by reference, see 3285.4) | Soil Description | Allowable Soil Bearing Pressure (psf)[1] | Blow Count ASTM D 1586-99 | Torque Probe[3] Value[4] (inch-pounds) |
| 1 | - | Rock or hard pan | 4,000+ | | |
| 2 | GW, GP, SW, SP, GM, SM | Sandy gravel and gravel; very dense and/or cemented sands; course gravel/cobbles; preloaded silts, clays and coral | 2,000 | 40+ | More than 550 |
| 3 | GC, SC, ML, CL | Sand; silty sand; clayey sand; silty gravel; medium dense course sands; sandy gravel; and very stiff silt, sand clays | 1,500 | 24-39 | 351-550 |
| 4A | CG, MH[2] | Loose to medium dense sands; firm to stiff clays and silts; alluvial fills | 1,000 | 18-23 | 276 – 350 |
| 4B | CH, MH[2] | Loose sands; firm clays; alluvial fills | 1,000 | 12-17 | 175-275 |
| 5 | OL, OH, PT | Uncompacted fill; peat; organic clays | Refer to 3285.202(e) | 0-11 | Less than 175 |

1. The values provided in this table have not been adjusted for overburden pressure, embedment depth, water table height or settlement problems.
2. For soils classified as CH or MH, without either torque probe values or blow count test results, selected anchors must be rated for a 4B soil.
3. The torque test probe is a device for measuring the torque value of soils to assist in evaluating the holding capacity of the soil in which the ground anchor is placed. The shaft must be of suitable length for the full depth of the ground anchor.
4. The torque value is a measure of the load resistance provided by the soil when subject to the turning or twisting force of the probe.

- Use default capacity. Use an allowable pressure of 1,500 psf, unless site-specific information requires the use of lower values based on soil classification and type according to Table 4.

Note that soil types may vary across a home site. In this case, the soil with the lowest bearing capacity should be assumed when designing the support system. Keep a record of the soil bearing capacity value; it will be used later to design the home's support system.



## Frost Line

Obtain the local design frost depth for footings from one of the following methods:

- The local authority having jurisdiction (LAHJ).
- Consult with a registered architect, registered engineer, or registered geologist.
- Use Table 5 with the site's Air Freezing Index (AFI). Interpolation is permitted.
- Use Figure 9 map.




Table 5. Design frost depth for footings

| Air-Freezing Index | Minimum Depth (in) | Air-Freezing Index | Minimum Depth (in) |
|---|---|---|---|
| 50 or lower | 3 | 2000 | 40 |
| 250 | 9 | 2500 | 45 |
| 350 | 12 | 3000 | 52 |
| 500 | 16 | 3500 | 57 |
| 1000 | 24 | 4000 | 62 |
| 1500 | 32 | 4250 | 65 |

The depth of interior pier footings may be taken as one half the depth required when interior footings are located within an enclosed space meeting all the following:

1. All vents in skirting are capable of closing at outdoor temperatures below 40 degrees F.
2. Skirting is insulated to a minimum R-5 when unadjusted frost depth is up to 45 inches and a minimum of R-10 when unadjusted frost depth exceeds 45 inches.
3. An approved ground vapor barrier is applied.

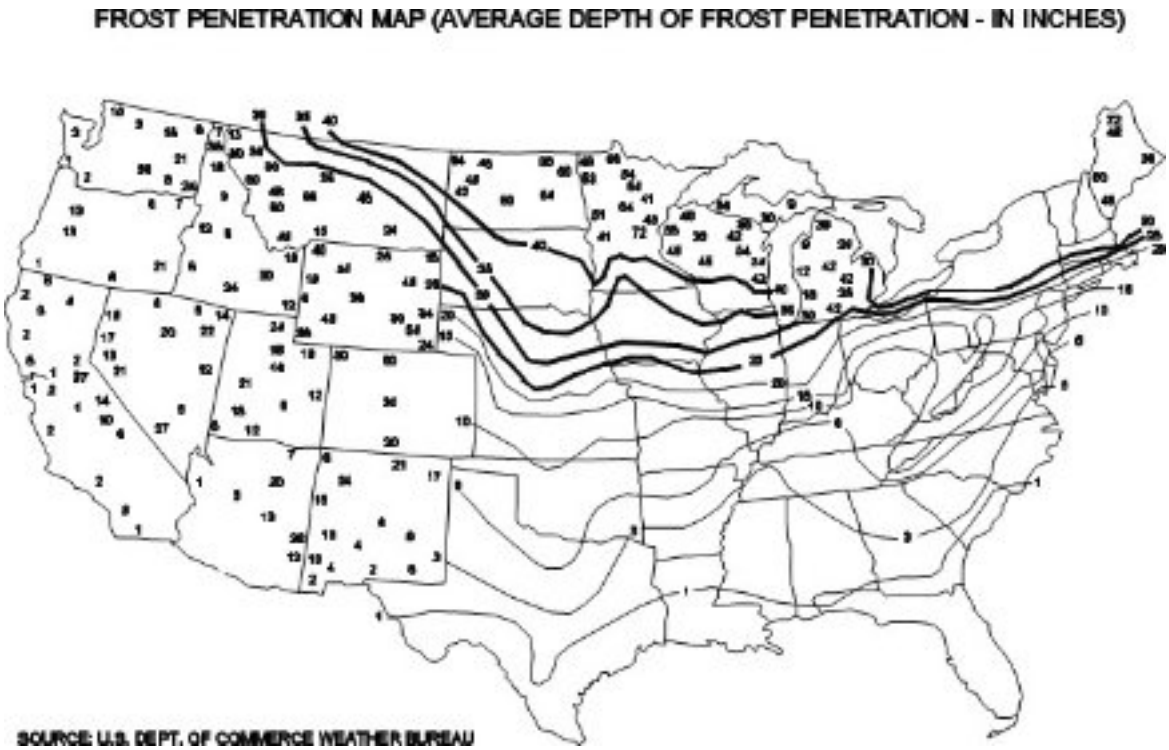



Figure 9. Frost Penetration Map

Will this installation use auger-type ground anchors or concrete anchors?

► **YES,** go to **STEP 6, Determine Ground Anchor Holding Capacity,** (p. 32).

► **NO,** go to **Construct Foundation,** (p. 50).

## Step 6. Determine Ground Anchor Holding Capacity

What type of support system will this installation use?

► **For pier and ground or concrete anchor**, go to **Install Stabilizing Systems,** (p. 93).

► **For load bearing perimeter wall**, go to **Construct Foundation,** (p. 50).

If the anchor holding strength of the soil is unknown test the soil per the requirements of the anchor manufacturer's installation instructions. Concrete anchors that are listed and certified by a professional engineer to have a working load capacity of 3150 lbs may be used as a substitute for the ground anchors specified within this manual when installed in accordance with manufacturer's installation instructions.

# INSTALL FOOTINGS

This chapter provides instructions for the design and construction of individual footings that transfer the load from a single pier to the ground. A footing and pier together (discussed in **Set the Home,** p. 54) is referred to as a "support". A footing may also be designed to carry the load of multiple piers (often called "strip" footings). This manual may be used to design unreinforced strip and slab footings. The foundation systems described in this manual have not been designed for flood resistance.

**Follow the Steps below:**


▼ **Step 1. Design Point Load Supports** (p. 33)
▼ **Step 2. Design Frame Supports** (Homes Without Perimeter Blocking) (p. 37)
▼ **Step 3. Design Frame And Perimeter Supports** (Homes With Perimeter Blocking) (p. 41)
▼ **Step 4. Select Footing Materials** (p. 45)
▼ **Step 5. Size Footings** (p. 45)
▼ **Step 6. Install Footings** (p. 48)


## Step 1. Design Point Load Supports

All homes will need supports, and therefore footings, under the frame, marriage line (for multi-section homes), exterior wall openings and other heavy point loads.

All pier locations required at the mating line, perimeter and any special pier support locations, as required by these instructions, will be identified from the factory by a pier tag, label, paint, or other means and must be visible after the home is installed. The pier designs, support loads, and footing construction shall be as indicated in the appropriate diagrams, tables, and instructions herein. Where perimeter piers are required along the exterior wall, alternate pier spacing may be used in lieu of the factory identified locations provided the instructions of this manual are satisfied in terms of allowable spacing, pier design, and footing size.

**PLEASE NOTE:** The manufacturer will not be responsible for damaged or removed pier tags. It is the responsibility of the installer to ensure that all piers are properly positioned in accordance with the tables & diagrams contained in these installation instructions.

Create a sketch of the home that includes the exterior walls, the frame I-beams and the marriage line(s), if a multi-section home. The sketch will be used in this chapter to locate each support and note the size of the corresponding footing. **Figure 10** is an example of such a completed support plan.

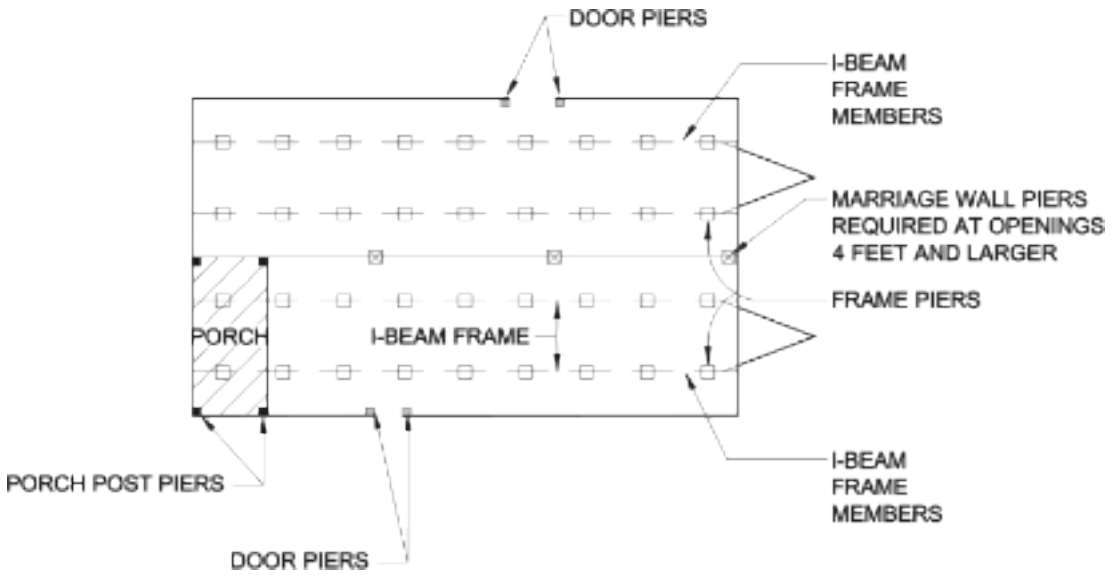



*Figure 10. Typical point-load support locations*

As the location and load for each support is determined, note it on the sketch. When selecting locations for supports, keep in mind that increasing the spacing between supports will increase the load on that support and the size of the required footing.

**Determine Locations**

Point loads exist where a bearing/structural weight is concentrated and transferred to the foundation at a specific point. Locate a support under each point load, including the following examples:

- Exterior doors in side walls at both sides of each door (blocking is not required at exterior doors in non-load bearing end walls).
- Additional factory installed floor joists may replace piers below openings larger than 48” if noted on other supplemental documents.
- Other exterior wall openings four feet and greater at both sides of each opening (including multiple windows that total four feet wide or more without intermediate supports, even if individual windows are less than four feet).
- Marriage line openings four feet or greater at both sides of each opening.
- Locations where through-the-rim crossover ducts penetrate the rim joist at the marriage line (unless otherwise noted in supplemental documents provided with the home or unless the home is constructed with a perimeter frame system).
- Marriage line columns.
- Load-bearing porch posts.
- On each side of factory installed fireplaces when located on the exterior sidewalls or marriage walls (except when supported by the steel header).

Adjustable outriggers may only be used to replace piers below exterior door or window openings less than 48” and below factory installed fireplaces located on the exterior sidewalls or marriage walls. Adjustable outriggers may only be used at these locations when they are installed according to the manufacturer’s installation instructions. Adjustable outriggers are not an acceptable replacement for perimeter supports on homes requiring perimeter blocking. Factory installed outriggers and crossmembers may also replace piers below exterior door or window openings less than 48”.

Mark the required point load support locations on the sketch. Supports are not required where the manufacturer has reinforced the floor (such as with additional outriggers or floor joists) and noted so in the documentation provided with the home.

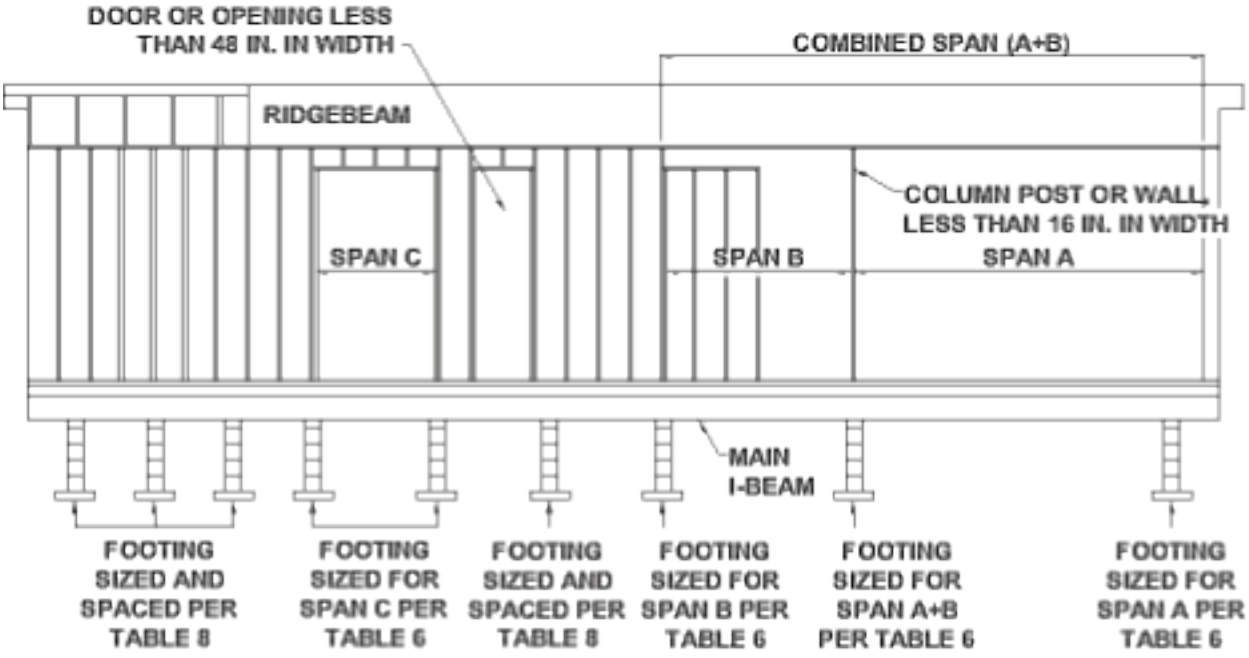



*Figure 11. Typical point-load support locations along the marriage line.*

### Calculate Loads

Use **Table 6** to determine the loads on point load supports (columns). For each support, find the table with the appropriate section width. Then, find the row with the appropriate roof load zone and the column corresponding to the span (see **Figure 11** for guidance on determining spans — if a support is shared by spans on both sides, add the respective loads together to arrive at the total load under that point). The number in the intersecting cell is the load. Interpolation is permitted.

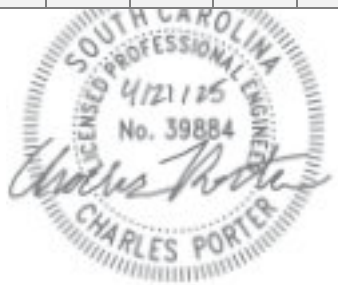
Note the required loads next to each point load support on the sketch.

Table 6. Point load on footings at marriage line openings (lbs)

| 20 ft Max. Home Width | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Roof Live Load (PSF) | Maximum Opening in Marriage Line (ft) | | | | | | | | | | | | | |
| | 4 | 8 | 12 | 14 | 16 | 18 | 20 | 24 | 28 | 32 | 36 | 40 | 44 | 48 |
| 20 | 1000 | 1600 | 2200 | 2500 | 2800 | 3100 | 3400 | 4000 | 4600 | 5200 | 5800 | 6400 | 7000 | 7600 |
| 30 | 2000 | 2800 | 3600 | 4000 | 4400 | 4800 | 5200 | 6000 | 6800 | 7600 | 8400 | 9200 | 10000 | 10800 |
| 40 | 2400 | 3400 | 4400 | 4900 | 5400 | 5900 | 6400 | 7400 | 8400 | 9400 | 10400 | 11400 | 12400 | 13400 |
| 60 | 3200 | 4600 | 6000 | 6700 | 7400 | 8100 | 8800 | 10200 | 11600 | 13000 | 14400 | 15800 | 17200 | 18600 |
| 80 | 4000 | 5800 | 7600 | 8500 | 9400 | 10300 | 11200 | 13000 | 14800 | 16600 | 18400 | 20200 | 22000 | 23800 |
| 100 | 4800 | 7000 | 9200 | 10300 | 11400 | 12500 | 13600 | 15800 | 18000 | 20200 | 22400 | 24600 | 26800 | 29000 |
| 120 | 5600 | 8200 | 10800 | 12100 | 13400 | 14700 | 16000 | 18600 | 21200 | 23800 | 26400 | 29000 | 31600 | 34200 |

| 24 ft Double Section /36 ft Triple Section Max. Home Width | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Roof Live Load (PSF) | Maximum Opening in Marriage Line (ft) | | | | | | | | | | | | | |
| | 4 | 8 | 12 | 14 | 16 | 18 | 20 | 24 | 28 | 32 | 36 | 40 | 44 | 48 |
| 20 | 1120 | 1840 | 2560 | 2920 | 3280 | 3640 | 4000 | 4720 | 5440 | 6160 | 6880 | 7600 | 8320 | 9040 |
| 30 | 2320 | 3280 | 4240 | 4720 | 5200 | 5680 | 6160 | 7120 | 8080 | 9040 | 10000 | 10960 | 11920 | 12880 |
| 40 | 2800 | 4000 | 5200 | 5800 | 6400 | 7000 | 7600 | 8800 | 10000 | 11200 | 12400 | 13600 | 14800 | 16000 |
| 60 | 3760 | 5440 | 7120 | 7960 | 8800 | 9640 | 10480 | 12160 | 13840 | 15520 | 17200 | 18880 | 20560 | 22240 |
| 80 | 4720 | 6880 | 9040 | 10120 | 11200 | 12280 | 13360 | 15520 | 17680 | 19840 | 22000 | 24160 | 26320 | 28480 |
| 100 | 5680 | 8320 | 10960 | 12280 | 13600 | 14920 | 16240 | 18880 | 21520 | 24160 | 26800 | 29440 | 32080 | 34720 |
| 120 | 6640 | 9760 | 12880 | 14440 | 16000 | 17560 | 19120 | 22240 | 25360 | 28480 | 31600 | 34720 | 37840 | 40960 |

| 28 ft Double Section /42 ft Triple Section Max. Home Width | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Roof Live Load (PSF) | Maximum Opening in Marriage Line (ft) | | | | | | | | | | | | | |
| | 4 | 8 | 12 | 14 | 16 | 18 | 20 | 24 | 28 | 32 | 36 | 40 | 44 | 48 |
| 20 | 1240 | 2080 | 2920 | 3340 | 3760 | 4180 | 4600 | 5440 | 6280 | 7120 | 7960 | 8800 | 9640 | 10480 |
| 30 | 2640 | 3760 | 4880 | 5440 | 6000 | 6560 | 7120 | 8240 | 9360 | 10480 | 11600 | 12720 | 13840 | 14960 |
| 40 | 3200 | 4600 | 6000 | 6700 | 7400 | 8100 | 8800 | 10200 | 11600 | 13000 | 14400 | 15800 | 17200 | 18600 |
| 60 | 4320 | 6280 | 8240 | 9220 | 10200 | 11180 | 12160 | 14120 | 16080 | 18040 | 20000 | 21960 | 23920 | 25880 |
| 80 | 5440 | 7960 | 10480 | 11740 | 13000 | 14260 | 15520 | 18040 | 20560 | 23080 | 25600 | 28120 | 30640 | 33160 |
| 100 | 6560 | 9640 | 12720 | 14260 | 15800 | 17340 | 18880 | 21960 | 25040 | 28120 | 31200 | 34280 | 37360 | 40440 |
| 120 | 7680 | 11320 | 14960 | 16780 | 18600 | 20420 | 22240 | 25880 | 29520 | 33160 | 36800 | 40440 | 44080 | 47720 |



| 32 ft Double Section /48 ft Triple Section Max. Home Width | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Roof Live Load (PSF) | Maximum Opening in Marriage Line (ft) | | | | | | | | | | | | | |
| | 4 | 8 | 12 | 14 | 16 | 18 | 20 | 24 | 28 | 32 | 36 | 40 | 44 | 48 |
| 20 | 1330 | 2260 | 3190 | 3655 | 4120 | 4585 | 5050 | 5980 | 6910 | 7840 | 8770 | 9700 | 10630 | 11560 |
| 30 | 2880 | 4120 | 5360 | 5980 | 6600 | 7220 | 7840 | 9080 | 10320 | 11560 | 12800 | 14040 | 15280 | 16520 |
| 40 | 3500 | 5050 | 6600 | 7375 | 8150 | 8925 | 9700 | 11250 | 12800 | 14350 | 15900 | 17450 | 19000 | 20550 |
| 60 | 4740 | 6910 | 9080 | 10165 | 11250 | 12335 | 13420 | 15590 | 17760 | 19930 | 22100 | 24270 | 26440 | 28610 |
| 80 | 5980 | 8770 | 11560 | 12955 | 14350 | 15745 | 17140 | 19930 | 22720 | 25510 | 28300 | 31090 | 33880 | 36670 |
| 100 | 7220 | 10630 | 14040 | 15745 | 17450 | 19155 | 20860 | 24270 | 27680 | 31090 | 34500 | 37910 | 41320 | 44730 |
| 120 | 8460 | 12490 | 16520 | 18535 | 20550 | 22565 | 24580 | 28610 | 32640 | 36670 | 40700 | 44730 | 48760 | 52790 |

Determine from the data plate and/or labels along the perimeter if the home requires perimeter blocking.

► If perimeter blocking is NOT required, go to Step 2, design frame supports (Homes Without Perimeter Blocking), (p. 37).

► If perimeter blocking is required, go to Step 3, design frame and perimeter supports (Homes With Perimeter Blocking), (p.41).

## Step 2. Design Frame Supports (Homes Without Perimeter Blocking)

### Determine Locations

All homes require regularly spaced supports along all main frame I-beams. Select spacing between supports and sketch them on the support plan. Keep in mind that frame supports under homes with 8” deep I-beams may be no more than eight feet apart. Those under homes with 10” or 12” deep I-beams may be no more than 10 feet apart. Generally, greater distances between supports will require larger footings. Figure 12 also shows typical frame support locations.



Spacing frame supports. There must be a support pier located near the end of each I-beam such that there is no more than 24 inches as measured from the outside edge of the floor to the center of the pier.



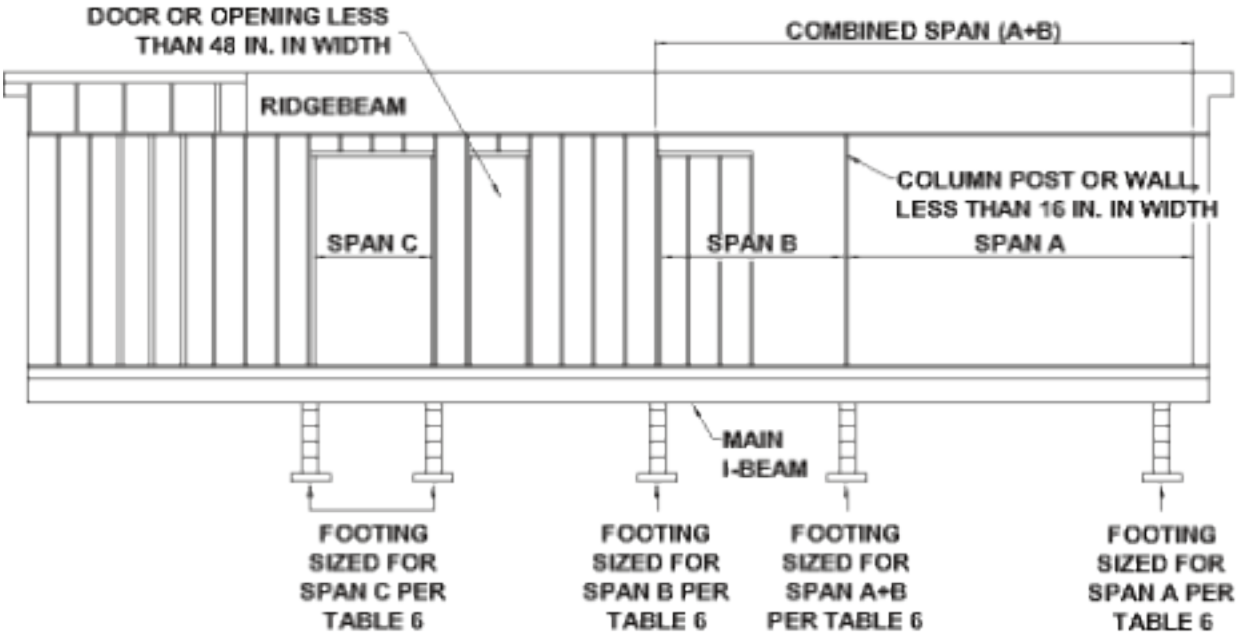



*Figure 12. Typical marriage line support locations for homes not requiring perimeter supports*

### Calculate Loads

Use **Table 7** to determine the loads on frame supports. Find the chart with the appropriate sidewall eave overhang. Then, find the column with the appropriate roof load zone and section width. Find the row corresponding to the selected support spacing. The number in the intersecting cell is the load.

Loads on all frame supports can be assumed to be equal if support spacing is equal. However, if different support spacings are used then each support with a different spacing should be calculated separately.

Interpolation between values is permitted on **Table 6**, **Table 7**, and **Table 8**.

Note the location and load required of each support on the sketch.

(Width listed in each column of the following charts is the overall width of the home. Eg. 16 ft is a single section, 32 ft is a double section and 48 ft is a triple section).

Table 7. Load on frame pier footings for homes not requiring perimeter blocking except at openings (lbs)

| Roof Load Zone and Max. Home Width (1-1/2" Max. Sidewall Eave Overhang) | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Pier Spacing | South (20 psf) | | | | | Middle (30 psf) | | | | | North (40 psf) | | | | |
| | 10 ft | 12 ft | 14 ft | 16 ft | 18 ft | 10 ft | 12 ft | 14 ft | 16 ft | 18 ft | 10 ft | 12 ft | 14 ft | 16 ft | 18 ft |
| Up to 4' | 2195 | 2519 | 2843 | 3140 | 3410 | 2400 | 2764 | 3128 | 3462 | 3765 | 2605 | 3009 | 3413 | 3783 | 4120 |
| > 4' to 6' | 3093 | 3579 | 4065 | 4510 | 4915 | 3400 | 3946 | 4492 | 4993 | 5448 | 3708 | 4314 | 4920 | 5475 | 5980 |
| > 6' to 8' | 3990 | 4638 | 5286 | 5666 | 6420 | 4400 | 5128 | 5856 | 6523 | 7130 | 4810 | 5618 | 6426 | 7167 | 7840 |
| > 8' to 10' | 4888 | 5698 | 6508 | 7250 | 7925 | 5400 | 6310 | 7220 | 8054 | 8813 | 5913 | 6923 | 7933 | 8858 | 9700 |

| Roof Load Zone and Max. Home Width (6" Max. Sidewall Eave Overhang) | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Pier Spacing | South (20 psf) | | | | | Middle (30 psf) | | | | | North (40 psf) | | | | |
| | 10 ft | 12 ft | 14 ft | 16 ft | 18 ft | 10 ft | 12 ft | 14 ft | 16 ft | 18 ft | 10 ft | 12 ft | 14 ft | 16 ft | 18 ft |
| Up to 4' | 2240 | 2564 | 2888 | 3185 | 3455 | 2460 | 2824 | 3188 | 3522 | 3825 | 2680 | 3084 | 3488 | 3858 | 4195 |
| > 4' to 6' | 3160 | 3646 | 4132 | 4578 | 4983 | 3490 | 4036 | 4582 | 5083 | 5538 | 3820 | 4426 | 5032 | 5588 | 6093 |
| > 6' to 8' | 4080 | 4728 | 5376 | 5750 | 6510 | 4520 | 5248 | 5976 | 6643 | 7250 | 4960 | 5768 | 6576 | 7317 | 7990 |
| > 8' to 10' | 5000 | 5810 | 6620 | 7363 | 8038 | 5550 | 6460 | 7370 | 8204 | 8963 | 6100 | 7110 | 8120 | 9046 | 9888 |

| Roof Load Zone and Max. Home Width (12" Max. Sidewall Eave Overhang) | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Pier Spacing | South (20 psf) | | | | Middle (30 psf) | | | | North (40 psf) | | | |
| | 10/20 ft | 12/24 ft | 14/28/ 42 ft | 16/32/ 48 ft | 10/20 ft | 12/24 ft | 14/28/ 42 ft | 16/32/ 48 ft | 10/20 ft | 12/24 ft | 14/28/ 42 ft | 16/32/ 48 ft |
| Up to 4' | 2300 | 2624 | 2948 | 3191 | 2540 | 2904 | 3268 | 3541 | 2780 | 3184 | 3588 | 3891 |
| > 4' to 6' | 3250 | 3736 | 4222 | 4587 | 3610 | 4156 | 4702 | 5112 | 3970 | 4576 | 5182 | 5637 |
| > 6' to 8' | 4200 | 4848 | 5496 | 5757 | 4680 | 5408 | 6136 | 6682 | 5160 | 5968 | 6776 | 7382 |
| > 8' to 10' | 5150 | 5960 | 6770 | 7378 | 5750 | 6660 | 7570 | 8253 | 6350 | 7360 | 8370 | 9128 |

| Roof Load Zone and Max. Home Width (24" Max. Sidewall Eave Overhang) | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Pier Spacing | South (20 psf) | | | | Middle (30 psf) | | | | North (40 psf) | | | |
| | 20 ft | 24 ft | 28/42 ft | 32/48 ft | 20 ft | 24 ft | 28/42 ft | 32/48 ft | 20 ft | 24 ft | 28/42 ft | 32/48 ft |
| Up to 4' | 2420 | 2744 | 3068 | 3311 | 2700 | 3064 | 3428 | 3701 | 2980 | 3384 | 3788 | 4091 |
| > 4' to 6' | 3430 | 3916 | 4402 | 4767 | 3850 | 4396 | 4942 | 5352 | 4270 | 4876 | 5482 | 5937 |
| > 6' to 8' | 4440 | 5088 | 5736 | 6222 | 5000 | 5728 | 6456 | 7002 | 5560 | 6368 | 7176 | 7782 |
| > 8' to 10' | 5450 | 6260 | 7070 | 7678 | 6150 | 7060 | 7970 | 8653 | 6850 | 7860 | 8870 | 9628 |



## Calculate Loads

Use Table 8 to determine the loads on supports below openings in the sidewall when perimeter blocking is not required. Find the row with the appropriate opening span. Then, find the column with the appropriate floor width. The number in the intersecting cell is the load.

Table 8. Load on pier footings at openings along the sidewall (lbs)

| Roof Load Zone and Max. Home Width (24" Max. Sidewall Eave Overhang) | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Pier Spacing | South (20 psf) | | | | | Pier Spacing | South (20 psf) | | | | |
| | 10/20 ft | 12/24 ft | 14/28/42 ft | 16/32/48 ft | 18 ft | | 10/20 ft | 12/24 ft | 14/28/42 ft | 16/32/48 ft | 18 ft |
| Up to 3' | 1025 | 1100 | 1175 | 1244 | 1306 | >5' to 6' | 1400 | 1520 | 1640 | 1750 | 1850 |
| > 3' to 4' | 1150 | 1240 | 1330 | 1413 | 1488 | >6' to 8' | 1650 | 1800 | 1950 | 2088 | 2213 |
| > 4' to 5' | 1275 | 1380 | 1485 | 1581 | 1669 | >8' to 10' | 1900 | 2080 | 2260 | 2425 | 2575 |

| Pier Spacing | Middle (30 psf) | | | | | Pier Spacing | Middle (30 psf) | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | 10/20 ft | 12/24 ft | 14/28/42 ft | 16/32/48 ft | 18 ft | | 10/20 ft | 12/24 ft | 14/28/42 ft | 16/32/48 ft | 18 ft |
| Up to 3' | 1200 | 1300 | 1400 | 1492 | 1575 | >5' to 6' | 1680 | 1840 | 2000 | 2147 | 2280 |
| > 3' to 4' | 1360 | 1480 | 1600 | 1710 | 1810 | >6' to 8' | 2000 | 2200 | 2400 | 2583 | 2750 |
| > 4' to 5' | 1520 | 1660 | 1800 | 1928 | 2045 | >8' to 10' | 2320 | 2560 | 2800 | 3020 | 3220 |

| Pier Spacing | North (40 psf) | | | | | Pier Spacing | North (40 psf) | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | 10/20 ft | 12/24 ft | 14/28/42 ft | 16/32/48 ft | 18 ft | | 10/20 ft | 12/24 ft | 14/28/42 ft | 16/32/48 ft | 18 ft |
| Up to 3' | 1375 | 1500 | 1625 | 1740 | 1844 | >5' to 6' | 1960 | 2160 | 2360 | 2543 | 2710 |
| > 3' to 4' | 1570 | 1720 | 1870 | 2008 | 2133 | >6' to 8' | 2350 | 2600 | 2850 | 3079 | 3288 |
| > 4' to 5' | 1765 | 1940 | 2115 | 2275 | 2421 | >8' to 10' | 2740 | 3040 | 3340 | 3615 | 3865 |

► go to Step 4. select footing materials (p. 45)



## Step 3. Design Frame and Perimeter Supports (Homes With Perimeter Blocking)

### Determine Locations

Depending on design and location, some homes require regularly spaced perimeter supports along all of the sidewalls and marriage walls in addition to frame supports. If required, perimeter support locations will be identified by labels attached along the bottom of the sidewalls and marriagewalls and noted on the Data Plate. A pier support will be required at each of the label locations. Additionally, perimeter support locations may be identified by a white stripe or mark beneath the home on the bottom board material. Perimeter blocking supports must be placed no further than 8 feet on-center.

If required, perimeter supports are only needed on bearing walls. For 20 psf roof live load, perimeter support is only required at exterior doors and other openings 48” and larger, unless noted otherwise. Supports may be added at each corner of each endwall for leveling purposes, but are not required. Bearing walls are those walls that support the ends of roof trusses or rafters (typically sidewalls and marriage walls but not end walls of main units or sidewalls of tag units).

To minimize the number of required perimeter supports, space them evenly between point load supports as shown in **Figure 13** (but not under open spans). These figures identify typical support locations for homes requiring perimeter supports.

**Spacing frame supports.** There must be a support pier located near the end of each I-beam such that there is no more than 24 inches as measured from the outside edge of the floor to the center of the pier.

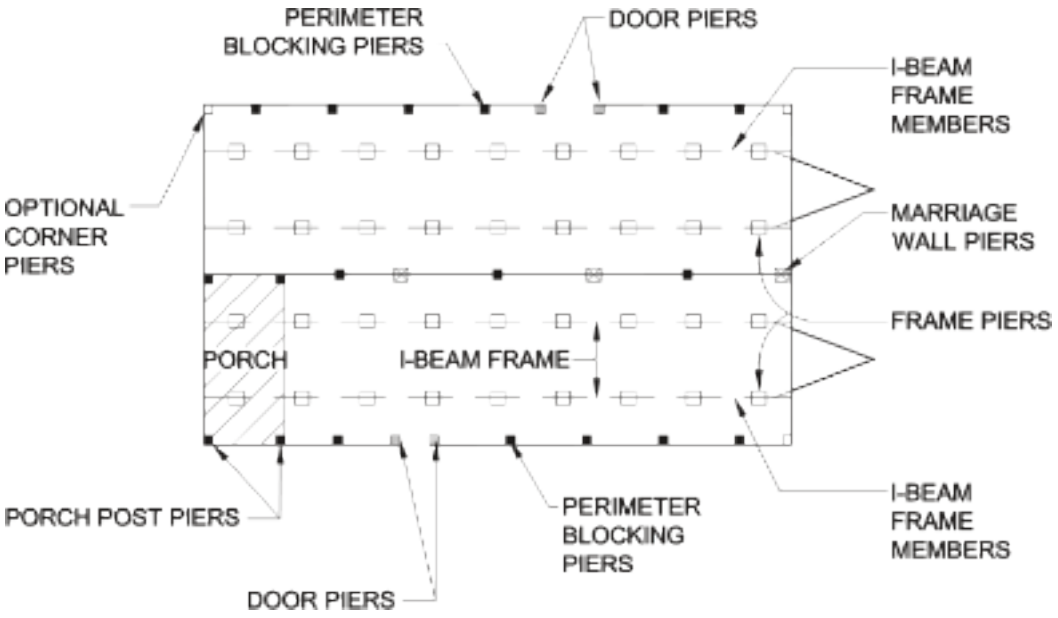



*Figure 13. Typical support locations for homes requiring perimeter supports*

**Calculate Loads**

Use **Table 9** to determine the loads on frame and perimeter supports for homes requiring perimeter blocking. Find the chart with the appropriate sidewall eave overhang. Then, find the column with the appropriate roof load and section width. Find the group of rows corresponding to the selected support spacing. The values in the intersecting cells are the loads for the frame, perimeter and marriage line supports respectively.

Loads on supports of a given type (frame, perimeter or marriage) can be assumed to be equal if support spacing is equal. However, if different support spacings are used then each support with a different spacing should be calculated separately.

Loads for piers installed at 5, 7 and 9 feet on-center can be estimated by adding the loads for the higher and lower spacing and dividing in half (eg. the load for frame piers installed at 7 feet on-center for a 16 wide with a 1-1/2” overhang in a 20 psf Roof Load Zone would be: 2823 lbs + 3630 lbs = 6453 lbs / 2 = 3227 lbs).

Note the location and load required of each support on the sketch.

(Width listed in each column of the following charts is the overall width of the home. Eg. 16 ft is a single section, 32 ft is a double section and 48 ft is a triple section).

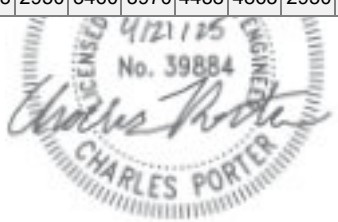


A perimeter support must be installed within 4 feet of column supports when the home is designated for perimeter blocking. The loads listed in Table 6 for homes greater than 20 psf roof load include the additional 4 foot span.

Table 9. Load on frame and perimeter pier footings for homes requiring perimeter blocking (lbs)

| Roof Load Zone and Max. Home Width (1-1/2" Max. Sidewall Eave Overhang) | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Pier Spacing | | South (20 psf) | | | | | Middle (30 psf) | | | | | North (40 psf) | | | | | North (60 psf) | | | | |
| | Location | 10 ft | 12 ft | 14 ft | 16 ft | 18 ft | 10 ft | 12 ft | 14 ft | 16 ft | 18 ft | 10 ft | 12 ft | 14 ft | 16 ft | 18 ft | 10 ft | 12 ft | 14 ft | 16 ft | 18 ft |
| Up to 4' | Frame | 1420 | 1624 | 1828 | 2015 | 2185 | 1420 | 1624 | 1828 | 2015 | 2185 | 1420 | 1624 | 1828 | 2015 | 2185 | 1420 | 1624 | 1828 | 2015 | 2185 |
| Up to 4' | Sidewall | 1175 | 1295 | 1415 | 1525 | 1625 | 1380 | 1540 | 1700 | 1847 | 1980 | 1585 | 1785 | 1985 | 2168 | 2335 | 1995 | 2275 | 2555 | 2812 | 3045 |
| > 4' to 6' | Frame | 1930 | 2236 | 2542 | 2823 | 3078 | 1930 | 2236 | 2542 | 2823 | 3078 | 1930 | 2236 | 2542 | 2823 | 3078 | 1930 | 2236 | 2542 | 2823 | 3078 |
| > 4' to 6' | Sidewall | 1563 | 1743 | 1923 | 2088 | 2238 | 1870 | 2110 | 2350 | 2570 | 2770 | 2178 | 2478 | 2778 | 3053 | 3303 | 2793 | 3213 | 3633 | 4018 | 4368 |
| > 6' to 8' | Frame | 2440 | 2848 | 3256 | 3630 | 3970 | 2440 | 2848 | 3256 | 3630 | 3970 | 2440 | 2848 | 3256 | 3630 | 3970 | 2440 | 2848 | 3256 | 3630 | 3970 |
| > 6' to 8' | Sidewall | 1950 | 2190 | 2430 | 2650 | 2850 | 2360 | 2680 | 3000 | 3293 | 3560 | 2770 | 3170 | 3570 | 3937 | 4270 | 3590 | 4150 | 4710 | 5223 | 5690 |
| > 8' to 10' | Frame | 2950 | 3460 | 3970 | 4438 | 4863 | 2950 | 3460 | 3970 | 4438 | 4863 | 2950 | 3460 | 3970 | 4438 | 4863 | 2950 | 3460 | 3970 | 4438 | 4863 |
| Pier Spacing | | North (80 psf) | | | | | North (100 psf) | | | | | North (120 psf) | | | | | | | | | |
| | Location | 10 ft | 12 ft | 14 ft | 16 ft | 18 ft | 10 ft | 12 ft | 14 ft | 16 ft | 18 ft | 10 ft | 12 ft | 14 ft | 16 ft | 18 ft | | | | | |
| Up to 4' | Frame | 1420 | 1624 | 1828 | 2015 | 2185 | 1420 | 1624 | 1828 | 2015 | 2185 | 1420 | 1624 | 1828 | 2015 | 2185 | | | | | |
| Up to 4' | Sidewall | 2405 | 2765 | 3125 | 3455 | 3755 | 2815 | 3255 | 3695 | 4098 | 4465 | 3225 | 3745 | 4265 | 4742 | 5175 | | | | | |
| > 4' to 6' | Frame | 1930 | 2236 | 2542 | 2823 | 3078 | 1930 | 2236 | 2542 | 2823 | 3078 | 1930 | 2236 | 2542 | 2823 | 3078 | | | | | |
| > 4' to 6' | Sidewall | 3408 | 3948 | 4488 | 4983 | 5433 | 4023 | 4683 | 5343 | 5948 | 6498 | 4638 | 5418 | 6198 | 6913 | 7563 | | | | | |
| > 6' to 8' | Frame | 2440 | 2848 | 3256 | 3630 | 3970 | 2440 | 2848 | 3256 | 3630 | 3970 | 2440 | 2848 | 3256 | 3630 | 3970 | | | | | |
| > 6' to 8' | Sidewall | 4410 | 5130 | 5850 | 6510 | 7110 | 5230 | 6110 | 6990 | 7797 | 8530 | 6050 | 7090 | 8130 | 9083 | 9950 | | | | | |
| > 8' to 10' | Frame | 2950 | 3460 | 3970 | 4438 | 4863 | 2950 | 3460 | 3970 | 4438 | 4863 | 2950 | 3460 | 3970 | 4438 | 4863 | | | | | |

| Roof Load Zone and Max. Home Width (6" Max. Sidewall Eave Overhang) | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Pier Spacing | | South (20 psf) | | | | | Middle (30 psf) | | | | | North (40 psf) | | | | | North (60 psf) | | | | |
| | Location | 10 ft | 12 ft | 14 ft | 16 ft | 18 ft | 10 ft | 12 ft | 14 ft | 16 ft | 18 ft | 10 ft | 12 ft | 14 ft | 16 ft | 18 ft | 10 ft | 12 ft | 14 ft | 16 ft | 18 ft |
| Up to 4' | Frame | 1420 | 1624 | 1828 | 2015 | 2185 | 1420 | 1624 | 1828 | 2015 | 2185 | 1420 | 1624 | 1828 | 2015 | 2185 | 1420 | 1624 | 1828 | 2015 | 2185 |
| Up to 4' | Sidewall | 1220 | 1340 | 1460 | 1570 | 1670 | 1440 | 1600 | 1760 | 1907 | 2040 | 1660 | 1860 | 2060 | 2243 | 2410 | 2100 | 2380 | 2660 | 2917 | 3150 |
| > 4' to 6' | Frame | 1930 | 2236 | 2542 | 2823 | 3078 | 1930 | 2236 | 2542 | 2823 | 3078 | 1930 | 2236 | 2542 | 2823 | 3078 | 1930 | 2236 | 2542 | 2823 | 3078 |
| > 4' to 6' | Sidewall | 1630 | 1810 | 1990 | 2155 | 2305 | 1960 | 2200 | 2440 | 2660 | 2860 | 2290 | 2590 | 2890 | 3165 | 3415 | 2950 | 3370 | 3790 | 4175 | 4525 |
| > 6' to 8' | Frame | 2440 | 2848 | 3256 | 3630 | 3970 | 2440 | 2848 | 3256 | 3630 | 3970 | 2440 | 2848 | 3256 | 3630 | 3970 | 2440 | 2848 | 3256 | 3630 | 3970 |
| > 6' to 8' | Sidewall | 2040 | 2280 | 2520 | 2740 | 2940 | 2480 | 2800 | 3120 | 3413 | 3680 | 2920 | 3320 | 3720 | 4087 | 4420 | 3800 | 4360 | 4920 | 5433 | 5900 |
| > 8' to 10' | Frame | 2950 | 3460 | 3970 | 4438 | 4863 | 2950 | 3460 | 3970 | 4438 | 4863 | 2950 | 3460 | 3970 | 4438 | 4863 | 2950 | 3460 | 3970 | 4438 | 4863 |
| Pier Spacing | | North (80 psf) | | | | | North (100 psf) | | | | | North (120 psf) | | | | | | | | | |
| | Location | 10 ft | 12 ft | 14 ft | 16 ft | 18 ft | 10 ft | 12 ft | 14 ft | 16 ft | 18 ft | 10 ft | 12 ft | 14 ft | 16 ft | 18 ft | | | | | |
| Up to 4' | Frame | 1420 | 1624 | 1828 | 2015 | 2185 | 1420 | 1624 | 1828 | 2015 | 2185 | 1420 | 1624 | 1828 | 2015 | 2185 | | | | | |
| Up to 4' | Sidewall | 2540 | 2900 | 3260 | 3590 | 3890 | 2980 | 3420 | 3860 | 4263 | 4630 | 3420 | 3940 | 4460 | 4937 | 5370 | | | | | |
| > 4' to 6' | Frame | 1930 | 2236 | 2542 | 2823 | 3078 | 1930 | 2236 | 2542 | 2823 | 3078 | 1930 | 2236 | 2542 | 2823 | 3078 | | | | | |
| > 4' to 6' | Sidewall | 3610 | 4150 | 4690 | 5185 | 5635 | 4270 | 4930 | 5590 | 6195 | 6745 | 4930 | 5710 | 6490 | 7205 | 7855 | | | | | |
| > 6' to 8' | Frame | 2440 | 2848 | 3256 | 3630 | 3970 | 2440 | 2848 | 3256 | 3630 | 3970 | 2440 | 2848 | 3256 | 3630 | 3970 | | | | | |
| > 6' to 8' | Sidewall | 4680 | 5400 | 6120 | 6780 | 7380 | 5560 | 6440 | 7320 | 8127 | 8860 | 6440 | 7480 | 8520 | 9473 | 10340 | | | | | |
| > 8' to 10' | Frame | 2950 | 3460 | 3970 | 4438 | 4863 | 2950 | 3460 | 3970 | 4438 | 4863 | 2950 | 3460 | 3970 | 4438 | 4863 | | | | | |



| Roof Load Zone and Max. Home Width (12" Max. Sidewall Eave Overhang) | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Pier Spacing | | South (20 psf) | | | | Middle (30 psf) | | | | North (40 psf) | | | | North (60 psf) | | | |
| | Location | 20 ft | 24/36 ft | 28/42 ft | 32/48 ft | 20 ft | 24/36 ft | 28/42 ft | 32/48 ft | 20 ft | 24/36 ft | 28/42 ft | 32/48 ft | 20 ft | 24/36 ft | 28/42 ft | 32/48 ft |
| Up to 4' | Frame | 1420 | 1624 | 1828 | 1981 | 1420 | 1624 | 1828 | 1981 | 1420 | 1624 | 1828 | 1981 | 1420 | 1624 | 1828 | 1981 |
| Up to 4' | Sidewall | 1280 | 1400 | 1520 | 1610 | 1520 | 1680 | 1840 | 1960 | 1760 | 1960 | 2160 | 2310 | 2240 | 2520 | 2800 | 3010 |
| Up to 4' | Marriagewall | 1760 | 2000 | 2240 | 2420 | 2160 | 2480 | 2800 | 3040 | 2560 | 2960 | 3360 | 3660 | 3360 | 3920 | 4480 | 4900 |
| > 4' to 6' | Frame | 1930 | 2236 | 2542 | 2772 | 1930 | 2236 | 2542 | 2772 | 1930 | 2236 | 2542 | 2772 | 1930 | 2236 | 2542 | 2772 |
| > 4' to 6' | Sidewall | 1720 | 1900 | 2080 | 2215 | 2080 | 2320 | 2560 | 2740 | 2440 | 2740 | 3040 | 3265 | 3160 | 3580 | 4000 | 4315 |
| > 4' to 6' | Marriagewall | 2440 | 2800 | 3160 | 3430 | 3040 | 3520 | 4000 | 4360 | 3640 | 4240 | 4840 | 5290 | 4840 | 5680 | 6520 | 7150 |
| > 6' to 8' | Frame | 2440 | 2848 | 3256 | 3562 | 2440 | 2848 | 3256 | 3562 | 2440 | 2848 | 3256 | 3562 | 2440 | 2848 | 3256 | 3562 |
| > 6' to 8' | Sidewall | 2160 | 2400 | 2640 | 2820 | 2640 | 2960 | 3280 | 3520 | 3120 | 3520 | 3920 | 4220 | 4080 | 4640 | 5200 | 5620 |
| > 6' to 8' | Marriagewall | 3120 | 3600 | 4080 | 4440 | 3920 | 4560 | 5200 | 5680 | 4720 | 5520 | 6320 | 6920 | 6320 | 7440 | 8560 | 9400 |
| > 8' to 10' | Frame | 2950 | 3460 | 3970 | 4353 | 2950 | 3460 | 3970 | 4353 | 2950 | 3460 | 3970 | 4353 | 2950 | 3460 | 3970 | 4353 |

| Pier Spacing | | North (80 psf) | | | | North (100 psf) | | | | North (120 psf) | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Location | 20 ft | 24 ft | 28/42 ft | 32/48 ft | 20 ft | 24 ft | 28/42 ft | 32/48 ft | 20 ft | 24 ft | 28/42 ft | 32/48 ft |
| Up to 4' | Frame | 1420 | 1624 | 1828 | 1981 | 1420 | 1624 | 1828 | 1981 | 1420 | 1624 | 1828 | 1981 |
| Up to 4' | Sidewall | 2720 | 3080 | 3440 | 3710 | 3200 | 3640 | 4080 | 4410 | 3680 | 4200 | 4720 | 5110 |
| Up to 4' | Marriagewall | 4160 | 4880 | 5600 | 6140 | 4960 | 5840 | 6720 | 7380 | 5760 | 6800 | 7840 | 8620 |
| > 4' to 6' | Frame | 1930 | 2236 | 2542 | 2772 | 1930 | 2236 | 2542 | 2772 | 1930 | 2236 | 2542 | 2772 |
| > 4' to 6' | Sidewall | 3880 | 4420 | 4960 | 5365 | 4600 | 5260 | 5920 | 6415 | 5320 | 6100 | 6880 | 7465 |
| > 4' to 6' | Marriagewall | 6040 | 7120 | 8200 | 9010 | 7240 | 8560 | 9880 | 10870 | 8440 | 10000 | 11560 | 12730 |
| > 6' to 8' | Frame | 2440 | 2848 | 3256 | 3562 | 2440 | 2848 | 3256 | 3562 | 2440 | 2848 | 3256 | 3562 |
| > 6' to 8' | Sidewall | 5040 | 5760 | 6480 | 7020 | 6000 | 6880 | 7760 | 8420 | 6960 | 8000 | 9040 | 9820 |
| > 6' to 8' | Marriagewall | 7920 | 9360 | 10800 | 11880 | 9520 | 11280 | 13040 | 14360 | 11120 | 13200 | 15280 | 16840 |
| > 8' to 10' | Frame | 2950 | 3460 | 3970 | 4353 | 2950 | 3460 | 3970 | 4353 | 2950 | 3460 | 3970 | 4353 |



## Step 4. Select Footing Material

Select one of the products and materials from Table 10 for the footings.

Table 10. Footing materials

| Material | Appropriate Use | Specification |
|---|---|---|
| Poured concrete | All soil types | Minimum 6” thick poured-in-place concrete pads, slabs, or ribbons with at least a 28 day compressive strength of 3,000 psi. Cast-in-place concrete footings may also require reinforcing steel based on acceptable engineering practice, the design loads, and site specific soil conditions. |
| Pre-cast concrete | All soil types | Minimum 4” thick nominal precast concrete pads meeting or exceeding ASTM C 90–02a, Standard Specification for Load Bearing Concrete Masonry Units, without reinforcement, with at least a 28 day compressive strength of 1,200 psi |
| ABS plastic | Per Pad Manufacturer’s Instructions | Use in accordance with the pad manufacturer’s instructions. Must be certified for use in the soil classification at the site, listed or labeled for the required load capacity. May be placed directly on soil or concrete. |
| Proprietary systems | Consult system manufacturer | Consult system manufacturer |

Will footings be concrete?

► YES, go to Step 5, size footing, (p. 45).

► NO, see footing system manufacturer’s instructions, then go to Set the Home (p. 54).

## Step 5. Size Footings

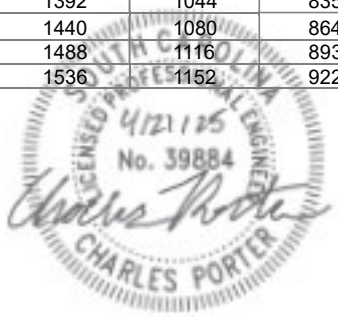
Once the load on the footing and the soil bearing capacity are known, calculate the size of each footing as follows:

1. From Table 14 (p.57) determine if the pier is to be of single stack blocks (8 inch x16 inch) or double stack blocks (16 inch x 16 inch).
2. Locate the group of columns in the Table 11 with the soil bearing capacity determined in Prepare the Site, Step 5. Determine soil bearing capacity and frost line (p. 29). Use the next lowest value if the exact value does not appear.
3. Find the row corresponding to the pier capacity required by Table 6, Table 7, Table 8 or Table 9. Then, read across the table to determine the minimum required footing area for the corresponding pier capacity and soil bearing capacity.
4. The required footing size and pier capacity may be changed by selecting different support spacing.



Table 11. Footing dimensions

| Pier Capacity (lbs) | Soil Bearing Capacity (PSF) | | | | | | Round Footing Diameter |
|---|---|---|---|---|---|---|---|
| | Required Footing Area (square inches) | | | | | | |
| | 1000 | 1500 | 2000 | 2500 | 3000 | 4000 | |
| 1000 | 144 | 128 | 128 | 128 | 128 | 128 | 18 inch |
| 1200 | 173 | 128 | 128 | 128 | 128 | 128 | |
| 1400 | 202 | 134 | 128 | 128 | 128 | 128 | |
| 1600 | 230 | 154 | 128 | 128 | 128 | 128 | |
| 1800 | 259 | 173 | 130 | 128 | 128 | 128 | |
| 2000 | 288 | 192 | 144 | 128 | 128 | 128 | |
| 2200 | 317 | 211 | 158 | 128 | 128 | 128 | |
| 2400 | 346 | 230 | 173 | 138 | 128 | 128 | |
| 2600 | 374 | 250 | 187 | 150 | 128 | 128 | |
| 2800 | 403 | 269 | 202 | 161 | 134 | 128 | |
| 3000 | 432 | 288 | 216 | 173 | 144 | 128 | |
| 3100 | 446 | 298 | 223 | 179 | 149 | 128 | |
| 3200 | 461 | 307 | 230 | 184 | 154 | 128 | |
| 3300 | 475 | 317 | 238 | 190 | 158 | 128 | |
| 3400 | 490 | 326 | 245 | 196 | 163 | 128 | |
| 3500 | 504 | 336 | 252 | 202 | 168 | 128 | |
| 3600 | 518 | 346 | 259 | 207 | 173 | 130 | |
| 3700 | 533 | 355 | 266 | 213 | 178 | 133 | |
| 3800 | 547 | 365 | 274 | 219 | 182 | 137 | |
| 3900 | 562 | 374 | 281 | 225 | 187 | 140 | |
| 4000 | 576 | 384 | 288 | 230 | 192 | 144 | |
| 4100 | 590 | 394 | 295 | 236 | 197 | 148 | |
| 4200 | 605 | 403 | 302 | 242 | 202 | 151 | |
| 4300 | 619 | 413 | 310 | 248 | 206 | 155 | |
| 4400 | 634 | 422 | 317 | 253 | 211 | 158 | |
| 4500 | 648 | 432 | 324 | 259 | 216 | 162 | |
| 4600 | 662 | 442 | 331 | 265 | 221 | 166 | |
| 4700 | 677 | 451 | 338 | 271 | 226 | 169 | |
| 4800 | 691 | 461 | 346 | 276 | 230 | 173 | |
| 4900 | 706 | 470 | 353 | 282 | 235 | 176 | |
| 5000 | 720 | 480 | 360 | 288 | 240 | 180 | |
| 5200 | 749 | 499 | 374 | 300 | 250 | 187 | |
| 5400 | 778 | 518 | 389 | 311 | 259 | 194 | |
| 5600 | 806 | 538 | 403 | 323 | 269 | 202 | |
| 5800 | 835 | 557 | 418 | 334 | 278 | 209 | |
| 6000 | 864 | 576 | 432 | 346 | 288 | 216 | |
| 6200 | 893 | 595 | 446 | 357 | 298 | 223 | |
| 6400 | 922 | 614 | 461 | 369 | 307 | 230 | |
| 6600 | 950 | 634 | 475 | 380 | 317 | 238 | |
| 6800 | 979 | 653 | 490 | 392 | 326 | 245 | |
| 7000 | 1008 | 672 | 504 | 403 | 336 | 252 | |
| 7200 | 1037 | 691 | 518 | 415 | 346 | 259 | 24 inch |
| 7400 | 1066 | 710 | 533 | 426 | 355 | 266 | |
| 7600 | 1094 | 730 | 547 | 438 | 365 | 274 | |
| 7800 | 1123 | 749 | 562 | 449 | 374 | 281 | |
| 8000 | 1152 | 768 | 576 | 461 | 384 | 288 | |
| 8500 | 1224 | 816 | 612 | 490 | 408 | 306 | |
| 9000 | 1296 | 864 | 648 | 518 | 432 | 324 | |
| 9500 | 1368 | 912 | 684 | 547 | 456 | 342 | |
| 10000 | 1440 | 960 | 720 | 576 | 480 | 360 | |
| 10500 | 1512 | 1008 | 756 | 605 | 504 | 378 | |
| 11000 | 1584 | 1056 | 792 | 634 | 528 | 396 | |
| 11500 | 1656 | 1104 | 828 | 662 | 552 | 414 | |
| 12000 | 1728 | 1152 | 864 | 691 | 576 | 432 | |
| 12500 | 1800 | 1200 | 900 | 720 | 600 | 450 | |
| 13000 | 1872 | 1248 | 936 | 749 | 624 | 468 | 30in |
| 13500 | 1944 | 1296 | 972 | 778 | 648 | 486 | |
| 14000 | 2016 | 1344 | 1008 | 806 | 672 | 504 | |
| 14500 | 2088 | 1392 | 1044 | 835 | 696 | 522 | |
| 15000 | 2160 | 1440 | 1080 | 864 | 720 | 540 | |
| 15500 | 2232 | 1488 | 1116 | 893 | 744 | 558 | |
| 16000 | 2304 | 1536 | 1152 | 922 | 768 | 576 | |



| To calculate a square or rectangular footing: | To calculate a round footing: |
|---|---|
| Length (in) x Width (in) = Area | 3.14 x .25 x Diameter (in) x Diameter (in) = Area |
| Example: 16 in x 20 in = 320 sq. inches | Example: For a 22” diameter footing |
| | 3.14 x .25 x 22 in x 22 in = 380 sq. inches |

Design individual or unreinforced strip footings to comply with the following additional requirements (does not apply to unreinforced slab design):

- To keep footings directly under I-beams and other support points, size them slightly larger than the minimum required area to allow slight adjustment of the pier location during home installation.
- Design footings with a footing extension (projection beyond the base of the pier) no greater than the footing thickness (Figure 14). Increase footing thickness if necessary.

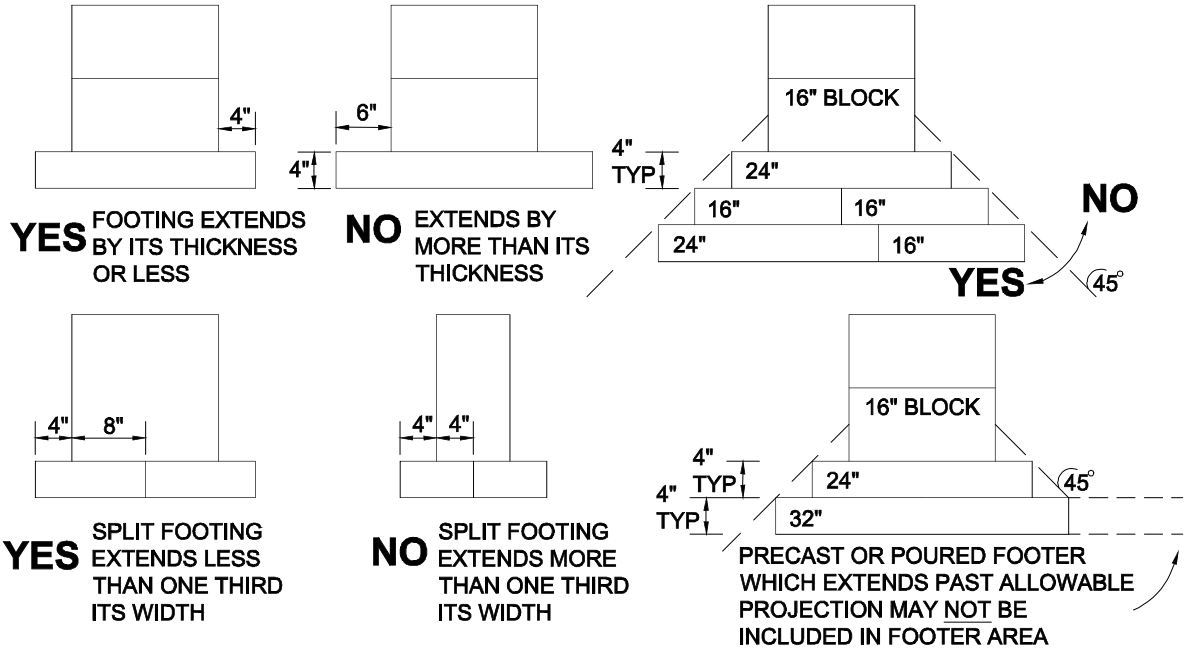



Figure 14. Maximum footing extensions

- The footing sizes shown are for square pads and are based on the surface area (square inches). Design non-square footings such that the area and thickness is equal to or greater than the area of the square footing shown in Table 11 and the distance from the edge of the pier to the edge of the footing is not more than the thickness of the footing. Footings may be either 4-inch precast or poured or a combination of both with a combined thickness complying with the requirements of Table 12.



Table 12. Footing thickness

| Poured in Place Footing for Single Stack Pier | | | Poured in Place Footing for Double Stack Pier | | |
|---|---|---|---|---|---|
| Min. Footing Thickness | Max Footing Dimension | Max. Area (sq. in) | Min. Footing Thickness | Max Footing Dimension | Max. Area (sq. in) |
| 6 | 28 x 20 | 560 | 6 | 28 x 28 | 784 |
| 8 | 32 x 24 | 768 | 8 | 32 x 32 | 1024 |
| 10 | 36 x 28 | 1008 | 10 | 36 x 36 | 1296 |
| 12 | 40 x 32 | 1280 | 12 | 40 x 40 | 1600 |
| | | | 14 | 44 x 44 | 1936 |
| | | | 16 | 48 x 48 | 2304 |

## Step 6. Install Footings

Construct the footings as follows:

- Maintain the distance between adjacent piers to within 10% of the tabulated spacing and so the average distance between piers is equal to or less than the tabulated spacing.
- Whenever possible, place pier supports directly under the required locations. If plumbing, electrical or mechanical equipment interferes, place supports no more than 12 inches in either direction of the support point.



Placing concrete anchors. If anchors will be placed in concrete follow instructions in Install Stabilizing System (p. 93) to determine anchor layout. Either place anchors immediately after the concrete has been poured or drill them in after the concrete has set.

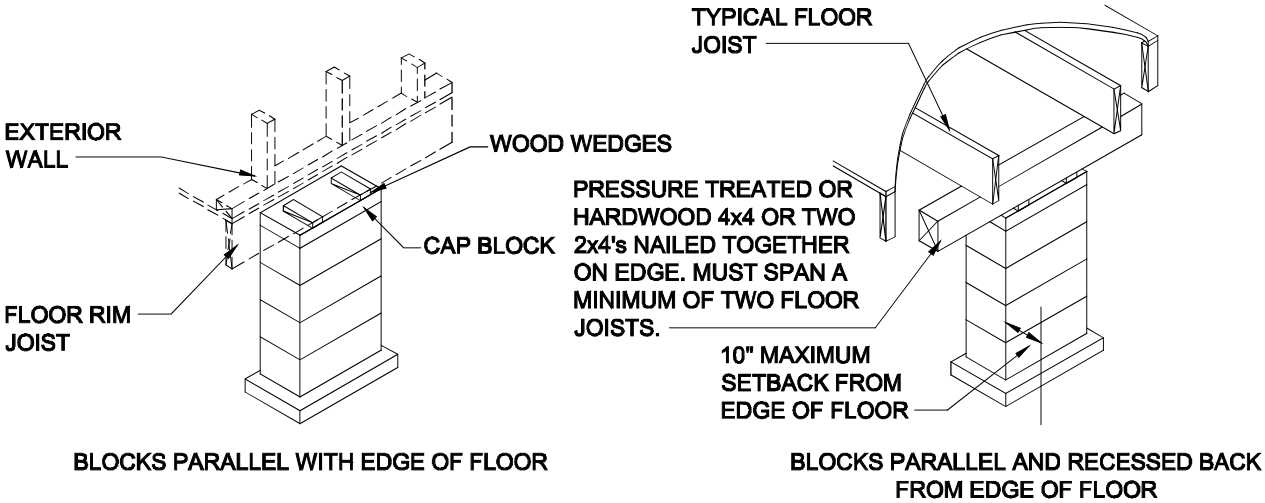



Figure 15. Perimeter supports



- If footings are rectangular, orient them so that the long side is perpendicular to the home's I-beam.
- Place the bottom of footings on undisturbed soil or fill compacted to at least 90% of its maximum relative density.
- In freezing climates protect footings from the effects of frost heave in accordance with any LAHJ requirements (see **Prepare the Site**, p. 25). Place the bottom of the footings below the frost line. Insulated foundations or other frost protection options are acceptable when designed by a registered engineer or registered architect. Monolithic slabs are allowed above frost depth when designed by a registered engineer or registered architect to resist the effects of frost heave. Anchorage requirements must be included with each registered engineer or registered architect design when the anchorage requirements listed in this manual cannot be accommodated. Prior to obtaining an alternative design contact the home building facility for available approved alternative designs or instructions for submitting an alternative design.
- Make sure the top surface of the footing is level, flat and smooth.

► go to **Set the Home** p. 54)

# CONSTRUCT FOUNDATION

**For Homes With Load Bearing Perimeter Wall:**

This chapter provides guidelines and recommendations for the design and construction of a basement or crawlspace foundation using a load bearing perimeter wall. A load bearing perimeter wall foundation system uses a wall along the outer edge of the home to support the home's outside walls. This perimeter support works with interior supports such as piers, columns and cross beams that support the home's frame and, if multi-section, marriage line.

**Follow the Steps below:**


▼ **Step 1. Obtain a Foundation Design** (p. 50)
▼ **Step 2. Excavate** (p. 50)
▼ **Step 3. Construct the Footing or Slab** (p. 51)
▼ **Step 4. Construct the Perimeter wall** (p. 51)
▼ **Step 5. Install Interior Supports** (p. 52)
▼ **Step 6. Waterproof Foundation Wall** (p. 53)
▼ **Step 7. Backfill and Grade** (p. 53)


## Step 1. Obtain A Foundation Design

If a load bearing perimeter wall foundation design has not been provided by the home manufacturer, it is the responsibility of the retailer to provide a design approved by an engineer or architect, licensed in the state where the home will be installed. The approved design must comply with the LAHJ regulations for foundation design, waterproofing and drainage, and the following:

- The foundation perimeter bearing wall must be supported with a concrete slab or continuous strip footing around the perimeter of the home. Interior piers must be supported by a slab or footings. If footings are used under interior piers, they may be designed as in **Design Frame and Perimeter Supports**, p. 41.
- Slabs must extend to the edges of the home. **IMPORTANT: Verify the dimensions of the actual floor width (eg. a 28' wide home does not measure 28 feet in width).**
- Footings and slabs must be protected from the effects of frost heave by extending the footings to or below the frost line or by using a frost protected shallow foundation design.

## Step 2. Excavate

Excavate for the foundation, properly disposing of the earth that is not needed for backfill or site grading purposes.

## Step 3. Construct the Footings or Slab

Construct the foundation according to the approved design, including the perimeter foundation wall, drainage system, footing(s) and/or slab.

## Step 4. Construct the Perimeter Wall

Unless the approved design requires otherwise, construct the perimeter wall with mortared and reinforced concrete blocks or reinforced poured-in-place concrete. Install reinforcement according to the approved design or LAHJ. Install ventilation and access openings according to the approved design, or if not specified, according to the requirements in **Complete Under the Home, STEP 3 INSTALL SKIRTING** (p. 139).

Where open slatted deck boards are used at recessed entries and porches, provisions must be made to ensure water is not permitted to drain into the area under the conditioned portion of the home. Any perimeter type skirting or foundation wall should be installed to follow the exterior of the wall of the home and permit the area beneath the porch to drain water away from the home.

When constructing pockets for a cross beam system, measure the beam depth and locate the pockets carefully. It is critical that the home's frame rests on top of the cross beam and the perimeter of the floor rests squarely on the foundation wall sill plate (**Figure 16**). Leave room for a two-inch nominal, hardwood spacer or steel plate spacer on top of the wall pockets (to prevent corrosion, the steel beams must not be in direct contact with concrete). Leave at least one inch for thermal expansion at the ends of the beams and maintain a minimum of two inches of bearing area for the beams in the pockets (yielding a minimum pocket depth of three inches).

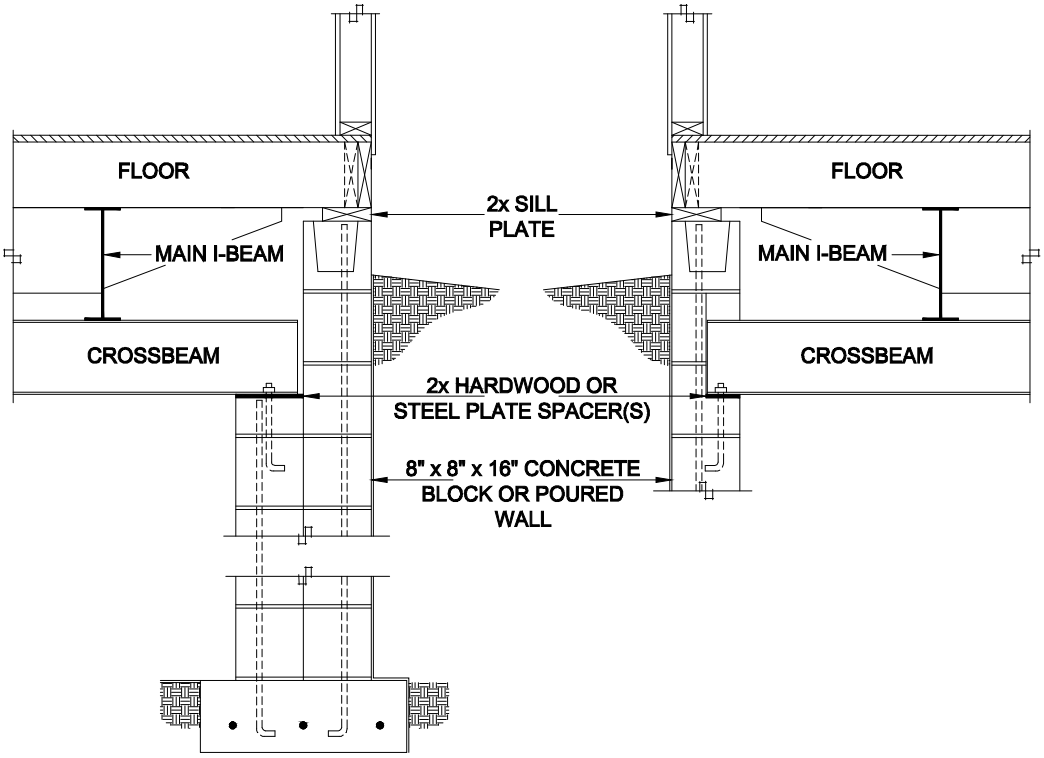



*Figure 16. Cross beam installation*

Bolt a pressure treated wood sill plate (minimum 2 x 6) to the top of the foundation wall. If the home's siding cannot be nailed through, use a 2 x 10 sill plate that extends into the foundation 1-1/4 inches (**Figure 17**). The home can then be connected to the foundation by fastening the sill plate into the floor joists from below. Connect the home to the foundation according to the approved design (See Step 1).

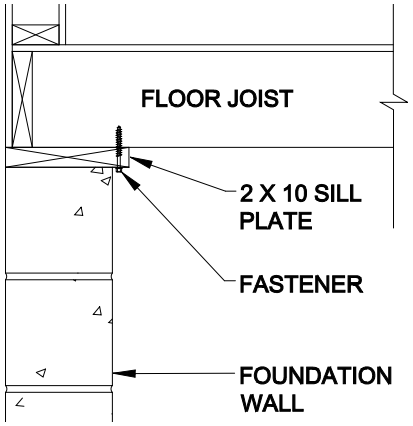



*Figure 17. Connection using 2 x 10 sill plate*

**Using engineered designs.** This section is NOT intended to provide a complete design for a buildable foundation. A complete design must be obtained that is suitable for the local area and sealed by a professional engineer or registered architect, licensed in the state. Prior to obtaining an alternative design contact the home building facility for available approved alternative designs.The manufacturer is capable of providing limited model specific foundation designs upon request.

**Foundation ready home.** Make sure that homes to be installed on a basement or a crawlspace have been ordered with a recessed frame or as a basement-ready frame system, where the frame is designed to avoid interference with the foundation wall.

**Level the wall.** Make sure the foundation is level and straight with no more than a 1/4 inch vertical variation over the entire foundation and no more than 1/8 inch vertical variation over any two-foot length.

**Check for plates.** When using a cross beam system, check and compensate for reinforcement plates that add thickness to the chassis beam at axle locations.

## Step 5. Install Interior Supports

Install piers, columns and cross beams to support the interior of the home according to the approved design.

## Step 6. Damp Proof Foundation Wall

Damp or water proof foundation walls as necessary according to local jurisdiction requirements.



**Backfilling.** Backfill against basement walls only after the home is connected to the foundation or the basement walls may deflect inward or collapse.

Is this a basement foundation?

► **YES,** go to **Set the Home,** (p. 54).

► **NO,** go to **step 7. backfill and grade,** (p.53)

## Step 7. Backfill and Grade

Backfill against the foundation wall to the height of the damp proofing. Take care to not damage the drainage system. Grade the fill as per **Prepare the Site** (p. 25).

Does the approved foundation design call for ground anchors?

► **YES**, go to **Install Stabilizing Systems**, (p. 93).

► **NO**, go to **Set the Home**, (p. 54).

# SET THE HOME

This chapter describes the process of installing the first section of the home (for single section homes this is the only section) onto the foundation.

**Follow the Steps below:**


▼ **Step 1. Prepare for Set** (p. 54)
▼ **Step 2. Position Home Section** (p. 55)
▼ **Step 3. Lift Home** (p. 55)
▼ **Step 4. Construct Piers** (p. 56)


## Step 1. Prepare for Set

Before beginning the home set, complete the following:

- Confirm that the site is properly cleared and graded (see **Prepare the Site**, p. 25).
- Ensure that the footings are in place and properly located.
- Install any utilities that will be difficult to install (e.g. those below grade beneath the home) after the home is in place.
- Secure or remove from the home and properly store all ship loose items (refer to shipping documents for items shipped with the home).
- Inspect the home interior, exterior and all provided materials, appliances and equipment. Immediately report any damage or shortages to the manufacturer.

For perimeter bearing wall foundations:

- Check that the actual length and width of the home matches the foundation walls.
- Check that the two main diagonal measurements of the foundation are equal.
- Check that the foundation walls and other support points are within 1/4 inch of level overall and within 1/8 inch of level within any four foot distance.
- For multi-section homes, check that each pair of diagonal measurements for each portion of the foundation corresponding to a home section are equal.
- For multi-section homes, find the electrical bonding lugs on the front or rear outriggers. Reverse them to the inside of the outrigger so they will be accessible after the home is placed on the foundation walls.

**Clearances under the home.** After the home is leveled, the resulting distance between the bottom of the entire chassis main frame beam and the ground must be no less than 12 inches.

**Utilize proper cribbing.** Homes weigh several tons. No one should be under the home (whether it is moving or stationary) unless proper cribbing is in place. Failure to utilize proper cribbing may result in serious injury or death.

## Step 2. Position Home Section

Position the home section in its final location (if possible, move the heaviest section of the home into place first). Then place materials needed to construct support piers near their final locations under the home as determined in **Install Footings,** (p. 33).

## Step 3. Lift Home

There are three primary methods available to place the home on the foundation: jacking, rolling and craning. Jacks, often with roller systems, are typically used for pier and anchor foundations; roller systems are commonly used for crawlspace foundations with load bearing perimeter walls; and cranes are most commonly used for basement foundations. Please scan the ***Lifting and Water Level QR code*** to view instructions for each primary lifting method and water level usage.

*Lifting and Water Level Instructions*

When jacks or rollers are used, comply with all safety precautions and the instructions outlined by the manufacturers of the equipment. Lifting the home involves potential risks and must be done with the utmost care and caution. Failure to follow warnings and procedures may result in serious injury or death.

No one should be under the home while it is suspended. Never put your hands between the home and the perimeter walls.

Have the interior foundation supports already been designed and installed as part of an approved load bearing perimeter wall foundation?

► **YES,** go to **Complete Multi-Section Set,** (p. 60) or go to **Connect Utilities,** (p. 119) for single section homes.

► **NO,** go to **step 4. Construct Piers,** (p. 56).

## Step 4. Construct Piers

For the side of the home section that is up on jacks, place piers on footings or pads following the home manufacturer's blocking plan (or tags). If no plan was provided, use the support plan developed in **Install Footings** (p. 33). Start at one end of the home section and work toward the other noting the required pier material specifications and procedure described below.

Construct piers so as to provide a stable foundation for the home using materials listed in the specifications box below and based on the location of the pier and its height as measured from the top of the footing, pad or grade to the top of the cap. The pier height can be measured from the lowest surrounding grade to the top of the cap when grade level is above the top of the footing. See **Table 14** for pier construction requirements.



**Designing piers.** Incorrect size, location or spacing of piers may result in serious structural damage to the home. Install piers at all required locations. Failure to do so may lead to sagging floors, walls and roofs, and could void the home's warranty.

*Table 13. Pier material minimum specifications*

| Component | Specification |
|---|---|
| **Concrete Block** | Nominal dimensions of at least 8" x 8" x 16"; confirming to ASTM designation C90 |
| **Caps** | Solid masonry (nominal 4" x 8" x 16" pre-cast concrete without reinforcement); treated or hardwood lumber (nominal 2" x 8" x 16"); or steel (minimum 1/2" thick, corrosion protected by a min. of a 10-mil coating of an exterior paint or equivalent) |
| **Spacers** | Nominal 2" thick hardwood boards or nominal 2" or 4" concrete blocks to fill vertical gaps. |
| **Shims (also called wedges)** | When required, nominal 4-inch by 6-inch by 1-inch (max. vertical height) wood shims used in pairs. Some states, counties, townships, and or municipalities may require the use of hardwood or treated lumber shims. Other listed shims may be used if installed in accordance with the listing (max load capacity). |
| **Commercial metal or pre-cast concrete piers** | Available in various sizes stamped with maximum load capacity and listed or labeled for the required vertical load capacity, and, where required by design, for the appropriate horizontal load capacity. Metal or other manufactured piers must be provided with protection against weather deterioration and corrosion at least equivalent to that provided by a coating of zinc on steel of .30 oz per sq. ft of surface coated. Manufactured pier heights must be selected so that the adjustable risers do not extend more than 2 inches. |
| **Pressure treated wood** | With a water borne preservative, in accordance with AWPA Standard U1–04 for Use Category 4B ground contact applications |

Table 14. Pier Configuration and Construction

| Pier location | Height | Configuration | Maximum offset top to bottom | Maximum Load (lbs) | |
|---|---|---|---|---|---|
| | | | | Without Mortar | With Mortar |
| Frame | Less than 36 in * | Single stack blocks with long side perpendicular to frame I-beam | ½” | 6,500 | 7,680 |
| | Between 36 in and 67 in | Double, interlocked blocks | 1” | 13,000 lbs. | 15,360 |
| | Between 36 in and 67 in | Triple, interlocked blocks | 1” | 17,280 lbs. | 23,034 |
| | Between 68 in and 108 in | Double, interlocked, reinforced blocks | 1” | NA | 39,500 |
| Perimeter | 54 in or less ** | Single stack blocks with long side parallel to perimeter rail (rim joist) | ½” | 6,500 | 7,680 |
| Marriage line | 54 in or less** | Single stack blocks with long side perpendicular to the marriage line | ½” | 6,500 | 7,680 |

* Single stack piers may be constructed up to 54” max. height only when installed as perimeter and marriage line support piers.

** Cross reference maximum allowable pier height with maximum allowable floor height listed in frame tiedown charts. If maximum height listed in frame tiedown charts is exceeded, then designs must be provided by a registered professional engineer or registered architect.

Maximum horizontal offset of ½” allowed for pier heights up to 36” and a 1” offset allowed for pier heights between 36” and 67”.

1. Prepare footing surface. Make sure the footing surface upon which the pier sits is flat and smooth. Before placing the pier on the footing, clean dirt, rocks or other material off the surface of the footing. If the footing surface is uneven, create a level, flat surface by mortaring on the first block (or manufactured pier base) or by placing the first block (or manufactured pier base) on a layer of premix dry sand mortar.
2. Stack blocks. Stack concrete blocks with their hollow cells aligned vertically. When piers are constructed of blocks stacked side-by-side, orient each layer at right angles to the previous one (Figure 18) and plan blocks so that split caps will be perpendicular to the blocks they rest on and perpendicular to the main I-beam.



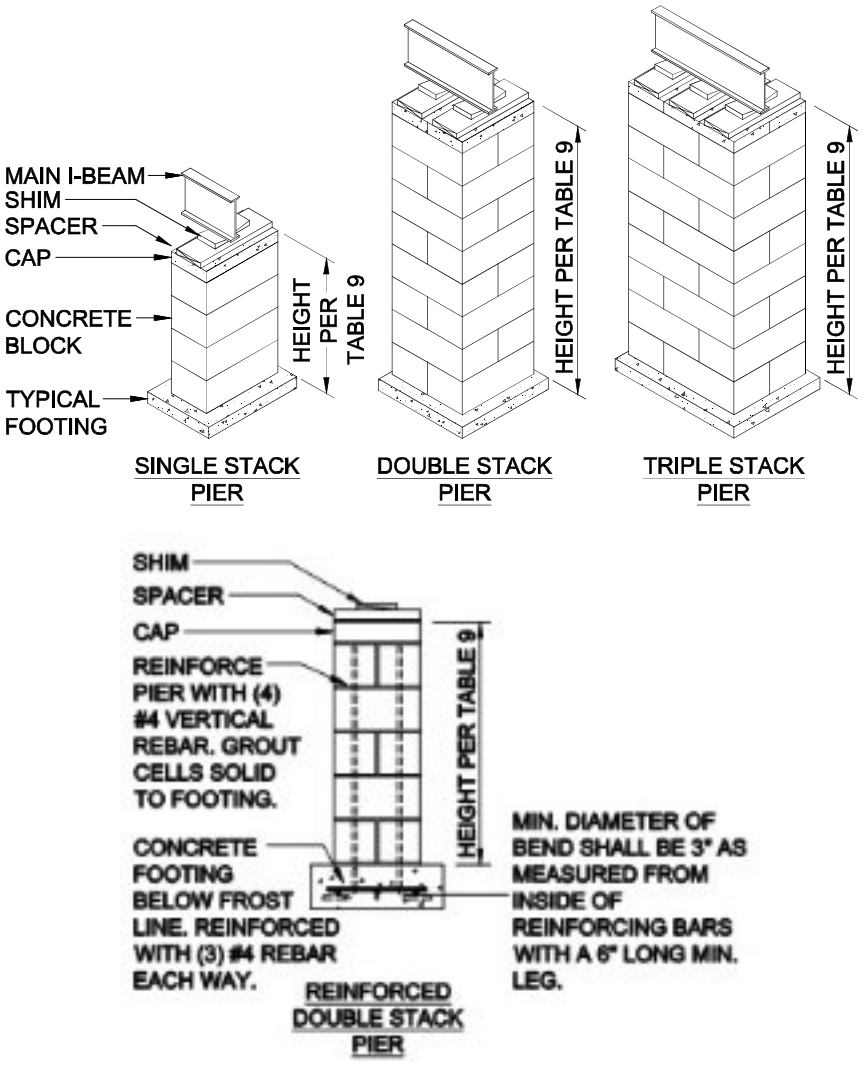



Figure 18. Frame pier construction

3. Cap piers. Place a cap on hollow block piers to evenly distribute the structural load. Use caps the same length and width as the piers they rest upon. When using split caps on double-stacked block piers, install the caps with the long dimension perpendicular to the joint in the blocks below and perpendicular to the main I-beam.

4. Install shims. Use shims to level the home and fill any gaps between the base of the I-beam and the top of the pier cap. When required, always use shims in pairs (Figure 19). Drive them in tightly so they do not occupy more than one inch of vertical space. When the space to be shimmed is greater than one inch and less than the minimum thickness of available caps or concrete blocks, use hardwood dimensional lumber (two inches maximum thickness) or 4" thick concrete block. For split caps, install shims and dimensional lumber/blocks over each individual cap. Two cap blocks may be used as the cap on double block piers provided the joint between the cap blocks is perpendicular to the joint between the open cell concrete blocks and is also perpendicular to the I-beam supported by the pier.

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**Curing time of mortar.** Where wet mortar is used to construct or level piers, allowed it to cure to at least 80% of strength capacity (usually requiring 96 hours) before setting the home.

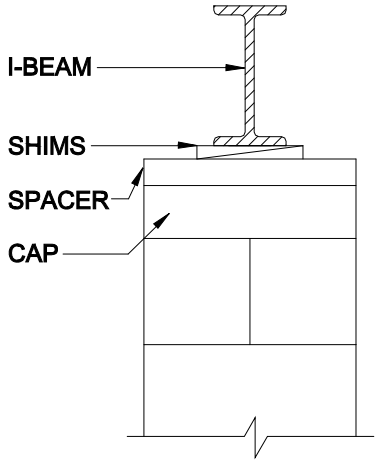



*Figure 19. Correct shim placement*

5. **Set up level.** Set up a water level with the fluid level at the desired height of the main piers.Carefully lower the side of the section down onto the leveled piers, adjusting the final height with shims.

### Using a Water Level

For instructions on how to use a water level please scan the ***Lifting and Water Level Instructions QR code*** *(p. 55)***.**

**Level the home.** The home is adequately leveled if there is no more than 1/4 inch difference between adjacent pier supports (frame or perimeter) and the exterior doors and windows of the home do not bind and can be properly operated. If differences in pier heights occur, drain lines should be inspected to correct reverse slope situations.

**Water level operation.** To operate the water level properly, both ends of the system must be open to the atmosphere and there must be approximately the same amount of fluid in the tubing at all times (within a few inches).

Is this a single-section home?

► **YES,** go to **Connect Utilities,** (p. 119).

► **NO,** go to **Complete Multi-Section Set,** (p. 60).

# COMPLETE MULTI-SECTION SET

This chapter covers the preparation and installation of additional home sections, including the structural connections between units, raising and fastening hinged roofs and fastening the home to a load bearing perimeter wall foundation.

**Follow the Steps below:**


▼ **Step 1. Install Marriage Line Anchors** (p. 60)
▼ **Step 2. Remove Protective Shipping Materials** (p. 60)
▼ **Step 3. Complete Hinged Roof** (p. 60)
▼ **Step 4. Repair or Install Marriage Line Gasket** (p. 61)
▼ **Step 5. Position Additional Home Sections** (p. 61)
▼ **Step 6. Connect Floors** (p. 62)
▼ **Step 7. Connect Roof** (p. 64)
▼ **Step 8. Connect Walls** (p.66)
▼ **Step 9. Attach Tag Units** (p. 68)
▼ **Step 10. Remove Temporary Items** (p. 68)
▼ **Step 11. Fasten Home to Foundation** (p. 69)


## Step 1. Install Marriage Line Anchors

If the home is in Wind Zone II or III, install ground anchors along the marriage line now; before mating sections are joined see **Install Stabilizing System** (p. 93). After installing marriage line anchors return to this point in **Complete Multi-Section Set**.

## Step 2. Remove Protective Shipping Materials

Remove all shipping protection and associated fasteners from both home sections to be joined, including plastic used to close up the open sides during transportation. Do not remove the temporary supports holding up the ceilings at major openings. Weather resistive barrier (such as Tyvek or other similar product) will be installed over the exterior wall OSB sheathing and under the shipping plastic. Be careful not to damage the weather resistive barrier when removing the shipping plastic.

## Step 3. Complete Hinged Roof

If the home has a hinged roof that has been folded down for shipping, refer to the Appendix (p. 8) and the appropriate setup addendum page for hinged truss installation information.



**Checklists for alternate construction**. If the serial number (see the data plate or the chassis front cross member) has the letters “AC” before or after it, then the Alternate Construction on-site check list supplied with the home must be completed and returned to the home manufacturer in a timely manner. If the AC checklist relates to the roof, then failure to do so may require future disassembly of the roof and further inspections.

## Step 4. Repair or Install Marriage Line Gasket

A continuous, non-porous gasket creating a permanent air barrier will be installed on at least one side of the marriage line; along the floor, end walls and ceiling (and marriage lines for any tag units). Inspect the gasket and repair any gaps or tears. If any voids are present in the marriage line gasket or if the gasket is damaged, seal the marriage line by installing an approved non-porous gasket or expanding foam. For homes with through-the-rim crossover ducts (see **Connect Crossovers**, p. 75) inspect and if necessary, repair gaskets around the rim joist duct openings using 3/4 inch thick fiberglass duct board or other material acceptable to the manufacturer. Ensure that duct openings are unobstructed.



**Checking through-the-rim-ducts.** Ensure that through-the-rim-duct connections are secure and tight after the home sections are together.

## Step 5. Position Additional Home Sections

Follow this procedure to install additional home sections:

1. **Remove obstructions.** Remove protruding nails and staples or anything else that will keep the home sections from fitting together snugly. If present, cut the temporary ceiling and floor plates at the edges of marriage line openings taking care not to damage ceiling or floor coverings or displace temporary marriage line support posts (these supports and the plates will be removed after the home sections have been structurally connected).
2. **Complete crossovers.** Before moving the two sections together, complete any crossover connections that require access from the open marriage line, including the attic duct connection (if present) and marriage wall interior electrical connections (see **Connect Crossovers**, p. 75)
3. **Position section.** Position the section as closely as possible (ideally within six inches) and line up with the previously set section. If using a mechanical positioning system or crane, follow the system manufacturer’s instructions or the crane operator’s directions.
4. **Construct piers.** With the outside walls of the home aligned, construct the piers for the home section according to the instructions in **Set the Home** (p. 54) before continuing to the steps below.

5. Level section. Lower the section onto the outside piers first, inside piers last. Before releasing the mechanical positioning system, check interior doorways and other openings for misalignments that may cause problems during trim-out. The floors should be flush, level and tight and the roof section should have little, if any, gap at the top of the marriage line.
6. Exterior gaps. Exterior close up gaps are open air spaces between mating line of multi-section homes which allow exterior air to enter the home between sections. Upon completion of exterior close-up no gaps are permitted between structural elements being connected. Prior to completion of the exterior close-up, gaps that do not exceed one inch are permitted between structural fasteners provided that the gaps are shimmed before completion of close-up and the marriage gasket provides a proper seal. Shims are only required at fasteners.
7. Shim fasteners. When the space between structural interconnection members of multi-sections exceeds ½", the space at fasteners shall be shimmed with dimensional lumber. The fastener length shall be increased to maintain fastener penetration into receiving member.

## Step 6. Connect Floors

Make floor structural connections according to the appropriate method described below. A sealing gasket shall be present between marriage line rim joists.

### Toed Fasteners through Bottom Board.

Make connections according to the fastener specifications in Table 15 and as shown on Figure 20. Spacing indicated in Table 15 is on-center, both sides of marriage line. Fasteners on each side of marriage line shall be staggered and offset using the spacing distance that is required by Figure 21. Repair any tears or holes in the bottom board after installation of fasteners.

Table 15. Floor connection fastening specifications

| Fastener | | Spacing | | |
|---|---|---|---|---|
| Type | Size | Wind Zone I | Wind Zone II | Wind Zone III |
| Lag screw with washer | 5/16" x 4"* | 32 in. | 16 in. | 16 in. |

* Increase fastener lengths by 3 inches for double rim joists.

If the home has a through-the-rim crossover duct, install one additional fastener at each side of the duct opening. Make sure to repair any openings in the bottom board with tape specially made for that purpose (may be provided).



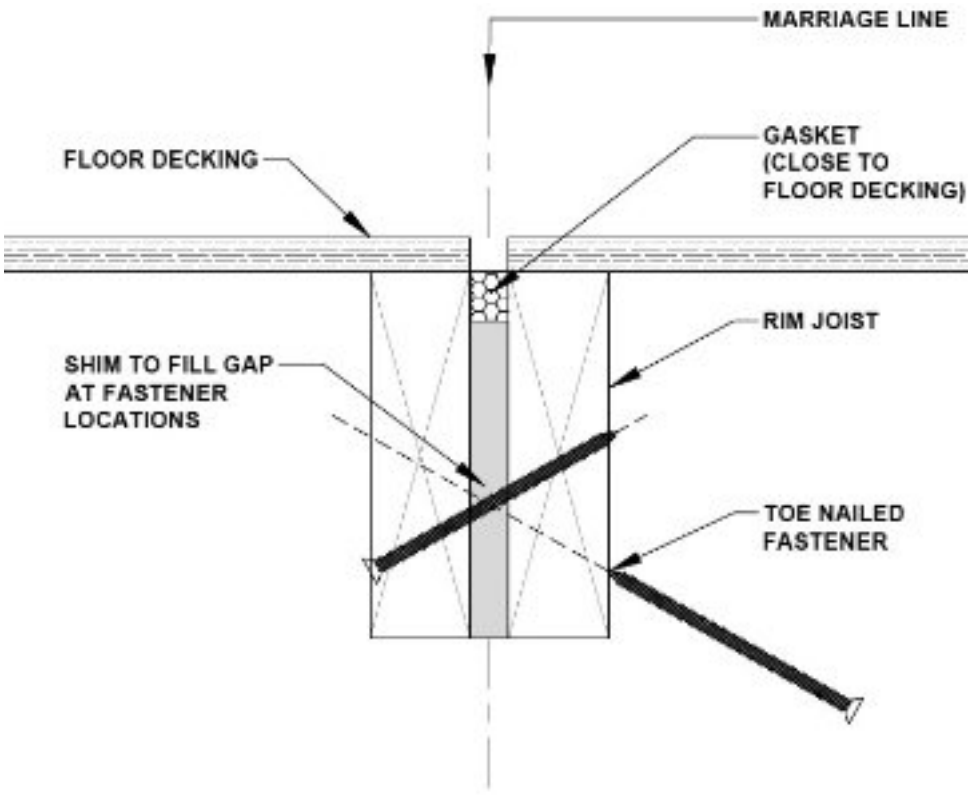



Figure 20. Floor connection through bottom board

Stagger fasteners on either side of marriage line and offset them by half the spacing distance (Figure 21).

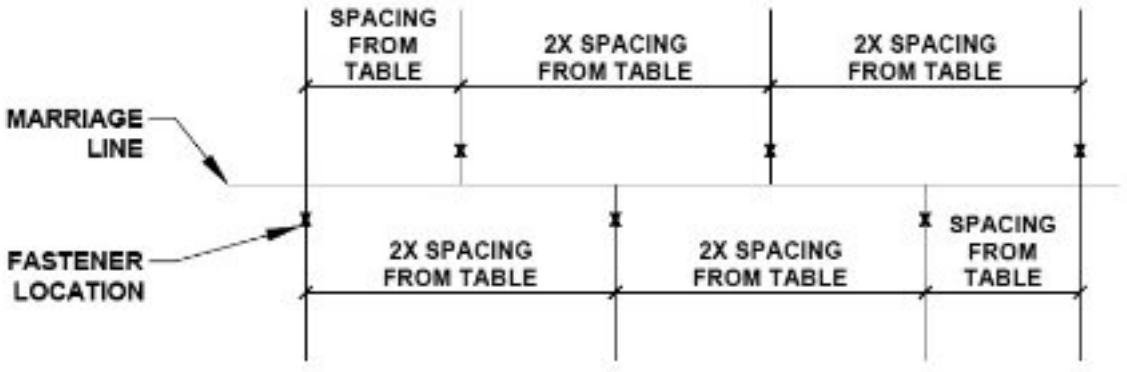



Figure 21. Staggered offset fastening along marriage line



## Step 7. Connect Roof

If the marriage line along the roof is not snug, position jacks every 20 feet or less along the outside I-beam and uniformly lift the section until the roof area is tight.

Check to make sure the ceiling joint is flush before installing the connections. If not, use a jack and tee to raise whichever ceiling is low, starting at the front and working to the rear of the home. Fasten the roof along the marriage line using one of the methods below for either double or triple section homes.




### Double Section Homes

Make roof structural connections in double section homes according to the one of the methods described below.

#### Method 1: Install Toed Fasteners to Connect Ridge Boards

Install fasteners through the ridge boards as shown in Figure 22 using the fastener spacing and specifications in Table 16. Stagger fasteners on each side of the marriage line and offset them by the spacing distances shown in Figure 21. Spacing indicated is on-center, both sides of ridge. Ensure fasteners penetrate the ridge boards by a minimum of 1-1/2 inches both sides of ridge.



Closing the roof gap. Level and fasten the marriage line joint at the floor before jacking to tighten a roof gap. DO NOT use roof fasteners to close any gaps. Separation of the ridge beams and trusses may occur.

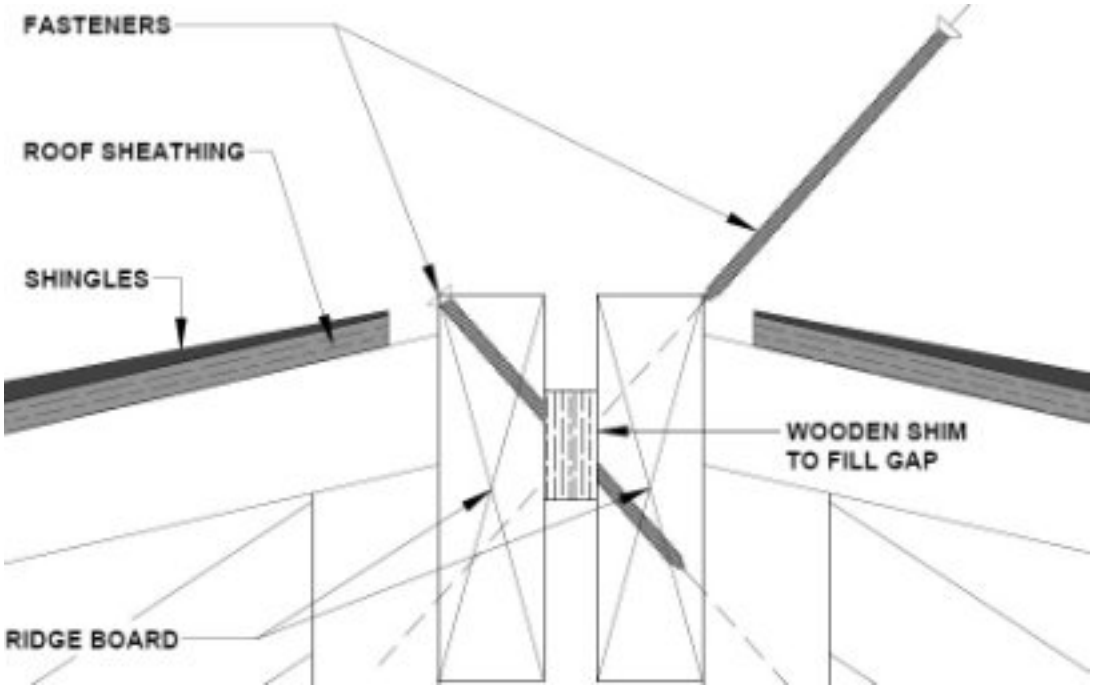



Figure 22. Fasteners toed through ridge boards with shims to fill gap

Table 16. Roof connections specifications

| Fastener | | Spacing | | |
|---|---|---|---|---|
| Type | Size | Wind Zone I | Wind Zone II | Wind Zone III |
| Wood screw | #10 x 5” | 14 in. | 8 in. | 6 in. |

In Wind Zones II and III install straps as follows:

- Select metal straps made of minimum 1-1/2” wide x 26 ga 18” long galvanized steel.
- Straps shall be applied on every other truss for Wind Zone II and Wind Zone III.
- All strap fasteners must be 1-1/4” minimum length.

Table 17. Fasteners per strap end

| Fastener type | Fasteners per strap end |
|---|---|
| 0.131” roofing nails | 5 |
| 0.120” roofing nails | 6 |
| 14 ga staples | 9 |
| 15 ga staples | 10 |
| 16 ga staples | 12 |

Applicable Configurations:

- 10’-18’ box width.
- 9’ maximum sidewalls.
- 4.36/12 maximum roof pitch.
- 16” maximum sidewall eave overhang.

### Method 2: Hinged Roof Homes with a Separate Ridge Assembly

For hinged roof homes with a separate ridge assembly, install and fasten the assembly per the appropriate section in the Appendix (p. 8) and the appropriate setup addendum page.



## Step 8. Connect Walls

Once the home is secured along the marriage line floor and roof, secure end walls, interior doorways and marriage line partitions according to the appropriate method described below.

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### End walls

If sheathing is installed in manufacturing facility, toe fasteners through sheathing from the exterior at an approximately 30 degree angle (+/- 5 degrees) (Figure 23) following the wood screw fastener specification and spacing requirements in Table 18. If the manufacturing facility omits the sheathing and ships it loose for site installation connect the end walls following the lag screw with washer fastener specification and spacing requirements in Table 18. Refer to the applicable setup addendum page for sheathing installation instructions.

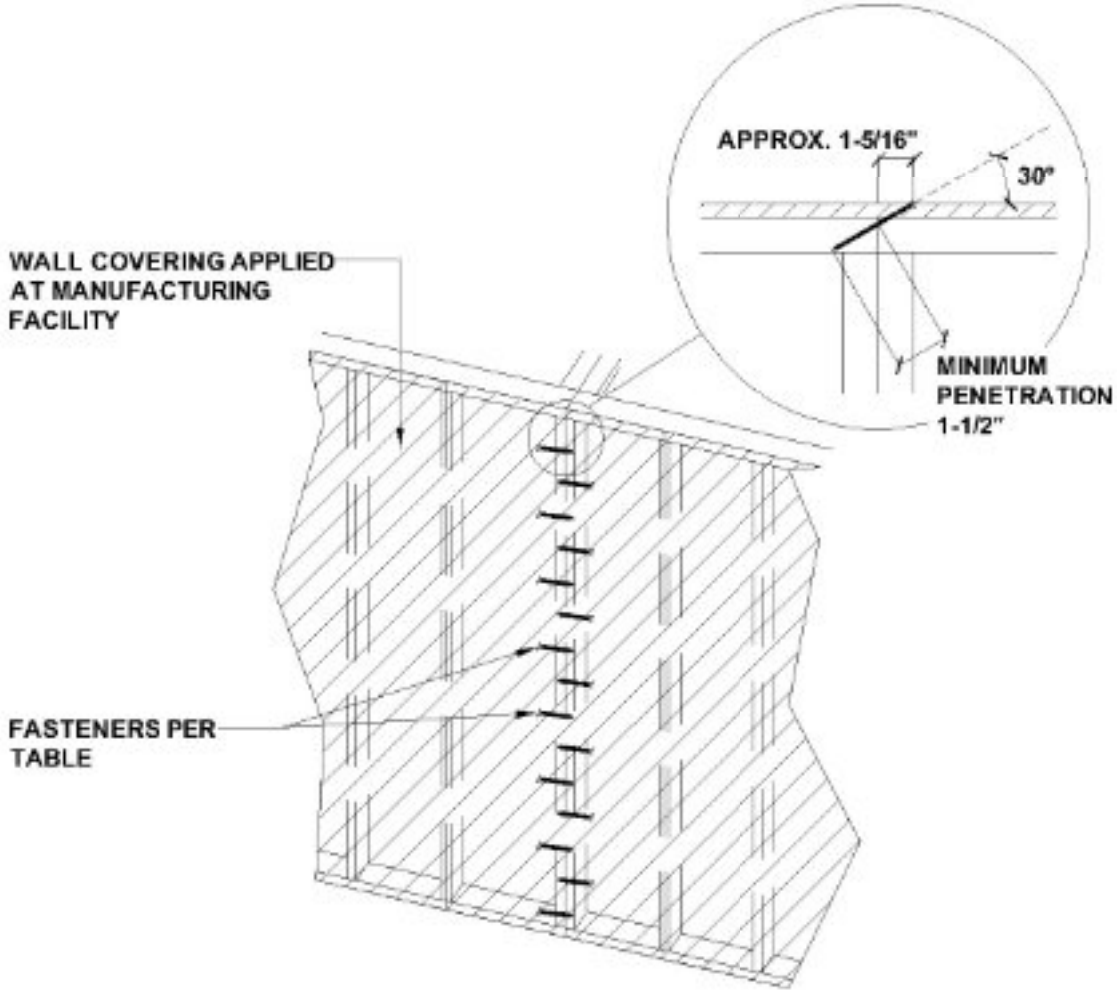



Figure 23. End wall connection

Table 18. End wall fastener specifications

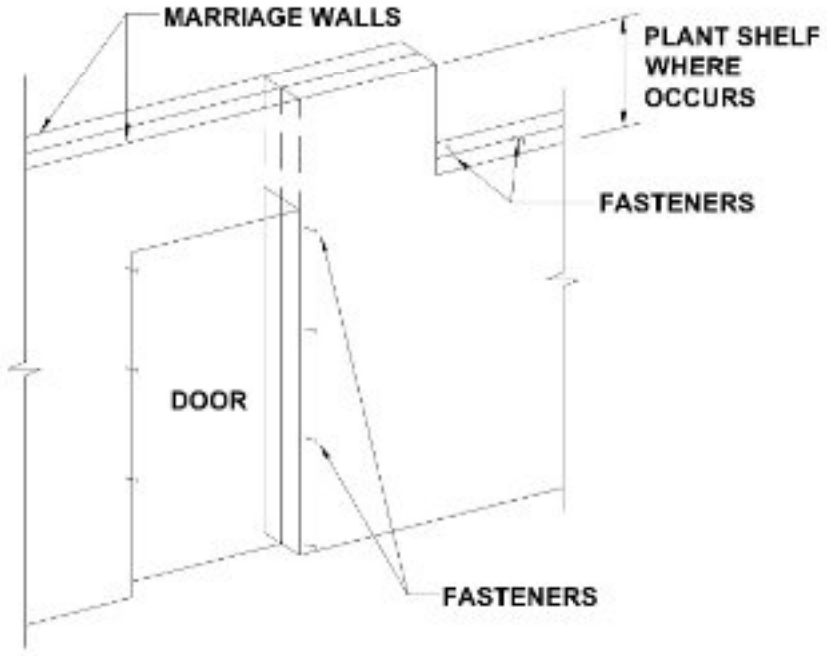
| Fastener type | Size | Spacing |
|---|---|---|
| Wood screw | #10 x 4” | 8in. o.c.<br>(toed through sheathing) |
| Wood screw | #10 x 4” | 8in. o.c.<br>(omitted sheathing, non-toed) |
| Lag screw with washer | 5/16” x 4” | 48in. o.c.<br>(omitted sheathing, non-toed) |

Marriage Wall Columns, Openings, and Interior Partitions (optional)

Make connections inside the home along the marriage walls (Figure 24 and Figure 25) by installing fasteners according to the specifications and spacing requirements in Table 19. Toe screw 2 x 4 or 2 x 6 studs; through-screw 2 x 3 studs. For securement of interior partition walls follow the procedure outlined in End walls (p. 66) using the specifications and spacing requirements in Table 19 below.

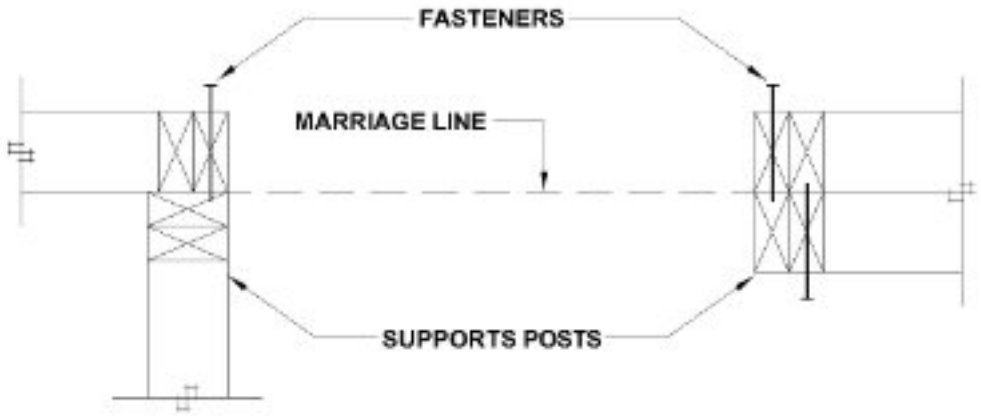
Table 19. Marriage wall connection specifications

| Fastener type | Size | Spacing |
|---|---|---|
| Wood screw | #8 x 4” | 16 in. o.c. |




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Figure 24. Marriage wall connections




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Figure 25. Marriage line column connections

Do additional sections (non tag) remain to be positioned?

► YES, go to step 1. Install Marriage Line Ground Anchors, (p. 60)

► NO, continue below.

Is there a tag unit?

► YES, go to step 9. Attach Tag Units (p. 68)

► NO, continue below.

Is there a hinged roof?

► YES, go to step 3. Complete Hinged Roof, (p. 60)

► NO, go to step 9. Attach tag units, (p. 68)

## Step 9. Attach Tag Units – See Appendix QR Code (p. 8)

## Step 10. Remove Temporary Items

Once the home is properly supported and the marriage line connections are completed, remove the temporary ridge beam supports and wall/ceiling plates used to brace the ridge beam during shipment. Take care not to damage the ceiling.

Is the home supported on a load bearing perimeter wall?

► YES, go to step 11. fasten home to foundation, (p. 69)

► NO, go to Complete Roof and Exterior Walls (p. 70).

## Step 11. Fasten Home to Foundation

Fasten the home to the foundation according to the fastening schedule provided in **Construct Foundation, STEP 1. OBTAIN A FOUNDATION design** (p. 50).



**Cutting the chassis.** Do not cut, notch, bend or alter in any manner beams, cross-members and other parts of the steel chassis.

► go to **Complete Roof** and **Exterior Walls** (p. 70).

# COMPLETE ROOF AND EXTERIOR WALLS

This chapter covers closing up and weatherproofing the home by completing the roofing and siding.

**Follow the Steps below:**


▼ **Step 1. Complete Roof** (p. 70)
▼ **Step 2. Complete Tag Unit Roof** (p. 73)
▼ **Step 3. Complete Side Walls** (p. 73)


## Step 1. Complete Roof

### Ridge Closeup

For multi-section homes, the first step in completing the exterior is sealing the roof along the ridge line (**Figure 26**). For homes with asphalt shingles, follow the procedure below. For homes with metal or other roofing materials, follow the instructions that come with the roofing materials or provided as a supplement to this manual.

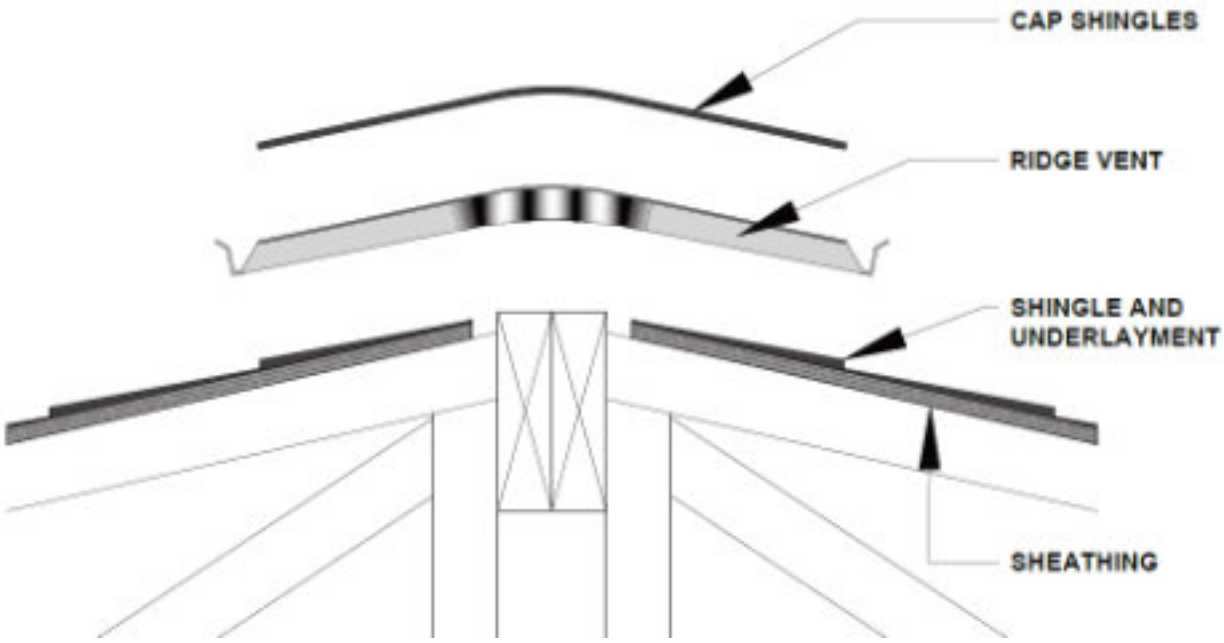



*Figure 26. Shingle installation at ridge*

**Weatherproofing.** It is vitally important to close up the home quickly to protect the interior from damage due to inclement weather.

1. **Install underlayment.** Sheathing must be fastened with an 8d nail at 6 inches on-center along the edges and 12 inches on-center in the field. Seams of field installed sheathing must be offset 16 inches from the seams on factory installed sheathing. Install 15# felt or equivalent continuously along the length of the ridge, covering all exposed sheathing and overlapping sheathing joints by at least five inches on each side. Fasten using 1” x 1” x 16 ga galvanized staples. For Wind Zones II and III, also apply a six inch wide strip of roofing cement along each side of ridge under the underlayment.
2. **Install shingles.** If shingles have been left off at the ridge line for site installation, install them now using 12 ga x 1-1/4” long, 3/8” diameter head roof nails or 16 ga 1” crown x 1” length staples. For Wind Zone I, fasten at 5/8 inch above each tab cutout slot and one at each end of the shingle one inch in from the edge (four fasteners for a three-tab shingle) (**Figure 27**). For Wind Zones II and III, use two fasteners 5/8 inch above and on either side of the tab slots and one at each end of the shingle one inch in from the edge (six fasteners for a three-tab shingle). Do not fasten through the shingle tar line.



**Removing shipping protection.** Remove shipping protection from the roof prior to completing roofing. Seal all holes in shingles resulting from shipping protection removal. It is also recommended that the top layer of shingles be lifted and the sealant applied to any holes in the second layer of shingles. **Refer to Warning on following page for acceptable types of sealants.**

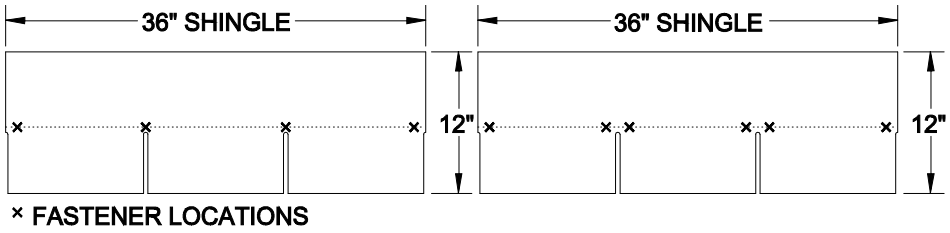



*Figure 27. Shingle fastener locations for Wind Zone I (left) and Wind Zones II and III (right).*

3. **Install underlayment.** Install 15# felt or equivalent underlayment that is at least 10 inches wide continuously along the ridge. Fasten with 1” x 1” x 16 ga galvanized staples. For Wind Zones II and III, also apply a six inch wide strip of roofing cement on both sides of the ridge under the underlayment.
4. **Install shingle cap**. Starting at the opposite end of the home from the prevailing wind, install ridge cap shingles provided by the manufacturer or use 12” x 12” shingles (36” shingles cut into three equal pieces indicated by the dotted line in **Figure 28**). Install using 12 ga x 1-1/4” long, 3/8” diameter head roofing nails spaced 5-5/8 inches from bottom edge and 1/2 inch to 1-1/2 inches in from both edges.

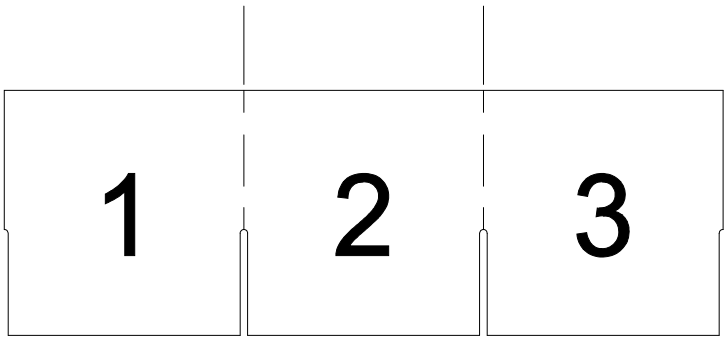



*Figure 28. Shingle cut into thirds*

5. **Ridge Vent**. For proper alignment – pre-nail through ridge vent holes at ends of each section. The felt paper underlayment should be folded back onto the roof decking and trimmed along the edge of the roof decking at the peak so that it does not obstruct the opening at the mate line. When using standard flat 3-tab shingles, caulking is not required under the flange of the ridge vent. Prior to applying vent to dimensional or architectural shingles on new construction, caulk between low areas of shingle and flange of vent. Before fastening vent, make sure filter is secured between shingles and vent. When installing vent in cold weather, leave an 1/8" gap between sections to allow for warm weather expansion.



**Exposed shingle fasteners.** Covering exposed fasteners with tar or cement is not an acceptable method of repair and is not covered under warranty by shingle manufacturers. Over time, the sun will dry out tar or shingle cement which may lead to future roof leakage. If any exposed fasteners are present after shingle installation the entire shingle must be removed and a new shingle must be reinstalled with properly installed fasteners.

**Installing a ridge vent.** If a ridge vent is to be installed, follow the ridge vent manufacturer's instructions provided with the material or as an addendum to this manual in lieu of underlayment over ridge line and shingle cap.

**ACETOXY TYPE SILICONES ARE NOT TO BE USED.** This type of silicone will weaken or melt ashphalt shingles. The Alcoxy or Neutral Cure type silicones will not melt asphalt and are the proper silicones to be used in roof applications. All sealants used in shingle and roof applications, excluding underlayment securement, must be an Alcoxy or Neutral Cure type silicone OR meet the ASTM D 4586 standard. Asphalt/Asbestos based sealants, such as Black Tar, must meet the ASTM D 4586 standard. Shingle underlayment sealants are required in Wind Zone II and III applications and must meet the ASTM 3019 standard.

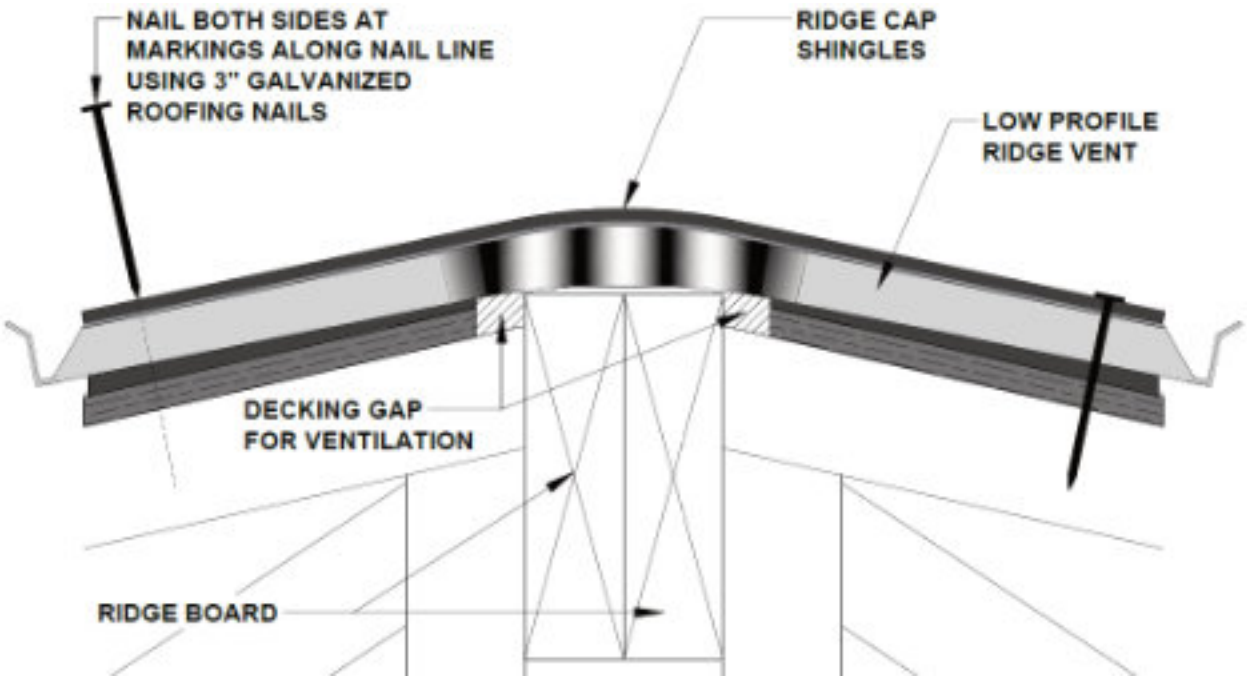



*Figure 29. Ridge vent installation*

**Complete Triple Section Roofs**

Complete roofing along triple section home marriage lines according to the appropriate methods of the Appendix QR code (p. 8) based on the construction of the home.

## Step 2. Complete Tag Unit Roof – Scan Appendix QR Code (p. 8)

If the home has a tag unit, complete roofing for this unit now. The process for completing the roof is different for flush and offset roofs. Follow the instructions in the appropriate section of the appendix QR code.

## Step 3. Complete Side Walls

Siding necessary to complete the exterior has been provided with the home. Follow the siding manufacturer's instructions (found on or with the packaging or as an addendum to this manual) and to complete the exterior siding as follows:

1. **Remove shipping protection.** Remove temporary shipping protection from walls.
2. **Complete crossovers.** Complete any crossover connections in the walls, including: electrical, stereo speaker, doorbell, telephone and intercom wires.
3. **Install siding.** Fasten siding only at stud locations, avoiding electrical wires that are present in the walls.
4. **Fasteners.** Fasteners must be installed as described in the manufacturer's installation instructions or DAPIA approved test reports for the siding.

5. **Install close-up strips.** If siding has been installed on the end walls at the factory, fasten close-up strips securely along both edges and seal the edges with a waterproof sealant.
6. **Install trim.** Install any matching trim required to complete the installation.
7. **Seal penetrations.** With a waterproof sealant, seal any penetrations in the siding that may have been caused by temporary shipping protection.

**Covering the HUD label.** Do not cover the HUD label on the exterior of the home.



**Removing shipping protection.** Remove temporary shipping protection from walls before installing siding or serious moisture damage may result. Weather resistive barrier (such as Tyvek or other similar product) will be installed over the exterior wall OSB sheathing and under the shipping plastic. Be careful not to damage the weather resistive barrier when removing the shipping plastic.

► go to **Connect Crossovers** (p. 75).

# CONNECT CROSSOVERS

This chapter covers crossover connections between units of multi-section homes, including ducts and electrical, water, waste, gas, telephone and cable TV connections.

**Follow the Steps below:**


▼ **Step 1. Connect Ducts** (p. 75)
▼ **Step 2. Connect Electrical Crossovers** (p. 80)
▼ **Step 3. Install Electrical Bonding** (p. 84)
▼ **Step 4. Connect Water Lines** (p. 85)
▼ **Step 5. Connect Drain, Waste and Vent Lines** (p. 87)
▼ **Step 6. Connect Gas Lines** (p. 90)
▼ **Step 7. Connect Telephone and Cable tv Wiring** (p. 90)


## Step 1. Connect Ducts

There are three main types of duct crossover connections. Based on the location of the duct, follow the installation steps on the page indicated below:

- Under the floor (p. 76).
- In the roof cavity (p. 79).
- In floor, through-the-rim joist (p. 79)

To prevent air leakage, seal all ductwork connections, including duct collars using one or more of the following materials:

- Galvanized metal straps in combination with galvanized sheet metal screws.
- For rigid air ducts and connectors, tape and mastics listed to UL 181A.
- For flexible air ducts and connectors, tape and mastics listed to UL 181B.

### Under Floor Flexible Crossover Duct

When heating or cooling equipment is installed in the home, the flexible crossover duct is provided by the manufacturer. In all cases the crossover duct must be listed for exterior use and should be wrapped with insulation of at least R-8 under a vapor barrier with a perm rating of not greater than one.

There are four common configurations of under floor crossover ducts depending on the number of home sections and the furnace/air handler location. See **Table 20** to locate the appropriate figure.

*Table 20. Under floor duct configurations*

| Furnace location | Two home sections | Three home sections |
|---|---|---|
| Furnace over trunk duct | See Figure 30 | See Figure 32 |
| Furnace offset from trunk duct | See Figure 31 | See Figure 33 |

**Access for service.** Whenever possible maintain access to connection areas for future maintenance.

**Make tight connections.** Permanent, durable and tight crossover duct connections are critical to the proper performance of the home. Leaky ducts can result in severe moisture problems in the home, discomfort from rooms not receiving the proper amount of conditioned air and high utility bills from wasted heating and/or cooling energy.

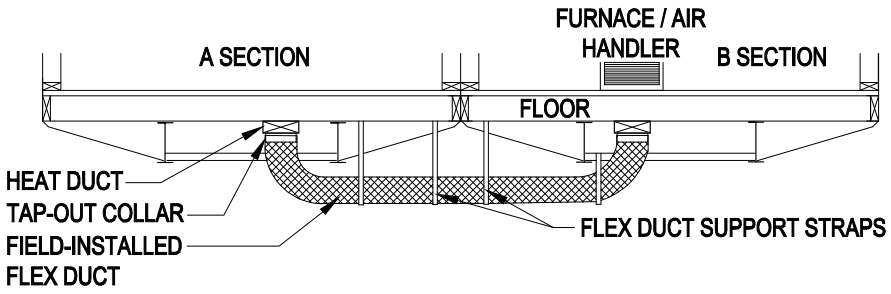



*Figure 30. Furnace over trunk duct, two home sections*

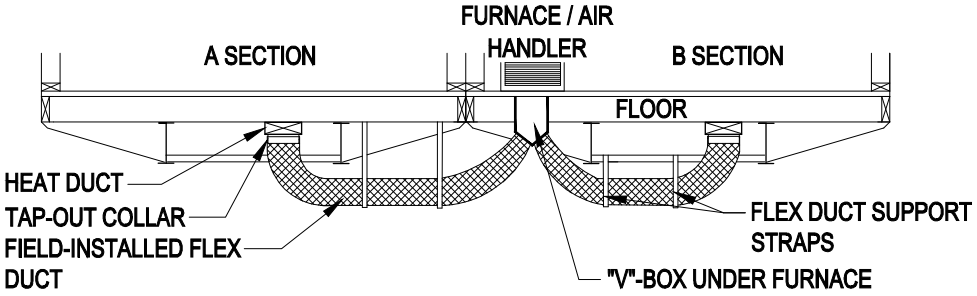



*Figure 31. Furnace offset from trunk duct, two home sections*

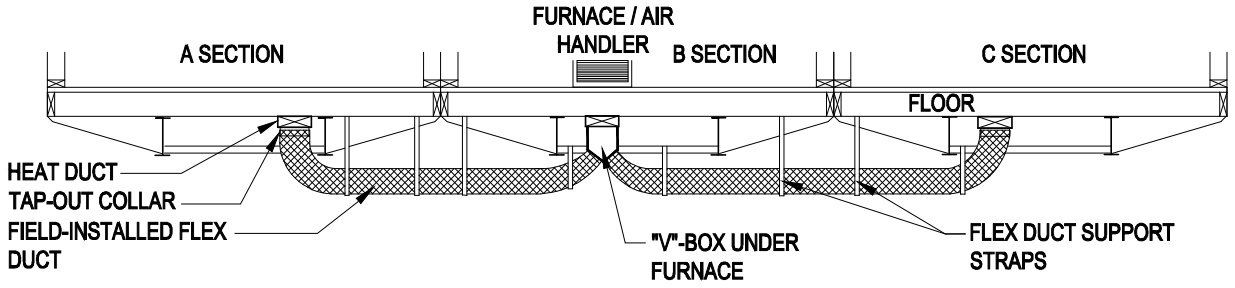



*Figure 32. Furnace over trunk duct, three home sections*

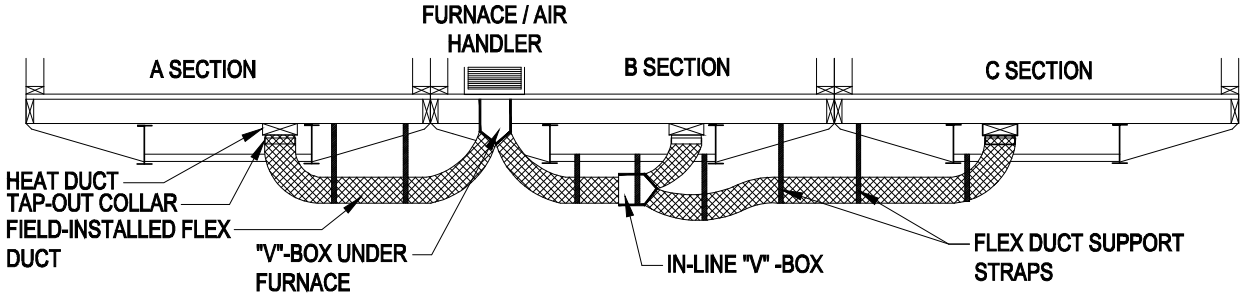



*Figure 33. Furnace offset from trunk duct, three home section*

### Under Floor Flexible Crossover Duct

For under floor flexible crossover ducts follow the steps below:

1. **Locate collars.** Locate the metal crossover collars (or V-box) connected to the main trunk duct (or furnace) under the home and remove temporary shipping protection.
2. **Install inner duct.** Apply mastic completely over inner liner and collar/V-box. Slide the crossover duct inner liner over the crossover collar/V-box as far as it will go. Install a large nylon zip tie over the inner liner just above the “ridge” around the crossover collar/V-box. Apply mastic completely over inner liner and collar/V-box.
3. **Connect duct insulation.** Bring the duct insulation up over the zip tie and above the home’s bottom board into the floor cavity. Temporarily duct tape it against the base of the trunk duct/V-box.
4. **V-box Insulation.** Verify that the V-box has been insulated with R-8 minimum.
5. **Pull duct wrap.** Pull the crossover duct outer wrap over the top of the insulation and temporarily secure it to the trunk duct/V-box with duct tape.
6. **Install zip tie.** Feel for the nylon zip tie that was installed over the inner liner. Place another nylon zip tie just under the first one to permanently secure the crossover duct insulation and outer wrap, making sure all of the insulation is inside the outer wrap.
7. **Trim duct.** Trim the crossover duct to length such that the installed duct will be straight with no kinks or unnecessary bends.

8. **Connect other end.** Follow the same procedure (steps 1 through 5) to connect the opposite end of the crossover duct and any other crossover ducts.
9. **Seal joints.** Seal the joints between the bottom board and the crossover duct with bottom board repair tape.
10. **Support duct.** Support the crossover duct(s) above the ground using nylon or galvanized metal straps and saddles spaced every 48 inches o.c. or less. Choose straps at least 1/2 inch wider than the spacing of the metal spirals encasing the crossover duct. Install the straps so they cannot slip between spirals. Secure metal straps with galvanized screws.

**Roof Cavity Crossover Duct**

For ducts installed in the roof cavity, follow the steps below:

1. **Access the duct.** Access the crossover location through an access panel in the ceiling or the open sides of the home before the sections are joined and remove any temporary shipping protection.
2. **Join ducts.** Using the provided flexible duct, join the distribution boxes in each section of the home as shown in **Figure 34**, cutting off any extra duct length to keep the duct as straight as possible. The duct must be joined in the center with the provided connecter.
3. **Fasten ducts.** At each connection point between ducts and distribution boxes or connectors, secure the inner duct liner with a nylon strap, apply mastic completely over the connection area, pull the duct insulation and outer liner over the connection area and secure them with a second nylon strap.
4. **Reinstall panel.** Reinstall and secure the access panel, if applicable.



**Cover exposed metal.** Completely cover all exposed metal connectors with insulation. Apply sealants and tapes only to surfaces that are dry and free of dust, dirt, and grease.

**Avoid ground contact.** Installed crossover ducts must not be in contact with the ground.

**Unobstructed airflow.** Excess length, kinks and bends in the crossover duct will restrict airflow and degrade the home's HVAC system performance.

**Compressed Duct.** Support the duct without compressing the insulation and restricting airflow.

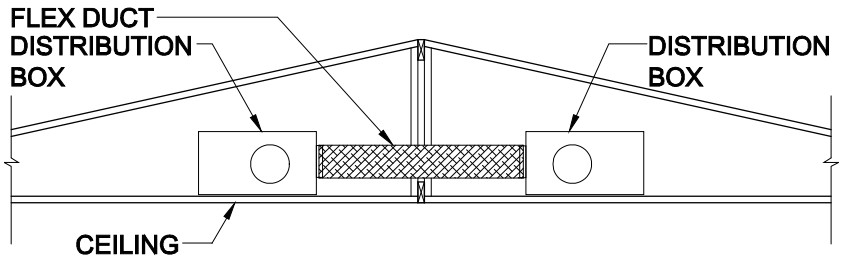



*Figure 34. Duct crossover located in the roof cavity*

### In the Floor Crossover Duct

Where one or more crossover ducts are built into the home's floor system, connect them either through or under the rim joist depending on the design of the home.

### Through the Rim Joist

With a through the rim joist design, the duct in each floor section terminates at an opening in the marriage line rim joist. Fixed through-the-rim crossovers employing a marriage line gasket such as in **Figure 35** were completed in **Complete Multi-Section Set, Step 4. Repair or install marriage line gasket** (p. 61) and require no additional work here. Connect other through-the-rim joist ducts using one of the following methods based on the design of the home.

### Method 1: Metal or Duct Board through Rim Joist without Sleeve

Connect ducts that pass through the rim joist (**Figure 35**) as follows:

1. **Remove Shipping Support.** If applicable, remove any strips of plywood, luan, or OSB that were factory installed as temporary shipping support for the ducts.
2. **Align Crossover duct locations.** Verify that when both halves of the home are installed that the crossover duct locations will align properly.
3. **Air tight Seal.** Verify that the gasket or duct board used to seal between both halves of the home is in good condition and will properly seal the duct system.
4. **Connect the Units.** Connect the floors using the procedure for connecting the floors (p. 62).

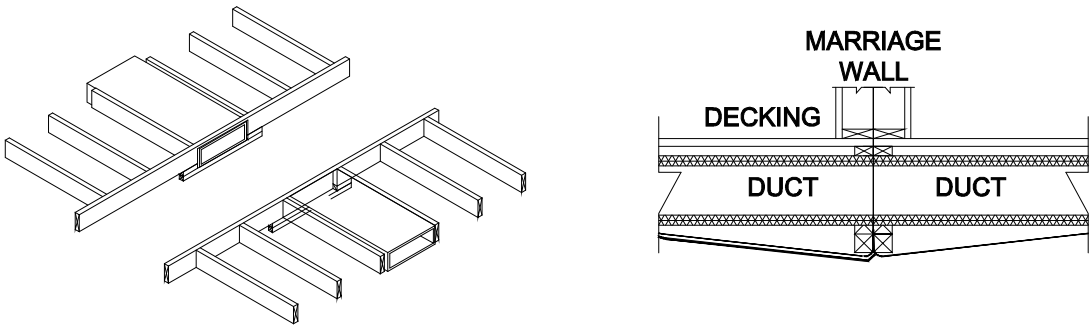



*Figure 35. In-floor duct connection through the rim joist without metal sleeve*

**Method 2: Duct Board with Sleeve**

Join duct board ducts with a metal sleeve as follows (**Figure 36**):

1. **Open bottom board.** On the section of the home with the furnace, cut the bottom board along the center line of the two floor joists on either side of the cross over duct starting at the marriage line and extending approximately three feet toward the center of the section.
2. **Open duct.** Create an opening in the duct by cutting the duct board as shown in **Figure 36**.
3. **Insert sleeve.** Insert the provided metal sleeve, centering it on the marriage line joint.
4. **Seal duct.** Close the bottom of the duct and seal it with tape specially made for that purpose (may be provided).
5. **Seal floor.** Replace the floor insulation to its original position and seal the bottom board tightly with tape specially made for that purpose.

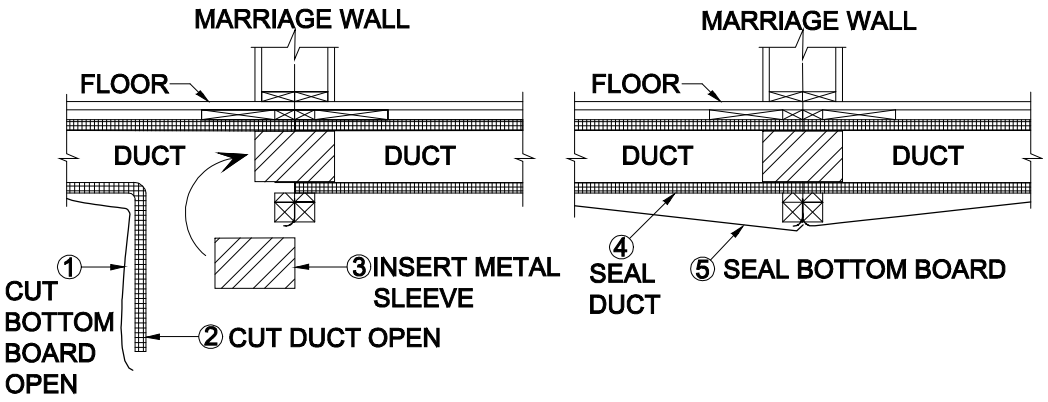



*Figure 36. In-floor duct with crossover using metal sleeve*

## Step 2. Connect Electrical Crossovers

Multi-section homes may have one or more electrical crossovers located in the wall(s) and/or floor(s) along the marriage line(s).

**Joining Wires**

Two types of connections may be present at these locations—snap connectors and junction boxes. Identify matching circuits if multiple circuits exist at a single crossover location. These will be coded for identification. Connect snap connectors according to the connector manufacturer's installation instructions, including fastener requirements.

Connect wires in junction boxes as follows (**Figure 37**):

1. **Pull wires.** Pull circuit wires into the junction box, sliding them through a romex connector and secure snugly. (**Figure 37**) Do not over-tighten.
2. **Strip wires.** Remove the outer jacket that holds the circuit wires together providing a minimum of four inches of free wire in the box.

3. **Connect wires.** Connect wires together matching like colors, using appropriately sized wire nuts. Use the ground wire to ground the junction box and/or cover plate(s), if metal. Junction boxes may contain single or multiple 15 or 20 amp circuits, or a single 240 volt appliance circuit.
4. **Replace cover.** Reposition the junction box cover and secure using machine (not sheet metal) screws.



**Use qualified electricians.** All electrical work must be performed by a qualified electrician and comply with the 2014 NEC.

**Disconnect power.** Turn off power to the home before making connections.

**Proper use of GFCI circuits.** Ensure Ground Fault Circuit Interrupted (GFCI) circuits are connected to the proper GFCI protected circuits from the power supply.

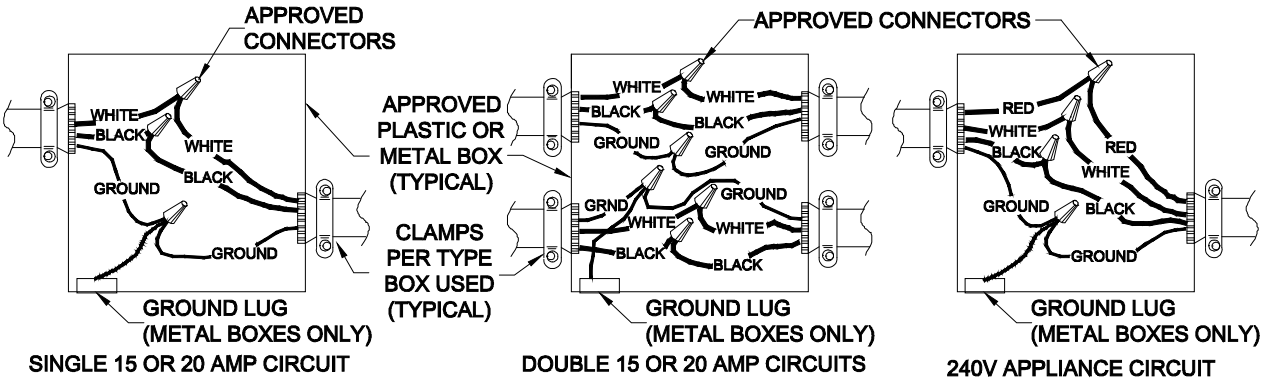



*Figure 37. Types of junction box wiring connections*

### Floor Crossover

When making electrical connection(s) in the floor, use one of the wiring options described below:

### Method 1: Access Panel

1. **Access wires.** Find the crossover location(s) and remove the access panel(s) if attached. If access panel openings are not provided, cut through bottom board to expose the wiring (**Figure 38**).
2. **Route wires.** Pass the wires through predrilled holes or notches in the rim joist or if there is a single bumped-out access panel as in **Figure 39**, then connect under the rim joists.
3. **Connect wires.** Connect wires via a junction box or snap-connector(s) as described above.
4. **Secure wires.** Secure wires with staples to adjacent joists or studs within eight inches of junction box or snap-connector(s).

5. **Install smash plates.** For notched perimeter joists, install steel wire protectors (smash plates).
6. **Seal bottom board.** Replace insulation and re-install access panels and/or seal the bottom board with tape specially made for that purpose (may be provided). The access panel(s) may be temporarily installed near the crossover location or shipped loose with the home.

**Using snap connectors.** Do not use oversized nails or drive nail heads into snap connectors. Some connectors are designed for one-time use only – a new connector must be used if they become separated.

**Protect cables.** Cover all cables with conduit or other suitable weather-resistant and protective material.

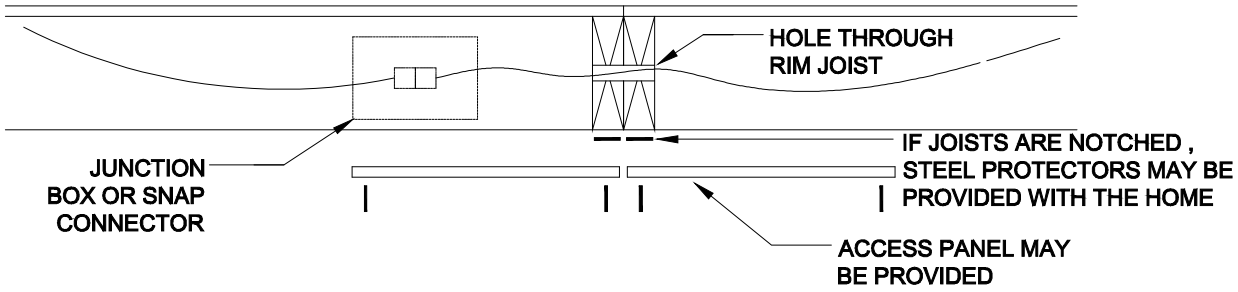



*Figure 38. Electrical crossover floor wires with flush access panels*

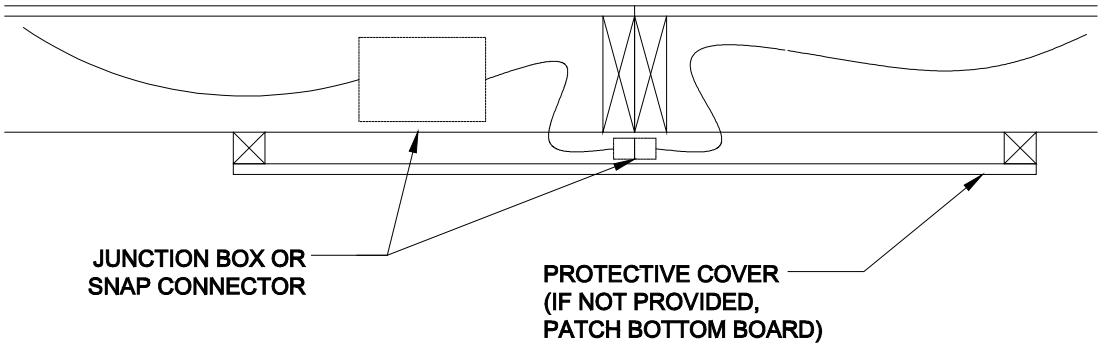



*Figure 39. Floor electrical crossover wires with bumped-out access panel*

**Method 2: Junction Box with Conduit**

1. **Access boxes.** If junction boxes are not exposed, access them inside the floor by cutting the bottom board or by removing the access panel(s). Remove the junction box covers.
2. **Connect wires.** If wire is coiled inside one junction box, insert it into flexible conduit and pass it under the rim joists to the opposing box and make the connection as described above. If no coiled wire is provided, install conduit with wire making connections in both boxes (**Figure 39**).
3. **Cover boxes.** Replace and secure covers on junction boxes.

4. **Seal floor.** Replace any displaced insulation and replace access panel or seal the bottom board with tape specially made for that purpose (may be provided).
5. **Variation to Method 2:** A junction box may be installed only on one side with the other side containing conduit behind an access panel. Bring the conduit to other side and make one connection in the junction box.

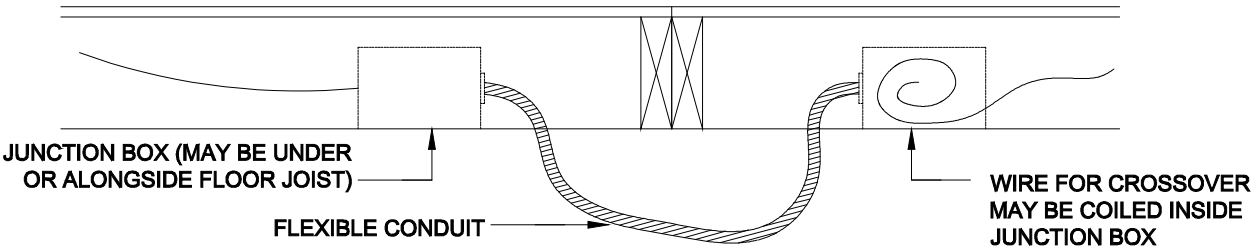



*Figure 40. Under the rim joist electrical crossover connection with conduit*

### Wall Crossovers

Connect electrical, phone, cable television and stereo speaker wires in marriage walls and/or partition and end walls.

If access panels into the marriage walls are not provided, then the connection should have been made prior to bringing the home sections together. If access panels are provided in marriage, end or partition walls, remove the panels, join the wires as described above using the provided snap-connector, junction box or at a receptacle, and re-attach the access panel (**Figure 41** for marriage walls and **Figure 42** for partitions and end walls).

**Avoid damaging crossover wires.** Carefully fold marriage wall crossover wires so they stay within a single bay and are not sandwiched between studs when the sections are pulled together.

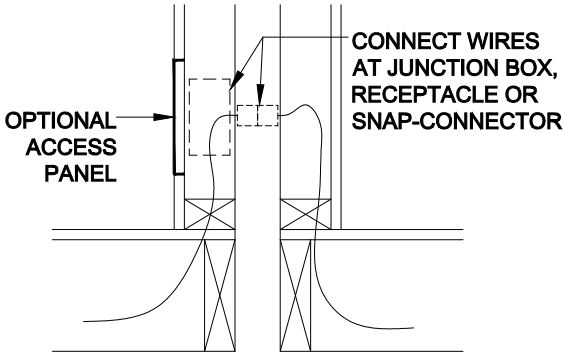



*Figure 41. Inside marriage wall crossover connection*

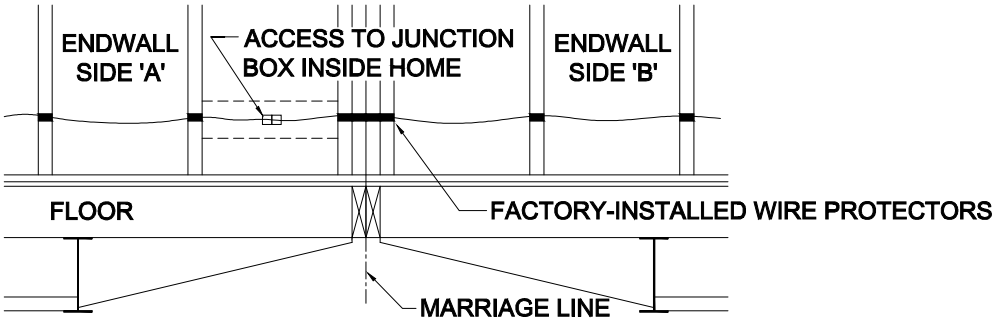



*Figure 42. Inside partition or end wall electrical crossover connection*

## Step 3. Install Electrical Bonding

To ensure all metal parts are effectively grounded, electrically bond all chassis together as follows (**Figure 43**):

1. **Find lugs.** Determine if solderless ground lugs are provided on the front or rear frame outriggers or headers.
2. **Attach wire.** If lugs are provided, uncoil the bonding wire (#8 minimum bare copper wire) from one side of the home and connect it to the lug provided on the opposing side using a paint penetrating star washer, tighten the set screw firmly on the wire and repeat for any additional home sections. Torque the set screw per the manufacturer's requirements.

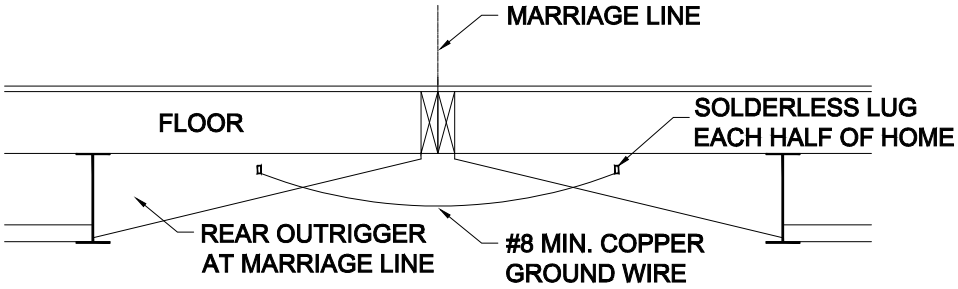



*Figure 43. Electrical bonding of multi-section homes*

3. **Attach strap.** If ground lugs and copper wire are not provided, attach the provided four inch bonding strap to each pair of adjacent chasses with two #8 x 3/4" self-tapping metal screws (one screw each side).

## Step 4. Connect Water Lines

Connect water lines inside the floor through access panels or below the bottom board as follows:

### Water Lines Accessed through Panels

1. **Remove panels.** Remove access panels from each home section.
2. **Remove caps.** Remove shipping caps from ends of water lines, if present.
3. **Pull lines.** Pull water lines through holes in rim joist or attach shipped loose flex connectors, if applicable (**Figure 44**).
4. **Connect pipes.** Connect threaded water lines using flexible pipe or a rigid connector line (if provided) and connector fittings (do not use lubricants or sealants).
5. **Test.** Test connections for leaks.
6. **Seal floor.** Securely replace insulation and access panels.



**Applying cement.** Follow cement manufacturer's instructions with respect to application and drying time. Allow cement to fully cure before filling pipes with water.

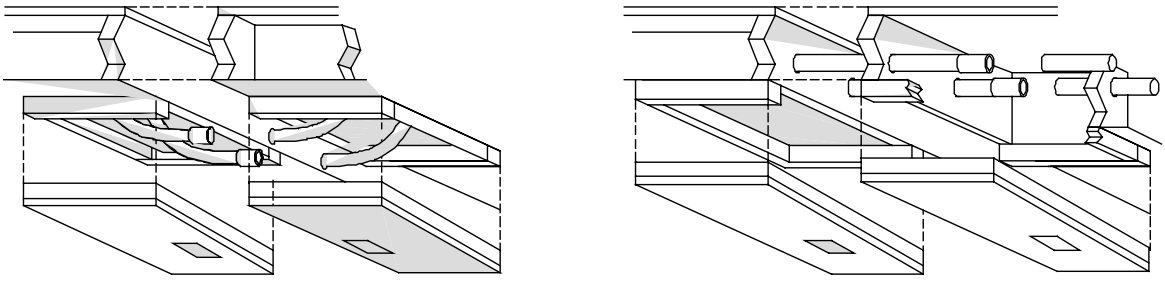


*Figure 44. Water line crossover with access panels*

**Water Lines Dropped below Bottom Board**

1. **Remove caps.** Remove protective shipping caps from ends of pipes and make sure pipe ends are clean and smooth.
2. **Connect pipes.** Connect threaded water lines using flexible pipe or a rigid connector line (if provided) and connector fittings (do not use lubricants or sealants) (**Figure 45**).
3. **Test.** Test connections for leaks.
4. **Protect pipes.** Wrap water lines with insulation and bottom board shipped loose with home or otherwise protect to prevent freezing. Tape bottom board using tape specially made for that purpose (may be provided) and staple with 7/16” x 1/2” staples at four inches o.c. around bottom board using a divergent (stitch) stapler or equivalent. Generally, only insulation is necessary to protect water supply lines from freezing when the home is skirted. Some homeowners may desire to protect their water supply lines with a heat tape. This tape must be approved for manufactured home use by a nationally recognized testing agency and be installed in compliance with manufacturer’s instructions. An electrical outlet has been provided under the home for the heat tape. This outlet is protected by a Ground Fault Circuit Interrupter and should not be used for any other purpose.

**Choosing cement type.** Use the proper cement for water lines as it may differ from the cement used for the DWV system.

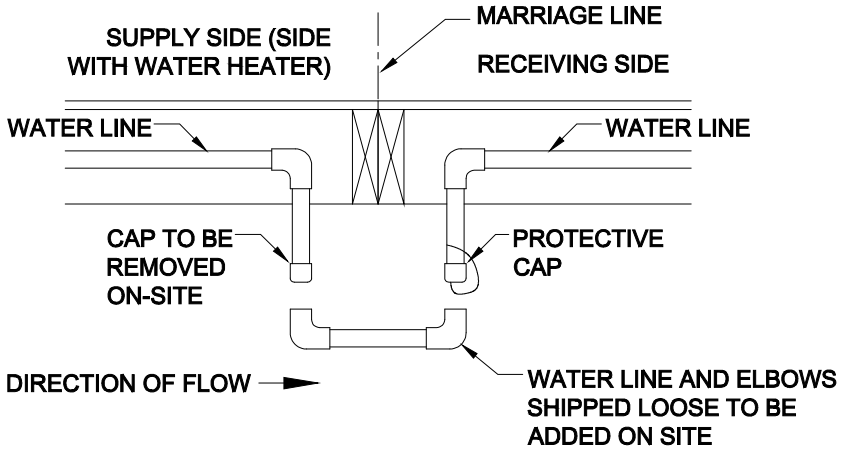



*Figure 45. Water line crossover through bottom board*

## Step 5. Connect Drain, Waste and Vent Lines

Complete portions of the drain, waste and vent (DWV) system that are below the floor as follows:

1. **Remove caps.** Remove shipping covers from pipes extending through the bottom board; inspect pipes and fittings and clean them of dirt, obstructions and burrs.
2. **Assemble pipes.** Using the drain schematic drawing provided with the home, begin assembling the DWV system starting at the location farthest from the sewer/septic connection and working towards the outlet, fastening the pipe with cement or adjustable screw-clamp connectors, if provided (**Figure 46**). As the system is assembled, support the piping with temporary blocking. Unless otherwise noted on the schematic diagram, provide a minimum 1/4 inch per foot slope towards the sewer/septic using a plumber's level. Where a slope of 1/4 inch per foot cannot be maintained, use a minimum slope of 1/8 inch per foot and install of a full-size clean-out at the uppermost point of the run (**Figure 47**).



**Providing required clearances.** Provide the drain outlet with a minimum clearance of three inches in any direction from all parts of the structure or any appurtenances and with not less than 18 inches unrestricted clearance directly in front of the drain outlet. Provide any cleanouts with a minimum clearance of 12 inches directly in front of its opening.

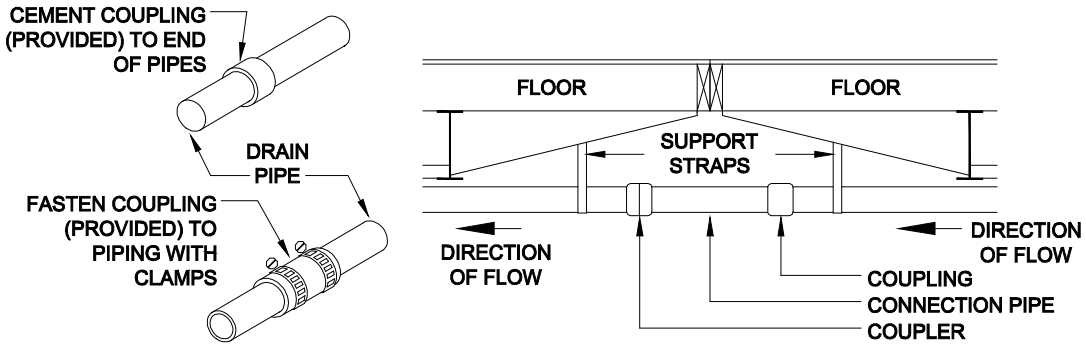



*Figure 46. Drain crossover connection*

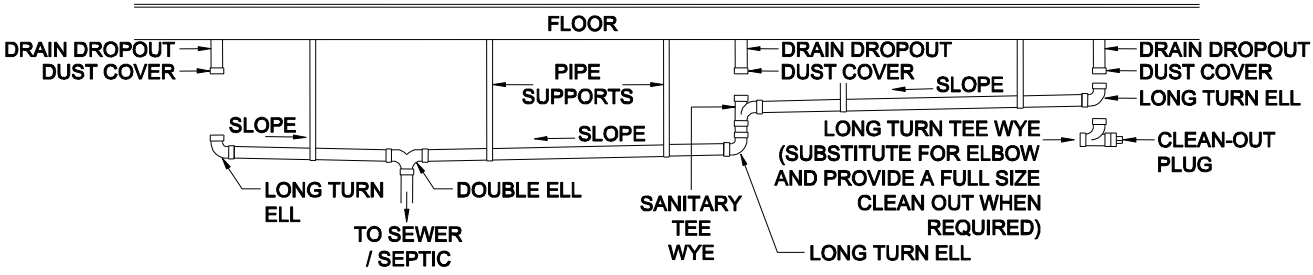



*Figure 47. DWV system*

3. **Test.** After all drain lines have been connected, conduct a two-part leakage test on the completed drainage system as follows:
    - **Part 1.** With all fixtures connected, and all tub and shower drains plugged, fill the system with water to the rim of the toilet bowl through a higher fixture. Release all trapped air, replace tub and shower plugs, backfill fixtures, and allow the system to stand at least 15 minutes. Check for leaks. Drain the system. If leaks are found, repair and retest.
    - **Part 2.** Plug all fixtures, sinks, showers, and tubs and fill with water. Release the water in all fixtures simultaneously to obtain the maximum possible drain piping flow. As water is draining, check for leaks. If any are found, repair and retest.
4. **Connect to outlet.** Connect the main drain line to the site sewer/septic hook-up, using an approved elastomer coupling or by other methods acceptable to the LAHJ (Figure 48).



**Choosing glues.** Use only solvents and glues compatible with the pipe (ABS or PVC). Follow manufacturer's instructions.

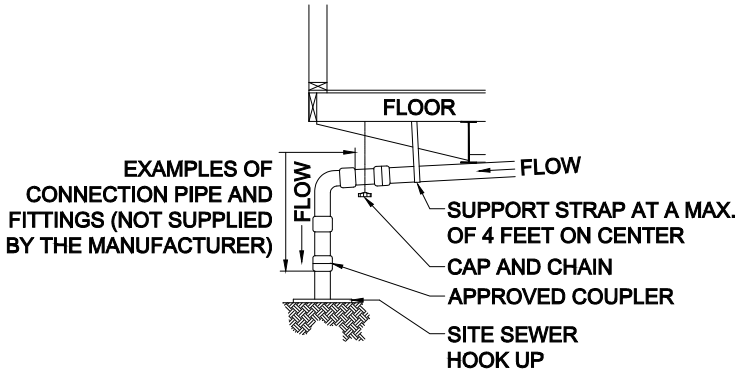



*Figure 48. DWV connection to sewer/septic*

5. **Install supports.** Install permanent drain line supports at a distance of four feet o.c. or less (see Figure 49). Alternate DWV pipe support may be used if approved by the local authority having jurisdiction.

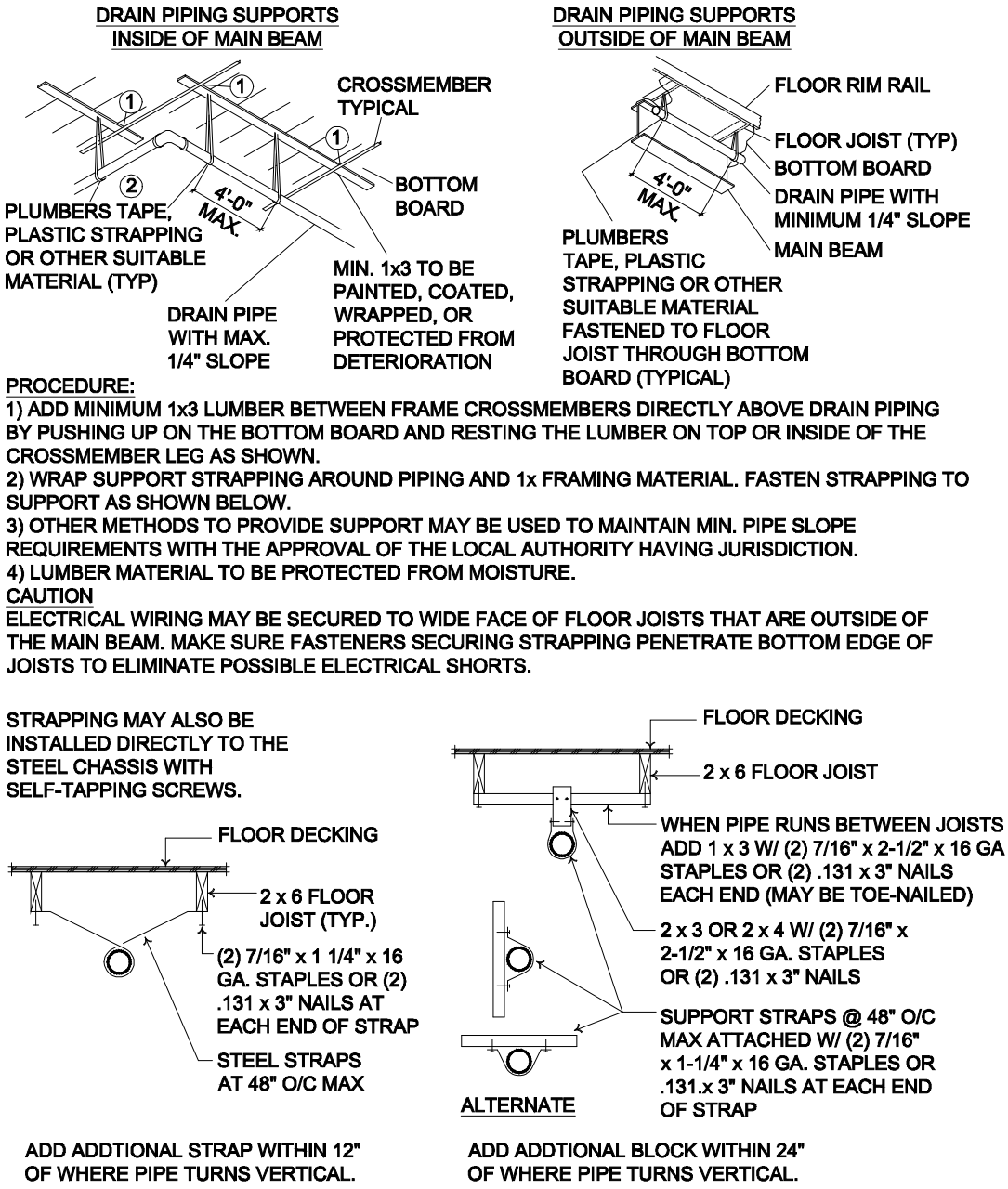



*Figure 49. DWV pipe support options*

6. **Insulate.** Replace all insulation and repair any tears or openings in bottom board.



**Protect pipes from freezing.** If the home is to be left unheated in cold weather, pour antifreeze solution into all drain traps, including sinks, tubs and toilets. Be sure that the antifreeze is safe for the fixtures and P-traps.

**Installing quick disconnect fittings.** Do not use tools, lubricants or sealants with quick disconnect fittings.

## Step 6. Connect Gas Lines

The gas crossover connection may use quick disconnect fittings or threaded connectors. Find the connection location below the floor at the marriage line and gather the connectors (they may be shipped loose if not present under the home).

For quick disconnect fittings (**Figure 50**), remove any dust caps and then with one hand, pull back on the quick disconnect device, snap it over the quick disconnect adaptor and release it to complete the connection.

For threaded connectors, remove the black cap and nipple (or any other plugs, such as a black iron plug) from both the supply and receiving sides and screw the connector onto the supply and return pipes as necessary.

Check for leaks before connecting to gas service (see **Connect Utilities, Step 3. Connect Gas Service,** p. 127).

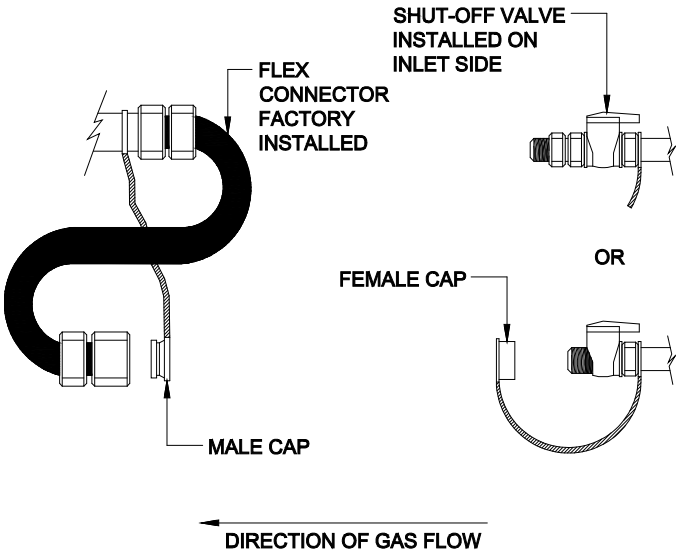



*Figure 50. Gas crossover connection*



**Gas Crossover.** A quick disconnect may be installed on the gas line crossover, but it may not replace the shut-off valve. A shut-off valve is required on the inlet side of the gas line crossover.
**Gas test precautions.** Do not connect to gas service until tests described in Connect Utilities have been successfully completed.

## Step 7. Communication Wiring

Install communication wiring in accordance with LAHJ, NEC, and NFPA No.70 2005. Seal bottom boards accordingly to hide exposed wire.

► go to **Complete the Interior** (p.91)

# COMPLETE THE INTERIOR

This chapter covers the completion of the home's interior finishes including finishing walls, ceilings, flooring, trim and miscellaneous items.

**Follow the Steps below:**


▼ **Step 1. Align Marriage Walls** (p. 91)
▼ **Step 2. Finish Gypsum Board** (p. 91)
▼ **Step 3. Complete Trim** (p. 92)
▼ **Step 4. Install Ship Loose Items** (p. 92)


## Step 1. Align Marriage Walls

Align and secure walls at marriage line openings as follows:

1. **Align walls.** Align walls and clamp in place.
2. **Fasten walls.** Secure walls together when necessary.
3. **Fill gaps.** Fill any remaining gaps up to 1 inch with wood or sheathing material. If gaps over 1 inch are present the home should be repositioned accordingly.

## Step 2. Finish Gypsum Board

Finish all unfinished gypsum board walls and ceilings as follows:

1. **Install panels.** Install ship loose gypsum panels using a 1/4 inch diameter bead of polyvinyl acetate (PVA) adhesive on all framing members and minimum 1-1/2 inch long drywall screws, nails or staples at six inches o.c. along panel edges and 12 inches o.c. in the field into framing members. Adhesive is not required when installing gypsum pieces used for trim or close-up materials.
2. **Mud seams.** Mud and tape all seams and corners, filling all fastener depressions. Follow mud manufacturer's directions.
3. **Paint.** When the final coat of mud is dry, sand, prime and paint all unfinished gypsum board to match existing paint color and finish texture.



**Fastening gypsum.** When attaching gypsum board, depress, but do not break the paper face with the fastener. Breaking the paper will weaken the connection.

**Using alternative materials.** Obtain the home manufacturer's approval before using interior finish materials other than those provided with the home.

## Step 4. Complete Trim

Using fine gauge wire staples or pin nails install ship loose molding and wainscot paneling to finish trimming out ceilings, marriage line walls, front and rear end walls and passageway doors where necessary.

## Step 5. Install Ship Loose Items

Remove all strapping, blocking and packaging from appliances, windows and doors. Install any drapes, mini-blinds, mirrors, door stops, closet shelves and hardware per the product manufacturer's installation instructions.

Carpet Manufacturers will not warranty their product if carpet seam sealer is not used at the seam.

Proceed based on the foundation type:

► **Pier and ground anchor,** go to **Install Stabilizing Systems,** (p. 93)

► **Load bearing perimeter wall,** go to **Connect Utilities** (p. 119).

# INSTALL STABILIZING SYSTEMS

This chapter covers the design and installation of the stabilizing system which secures the home against lateral and upward forces caused by wind. The system covered here uses earth (or ground) anchors and steel straps connected to the home's longitudinal steel beams and/or exterior walls. Stabilizer plates may also be used to prevent the anchor head from moving laterally in the ground. An anchor, strap and stabilizer plate (if used) together are referred to as a tie down.

**Follow the Steps below:**


▼ **Step 1. Determine Anchor Locations** (p. 93)
▼ **Step 2. Determine Tiedown Configuration** (p. 112)
▼ **Step 3. Select Anchors** (p. 112)
▼ **Step 4. Install Anchors** (p. 113)
▼ **Step 5. Install Straps** (p. 115)
▼ **Step 6. Tighten and Adjust Straps** (p. 118)


## Step 1. Determine Anchor Locations

Create a sketch of the home plan showing the exterior walls, marriage line(s) (if any) and frame I-beams (this will be similar to the base sketch created in **Install Footings**, p. 33).

See **Table 21** for a list of anchor locations, types and where they are required. Page numbers where the requirements are provided are noted in the last column.

As each anchor location is determined, mark it on the sketch, noting important dimensions such as spacing between anchors. When complete, this will be the home's tie down plan **Figure 51** and **Figure 52**.

*Table 21. Anchor Location Types*

| Location | Type | Wind Zone I | Wind Zones II and III | See page |
|---|---|---|---|---|
| **Sidewall** | Frame | Yes | Yes | 94, 95 |
| | Vertical | No | Yes | 108 |
| **Longitudinal** | Frame | Yes | Yes | 108 |
| | Vertical | No | No | - |
| **Marriage line** | Vertical | No | Yes | 111 |
| **Tag Unit** | Frame | Yes | Yes | 112 |
| | Vertical | No | Yes | - |
| **Porch Post** | Vertical | Yes | Yes | 112 |
| **Offset Unit** | | Yes | Yes | 112 |

[1.] Install marriage line anchors prior to moving the home over the top of the anchor locations and then return to Set the Home or Complete Multi Section Set respectively.

**Need for a stabilizing system.** The home must be secured against the wind by the use of an anchor assembly or an alternative foundation system. Where site or other conditions prohibit the use of the manufacturer's instructions, a registered engineer or registered architect must design the stabilizing system. Alternate foundation designs must be approved by the manufacturer and DAPIA. Refer to page 9 for directions for obtaining available approved designs.

**Figure 51** and **Figure 52** illustrate typical anchor locations for a double section home in Wind Zone I and Wind Zones II and III respectively.

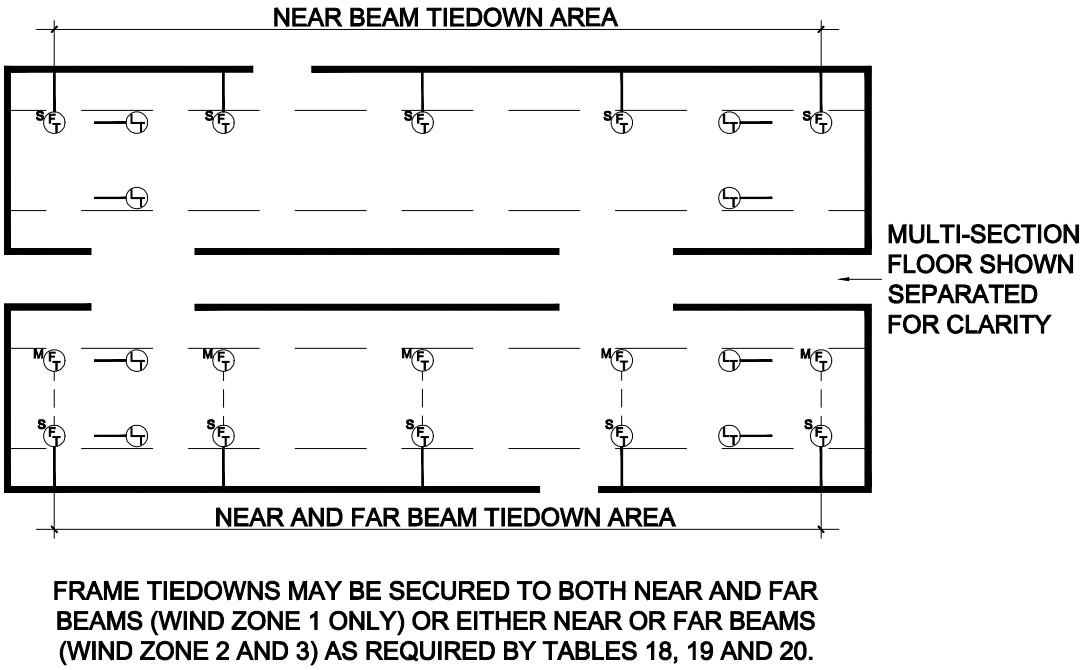



*Figure 51. Typical anchor locations for a double section home in Wind Zone I*

**Flood and seismic forces.** The stabilizing system requirements in this chapter do not consider flood or seismic loads and are not intended for use in flood or seismic hazard areas. In those areas a registered engineer or registered architect must design the stabilizing system. Alternate foundation designs must be approved by the manufacturer and DAPIA.

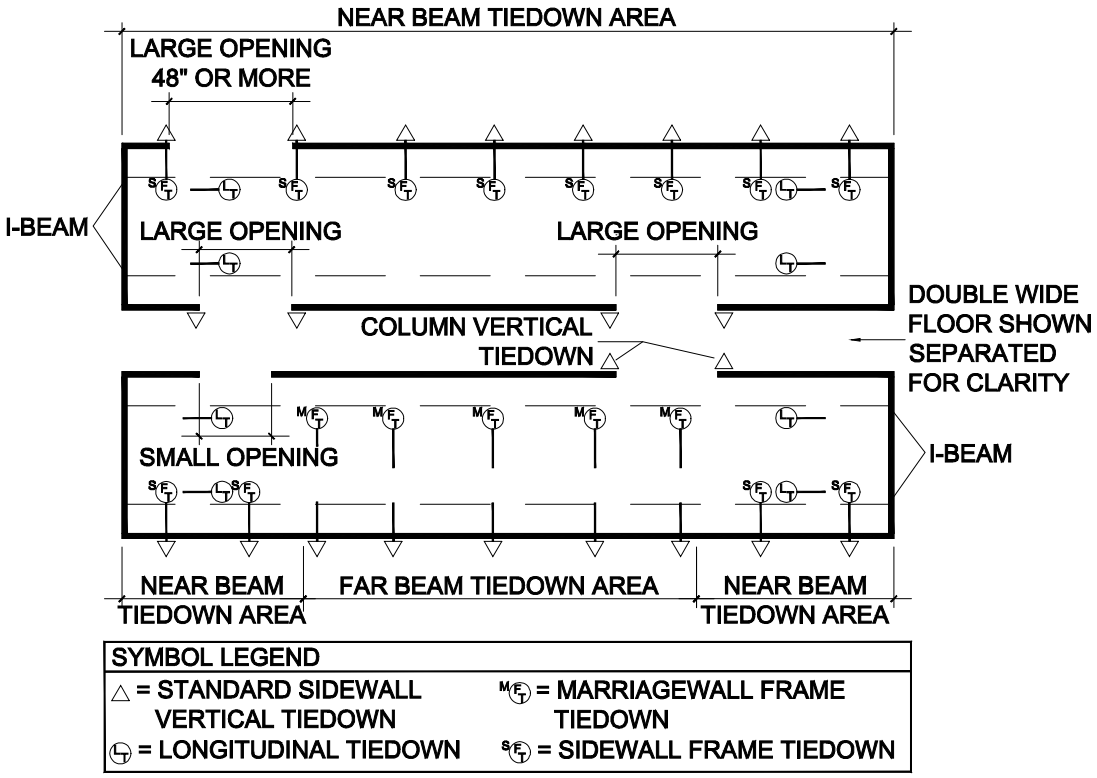



*Figure 52. Typical anchor locations for a double section home in Wind Zones II or III*

### Sidewall Frame Anchors

Use **Tables 28-48** to determine the spacing between anchors for Wind Zones I, II and III, respectively. Spacing requirements will vary depending on the type of home (single or multi-section), the slope of the roof, the width of the floor for each section, the sidewall height, I-beam spacing and the height from the ground to the strap attachment point. Determine the values for the home. Using the table for the appropriate wind zone, determine the column and row that corresponds to the characteristics of the home. The value on the tables is the maximum distance between anchors. **Keep in mind that sidewall frame tie downs must be located no more than two feet from each end of home.**

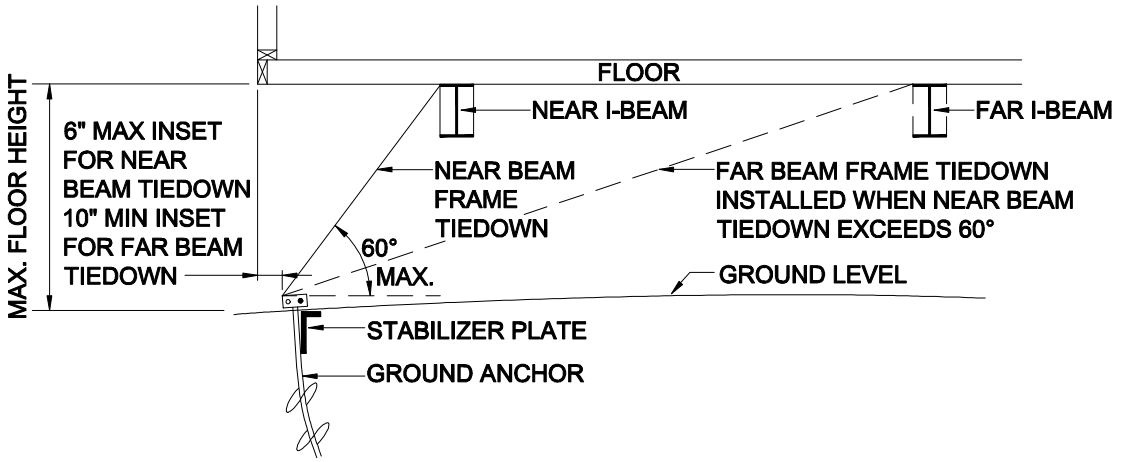
**Maximum spacing requirements.** The LAHJ may have anchor spacing requirements that supersede the values provided in this manual.

## Frame Tiedown Spacing

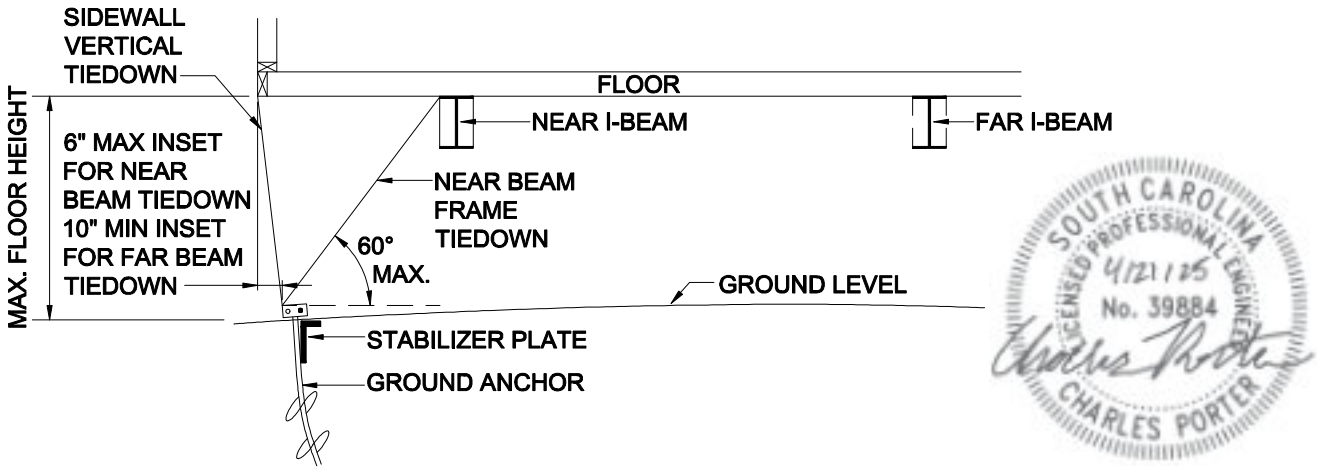

| Wind Zone 1 Tiedown Spacing | | Page No. |
|---|---|---|
| Table 22 | Single Section Roof Pitch 4.36/12 Max. | 99 |
| Table 23 | Multi Section Roof Pitch 4.36/12 Max. | 99 |
| Table 24 | Triple Section Roof Pitch 4.36/12 Max. | 100 |
| **Wind Zone 1 Tiedown Spacing – High Pitch Truss** | | |
| Table 25 | Multi Section Roof Pitch 5/12 Max. | 100 |
| Table 26 | Triple Section Roof Pitch 5/12 Max. | 101 |
| Table 27 | Multi Section Roof Pitch 6/12 Max. | 101 |
| Table 28 | Triple Section Roof Pitch 6/12 Max. | 101 |
| Table 29 | Multi Section Roof Pitch 7/12 Max. | 102 |
| Table 30 | Triple Section Roof Pitch 7/12 Max. | 102 |
| **Wind Zone 2 Near I-Beam Tiedown Spacing** | | |
| Table 31 | Single Section Roof Pitch 4.36/12 Max. | 103 |
| Table 32 | Multi Section Roof Pitch 4.36/12 Max. | 103 |
| **Wind Zone 2 Far I-Beam Tiedown Spacing** | | |
| Table 33 | Single Section Roof Pitch 4.36/12 Max. | 104 |
| Table 34 | Multi Section Roof Pitch 4.36/12 Max. | 104 |
| **Wind Zone 2 Near I-Beam Tiedown Spacing – High Pitch Truss** | | |
| Table 35 | Multi Section Roof Pitch 5/12 Max. | 105 |
| **Wind Zone 2 Far I-Beam Tiedown Spacing – High Pitch Truss** | | |
| Table 36 | Multi Section Roof Pitch 5/12 Max. | 105 |
| Table 37 | Multi Section Roof Pitch 6/12 Max. | 105 |
| Table 38 | Multi Section Roof Pitch 7/12 Max. | 106 |
| **Wind Zone 3 Near I-Beam Tiedown Spacing** | | |
| Table 39 | Single Section Roof Pitch 4.36/12 Max. | 106 |
| Table 40 | Multi Section Roof Pitch 4.36/12 Max. | 107 |
| **Wind Zone 3 Far I-Beam Tiedown Spacing** | | |
| Table 41 | Single Section Roof Pitch 4.36/12 Max. | 107 |
| Table 42 | Multi Section Roof Pitch 4.36/12 Max. | 108 |

**Tiedown Limitations and Specifications**

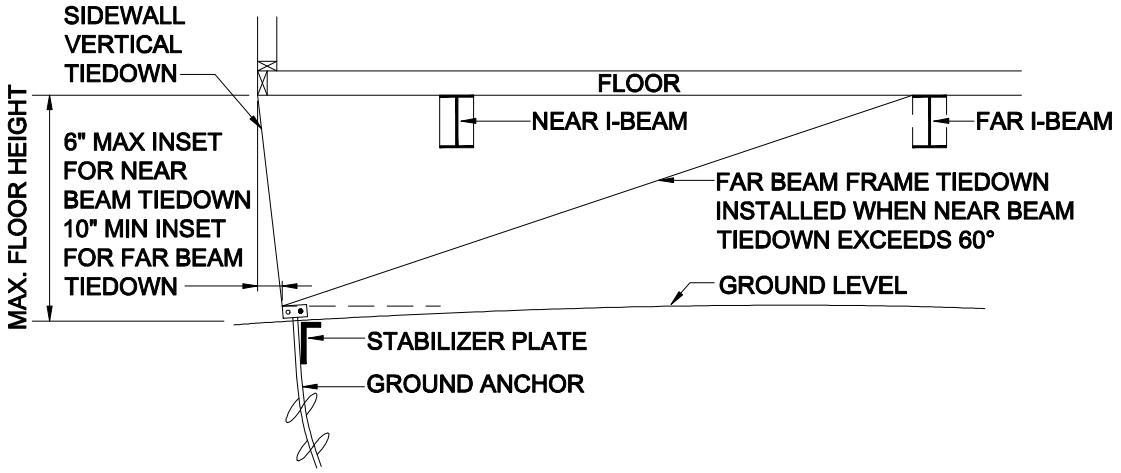
- Anchors shall be certified for site conditions including soil type for design capacity of 3150 lbs. with resistant pull load applied at a minimum 30 degree angle from horizontal.
- Anchors may be inset from the edge of the sidewall 6” maximum for a near beam set and 10” minimum for a far beam set.
- The max floor widths listed in the following charts correspond to a range of widths: Max floor widths should be rounded down to the next highest width if in between values (i.e. 11’ home to use values for 10’ max floor width).
- The maximum overhang allowed for a single section is 6” and the maximum overhang allowed for a multi section is 12”.
- The following frame tiedown charts list the maximum floor height measured from the top of the grade to the connection point of the tiedown strap to the I-beam. Cross reference the maximum allowable pier height (**Table 14**) with maximum allowable floor height listed in the frame tiedown charts. If maximum height listed in frame tiedown charts is exceeded, then designs must be provided by a registered professional engineer or registered architect.
- Contact the manufacturer if the home you are attempting to set has a floor width, I-beam spacing, wall height or roof pitch that is not shown in any of the following charts (or listed as NA).
- The “ * “ listed after any tiedown spacing signifies that a 60 degree angle has been exceeded and an additional strap must be added to the far beam.
- Ground anchors must not be spaced closer than the minimum spacing permitted by the listing or certification. Verify that sidewall vertical tiedown bracket spacing meets the requirements for Wind Zone 2 & 3 homes. Bracket spacing may be increased or additional brackets may be added as required.




Wind Zone 1 Frame Tiedown: When the angle of the near beam frame tiedown strap exceeds 60 degrees the far beam frame tiedown strap is installed in addition to the near beam strap.




Wind Zone 2 & 3 Near Beam Frame Tiedown: When the angle of the near beam frame tiedown strap exceeds 60 degrees the tiedown strap must be installed to the far beam frame.




Wind Zone 2 & 3 Far Beam Frame Tiedown

Figure 53. Tiedown methods for Wind Zones I, II, and III

Table 22. Single section – wind zone 1 frame tiedown spacing (ft) – max. Roof pitch 4.36/12

| Max. Wall Height | Max. Floor Height | 10 ft Max. Width I-Beam Spacing | | 12 ft Max. Width I-Beam Spacing | | 14 ft Max. Width I-Beam Spacing | | 16 ft Max. Width I-Beam Spacing | | 18 ft Max. Width I-Beam Spacing |
|---|---|---|---|---|---|---|---|---|---|---|
| | | 95.5” | 99.5” | 95.5” | 99.5” | 95.5” | 99.5” | 95.5” | 99.5” | 99.5” |
| 7 ft | 25” | 6’-0”* | 4’-0”* | 12’-0” | 12’-0” | 12’-0” | 12’-0” | 12’-0” | 12’-0” | 12’-0” |
| | 48” | NA | NA | 9’-2”* | 8’-3”* | 12’-0” | 12’-0” | 12’-0” | 12’-0” | 12’-0” |
| | 67” | NA | NA | 4’-7”* | 4’-1”* | 7’-2”* | 6’-8”* | 9’-3” | 9’-0” | 10’-9” |
| | 80” | NA | NA | NA | NA | 6’-2”* | 5’-8”* | 8’-2”* | 7’-10”* | 9’-7” |
| 8 ft | 25” | 5’-0”* | NA | 12’-0” | 11’-10” | 12’-0” | 12’-0” | 12’-0” | 12’-0” | 12’-0” |
| | 48” | NA | NA | 7’-9”* | 7’-0”* | 12’-0” | 11’-6” | 12’-0” | 12’-0” | 12’-0” |
| | 67” | NA | NA | 4’-1”* | NA | 6’-4”* | 6’-0”* | 8’-3” | 8’-0” | 9’-6” |
| | 80” | NA | NA | NA | NA | 5’-6”* | 5’-1”* | 7’-3”* | 7’-0”* | 8’-6” |
| 9 ft | 25” | 4’-5”* | NA | 11’-1” | 10’-3” | 12’-0” | 12’-0” | 12’-0” | 12’-0” | 11’-2” |
| | 48” | NA | NA | 6’-8”* | 6’-0”* | 10’-4” | 9’-10” | 12’-0” | 12’-0” | 12’-0” |
| | 67” | NA | NA | NA | NA | 5’-9”* | 5’-5”* | 7’-5” | 7’-2” | 8’-7” |
| | 80” | NA | NA | NA | NA | 4’-11”* | 4’-7”* | 6’-6”* | 6’-3”* | 7’-8” |

Table 23. Multi section – wind zone 1 frame tiedown spacing (ft) – max. Roof pitch 4.36/12

| Max. Wall Height | Max. Floor Height | 20 ft Max. Width I-Beam Spacing 95.5”-99.5” | 24 ft Max. Width I-Beam Spacing 95.5”-99.5” | 28 ft Max. Width I-Beam Spacing 95.5”-99.5” | 32 ft Max. Width I-Beam Spacing 95.5”-99.5” |
|---|---|---|---|---|---|
| 7 ft | 25” | 12’-0”* | 12’-0” | 12’-0” | 12’-0” |
| | 48” | 12’-0”* | 12’-0”* | 12’-0” | 12’-0” |
| | 67” | 9’-8”* | 9’-5”* | 9’-2”* | 9’-0” |
| | 80” | 9’-6”* | 9’-4”* | 9’-2”* | 9’-0”* |
| 8 ft | 25” | 12’-0”* | 12’-0” | 12’-0” | 12’-0” |
| | 48” | 12’-0”* | 12’-0”* | 12’-0” | 12’-0” |
| | 67” | 8’-7”* | 8’-4”* | 8’-2”* | 8’-0” |
| | 80” | 8’-6”* | 8’-4”* | 8’-2”* | 8’-1”* |
| 9 ft | 25” | 12’-0”* | 12’-0” | 12’-0” | 12’-0” |
| | 48” | 12’-0”* | 12’-0”* | 12’-0” | 12’-0” |
| | 67” | 7’-1”* | 8’-8”* | 9’-10”* | 7’-2” |
| | 80” | 6’-2”* | 7’-9”* | 8’-11”* | 9’-10”* |

The “ * ” listed after any tiedown spacing signifies that a 60 degree angle has been exceeded and an additional strap must be added to the far beam.



Table 24. Multi section – wind zone 1 frame tiedown spacing (ft) – max. Roof pitch 4.36/12

| Max. Wall Height | Max. Floor Height | 36 ft Max. Width I-Beam Spacing 95.5”-99.5” | 42 ft Max. Width I-Beam Spacing 95.5”-99.5” | 48 ft Max. Width I-Beam Spacing 95.5”-99.5” |
|---|---|---|---|---|
| 7 ft | 25” | 12’-0” | 12’-0” | 12’-0” |
| | 48” | 12’-0”* | 12’-0” | 12’-0” |
| | 67” | 9’-5”* | 9’-2”* | 9’-0” |
| | 80” | 9’-4”* | 9’-2”* | 9’-1”* |
| 8 ft | 25” | 12’-0” | 12’-0” | 12’-0” |
| | 48” | 12’-0”* | 12’-0” | 12’-0” |
| | 67” | 8’-5”* | 8’-2”* | 8’-0” |
| | 80” | 8’-4”* | 8’-2”* | 8’-1”* |
| 9 ft | 25” | 12’-0” | 12’-0” | 12’-0” |
| | 48” | 12’-0”* | 12’-0” | 12’-0” |
| | 67” | 7’-6”* | 7’-5”* | 7’-2” |
| | 80” | 7’-6”* | 7’-4”* | 7’-3”* |

Table 25. Multi section – wind zone 1 frame tiedown spacing (ft) – max. Roof pitch 5/12

| Max. Wall Height | Max. Floor Height | 20 ft Max. Width I-Beam Spacing 95.5”-99.5” | 24 ft Max. Width I-Beam Spacing 95.5”-99.5” | 28 ft Max. Width I-Beam Spacing 95.5”-99.5” | 32 ft Max. Width I-Beam Spacing 95.5”-99.5” |
|---|---|---|---|---|---|
| 8 ft | 25” | 12’-0”’* | 8’-3’’ | 11’-0’’ | 12’-0” |
| | 48” | 10’-0”’* | 11’-4”’* | 7’-5’’ | 9’-1’’ |
| | 67” | 5’-3”’* | 6’-1”’* | 6’-5”’* | 4’-6’’ |
| | 80” | 4’-7”’* | 5’-4”’* | 5’-10”’* | 6’-2”’* |
| 9 ft | 25” | 12’-0”’* | 7’-6’’ | 9’-11’’ | 10’-11’’ |
| | 48” | 9’-0”’* | 10’-2”’* | 6’-9’’ | 8’-3’’ |
| | 67” | 4’-10”’* | 5’-8”’* | 6’-1”’* | 4’-2’’ |
| | 80” | 4’-3”’* | 5’-0”’* | 5’-6”’* | 5’-9”’* |

The “ * ” listed after any tiedown spacing signifies that a 60 degree angle has been exceeded and an additional strap must be added to the far beam.



Table 26. Multi section – wind zone 1 frame tiedown spacing (ft) – max. roof pitch 5/12

| Max. Wall Height | Max. Floor Height | 36 ft Max. Width I-Beam Spacing 95.5”-99.5” | 42 ft Max. Width I-Beam Spacing 95.5”-99.5” | 48 ft Max. Width I-Beam Spacing 95.5”-99.5” |
|---|---|---|---|---|
| 8 ft | 25” | 8’-3’’ | 10’-9’’ | 11’-7’’ |
| | 48” | 11’-9’’* | 7’-4’’* | 8’-9’’ |
| | 67” | 5’-5’’* | 5’-7’’* | 5’-7’’* |
| | 80” | 4’-11’’* | 5’-1’’* | 5’-3’’* |
| 9 ft | 25” | 7’-6’’ | 9’-9’’ | 10’-7’’ |
| | 48” | 10’-7’’* | 6’-7’’ | 8’-0’’ |
| | 67” | 5’-2’’* | 5’-4’’* | 5’-4’’* |
| | 80” | 4’-7’’* | 4’-10’’* | 5’-0’’* |

Table 27. Multi section – wind zone 1 frame tiedown spacing (ft) – max. roof pitch 6/12

| Max. Wall Height | Max. Floor Height | 20 ft Max. Width I-Beam Spacing 95.5”-99.5” | 24 ft Max. Width I-Beam Spacing 95.5”-99.5” | 28 ft Max. Width I-Beam Spacing 95.5”-99.5” | 32 ft Max. Width I-Beam Spacing 95.5”-99.5” |
|---|---|---|---|---|---|
| 8 ft | 25” | 12’-0’’* | 7’-6’’ | 9’-9’’ | 10’-7’’ |
| | 48” | 9’-2’’* | 10’-2’’* | 6’-7’’ | 8’-0’’ |
| | 67” | 4’-11’’* | 5’-8’’* | 6’-0’’* | 4’-1’’* |
| | 80” | 4’-3’’* | 5’-0’’* | 5’-5’’* | 5’-8’’* |
| 9 ft | 25” | 11’-1’’* | 6’-9’’ | 9’-0’’ | 9’-9’’ |
| | 48” | 8’-4’’* | 9’-3’’* | 6’-0’’* | 7’-4’’ |
| | 67” | 4’-7’’* | 5’-4’’* | 5’-8’’* | 5’-9’’* |
| | 80” | 4’-0’’* | 4’-9’’* | 5’-2’’* | 5’-4’’* |

Table 28. Multi section – wind zone 1 frame tiedown spacing (ft) – max. roof pitch 6/12

| Max. Wall Height | Max. Floor Height | 36 ft Max. Width I-Beam Spacing 95.5”-99.5” | 42 ft Max. Width I-Beam Spacing 95.5”-99.5” | 48 ft Max. Width I-Beam Spacing 95.5”-99.5” |
|---|---|---|---|---|
| 8 ft | 25” | 7’-1’’ | 9’-2’’ | 9’-9’’ |
| | 48” | 10’-1’’* | 6’-2’’ | 7’-4’’ |
| | 67” | 5’-0’’* | 5’-1’’* | 5’-1’’* |
| | 80” | 4’-6’’* | 4’-8’’* | 4’-9’’* |
| 9 ft | 25” | 6’-6’’ | 8’-5’’ | 9’-0’’ |
| | 48” | 9’-3’’* | 5’-8’’* | 6’-9’’ |
| | 67” | 4’-9’’* | 4’-10’’* | 4’-10’’* |
| | 80” | 4’-3’’* | 4’-5’’* | 4’-6’’* |

The “ * ” listed after any tiedown spacing signifies that a 60 degree angle has been exceeded and an additional strap must be added to the far beam.

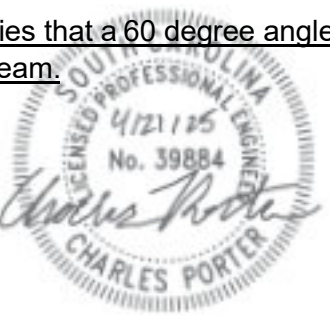


Table 29. Multi section – wind zone 1 frame tiedown spacing (ft) – max. roof pitch 7/12

| Max. Wall Height | Max. Floor Height | 20 ft Max. Width I-Beam Spacing 95.5”-99.5” | 24 ft Max. Width I-Beam Spacing 95.5”-99.5” | 28 ft Max. Width I-Beam Spacing 95.5”-99.5” | 32 ft Max. Width I-Beam Spacing 95.5”-99.5” |
|---|---|---|---|---|---|
| 8 ft | 25” | 11’-3’’* | 6’-10’’ | 8’-10’’ | 9’-6’’ |
| | 48” | 8’-5’’* | 9’-3’’* | 6’-0’’ | 7’-2’’ |
| | 67” | 4’-8’’* | 5’-4’’* | 5’-7’’* | 5’-8’’* |
| | 80” | 4’-0’’* | 4’-9’’* | 5’-1’’* | 5’-3’’* |
| 9 ft | 25” | 10’-3’’* | 6’-3’’ | 8’-1’’ | 8’-9’’ |
| | 48” | 7’-8’’* | 8’-6’’* | 5’-6’’* | 6’-7’’ |
| | 67” | 4’-4’’* | 5’-0’’* | 5’-4’’* | 5’-4’’* |
| | 80” | NA | 4’-5’’* | 4’-10’’* | 5’-0’’* |

Table 30. Multi section – wind zone 1 frame tiedown spacing (ft) – max. roof pitch 7/12

| Max. Wall Height | Max. Floor Height | 36 ft Max. Width I-Beam Spacing 95.5”-99.5” | 42 ft Max. Width I-Beam Spacing 95.5”-99.5” | 48 ft Max. Width I-Beam Spacing 95.5”-99.5” |
|---|---|---|---|---|
| 8 ft | 25” | 6’-3’’ | 7’-11’’ | 8’-4’’ |
| | 48” | 8’-10’’* | 5’-4’’ | 6’-4’’ |
| | 67” | 4’-7’’* | 4’-9’’* | 4’-7’’* |
| | 80” | 4’-2’’* | 4’-3’’* | 4’-4’’* |
| 9 ft | 25” | 5’-9’’ | 7’-4’’ | 7’-9’’ |
| | 48” | 8’-2’’* | 5’-0’’ | 5’-11’’ |
| | 67” | 4’-5’’* | 4’-6’’* | 4’-5’’* |
| | 80” | 4’-0’’* | 4’-1’’* | 4’-2’’* |

The “ * ” listed after any tiedown spacing signifies that a 60 degree angle has been exceeded and an additional strap must be added to the far beam.



Table 31. Single section – wind zone 2 near I-Beam frame tiedown spacing (ft) – max. roof pitch 4.36/12

| Max. Wall Height | Max. Floor Height | 12 ft Max. Width I-Beam Spacing | | 14 ft Max. Width I-Beam Spacing | | 16 ft Max. Width I-Beam Spacing | | 18 ft Max. Width I-Beam Spacing |
|---|---|---|---|---|---|---|---|---|
| | | 95.5” | 99.5” | 95.5” | 99.5” | 95.5” | 99.5” | 99.5” |
| 7 ft | 25” | 7’-3’’ | 6’-9’’ | 9’-3” | 9’-3” | 9’-7’’ | 9’-7’’ | 9’-11’’ |
| | 48” | 4’-4” | 4’-0’’ | 6’-9” | 6’-5” | 8’-6’’ | 8’-3’’ | 9’-1’’ |
| | 67” | NA | NA | 4’-1” | NA | 5’-4’’ | 5’-2” | 6’-2” |
| | 80” | NA | NA | NA | NA | 4’-8’’ | 4’-6’’ | 5’-6” |
| 8 ft | 25” | 6’-4” | 5’-10” | 7’-11’’ | 7’-11’’ | 8’-3’’ | 8’-3’’ | 8’-6’’ |
| | 48” | NA | NA | 5’-10’’ | 5’-6” | 7’-4’’ | 7’-2’’ | 7’-10’’ |
| | 67” | NA | NA | NA | NA | 4’-9’’ | 4’-7’’ | 5’-6’’ |
| | 80” | NA | NA | NA | NA | 4’-2’’ | 4’-0’’ | 4’-10’’ |
| 9 ft | 25” | 5’-7” | 5’-1” | 6’-11’’ | 6’-11’’ | 7’-2’’ | 7’-2’’ | 7’-5’’ |
| | 48” | NA | NA | 5’-1” | 4’-10’’ | 6’-5’’ | 6’-3’’ | 6’-10’’ |
| | 67” | NA | NA | NA | NA | 4’-3’’ | 4’-1’’ | 4’-11’’ |
| | 80” | NA | NA | NA | NA | NA | NA | 4’-5’’ |

Table 32. Multi section – wind zone 2 near I-Beam frame tiedown spacing (ft) – max. roof pitch 4.36/12

| Max. Wall Height | Max. Floor Height | 24 ft Max. Width I-Beam Spacing | | 28 ft Max. Width I-Beam Spacing | | 32 ft Max. Width I-Beam Spacing | |
|---|---|---|---|---|---|---|---|
| | | 95.5” | 99.5” | 95.5” | 99.5” | 95.5” | 99.5” |
| 7 ft | 25” | 9’-5” | 8’-9” | 11’-5” | 11’-4” | 9’-2” | 9’-6” |
| | 48” | 5’-8” | 5’-1” | 9’-0” | 8’-7” | 10’-10” | 10’-9” |
| | 67” | NA | NA | 4’-1” | NA | 5’-4” | 5’-2” |
| | 80” | NA | NA | NA | NA | 4’-8” | 4’-6” |
| 8 ft | 25” | 7’-10” | 7’-3” | 10’-0” | 10’-0” | 9’-10” | 9’-11” |
| | 48” | 4’-9” | 4’-3” | 7’-6” | 7’-1” | 9’-6” | 9’-6” |
| | 67” | NA | NA | NA | NA | 4’-9” | 4’-7” |
| | 80” | NA | NA | NA | NA | 4’-2” | 4’-0” |
| 9 ft | 25” | 6’-9” | 6’-3” | 8’-9” | 8’-9” | 8’-10” | 8’-10” |
| | 48” | 4’-0” | NA | 6’-4” | 6’-0” | 8’-3” | 8’-0” |
| | 67” | NA | NA | NA | NA | 4’-3” | 4’-1” |
| | 80” | NA | NA | NA | NA | NA | NA |



Table 33. Single section – wind zone 2 far I-Beam frame tiedown spacing (ft) – max. roof pitch 4.36/12

| Max. Wall Height | Max. Floor Height | 10 ft Max. Width I-Beam Spacing 95.5”-99.5” | 12 ft Max. Width I-Beam Spacing 95.5”-99.5” | 14 ft Max. Width I-Beam Spacing 95.5”-99.5” | 16 ft Max. Width I-Beam Spacing 95.5”-99.5” | 18 ft Max. Width I-Beam Spacing 99.5” |
|---|---|---|---|---|---|---|
| 7 ft | 25” | 5’-9’’ | 5’-11’’ | 6’-1’’ | 6’-2’’ | 6’-4’’ |
| | 48” | 5’-9’’ | 5’-11’’ | 6’-1’’ | 6’-2’’ | 6’-4’’ |
| | 67” | 5’-9’’ | 5’-11’’ | 6’-1’’ | 6’-2’’ | 6’-4’’ |
| | 80” | 5’-9’’ | 5’-11’’ | 6’-1’’ | 6’-2’’ | 6’-4’’ |
| 8 ft | 25” | 4’-9’’ | 5’-0’’ | 5’-2’’ | 5’-4’’ | 5’-5’’ |
| | 48” | 4’-9’’ | 5’-0’’ | 5’-2’’ | 5’-4’’ | 5’-5’’ |
| | 67” | 4’-9’’ | 5’-0’’ | 5’-2’’ | 5’-4’’ | 5’-5’’ |
| | 80” | 4’-9’’ | 5’-0’’ | 5’-2’’ | 5’-4’’ | 5’-5’’ |
| 9 ft | 25” | 4’-0’’ | 4’-2’’ | 4’-5’’ | 4’-7’’ | 4’-9’’ |
| | 48” | 4’-0’’ | 4’-2’’ | 4’-5’’ | 4’-7’’ | 4’-9’’ |
| | 67” | 4’-0’’ | 4’-2’’ | 4’-5’’ | 4’-7’’ | 4’-9’’ |
| | 80” | 4’-0’’ | 4’-2’’ | 4’-5’’ | 4’-7’’ | 4’-9’’ |

Table 34. Multi section – wind zone 2 far I-Beam frame tiedown spacing (ft) – max. roof pitch 4.36/12

| Max. Wall Height | Max. Floor Height | 20 ft Max. Width I-Beam Spacing 95.5”-99.5” | 24 ft Max. Width I-Beam Spacing 95.5”-99.5” | 28 ft Max. Width I-Beam Spacing 95.5”-99.5” | 32 ft Max. Width I-Beam Spacing 95.5”-99.5” |
|---|---|---|---|---|---|
| 7 ft | 25” | 8’-5’’ | 7’-3” | 6’-5’’ | 5’-8” |
| | 48” | 10’-5’’ | 8’-6’’ | 7’-2” | 6’-2” |
| | 67” | 7’-10’’ | 7’-10’’ | 7’-9” | 7’-7” |
| | 80” | 7’-7’’ | 7’-7” | 7’-7” | 7’-6’’ |
| 8 ft | 25” | 7’-8’’ | 6’-9” | 6’-0’’ | 5’-5’’ |
| | 48” | 9’-5’’ | 8’-1’’ | 6’-11’’ | 6’-0” |
| | 67’ | 7’-0’’ | 7’-0’’ | 7’-0’’ | 6’-11” |
| | 80” | 6’-9’’ | 6’-10” | 6’-10’’ | 6’-9” |
| 9 ft | 25” | 6’-11’’ | 6’-3’’ | 5’-8” | 5’-2’’ |
| | 48” | 8’-3” | 7’-7’’ | 6’-8” | 5’-10” |
| | 67” | 6’-3” | 6’-4” | 6’-4” | 6’-3” |
| | 80” | 6’-1” | 6’-2’’ | 6’-2” | 6’-2” |



Table 35. Multi section – wind zone 2 near I-Beam frame tiedown spacing (ft) – max. roof pitch 5/12

| Max. Wall Height | Max. Floor Height | 28 ft Max. Width I-Beam Spacing | | 32 ft Max. Width I-Beam Spacing | |
|---|---|---|---|---|---|
| | | 95.5” | 99.5” | 95.5” | 99.5” |
| 8 ft | 25” | 5’-2” | 5’-0” | 5’-1” | 5’-1” |
| | 48” | NA | NA | 4’-3” | 4’-1” |
| | 67” | NA | NA | NA | NA |
| | 80” | NA | NA | NA | NA |
| 9 ft | 25” | 4’-9” | 4’-8” | 4’-9” | 4’-9” |
| | 48” | NA | NA | NA | NA |
| | 67” | NA | NA | NA | NA |
| | 80” | NA | NA | NA | NA |

Table 36. Multi section – wind zone 2 far I-Beam frame tiedown spacing (ft) – max. Roof pitch 5/12

| Max. Wall Height | Max. Floor Height | 20 ft Max. Width I-Beam Spacing 95.5”-99.5” | 24 ft Max. Width I-Beam Spacing 95.5”-99.5” | 28 ft Max. Width I-Beam Spacing 95.5”-99.5” | 32 ft Max. Width I-Beam Spacing 95.5”-99.5” |
|---|---|---|---|---|---|
| 8 ft | 25” | 4’-10” | 4’-4” | 4’-0” | NA |
| | 48” | 5’-7” | 5’-4” | 4’-10” | 4’-4” |
| | 67” | 4’-7” | 4’-4” | 4’-2” | 4’-0” |
| | 80” | 4’-5” | 4’-3” | 4’-1” | 4’-0” |
| 9 ft | 25” | 4’-4” | 4’-0” | NA | NA |
| | 48” | 5’-1” | 4’-11” | 4’-6” | 4’-1” |
| | 67” | 4’-3” | 4’-1” | NA | NA |
| | 80” | 4’-1” | 4’-0” | NA | NA |

Table 37. Multi section – wind zone 2 far I-Beam frame tiedown spacing (ft) – max. roof pitch 6/12

| Max. Wall Height | Max. Floor Height | 20 ft Max. Width I-Beam Spacing 95.5”-99.5” | 24 ft Max. Width I-Beam Spacing 95.5”-99.5” | 28 ft Max. Width I-Beam Spacing 95.5”-99.5” | 32 ft Max. Width I-Beam Spacing 95.5”-99.5” |
|---|---|---|---|---|---|
| 8 ft | 25” | 4’-5” | 4’-0” | NA | NA |
| | 48” | 5’-2” | 4’-11” | 4’-6” | 4’-0” |
| | 67” | 4’-3” | 4’-1” | NA | NA |
| | 80” | 4’-2” | 4’-0” | NA | |
| 9 ft | 25” | 4’-0” | NA | NA | |
| | 48” | 4’-9” | 4’-6” | 4’-2” | |
| | 67” | 4’-0” | NA | NA | |
| | 80” | NA | NA | NA | |

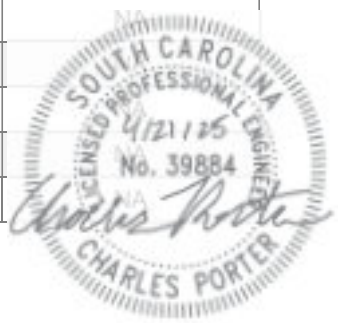


Table 38. Multi section – wind zone 2 far I-Beam frame tiedown spacing (ft) – max. Roof pitch 7/12

| Max. Wall Height | Max. Floor Height | 20 ft Max. Width I-Beam Spacing 95.5”-99.5” | 24 ft Max. Width I-Beam Spacing 95.5”-99.5” | 28 ft Max. Width I-Beam Spacing 95.5”-99.5” | 32 ft Max. Width I-Beam Spacing 95.5”-99.5” |
|---|---|---|---|---|---|
| 8 ft | 25” | 4’-1” | NA | NA | NA |
| | 48” | 4’-10” | 4’-6” | 4’-1” | NA |
| | 67” | 4’-0” | NA | NA | NA |
| | 80” | NA | NA | NA | NA |

Table 39. Single section – wind zone 3 near I-beam frame tiedown spacing (ft) – max. Roof pitch 4.36/12

| Max. Wall Height | Max. Floor Height | 12 ft Max. Width I-Beam Spacing | | 14 ft Max. Width I-Beam Spacing | | 16 ft Max. Width I-Beam Spacing | | 18 ft Max. Width I-Beam Spacing |
|---|---|---|---|---|---|---|---|---|
| | | 95.5” | 99.5” | 95.5” | 99.5” | 95.5” | 99.5” | 99.5” |
| 7 ft | 25” | 5’-10’’ | 5’-4’’ | 7’-3” | 7’-3” | 7’-6’’ | 7’-6’’ | 7’-9’’ |
| | 48” | NA | NA | 5’-4” | 5’-1” | 6’-9’’ | 6’-7’’ | 7’-1’’ |
| | 67” | NA | NA | NA | NA | 4’-5” | 4’-3” | 5’-1” |
| | 80” | NA | NA | NA | NA | NA | NA | 4’-6” |
| 8 ft | 25” | 5’-1” | 4’-8” | 6’-3’’ | 6’-3’’ | 6’-6’’ | 6’-6’’ | 6’-8’’ |
| | 48” | NA | NA | 4’-8’’ | 4’-5” | 5’-10’’ | 5’-8’’ | 6’-2’’ |
| | 67” | NA | NA | NA | NA | NA | NA | 4’-7’’ |
| | 80” | NA | NA | NA | NA | NA | NA | 4’-0’’ |
| 9 ft | 25” | 4’-6” | 4’-2” | 5’-6’’ | 5’-6’’ | 5’-8’’ | 5’-9’’ | 5’-11’’ |
| | 48” | NA | NA | 4’-1” | NA | 5’-2’’ | 5’-0’’ | 5’-5” |
| | 67” | NA | NA | NA | NA | NA | NA | 4’-1’’ |
| | 80” | NA | NA | NA | NA | NA | NA | NA |



Table 40. Multi section – wind zone 3 near I-Beam frame tiedown spacing (ft) – max. roof pitch 4.36/12

| Max. Wall Height | Max. Floor Height | 24 ft Max. Width I-Beam Spacing | | 28 ft Max. Width I-Beam Spacing | | 32 ft Max. Width I-Beam Spacing | |
|---|---|---|---|---|---|---|---|
| | | 95.5” | 99.5” | 95.5” | 99.5” | 95.5” | 99.5” |
| 7 ft | 25” | 7’-1” | 6’-7” | 8’-3” | 8’-3” | 6’-5” | 6’-7” |
| | 48” | 4’-3” | NA | 6’-8” | 6’-5” | 7’-9” | 7’-9” |
| | 67” | NA | NA | NA | NA | 4’-5” | 4’-3” |
| | 80” | NA | NA | NA | NA | NA | NA |
| 8 ft | 25” | 6’-0” | 5’-7” | 7’-4” | 7’-4” | 6’-9” | 7’-1” |
| | 48” | NA | NA | 5’-8” | 5’-4” | 7’-0” | 7’-0” |
| | 67” | NA | NA | NA | NA | NA | NA |
| | 80” | NA | NA | NA | NA | NA | NA |
| 9 ft | 25” | 5’-3” | 4’-10” | 6’-7” | 6’-7” | 6’-6” | 6’-6” |
| | 48” | NA | NA | 4’-10” | 4’-7” | 6’-3” | 6’-1” |
| | 67” | NA | NA | NA | NA | NA | NA |
| | 80” | NA | NA | NA | NA | NA | NA |

Table 41. Single section – wind zone 3 far I-Beam frame tiedown spacing (ft) – max. Roof pitch 4.36/12

| Max. Wall Height | Max. Floor Height | 10 ft Max. Width I-Beam Spacing 95.5”-99.5” | 12 ft Max. Width I-Beam Spacing 95.5”-99.5” | 14 ft Max. Width I-Beam Spacing 95.5”-99.5” | 16 ft Max. Width I-Beam Spacing 95.5”-99.5” | 18 ft Max. Width I-Beam Spacing 99.5” |
|---|---|---|---|---|---|---|
| 7 ft | 25” | 4’-5’’ | 4’-6’’ | 4’-8’’ | 4’-9’’ | 4’-10” |
| | 48” | 4’-5’’ | 4’-6’’ | 4’-8’’ | 4’-9’’ | 4’-10” |
| | 67” | 4’-5’’ | 4’-6’’ | 4’-8’’ | 4’-9’’ | 4’-10” |
| | 80” | 4’-5’’ | 4’-6’’ | 4’-8’’ | 4’-9’’ | 4’-10” |
| 8 ft | 25” | NA | NA | 4’-0’’ | 4’-1” | 4’-3’’ |
| | 48” | NA | NA | 4’-0’’ | 4’-1” | 4’-3’’ |
| | 67” | NA | NA | 4’-0’’ | 4’-1” | 4’-3’’ |
| | 80” | NA | NA | 4’-0’’ | 4’-1” | 4’-3’’ |
| 9 ft | 25” | NA | NA | NA | NA | NA |
| | 48” | NA | NA | NA | NA | NA |
| | 67” | NA | NA | NA | NA | NA |
| | 80” | NA | NA | NA | NA | NA |



Table 42. Multi section – wind zone 3 far I-Beam frame tiedown spacing (ft) – max. Roof pitch 4.36/12

| Max. Wall Height | Max. Floor Height | 20 ft Max. Width I-Beam Spacing 95.5”-99.5” | 24 ft Max. Width I-Beam Spacing 95.5”-99.5” | 28 ft Max. Width I-Beam Spacing 95.5”-99.5” | 32 ft Max. Width I-Beam Spacing 95.5”-99.5” |
|---|---|---|---|---|---|
| 7 ft | 25” | 5’-9’’ | 5’-1” | 4’-6’’ | 4’-0” |
| | 48” | 6’-11’’ | 5’-10’’ | 5’-0” | 4’-5” |
| | 67” | 6’-3’’ | 6’-2’’ | 6’-1” | 5’-9” |
| | 80” | 6’-0’’ | 6’-0” | 6’-0” | 5’-10’’ |
| 8 ft | 25” | 5’-3’’ | 4’-9” | 4’-3’’ | NA |
| | 48” | 6’-6’’ | 5’-7’’ | 4’-10’’ | 4’-3” |
| | 67’ | 5’-7’’ | 5’-7’’ | 5’-6’’ | 5’-5” |
| | 80” | 5’-5’’ | 5’-5” | 5’-5’’ | 5’-4” |
| 9 ft | 25” | 4’-10’’ | 4’-5’’ | 4’-1” | NA |
| | 48” | 6’-0” | 5’-3’’ | 4’-8” | 4’-2” |
| | 67” | 5’-0” | 5’-0” | 5’-0” | 5’-0” |
| | 80” | 4’-10” | 4’-11’’ | 4’-11” | 4’-11” |

### Sidewall Vertical Anchors

Homes designed for Wind Zones II and III also require vertical tie downs along the sidewalls (Figure 60). The vertical tie down brackets will be factory installed. Vertical and frame sidewall tie downs may connect to one double-headed anchor or each to its own dedicated anchor per the spacing requirements listed in Tables 37-48. If additional brackets are required to be added or existing brackets are required to be relocated due to interferences or site conditions, install Mastercraft #5705 or steel angle brackets and straps as shown in Figure 60.

### Longitudinal Frame Anchors

Use Table 43 to determine the number of longitudinal frame anchors required at each end of the home. The longitudinal frame anchors may be factory installed, site installed or connected to a crossmember within 3” of the main I-beam.

NOTE:

Friction is assumed to contribute to the resistance in the longitudinal direction when piers are no more than 64” high. Friction from a single block pier is assumed for piers less than 44” high. Friction from a double block pier is assumed for piers between 44” and 64” high.



Table 43. Wind Zone 1 Longitudinal Frame Tiedown Quantities (Quantity Each End of Home)

| Max. Roof Pitch | Max. Pier Height | Single Section (18 ft Max. Floor Width) Minimum Unit Lengths | | | | | | | | Multi Section (32 ft Max. Floor Width) Minimum Unit Lengths | | | | | | | | Triple Section (48 ft Max. Floor Width) Minimum Unit Lengths | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 36’ | 42’ | 48’ | 54’ | 60’ | 66’ | 72’ | 80’ | 36’ | 42’ | 48’ | 54’ | 60’ | 66’ | 72’ | 80’ | 36’ | 42’ | 48’ | 54’ | 60’ | 66’ | 72’ | 80’ |
| 4.36 | 24” | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 3 | 2 | 1 | 0 | 0 | 0 | 0 | 0 |
| | 36” | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 3 | 2 | 2 | 1 | 1 | 1 | 0 | 0 | 4 | 4 | 3 | 3 | 2 | 2 | 1 | 1 |
| | 44” | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| | 52” | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 2 | 1 | 0 | 0 | 0 | 0 | 0 | 0 |
| | 64” | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 2 | 2 | 1 | 1 | 1 | 0 | 0 | 0 | 4 | 3 | 3 | 2 | 1 | 1 | 0 | 0 |
| 7 | 24” | | | | | | | | | 3 | 2 | 1 | 0 | 0 | 0 | 0 | 0 | 4 | 3 | 2 | 1 | 1 | 0 | 0 | 0 |
| | 36” | | | | | | | | | 4 | 3 | 2 | 2 | 2 | 1 | 1 | 1 | 6 | 5 | 4 | 4 | 3 | 3 | 2 | 2 |
| | 44” | | | | | | | | | 2 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 2 | 1 | 0 | 0 | 0 | 0 | 0 | 0 |
| | 52” | | | | | | | | | 2 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 3 | 2 | 1 | 0 | 0 | 0 | 0 | 0 |
| | 64” | | | | | | | | | 3 | 2 | 2 | 2 | 1 | 1 | 0 | 0 | 5 | 4 | 4 | 3 | 3 | 2 | 1 | 1 |

Table 44. Pier height greater than 64” – wind zone 1 longitudinal frame tiedown quantities (quantity each end of home)

| Max. Wall Height | Max. Roof Pitch | Max. Home Width | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Single Section | | | | | Multi Section | | | | Triple | |
| | | 10 ft | 12 ft | 14 ft | 16 ft | 18 ft | 20 ft | 24 ft | 28 ft | 32 ft | 36 ft | 48 ft |
| 7 ft | 4.36 | 1 | 2 | 2 | 2 | 2 | 3 | 3 | 4 | 4 | 5 | 6 |
| 7.5 ft | 4.36 | 1 | 2 | 2 | 2 | 2 | 3 | 3 | 4 | 4 | 5 | 7 |
| 8 ft | 4.36 | 2 | 2 | 2 | 2 | 2 | 3 | 3 | 4 | 4 | 5 | 7 |
| | 5 | NA | NA | NA | NA | NA | 3 | 3 | 4 | 4 | 5 | 7 |
| | 6 | NA | NA | NA | NA | NA | 3 | 3 | 4 | 5 | 5 | 8 |
| | 7 | NA | NA | NA | NA | NA | 3 | 4 | 4 | 5 | 6 | 8 |
| 9 ft | 4.36 | 2 | 2 | 2 | 2 | 3 | 3 | 3 | 4 | 5 | 5 | 7 |
| | 5 | NA | NA | NA | NA | NA | 3 | 4 | 4 | 5 | 6 | 8 |
| | 6 | NA | NA | NA | NA | NA | 3 | 4 | 4 | 5 | 6 | 8 |
| | 7 | NA | NA | NA | NA | NA | 3 | 4 | 5 | 5 | 6 | 9 |
| 10 ft | 4.36 | 2 | 2 | 2 | 3 | 3 | 3 | 4 | 4 | 5 | 6 | 8 |
| | 5 | NA | NA | NA | NA | NA | 3 | 4 | 4 | 5 | 6 | 8 |
| | 6 | NA | NA | NA | NA | NA | 3 | 4 | 5 | 5 | 6 | 9 |
| | 7 | NA | NA | NA | NA | NA | 3 | 4 | 5 | 6 | 6 | 9 |



Table 45. Wind zone 2 longitudinal frame tiedown quantities (each end of home)

| Max. Wall Height | Max. Roof Pitch | Max. Home Width | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Single Section | | | | | Multi Section | | | | Triple | |
| | | 10 ft | 12 ft | 14 ft | 16 ft | 18 ft | 20 ft | 24 ft | 28 ft | 32 ft | 36 ft | 48 ft |
| 7 ft | 3 | 2 | 2 | 3 | 3 | 3 | 4 | 5 | 5 | 6 | 8 | 9 |
| | 4.36 | 2 | 2 | 3 | 3 | 3 | 4 | 5 | 6 | 7 | 9 | 10 |
| 7.5 ft | 3 | 2 | 2 | 3 | 3 | 3 | 4 | 5 | 6 | 6 | 9 | 10 |
| | 4.36 | 2 | 3 | 3 | 3 | 4 | 4 | 5 | 6 | 7 | 9 | 11 |
| 8 ft | 3 | 2 | 3 | 3 | 3 | 4 | 4 | 5 | 6 | 7 | 9 | 10 |
| | 4.36 | 2 | 3 | 3 | 3 | 4 | 4 | 5 | 6 | 7 | 10 | 11 |
| | 5 | NA | NA | NA | NA | NA | 4 | 5 | 6 | 7 | 10 | 12 |
| | 6 | NA | NA | NA | NA | NA | 5 | 6 | 7 | 8 | 11 | 12 |
| | 7 | NA | NA | NA | NA | NA | 5 | 6 | 7 | 8 | 12 | 13 |
| 9 ft | 3 | 2 | 3 | 3 | 4 | 4 | 4 | 5 | 6 | 7 | 10 | 11 |
| | 4.36 | 2 | 3 | 3 | 4 | 4 | 5 | 6 | 7 | 8 | 11 | 12 |
| | 5 | NA | NA | NA | NA | NA | 5 | 6 | 7 | 8 | 11 | 13 |
| | 6 | NA | NA | NA | NA | NA | 5 | 6 | 7 | 8 | 12 | 13 |
| | 7 | NA | NA | NA | NA | NA | 5 | 6 | 7 | 9 | 12 | 14 |
| 10 ft | 3 | 3 | 3 | 3 | 4 | 4 | 5 | 6 | 7 | 8 | 10 | 12 |
| | 4.36 | 3 | 3 | 4 | 4 | 4 | 5 | 6 | 7 | 8 | 11 | 13 |
| | 5 | NA | NA | NA | NA | NA | 5 | 6 | 7 | 9 | 12 | 13 |
| | 6 | NA | NA | NA | NA | NA | 5 | 6 | 8 | 9 | 12 | 14 |
| | 7 | NA | NA | NA | NA | NA | 5 | 7 | 8 | 9 | 13 | 15 |



Table 46. Wind zone 3 longitudinal frame tiedown quantities (each end of home)

| Max. Wall Height | Max. Roof Pitch | Max. Home Width | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Single Section | | | | | Multi Section | | | | Triple | |
| | | 10 ft | 12 ft | 14 ft | 16 ft | 18 ft | 20 ft | 24 ft | 28 ft | 32 ft | 36 ft | 48 ft |
| 7 ft | 3 | 2 | 3 | 3 | 4 | 4 | 4 | 5 | 6 | 7 | 10 | 11 |
| | 4.36 | 2 | 3 | 3 | 4 | 4 | 5 | 6 | 7 | 8 | 11 | 12 |
| 7.5 ft | 3 | 2 | 3 | 3 | 4 | 4 | 5 | 6 | 7 | 8 | 10 | 12 |
| | 4.36 | 3 | 3 | 3 | 4 | 4 | 5 | 6 | 7 | 8 | 11 | 13 |
| 8 ft | 3 | 3 | 3 | 3 | 4 | 4 | 5 | 6 | 7 | 8 | 11 | 12 |
| | 4.36 | 3 | 3 | 4 | 4 | 4 | 5 | 6 | 7 | 9 | 12 | 13 |
| | 5 | NA | NA | NA | NA | NA | 5 | 6 | 8 | 9 | 12 | 14 |
| | 6 | NA | NA | NA | NA | NA | 5 | 7 | 8 | 9 | 13 | 15 |
| | 7 | NA | NA | NA | NA | NA | 6 | 7 | 8 | 10 | 14 | 16 |
| 9 ft | 3 | 3 | 3 | 4 | 4 | 5 | 5 | 6 | 8 | 9 | 12 | 13 |
| | 4.36 | 3 | 3 | 4 | 4 | 5 | 6 | 7 | 8 | 9 | 13 | 14 |
| | 5 | NA | NA | NA | NA | NA | 6 | 7 | 8 | 9 | 13 | 15 |
| | 6 | NA | NA | NA | NA | NA | 6 | 7 | 9 | 10 | 14 | 16 |
| | 7 | NA | NA | NA | NA | NA | 6 | 7 | 9 | 10 | 15 | 17 |
| 10 ft | 3 | 3 | 4 | 4 | 5 | 5 | 6 | 7 | 8 | 9 | 13 | 14 |
| | 4.36 | 3 | 4 | 4 | 5 | 5 | 6 | 7 | 9 | 10 | 14 | 15 |
| | 5 | NA | NA | NA | NA | NA | 6 | 7 | 9 | 10 | 14 | 16 |
| | 6 | NA | NA | NA | NA | NA | 6 | 8 | 9 | 11 | 15 | 17 |
| | 7 | NA | NA | NA | NA | NA | 6 | 8 | 10 | 11 | 16 | 18 |

### Marriage Line Vertical Anchors (Wind Zones II and III Only)

In Wind Zones II and III, marriage line anchors are required at each column along the marriage wall.

There may be manufacturer-installed brackets indicating required tie down locations (may be identified by tags or paint). If brackets are not present, then an alternative acceptable connection method, such as steel angles (provided by the manufacturer) must be used (Figure 60). If necessary to avoid interference with piers, the tie down location may be offset horizontally from the column by a maximum of 12 inches.



When to install marriage line anchors. Anchors along the marriage line must be installed prior to the second half of a multi-section home being set.



### Tag Unit Frame and Vertical Anchors

Tag unit anchoring is not covered in this manual. The required anchoring details will accompany homes with tag units.

### Porch Post Anchors

Each post that requires an anchor will have a tiedown bracket attached from the factory and be designated by a pier label. Homes with roof pitches of 4.36/12 or less do not require frame tiedowns below a full width endwall porch. Required frame and vertical tiedowns in Wind Zone 2 and 3 may begin on-center spacing at the corner porch post/endwall location. This will also satisfy the Sidewall Frame Anchor requirement of locating one “no more than two feet from each end of home” stated on page 90. Homes with roof pitches above 4.36/12 must also contain frame tiedowns below endwall porches and can be combined with vertical tiedown brackets at porch posts.

### Offset Unit Anchors

Anchors installed on offset units will be installed per the standard tiedown charts unless otherwise instructed by details that accompany the home.

## Step 2. Determine Tiedown Configuration

Tiedown spacings have been provided when the tie down strap is connected to the near I-beam and/or when the strap is connected to the far I-beam.

## Step 3. Select Anchors

Use the torque probe results from **Prepare the Site, Step 6. Determine ground anchor holding capacity** (p. 32) and the anchor manufacturer instructions select the type and length of anchor to use. The installed ground anchor size (length) must be listed (i.e. approved) for the soil class.

Make sure the anchor is of sufficient length such that the top of the helix is below the frost line. Select a shaft diameter sufficient to resist excessive torsion, “ring-off” (when the helix or anchor head separates from shaft) or shaft splitting. Consult the anchor supplier for guidance.

Review all of **Step 4. Install Anchors** (p. 113) and **Step 5. Install Straps** (p. 115) before making final anchor selections to determine when single headed vs. double headed anchors should be used. Follow the specifications in **Table 47** when selecting stabilizing system components.

*Table 47. Anchor types*

| Anchor type | Anchor (soil) class |
|---|---|
| Helix soil anchor | 2, 3, 4A |
| Rock anchor | 1 (for use in solid rock only) |
| Concrete anchor | Concrete ONLY |

*Table 48. Anchor System Materials specifications*

| Component | Specification |
|---|---|
| **Anchors** | Anchors must be tested and listed to resist a minimum ultimate load of 4,725 lbs and a minimum allowable working load of 3,150 lbs or ultimate and corresponding working load limited by soil conditions and anchor length. The working load is the maximum load the designer can use. Anchors that are tested and listed to exceed the 3150 lbs design load may be considered. Ground anchors must be provided with protection against weather deterioration and corrosion at least equivalent to that provided by a coating of zinc on steel of not less than 0.030 oz per sq ft of surface coated. |
| **Straps** | Straps must be minimum 1-1/4" x 0.035" zinc-coated (0.030 oz per sq ft) steel strapping conforming to ASTM D3953-97, Type 1, Grade 1, Finish B with a minimum allowable working load capacity of 3,150 lbs and a minimum ultimate load of 4,725 lbs. Slit or cut edges of zinc-coated strapping do not need to be zinc coated. |
| **Stabilizer plates** | The size and type of stabilizer plate, if required by the ground anchor manufacturer, will be specified in the anchor manufacturer's instructions. Stabilizer plates must be provided with protection against weather deterioration and corrosion at least equivalent to that provided by a coating of zinc on steel of not less than 0.30 oz per sq ft of surface coated. Alternatively, ABS stabilizer plates may be used when listed and certified for such use. |

## Step 4. Install Anchors

Before beginning anchor installation, check for obstructions under the home such as piers and frame members that may interfere with the tie down strapping. Check with utility companies to determine the location of underground utilities, such as electrical and phone lines, and water, sewer and gas pipes, that may be buried in potential anchor locations. Also check for homeowner-installed wires and pipes, such as those connecting exterior lighting or sheds to the home. These must also be avoided.

### Frame Anchors

Frame anchors can be installed in two ways, in-line and against a stabilizer plate. The two methods are discussed below:

### In-Line Configuration

The in-line configuration **(Figure 54**) for homes can be used in Wind Zone I only. Typically, in-line anchors are used under high homes where the anchors can be installed from under the home after the home is set. In-line anchors can also be installed before the home is set, however precisely aligning the anchor with the home both vertically and horizontally is difficult. Using swivel connectors for the strap to beam connection can provide some horizontal flexibility.

To install in-line frame anchors, drive the anchor into the ground at an angle and location such that a straight line can be drawn from the tip of the anchor through the anchor head and to the connection point on the I-beam (**Figure 54**).

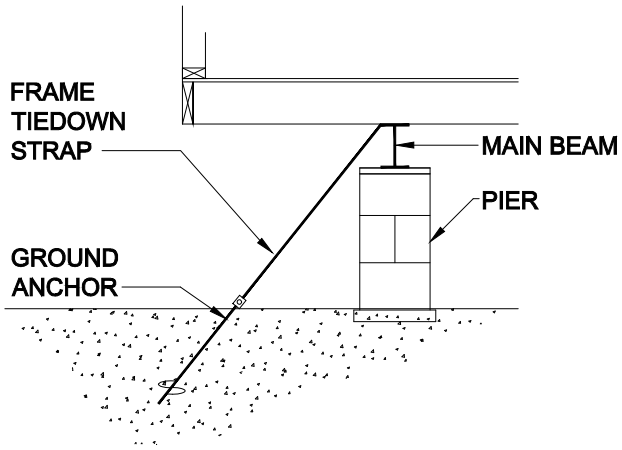



*Figure 54. In-line anchor configuration*

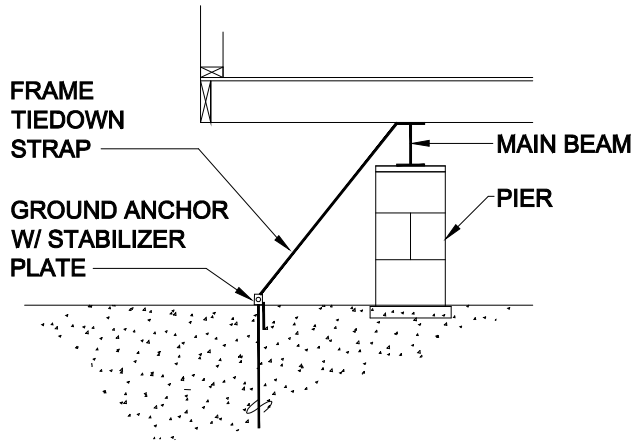



*Figure 55. Stabilizer plate configuration*

**Anchor alignment.** Properly aligning the anchor is critical to performance.

### Stabilizer Plate Configuration

Stabilizer plate configurations are suitable for homes in all wind zones. Anchors may be installed after the home is set. A stabilizer device, typically an ABS or metal plate, is used to prevent the top of the anchor from slicing through the soil when the load is applied. Stabilizer plates are available in a variety of widths. Choose the widest plate that can be driven into the soil to maximize resistance to movement. The LAHJ may have stabilizer plate requirements.

Install anchors with stabilizer plates as follows:

1. **Measure.** To determine the stabilizer plate location, measure from the top of the I-beam to the ground directly under it and then use the same measurement directly away from that point under the beam as shown in the ***Strapping and Anchoring QR code*** (p. 115). The anchor

and plate must be under the home, but within six inches of the exterior wall for a near beam set and ten inches minimum from the exterior wall for a far beam set. Be careful not to place in a location that will interfere with skirting.



*Strapping and Anchoring Instructions*

2. **Install anchor.** To assure that the anchor attachment point will end up at the stabilizer plate, start the anchor insertion approximately 12 inches back from the desired location if using a 48 inch long anchor, or approximately 16 inches if using a 60 inch long anchor. Install the anchor at about 10 degrees off vertical, with the head tilted away from the home. Install the anchor to a depth of approximately one half its length.
3. **Drive stabilizer plate.** Drive the stabilizer plate into the ground to its full depth at the point determined in Step 1.
4. **Complete anchor installation.** Screw the anchor the rest of the way into the ground. The finished anchor must be installed to its full depth. When the anchor strap is properly tensioned it will pull the anchor head and shaft into the stabilizer plate.

### Vertical Anchors

To install vertical anchors, screw the anchor into the ground directly under the strap attachment point on the home until the bottom of the anchor head is flush with the ground or no more than one inch above grade.

## Step 5. Install Straps

Follow the instructions below to connect straps from the home to sidewall frame, end wall frame and vertical anchors.

Always protect straps at sharp corners including around I-beams with radius clips or other methods (**Figure 56**). Radius clips may be fabricated from galvanized steel strap formed to fit around corners.

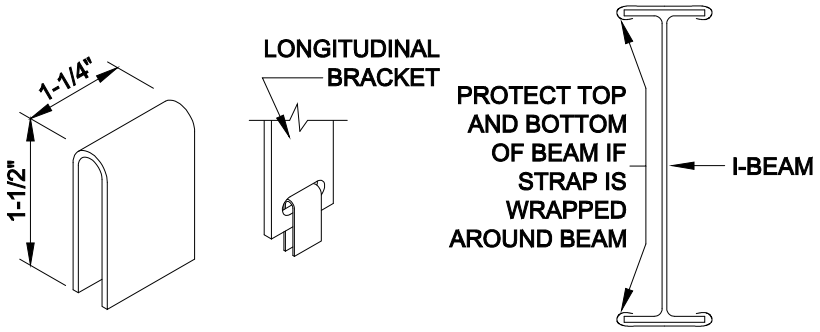



*Figure 56. Radius clips*

### Splicing Straps

Splicing may be required when a pre-cut strap is of insufficient length. Splices must be made by overlapping the straps by 12 inches, applying one splice clip from above and the other from below; use a crimping tool to tightly seal the splice clips (**Figure 57**). Do not run any portion of the splice through an anchor head bolt.

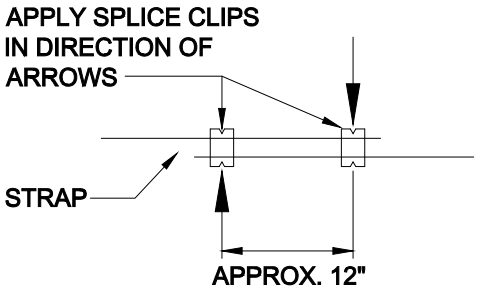



*Figure 57. Tie down strap splice*

### Frame Anchors

Install straps to frame anchors as follows:

1. **Connect strap to home.** Connect one end of the strap to the top of the I-beam using approved buckles or clips (swivel or hook clip preferred). When frame ties are connected to the bottom of the beam it must be within 3” of a crossmember.

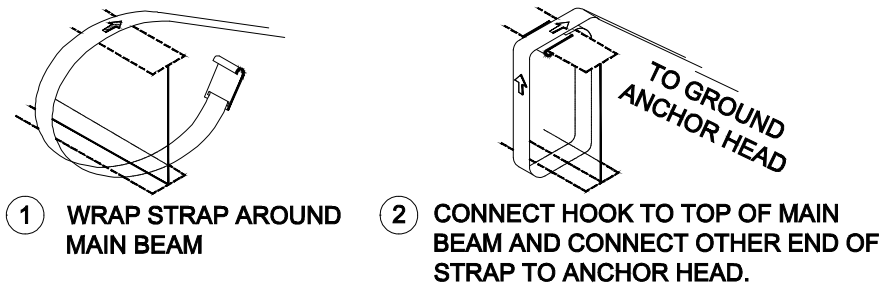



*Figure 58. Strap to beam connection*

2. **Connect strap to anchor**. Connect the other end of the strap to the split bolt in the anchor. Leave enough strap length to be able to make three complete turns, or minimum required per manufacturer's installation instructions, around the bolt before it becomes tight (approximately 2-1/2 inches per turn or 13 inches total). Fewer than three turns, or required per manufacturer's installation instructions, and the strap may not hold onto the bolt when force is applied. Conversely, too many turns may not fit within the U-channel of the anchor head. Follow the procedure outlined in the ***Strapping and Anchoring QR code*** (p. 115).
3. **Pretension anchor.** For anchors with stabilizer plates, pretension the anchor by pulling it up to the stabilizer plate using the strap and take-up bolt to move the anchor head. Continue pulling the strap until the plate moves a small amount (about 1/2 inch). This is called packing the plate and it will yield the strongest resistance (the bottom of the anchor head should be a maximum one inch above the top of the stabilizer plate).



**Anchor head location.** As the anchor is pulled up to meet the stabilizer plate, the head of the anchor will rise. In its final position, the bottom of the anchor head should be no more than 1/4 inch above the top of the stabilizer plate.

### Longitudinal Frame Anchors

Attach straps to the bracket welded by the manufacturer to the frame (**Figure 59**). If no brackets have been installed, use approved beam clamps designed specifically for this purpose, available from anchor suppliers or connect the strap to a spring hanger or a crossmember (within 3" of the main I-beam). Connect straps to anchors following same procedure as for sidewall frame anchors. Protection of the strap at sharp corners must be provided (p.116).

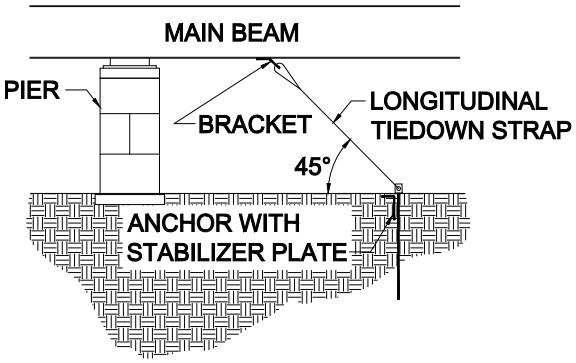



*Figure 59. Longitudinal frame anchor attachment method*

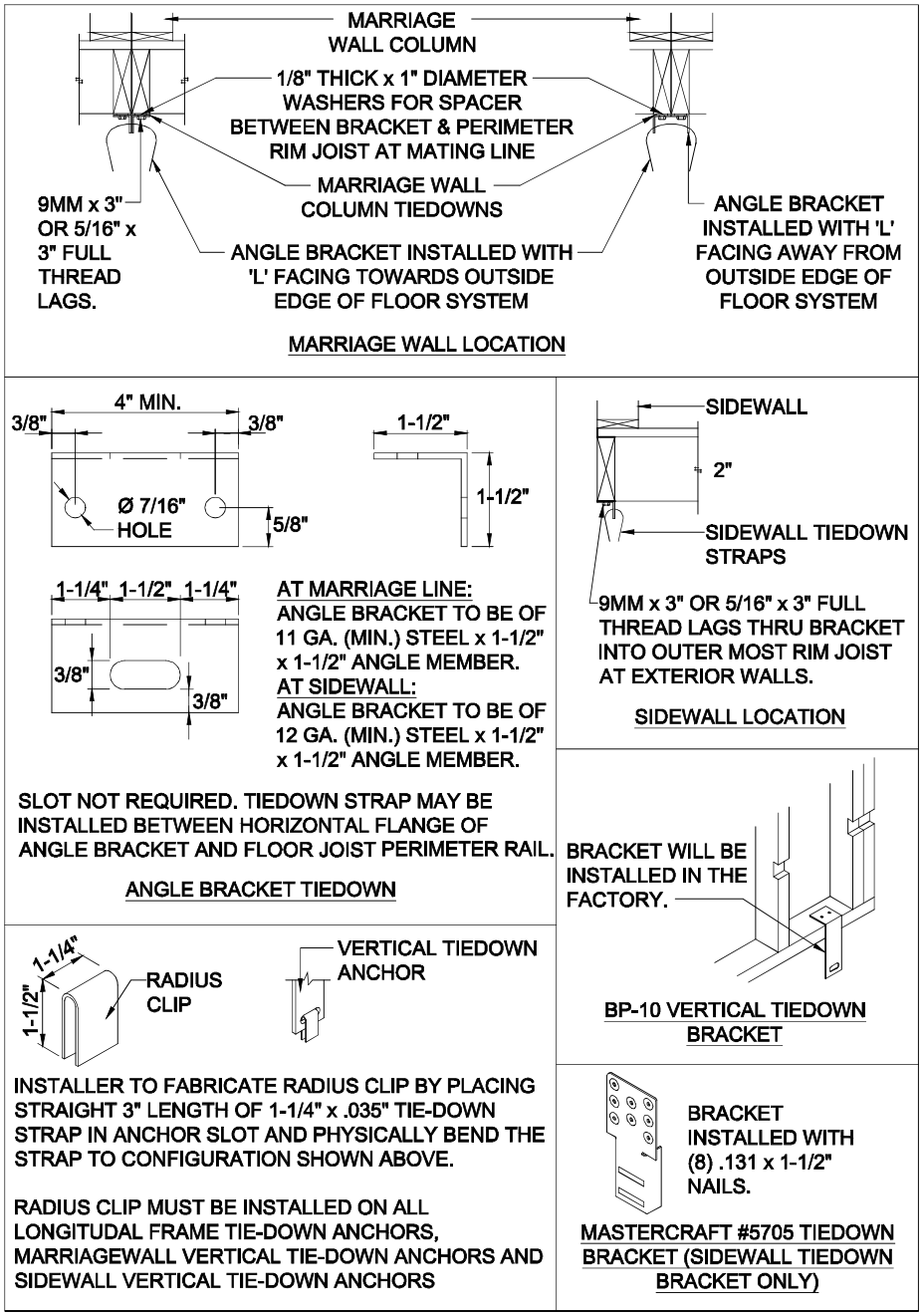






Figure 60. Sidewall and marriage line vertical tie down connections

## Step 6. Tighten and Adjust Straps

After all anchors have been installed and pre-tensioned, recheck all anchor straps to assure that they are tight and that the anchor shafts have remained in contact with the stabilizer plates. Do not over tension straps.

► go to Connect Utilities (p. 119)

# CONNECT UTILITIES

This chapter contains procedures and requirements for the connection and testing of utility hook-ups. Responsibility for making utility connections varies by location. Consult the LAHJ and the utility before connecting the home to any utilities.

**Follow the Steps below:**


▼ **Step 1. Connect Electrical Service** (p. 119)
▼ **Step 2. Connect Water Service** (p. 124)
▼ **Step 3. Connect Gas Service** (p. 127)
▼ **Step 4. Connect Oil Service** (p. 128)


## Step 1. Connect Electrical Service

The home is designed for connection to an electrical wiring system rated at 120/240 volt AC. Service connection requirements depend on whether the meter will be installed on a post or pole or mounted on the home more than 6 feet from the circuit breaker panel. It is recommended that this connection be performed by a licensed electrician.

### Meter Mounted on Post or Pole (or Mounted on the Home More Than 6 Feet from the Circuit Breaker Panel)

#### Feeder Wire and Equipment Sizes

The feeder must contain four continuous insulated, color-coded, feeder conductors, with one used as the equipment grounding conductor (**Figure 61**). The current rating (in amperes) of the home can be found on the tag located on the outside next to the feeder or service entrance, and on the electrical distribution panel. Using this information, determine the required feeder wire size from **Table 49**. These sizes are based on an ambient temperature of 86 degrees Fahrenheit and do not take voltage drop into consideration.

Acceptable conductor types are: RHH, RHW, RHW-2, THHN, THHW, THW, THW2, THWN, THWN-2, XHHW, XHHW-2, SE, USE, and USE-2.



**Special precautions when installing electrical service.** Installation of the electric power to the home can cause exposure to live electrical circuits. The neutral conductor must not be grounded in the distribution panel board. Exposure to live electrical circuits or improper grounding of the conductor in the panel board may result in severe shock or possible electrocution. A qualified installer must make the connections for the electric power.

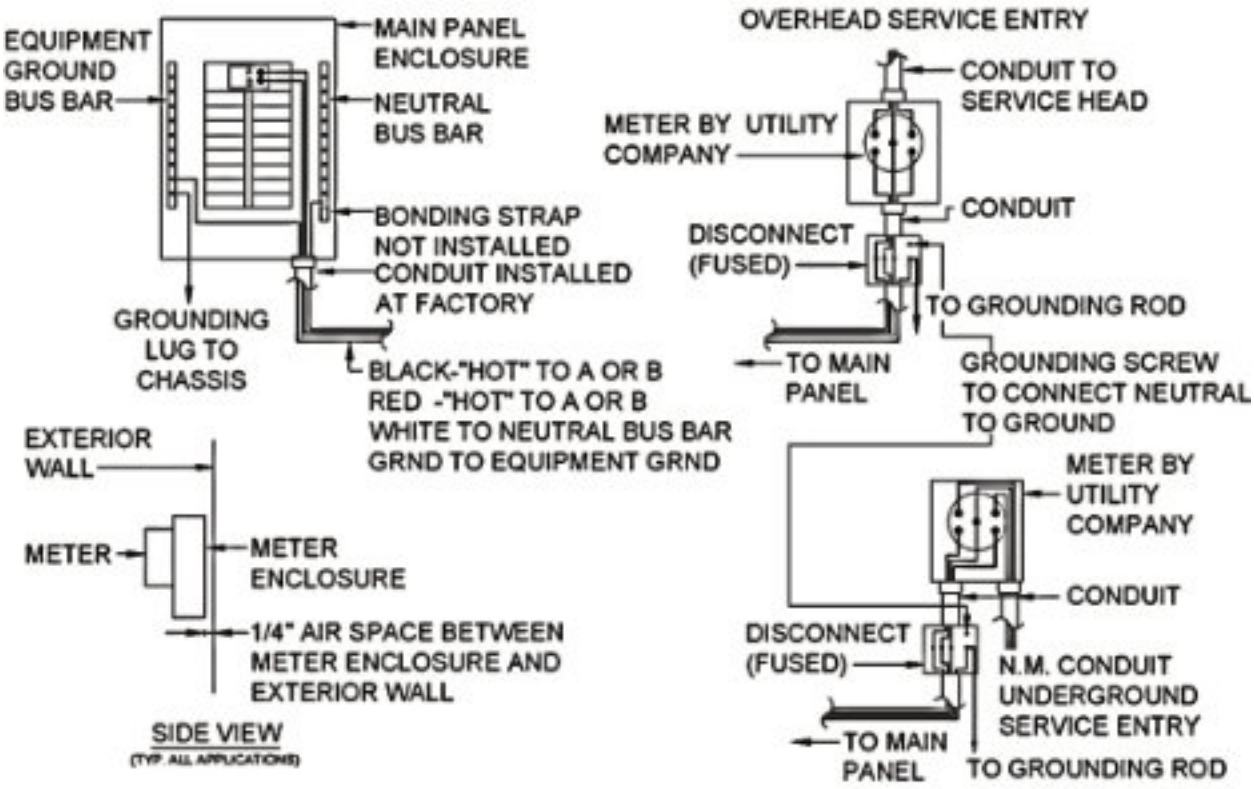



*Figure 61. Electrical feeder connection when meter is on a post or pole or mounted on the home more than 6 feet from the circuit breaker panel*

**Power supply.** A large enough power supply must be available at the site. An inadequate power supply may result in improper operation of and possible damage to motors and appliances. It may also increase electricity costs.

*Table 49. Electrical feeder wire and equipment sizes For Copper Conductors*

| Main Breaker Size in Panel Box (AMPS) | Max. Neutral Feeder Load (AMPS) | Minimum Size Raceway Conduit Diameter (in) | Copper Conductors | | | |
|---|---|---|---|---|---|---|
| | | | Red & Black (Power) (Cu) | White (Neutral) (Cu) | Green (Grounding) (Cu) | Bare Ground (Cu) |
| 50 | 40 | 1-1/4 | #4 AWG | #8 AWG | #10 AWG | #8 |
| 100 | 70 | 1-1/4 | #4 AWG | #4 AWG | #8 AWG | #8 |
| 125 | 100 | 1-1/4 | #2 AWG | #3 AWG | #6 AWG | #8 |
| 150 | 120 | 1-1/2 | #1 AWG | #2 AWG | #6 AWG | #6 |
| 200 | 175 | 2 | 2/0 AWG | 1/0 AWG | #6 AWG | #4 |
| 225 | 180 | 2 | 3/0 AWG | 3/0 AWG | #4 AWG | #4 |
| 400 | 320 | 3 | 400 kcmil | 400 kcmil | #3 AWG | 1/0 |



**Grounding the electrical system.** Do not provide electrical power until the grounding electrode is installed and connected. When the meter base is not on the house never use the neutral conductor of the feeder cable as a ground wire. Do not ground the neutral bar in the electrical distribution panel.

*Table 50. Electrical feeder wire and equipment sizes For Aluminum or Copper-Clad Aluminum Conductors*

| Main Breaker Size in Panel Box (AMPS) | Max. Neutral Feeder Load (AMPS) | Minimum Size Raceway Conduit Diameter (in) | Aluminum or Copper-Clad Aluminum Conductors | | | |
|---|---|---|---|---|---|---|
| | | | Red & Black (Power) (Al) | White (Neutral) (Al) | Green (Grounding) (Al) | Bare Ground (Al) |
| 50 | 40 | 1-1/4 | #2 AWG | #8 AWG | #8 AWG | #6 |
| 100 | 70 | 1-1/4 | #2 AWG | #2 AWG | #6 AWG | #6 |
| 125 | 100 | 2 | 1/0 AWG | #1 AWG | #4 AWG | #6 |
| 150 | 120 | 2 | 2/0 AWG | 1/0 AWG | #4 AWG | #4 |
| 200 | 175 | 2-1/2 | 4/0 AWG | 3/0 AWG | #4 AWG | #2 |
| 225 | 180 | 2-1/2 | 250 kcmil | 4/0 AWG | #2 AWG | 1/0 |
| 400 | 320 | 3-1/2 | 600 kcmil | 600 kcmil | #1 AWG | 3/0 |



Prior to energizing the home, turn off the water heater until it is completely filled with water.

### Grounding

The home must be properly grounded to protect the occupants. The only safe and approved method of grounding the home is through an electrically-isolated grounding bar in the home's distribution panel board. This grounds all non-current-carrying metal parts to the electrical system in the home at a single point. The ground conductor of the power supply feeder cable in turn connects the grounding bar to a good electrical ground back through the power supply system. Therefore, for 120/240 volt service a four wire power supply feeder cable is required. It is important to:

- Isolate (insulate) grounded circuit conductor (neutral or white wire) from the grounding conductors (green wires) and from equipment enclosures and other grounded parts.
- Isolate (insulate) the neutral circuit terminals in the distribution panel board and in ranges, clothes dryers, and counter-mounted cooking units from the equipment enclosure.

### Feeder Connections

Feeder connections are made from above or from below the home as follows:

- From above—mast weatherhead feeder. The routing, connection, and support of the service drop must meet local codes. Homes equipped this way contain all necessary conduits to the

electrical distribution panel. However, the four feeder conductors (not provided with the home) are installed on site. If the masthead is located above the roof overhang, allow a minimum clearance of eight feet above all roof points that the conductors pass over. There are two exceptions to this rule: (1) The vertical clearance may be reduced to three feet if the roof has a minimum slope of 4 in 12; and (2) The vertical clearance may be reduced to 18 inches if no more than four feet of service-drop conductors pass above the roof overhang, and if they terminate at a through-the-roof raceway or approved support. A minimum clearance must also be provided from the final grade to the service-drop conductors. This measurement may vary from 10 feet to 18 feet, depending on the types of traffic anticipated below the service drop (refer to the NEC). Unless impractical, locate service heads above the point of attachment of the service-drop conductors and make them rain-tight. If individual conductors do not extend downward, form drip loops.

- From below. A section of conduit is factory installed through the floor cavity. Install properly sized service entrance conductors from the main power supply to the panel board. Depending on the location of the main panel board inside the home. Refer to **Table 49** for the conductor requirements. The installer must provide the supply connection including the four feeder conductors, junction box and conduit connectors. Protect conductors emerging from the ground from a minimum of 18 inches below grade to eight feet above grade, or to the point of entrance to the home. The distance measured from the top surface of a buried cable, conduit, or raceway to the finished grade must meet the minimum burial requirements outlined in the NEC. Use a moisture-proof bushing at the end of the conduit from which the buried cable emerges.

**Meter Mounted on Home within 6 Feet of Circuit Breaker Panel**

If the meter is mounted on the home, the following requirements apply (refer to **Figure 62**):

- Use straps to support any conduit. Do not use the meter base equipment for support.
- Use exterior equipment and enclosures listed as weatherproof and entrance conductors listed for wet locations.
- The grounding bar may be installed separate from the neutral bar for purposes of testing the electrical system.
- The grounding bar may be isolated during the electrical check and re-attached, after the tests are completed.
- Check with the local electrical utility to verify meter base requirements and locations and distances for the main panel and meter box.
- The field installed meter base enclosure must be installed in accordance with its listing. Fasten securely to exterior wall studs and provide for a weather tight seal.
- All field work must be done by a licensed electrician or other person approved by the LAHJ.
- Check the local code for any requirements regarding the location of the meter base.

Note that bonding, screws, straps, or buses in the distribution panel board or in appliances have been removed and discarded at the manufacturing facility.

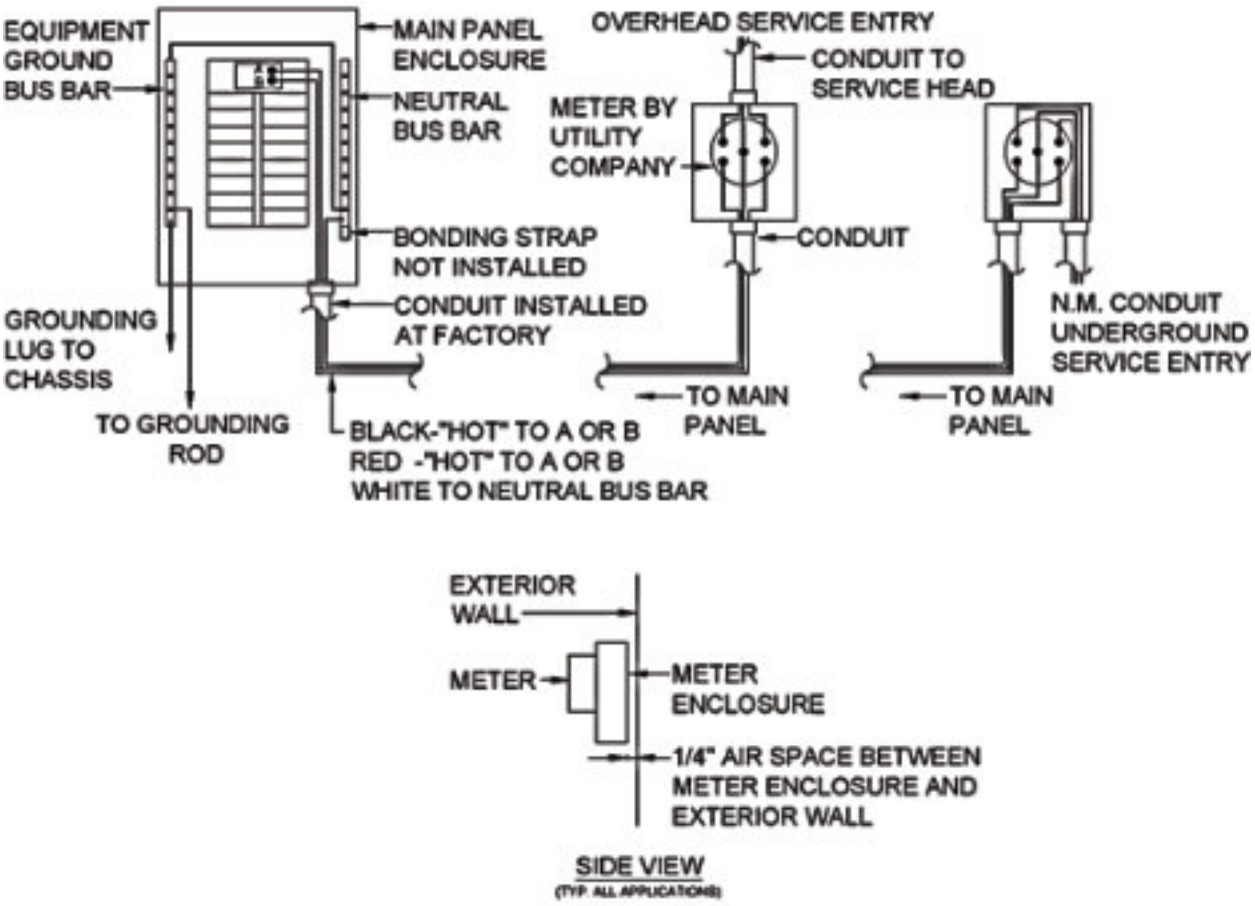



*Figure 62. Meter base wiring when the meter is on the home within 6 feet of Circuit Breaker Panel*

When a factory installed service meter base is provided on a home, a grounding electrode conductor and a ground wire must be installed according to the following specifications:

1. Grounding wire material and size per **Table 49** provided by the manufacturer. If manufacturer provides a minimum 1/2 inch EMT or conduit raceway, the wire is provided by the retailer for installation.
2. The clamp connecting the grounding wire to the electrode shall be suitable for direct burial and located flush or below ground level.
3. Use a 5/8 inch diameter by eight foot long iron electrode for grounding. Larger sizes may be required by LAHJ.
4. Drive the electrode to a depth of not less than eight feet so that at least eight feet of the electrode is in contact with the soil.
5. When rock is encountered, the electrode may be driven at an angle not to exceed 45 degrees from vertical or buried in a trench that is at least 2-1/2 feet deep.

## Step 2. Connect Water Service

### Connection

To connect the home's water system to the water source, identify the water inlet located under the home (usually below the water heater compartment or utility room) and follow the procedure described below (refer to **Figure 63**):

1. **Flush pipe.** Flush field installed water piping free of all debris prior to connection to the home's water inlet.
2. **Clean threads.** Ensure that pipe threads are clean.
3. **Install pressure-reducing valve.** If the local water supply exceeds 80 psi install a pressure-reducing valve.
4. **Connect valve.** Install a main shut-off valve between the water supply source and the home in an accessible location underneath or adjacent to the home. Select a full flow gate or ball shutoff valve, or equivalent valve. To prevent the possibility of fresh water contamination, install an anti-siphon valve on all field installed exterior faucets.



**Maximum water pressure.** The water system for the home was designed for a maximum inlet pressure of 80 psi.

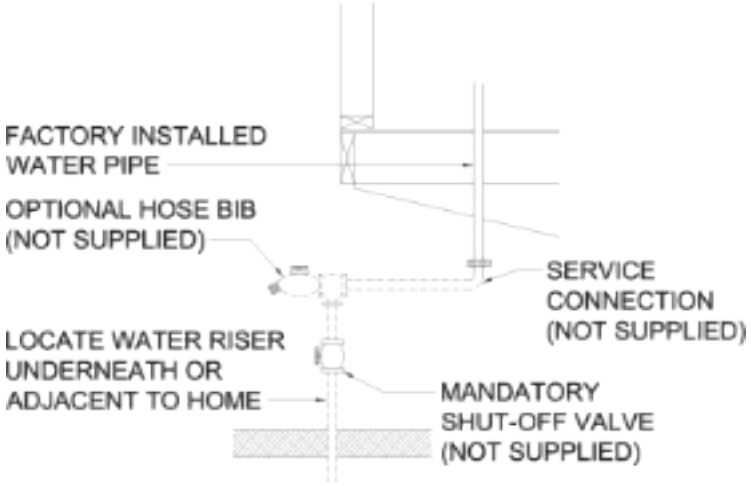



*Figure 63. Water system connection*

5. **Inspect water heater drip pan drain pipe.** Inspect the drain opening on the water heater to ensure that it is clear of any obstruction. Drain pipe cannot connect with the DWV line. Drain pan line must be separate from the T/P line.
6. **Inspect water heat T/P relief line.** Verify that the T/P relief line terminates 6" min and 24" max from the finish grade. Relief line must be directed down at the exit. T/P line must be separate from the drain pan line.

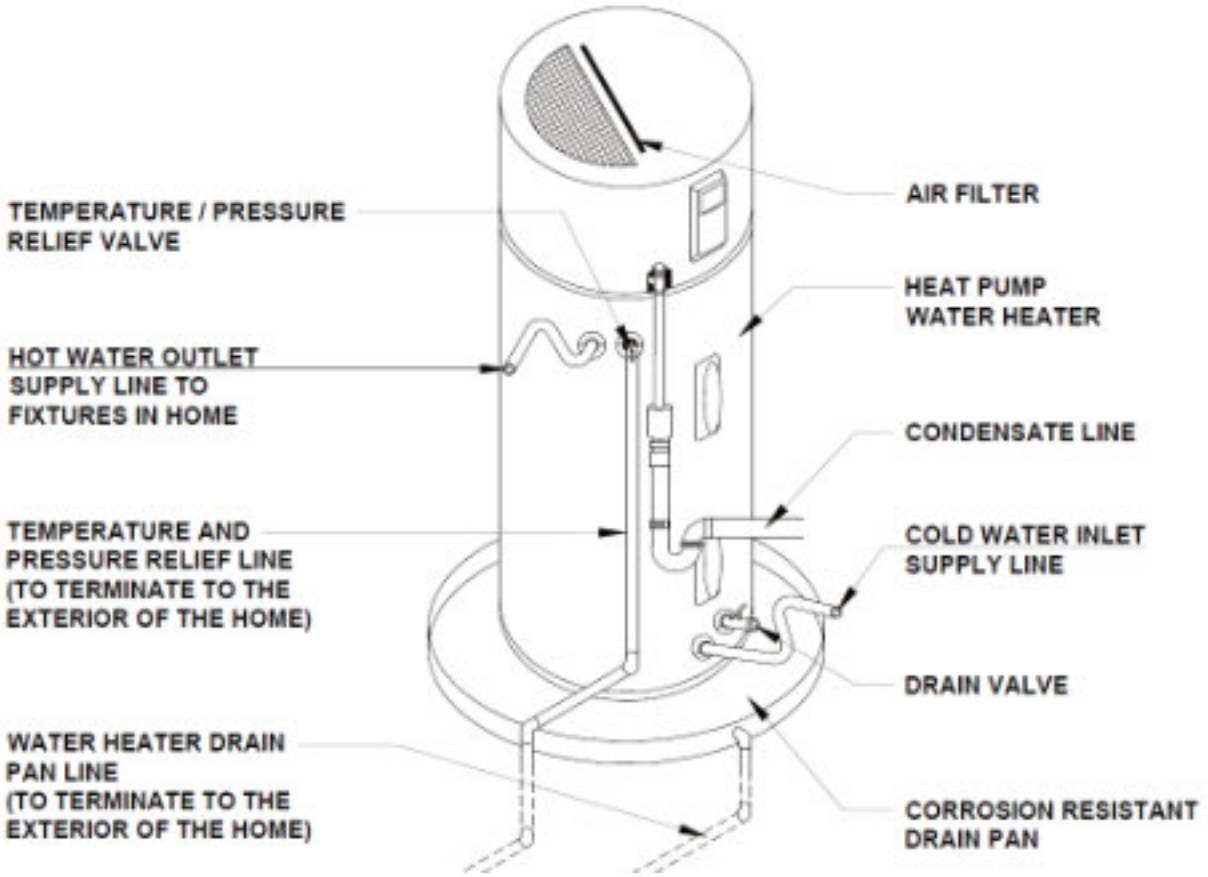



*Figure 64. Heat pump water heater*

7. **Insulate.** All exposed water supply lines are required to be protected with insulation or heat tape. Connect heat tape to the electrical outlet under the home near the water supply inlet. Heat tape must not be installed on the DWV pipe. Electrical outlet provided under the home must only be used for the heat tape connection because it is GFCI protected. In addition to insulating exposed water lines, all penetrations in the bottom board must be completely sealed, as outlined in **Step 1. Repair and Seal Bottom Board** (p. 138).

**Testing Water System**

After connecting the water lines check the water system for leaks using one of the procedures described in the ***Water System Testing Protocols QR code*** below. Before testing, close all water faucets, spigots, and toilet-tank float valves.

*Water System Testing Protocol*

**Selecting heat tape.** Use only pipe heating cable (tape) listed for manufactured homes and install it in accordance with the cable (tape) manufacturer installation instructions.

**Testing water lines.** Only use pneumatic (air) testing when hydrostatic testing is not practical. Air under pressure is explosive. Exercise extreme caution and notify all site personnel of the test. Wear protective eyewear and take precautions to prevent impact damage to the system while the test is in progress. Do not pneumatically test CPVC systems. Pneumatically test Flow Guard Gold systems only at low pressure levels (20 psi or less).



**Pressurizing water lines.** When pressurizing the water system, connect the pump to a location above a closed shut-off valve so as not to introduce pressure into the municipal water supply.

**Anti-Scald Valves.** Anti-Scald valves have been installed on all tubs, tub/showers and showers in the home. The valves are preset by the valve manufacturer to about 105°F (41°C). After the water lines have been flushed, the outlet temperature at each tub, tub/shower and shower must be tested to ensure that it does not exceed 120°F (49°C). Water should run for at least one minute on the hottest setting before taking the temperature reading. Some customers may desire temperatures higher than 105°F (41°C). Temperature may be adjusted using the instructions provided with the valve. In no case should the temperature exceed 120°F (49°C) as this may result in serious bodily harm and/or death.

## Step 3. Connect Gas Service

### Connection

If the home uses natural or liquid petroleum gas (LPG, also known as propane) for water or space heating, cooking or other appliances, follow the procedure described below:

1. **Inspect vents.** Assure that all exhaust vents on gas-fired equipment are securely connected and that roof jacks and stacks have not come loose during transit and they are properly installed.
2. **Review appliance instructions.** Review each appliance manufacturer's instructions before the home is connected to the gas supply. Most gas appliances are typically configured to operate on natural gas. If the gas supply will be LPG, consult the appliance manufacturer's instructions to determine what changes need to be made. For homes located above 3,000 feet in elevation, appliances may require a different orifice.
3. **Remove cap.** Remove the protective cap from home inlet pipe and install a full flow shut-off valve at the supply inlet (**Figure 65**).
4. **Install regulator.** The gas piping system is designed for a pressure that is at least seven inches of water column (4 oz. per sq in or 0.25 psi) but not more than 14 inches of water column (8 oz. per sq in or 0.5 psi). If gas from any supply source exceeds, or could exceed this pressure, install a regulator if required by the LAHJ.
5. **Connect supply.** Using matching threaded fittings connect the gas supply to the inlet side of the shut-off valve.
6. **Close valves.** Close all valves at appliances prior to opening the main supply valve.



**Installing gas lines.** Only qualified professionals may connect and test gas service.

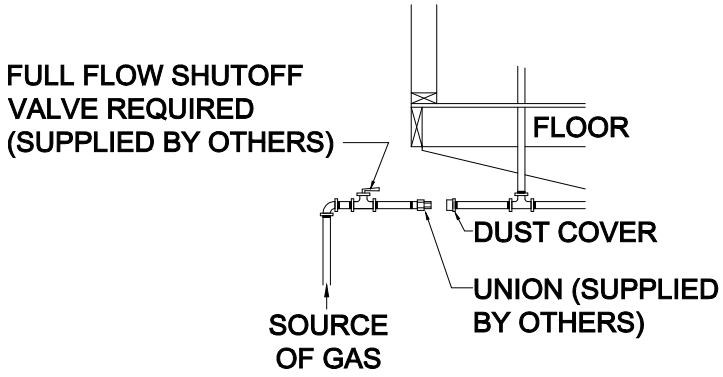



*Figure 65. Gas service connection*

### Testing Gas Piping System

Test the gas piping system in the following two ways: 1) piping only and 2) entire system. These test procedures are described in the ***Gas Piping Testing Protocols QR code*** below. Consult with the LAHJ for any additional testing or start-up requirements.

*Gas Piping System Test Protocol*

## Step 4. Connect Oil Service

Homes that are equipped with oil burning furnaces must have oil supply piping installed and tested on site by a qualified professional in accordance with NFPA 31, Standard for the Installation of Oil Burning Equipment, 2001 or the requirements of the LAHJ, whichever is more stringent. The home manufacturer does not supply oil piping or tanks.



**Fill gas water heaters.** Before lighting the pilot on a gas powered water heater, fill the tank with water. Failure to do so could damage the water heater.

### Oil Connection

Consult the furnace manufacturer's instructions for proper pipe sizing and installation procedures. Where piping is run through the bottom of the home, ensure all holes in the bottom board are sealed tight with foam, mastic, and/or tape specially made for that purpose and made rodent proof.

When equipping the home with an oil storage tank, comply with the following:

- Install the pipe with a gradual slope toward the fill end or drain plug (if so equipped) to facilitate pumping or draining of water and sludge.
- Provide a readily accessible approved manual shut-off valve at the outlet, installed to close against the supply.

- Equip the tank with an approved oil filter or strainer located downstream from the tank shut-off valve. Use a filter or strainer containing a sump with a drain to trap water.
- Equip underground tanks with a filler neck extending one foot above grade and a minimum 1-1/4 inch diameter vent pipe extending at least two feet above grade.
- Locate the tank to be accessible for service and inspection, and safe from fire and other hazards.
- If the tank is located inside a compartment of the home, provide ventilation at the bottom of the compartment to permit diffusion of vapors. If the tank is fixed to the home, provide for filling and draining from the outside.
- Insulate interior tanks from the structural members of the home. Provide tanks so installed with an outside fill and vent pipe and an approved liquid level gauge.
- Install tanks that feed vaporizing type oil furnaces so that oil flows by gravity. To achieve efficient gravity flow, make sure that the bottom of the tank is at least 18 inches above the furnace oil control level.
- Tanks for gun type oil furnaces (these furnaces include a fuel pump) may be installed above or below ground.

### Oil System Testing

Before operating the system, fill the tank to capacity with the fuel to be burned and visually check all joints in the system for leakage. Replace (do not repair) parts that leak.



**Testing oil tanks.** All oil storage tank and piping installations and tests must meet all applicable local regulations and should be made only by experienced, qualified personnel.

► go to **Prepare Appliances and Equipment** (p. 130)

# PREPARE APPLIANCES AND EQUIPMENT

This chapter provides instructions for installing and/or preparing appliances and other equipment. While the items below can be completed in any order, the last item, **Test Electrical System,** must be done last.

**Follow the Steps below:**


▼ **Install Air Conditioner or Heat Pump** (p. 130)
▼ **Prepare Heating System** (p. 131)
▼ **Prepare Clothes Dryer** (p. 131)
▼ **Prepare Smoke Alarms** (p. 132)
▼ **Prepare Fireplaces** (p. 133)
▼ **Prepare Kitchen and Bath Appliances / Fixtures** (p. 135)
▼ **Install Exterior Lighting** (p. 135)
▼ **Install Ceiling Fans and Lighting** (p. 136)
▼ **Test Electrical System** (p. 137)


## Install Air Conditioner or Heat Pump (if Applicable)

The HVAC design package should be included in a plastic sleeve attached to the furnace. Ensure the cooling coil and the outside unit to be installed matches the size listed on the Manual S page from the HVAC design package. Verify the equipment manufacturer's recommended settings for Blower Speed and Subcooling and adjust the furnace and outdoor unit accordingly. Refer to the HVAC Installation Guide included in the plastic sleeve attached to the furnace for more detailed installation instructions.

1. **Point of Source (POS) duct.** Ensure the POS duct extends to the furnace at the return air intake. The duct must be straight and should not have excess slack or kinks. For a downflow furnace the POS should extend to the top of the furnace and for an upflow furnace the POS duct should extend to the bottom of the furnace.

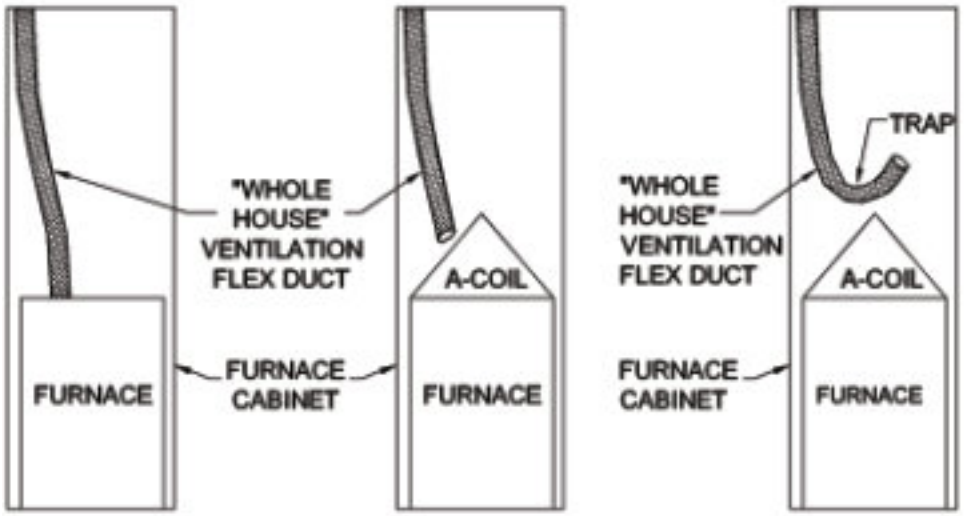



*Figure 66. Whole house ventilation flex duct in an electric down-flow furnace. From left to right: as prepared in the factory; proper installation with A-coil; improper installation with A-coil*

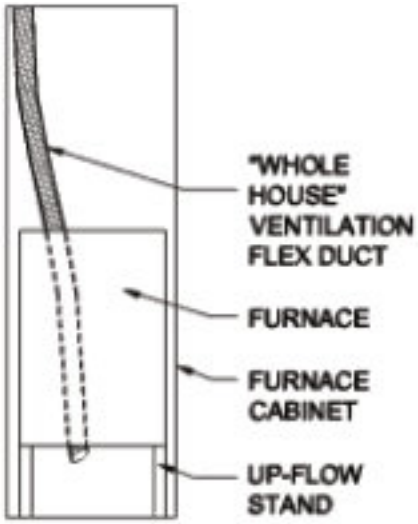



*Figure 67. Whole house ventilation flex duct in an electric up-flow furnace*

## Prepare Heating System

1. **Convert for LP gas.** If LP gas (propane) will be used, convert the appliance from natural gas to LP gas use. Conversion must be made by a qualified and (if required by the LAHJ) licensed technician.
2. **Install flue.** For combustion appliances, instal the flue roof cap and stack assembly as described in the manufacturer's installation instructions.

## Prepare Clothes Dryer

Equipment is to be installed per manufacturers specifications. Gas dryers are to be installed by licensed professionals. Dryer vents shall not terminate underneath the home. Verify the dryer vent is connected according to the specifications shown in (insert figure reference).

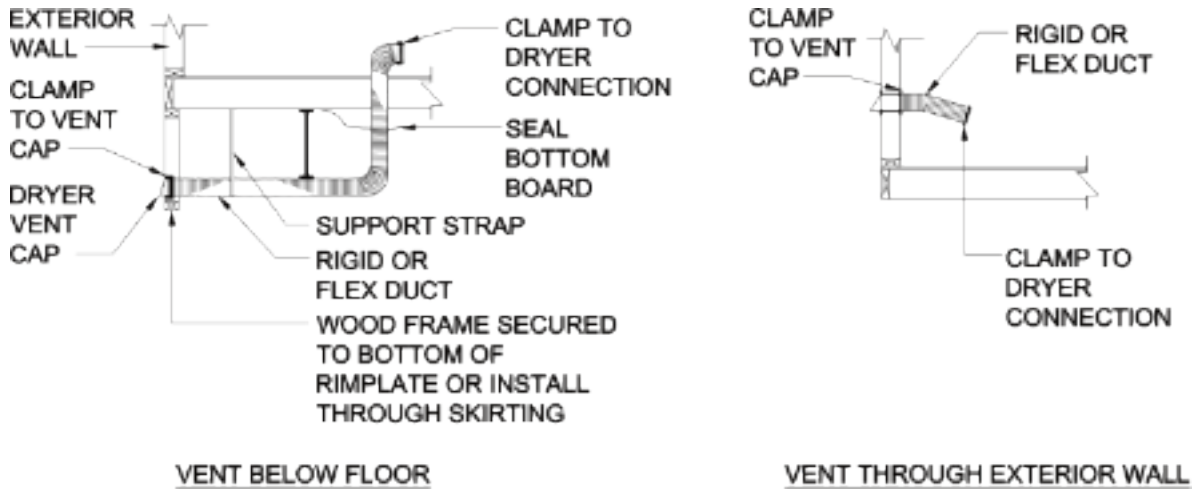



*Figure 68. Gas service connection*

## Prepare Smoke Alarms

The home has several factory installed smoke alarms that are wired to a 120 volt circuit. If the home was designed for placement on a basement, an additional alarm is provided for installation at a pre-wired location under the home. Connect the basement smoke alarm and test all alarms as follows:

1. **Check circuit.** Ensure that the batteries are installed and the electrical power to the home is activated and that the smoke alarm circuit is on.
2. **Test alarms.** Press the **"test"** button on the alarm and hold for 5 seconds (or until the alarm sounds). When the alarm begins to sound, release and confirm that each alarm in the home is sounding. Replace (with the same brand as those installed elsewhere in the home) any alarms that do not sound and retest.
3. **Disconnect AC.** With the AC current disconnected and batteries installed, test for DC operation following the same procedure as shown in step 1.

Both steps 1 and 2 above must be conducted on each alarm installed in the home. If any alarm fails to sound during the test, turn off the power at the main panel box and check the wiring. Make any necessary repairs and conduct a complete re-test. Repeat if necessary until all alarms are functioning properly.

## Prepare Fireplaces

Install chimneys, chimney flashing and roofing, fireplace combustion air inlets and hearths according to the manufacturer's instructions and the procedures described below.

### Chimneys

Fireplace and wood stoves may require on-site installation of additional sections of approved chimney pipe, a spark arrestor and a rain cap assembly. Follow the manufacturer's instructions and the procedures described below:

1. **Remove coverings.** Remove protective materials covering the roof flashing and any foreign material from the installed part of the chimney.
2. **Install chimney pipe.** Assemble and seal the chimney per the fireplace or wood stove manufacturer's installation instructions and if there is a conflict between the instructions and the figure, follow the instructions. To assure sufficient draft for proper operation, extend the chimney at least three feet above the highest point where it penetrates the roof and at least two feet higher than any surface within 10 feet of the chimney (**Figure 69**). Use additional section(s) of chimney pipe (not provided) if required by local code or if the site has obstructions within 10 feet of the chimney.

Fireplaces and wood stoves not provided by the home manufacturer, including chimneys and air inlets for fireplaces and wood stoves must be listed for use with manufactured homes and must be installed in accordance with their listings.

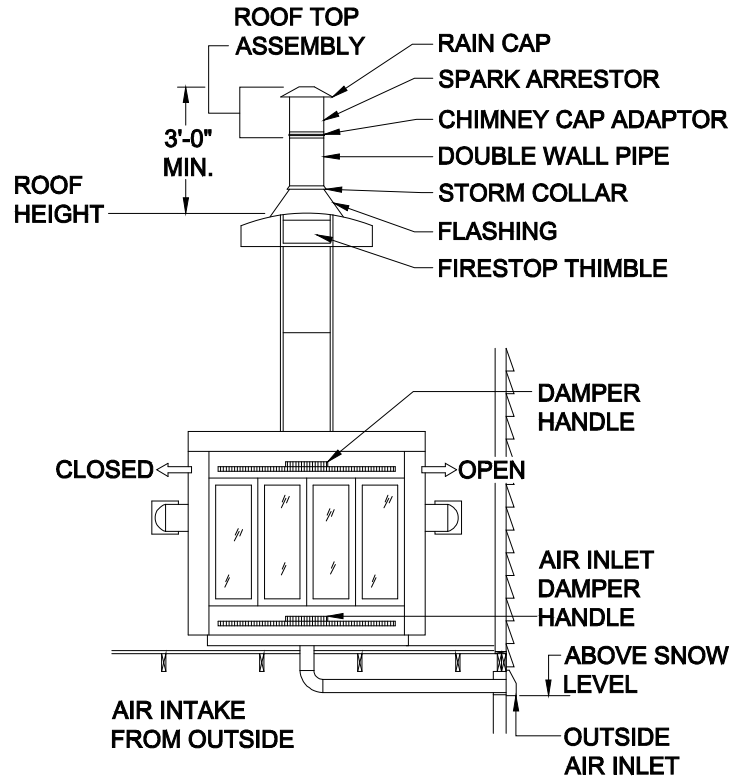



*Figure 69. Chimney and combustion air intake duct installation*

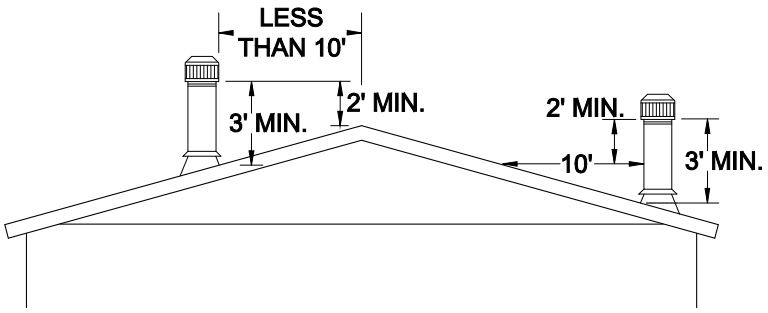



*Figure 70. Chimney clearance*

**Selecting the appropriate flue.** Wood fireplaces use larger diameter flues than gas fireplaces. Make sure flue matches the appliance type. Gas fireplaces cannot burn wood because the flue is too small and smoke will back up into the home.

3. **Install shingles.** Install shingles up to the edge of the flue cut-out in the roof deck. Secure shingles installed under the roof flashing with asphalt cement.
4. **Install flashing.** Place flashing over pipe section and shingles and set in asphalt cement. Secure flashing to roof deck at top two corners with roofing nails.
5. **Complete shingles.** Cut shingles in successive courses to fit around the pipe and embed them in asphalt cement where they overlap the flashing. Secure shingles with roofing nails through flashing and apply asphalt cement over nail heads. The completed installation should appear as shown in **Figure 71**, with the lower part of the flange overlapping the lower shingles and the side and upper shingles overlapping the flange.

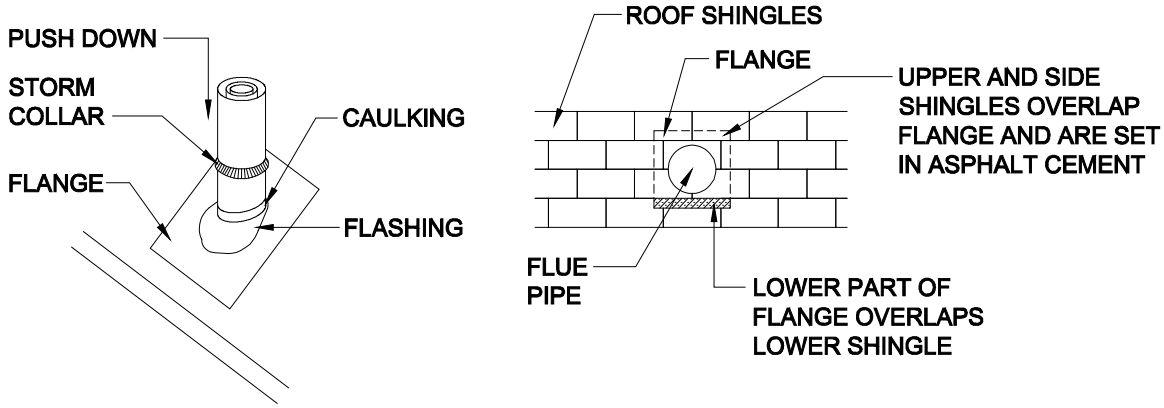



*Figure 71. Roof flashing and shingle installation around chimney*

### Combustion Air Inlets

Combustion air inlets provide combustion air through the floor or an exterior wall to a combustion appliance. If installed through an exterior wall, no further site installation is required. If through the

floor, extend the duct from its point just below the floor to the outside. Locate the outside air inlet above expected snow levels (contact the LAHJ for snow levels).

Follow the fireplace manufacturer's instructions (typically in the fireplace/stove or with the chimney parts).

## Prepare Kitchen and Bath Appliances / Fixtures

Install kitchen and bath appliances according to the manufacturer's instructions.

### Site-Installed Appliances and Fixtures

If sinks, tubs, showers or other fixtures or appliances are to be site-installed, follow the manufacturer's installation instructions. Use only products listed for use in manufactured homes and follow all applicable local codes.

## Install Exterior Lighting

Install exterior lighting according to **Figure 72** and the following:

1. **Remove cover.** Remove the screws and cover from electrical junction box.
2. **Install flash ring.** Place the non-combustible flash ring over the junction box.
3. **Connect wires.** Connect fixture wires to house wires in the box, black to black, white to white and equipment ground to equipment ground, using listed wire connectors. Push wires into the box.
4. **Connect fixtures.** Connect the fixture to the junction box or strap using screws provided with the light fixture.
5. **Weatherproof.** Weatherproof/caulk around the base of the fixture, leaving a small gap in the caulking on the bottom to permit drainage of water that may accumulate.
6. **Complete installation.** Install bulb and globe on the fixture and verify proper operation.



**Lighting installation.** A qualified electrician should install lighting. Before connecting lighting, disconnect power to the lighting circuit. Ground all exterior light fixtures.

**Using a non-combustible ring.** Install a non-combustible ring completely covering any combustible surfaces the fixtures may be mounted on (e.g. hard-board, clad wood and vinyl siding), or when ceiling material is exposed between the light fixture canopy and the junction box.

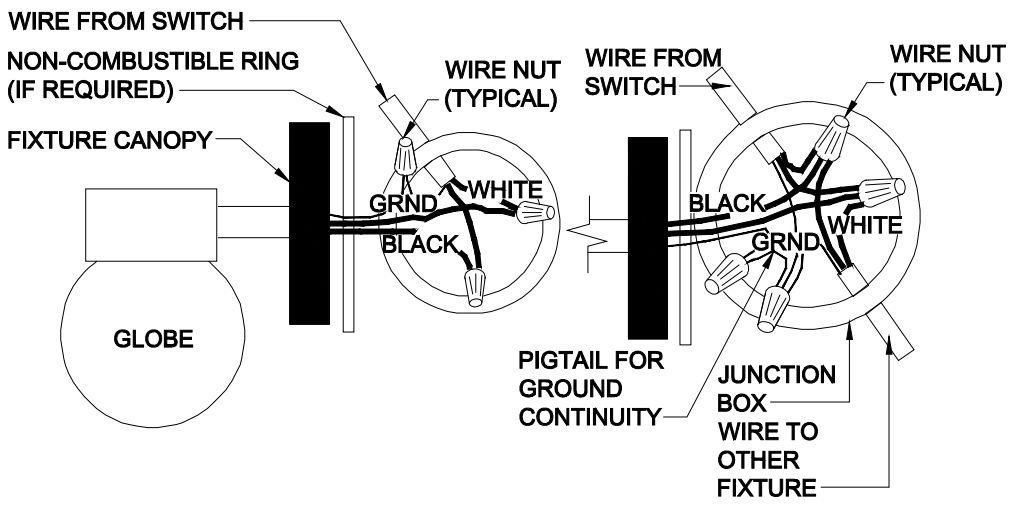



*Figure 72. Exterior lighting connection*

## Install Ceiling Fans/Lights

Install ceiling fans no closer than 36 inches, measured from the tip of the fan blade to the nearest side of the smoke alarm and with the trailing edges of the blades at least six feet-four inches above the finished floor. Comply with all unit manufacturer's instructions and the requirements below.

**Installing ceiling fans.** A qualified electrician should install lighting and fans. Before connecting the ceiling fan or light, disconnect power to the fan wires. Any unit installed outside of the home (such as in a porch ceiling) shall be listed for wet locations.

### Ceiling Fan/Light Mounted on Site-Installed Decorative Beam

Where the electrical box will be mounted in a site-installed decorative center beam attached to the ridge beam, connect the fixture as follows (see **Figure 73**).

**Choosing ceiling fan junction boxes.** Connect ceiling fans only to junction boxes listed and marked for ceiling fan application in accordance with Article 314.27(b) of 2014 NEC. Always ground metal junction boxes.**Selecting fan weight.** Do not use any ceiling fans or light fixtures that exceed the weight rating of the box (35 lbs unless otherwise noted).

**Grounding electrical devices.** Ground fans/lights using a fixture-grounding device or a fixture-grounding wire as specified in the manufacturer's instructions.

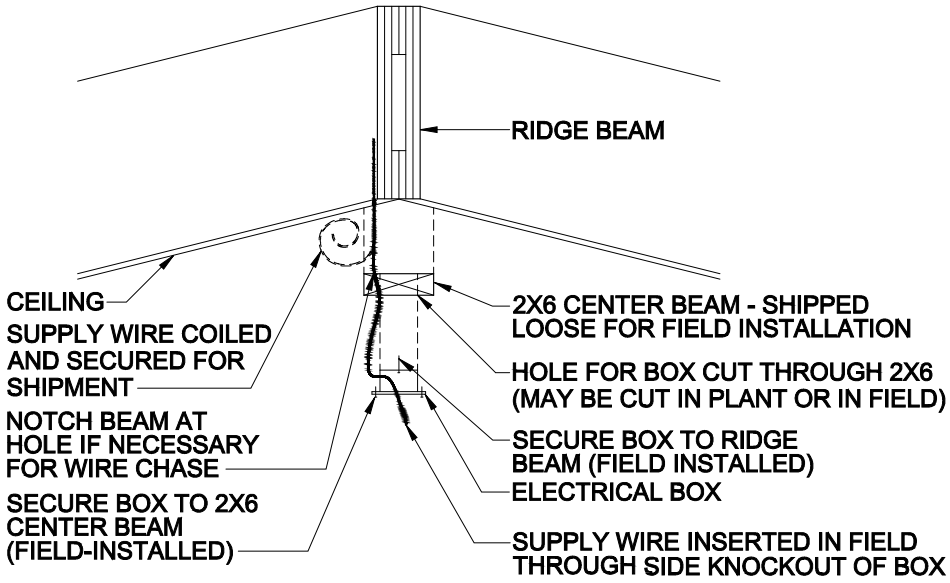



*Figure 73. Ceiling fan/light mounted to flush ridge beam*

## Test Electrical System

After completion of all electrical wiring and connections, including crossovers, appliances, lights, and ceiling fans, the home should be tested to ensure that no damage occurred during transit and that all electrical connections were properly performed. **These tests should be performed by qualified personnel familiar with the local codes and required test procedures.** Inspect and test the electrical system as follows:

1. **Fill water heater.** Fill water heater before turning on power to the home or switching on the circuit breaker.
2. **Test continuity.** Before turning on the electrical power to the home, conduct an electrical continuity test to ensure that exposed metallic parts of the home and the chassis are effectively bonded.
3. **Test operation.** After turning on the electrical power to the home, conduct operational tests of all devices to demonstrate that they are connected and in working order.
4. **Test polarity.** After turning on the electrical power to the home, conduct electrical polarity checks to determine that connections of electrical equipment installed or completed during installation have been made properly. Visual verification is an acceptable electrical polarity check for these on-site connections.

► go to **Complete Exterior Work** (p. 138).

# COMPLETE EXTERIOR WORK

This chapter covers sealing the bottom board, installing the ground cover and skirting, preparing wind protection shutters and completing site built structures.

**Follow the Steps below:**


▼ **Step 1. Repair and Seal Bottom Board** (p. 138)
▼ **Step 2. Install Ground Moisture Retarder** (p. 139)
▼ **Step 3. Install Skirting** (p. 139)
▼ **Step 4. Assemble Optional Wlnd Protection Shutters** (p. 140)
▼ **Step 5. Complete Site Built Structures** (p. 141)


## Step 1. Repair and Seal Bottom Board

Tears and openings in the bottom board can result from transportation or installation activities. Inspect for holes and gaps in the entire bottom board, especially areas around service penetrations, crossover connections, pipe and duct hangers, foundation elements and the perimeter of the floor. Using approved materials appropriate for the type of repair, repair the bottom board wherever torn or loosened as follows:

1. **Clean.** Prepare the bottom board by thoroughly cleaning its exterior surface. Ensure all areas to be resealed are free of debris to promote proper adhesion.
2. **Insulate.** Replace any missing insulation before closing and repairing the bottom board. Pay special attention to insulation gaps that may have been created at P-traps. Ensure insulation provides full coverage in the breached areas and add additional insulation as needed to maintain adequate coverage.
3. **Repair Opening.** Use a minimum of 6” wide Flex-Mend self-adhesive bottom board material (or equivalent) to cover all breaches in the belly board. Secure the Flex-Mend material around the perimeter with the divergent staples spaced 4” to 6” on average for support.



**Securing Flex-Mend repair tape**. A continuous and sealed bottom board is critical for home performance, energy efficiency, protection against moisture problems, prevention of pipe freezing and protection against insects and rodents. Tapes shall never be used alone to repair a large opening in the bottom board. Divergent staples must be used in tandem with tape to prevent future tear off.

**Using Divergent Staples**. Divergent staples are designed to secure material-to-material connections only. Do not attempt to fasten to a wooden structure of any kind.

## Step 2. Install Ground Moisture Retarder

If the space under the home is to be enclosed with skirting or other materials, a ground moisture retarder of a minimum six mil thick polyethylene sheeting or equivalent must be installed covering the ground under the home. Moisture retarders are not required in arid regions (less than 15 inches of rainfall annually) with dry soil conditions. If on-grade (surface) footings are used, install the ground moisture retarder prior to placing the footings, or install it around the footings after all other work under the home is complete.

Install the ground moisture retarder as follows:

1. **Apply sheeting.** Unroll the ground moisture overlapping joints in the sheeting a minimum of 12 inches and covering the entire area under the home except for areas under recessed entries, decks and porches.
2. **Repair tears**. Repair any large voids or tears in the retarder by patching with like material, maintaining a minimum 12 inch overlap, secured with tape or adhesive. Repair small voids and tears with tape, adhesive or per manufacturer's installation instructions.

## Step 3. Skirting (if Used)

Skirting is any structural or non-structural perimeter crawlspace enclosure. Complete site built structures (see **Step 5. Complete Site Built Structures**, p. 141) that abut the home (such as porches, attached garages and steps) prior to installing skirting.



**Fastening vinyl siding**. Do not install fasteners directly into vinyl siding. Allow for siding thermal expansion by pre-drilling minimum 1/2 inch diameter fastener holes or fastening skirting to a ledger under the home.

**Attaching vinyl skirting**. Attach skirting to the home, but allow for contraction and expansion characteristics of the skirting material.

**Avoid backfilling against skirting.** Do not backfill against non-structural skirting.

*Table 51. Skirting and ventilation specifications*

| Component | Specification |
|---|---|
| **Skirting** | Skirting must be of weather-resistant materials or provided with protection against weather deterioration at least equivalent to that provided by a coating of zinc on steel of not less than 0.30 oz per sq ft of surface coated. Skirting made from wood or wood products and used within six inches of the ground need to be made of materials naturally resistant to decay and termite infestation or pressure treated. |
| **Vents** | Ventilation openings must be covered for their full height and width with a perforated (1/4 inch maximum opening in any dimension) corrosion and weather resistant covering that is designed to prevent the entry of rodents. In areas subject to freezing, the coverings for the ventilation openings must have an operable damper, permitting them to be in the open or closed position depending on the weather. |

To design and install skirting, comply with the skirting manufacturer's instructions (if provided) and the following:

1. **Configure skirting.** Run the skirting along the perimeter of the home's heated, conditioned space. Do not enclose with skirting areas under recessed entries, porches or decks (whether constructed as part of the home or added on site) unless skirting is of the fully vented type and installed so as to allow water to freely flow out from under the home.
2. **Fasten skirting.** Attach it to the home in a manner that prevents water from being trapped between the siding or trim and the skirting. Allow for frost heave when installing skirting in areas subject to frost.
3. **Provide ventilation.** Unless the skirting has integral ventilation openings that meet the following ventilation requirements, install equally sized ventilation openings on at least two opposite sides of the foundation. Size ventilation area with vapor barriers equal to 1 square foot of free vent area per 1500 square foot of floor space.The ventilation area must be the net free area of the foundation vent, not the area of the foundation opening. Place vents as high above the ground as practical.

*Table 52. Foundation ventilation*

| One Square Foot of Vent per 150 Square Feet of Under-Floor Area | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Total Home Sq. Feet | Sq. Feet of Vents Required | Total Home Sq. Feet | Sq. Feet of Vents Required | Total Home Sq. Feet | Sq. Feet of Vents Required | Total Home Sq. Feet | Sq. Feet of Vents Required | Total Home Sq. Feet | Sq. Feet of Vents Required |
| **500** | 3.33 | **1000** | 6.67 | **1500** | 10.00 | **2000** | 13.33 | **2500** | 16.67 |
| **600** | 4.00 | **1100** | 7.33 | **1600** | 10.67 | **2100** | 14.00 | **2600** | 17.33 |
| **700** | 4.67 | **1200** | 8.00 | **1700** | 11.33 | **2200** | 14.67 | **2700** | 18.00 |
| **800** | 5.33 | **1300** | 8.67 | **1800** | 12.00 | **2300** | 15.33 | **2800** | 18.67 |
| **900** | 6.00 | **1400** | 9.33 | **1900** | 12.67 | **2400** | 16.00 | **2900** | 19.33 |

4. **Install access.** Provide an access opening not less than 18 inches wide and 24 inches high (minimum 3 square feet in area) and located so that any utility connections located under the home are accessible.
5. **Extend vents, drains and inlets.** Run appliance exhaust vents, combustion air inlets and air conditioner condensation drains through the skirting to the outside and terminate each as instructed in the sections of this manual corresponding to each appliance.

## Step 4. Assemble Optional Wind Protection Shutters

Prepare temporary protective window covers as required for use during severe wind storms and hurricanes**.** Reference the appropriate section in the Appendix (p. 8).

## Step 5. Complete Site Built Structures

Install site built structures such as steps, landings, garages, awnings, carports, breezeways, porches, decks, railings, sheds and utility rooms according to manufacturer's instructions (if any), in compliance with all local regulations including fire separation and electrical requirements, and according to the following:

- Do not obstruct any of the egress windows or the two required exit doors from the home.
- The addition must be entirely self-supported and cannot rely on the home for support (superficial connections are acceptable). The home's structural system is not designed to support the extra loads imposed by the addition.
- Do not damage the integrity of the home's structural or weatherproofing system. Seal any weatherproofing connections between the site built structure and the home and flash any roof connections.
- The home's structural system may not be cut or altered in any way. A registered engineer or architect shall approve any alterations or changes.
- Utilize only GFCI outlets for site built structures.
- Install and test smoke alarms in any site built structures according to local code.
- All joints between the home and the addition must be properly sealed so they are watertight.
- The home's mechanical system has been designed for the home itself and does not consider the heating or cooling of the addition.
- The addition must meet all local codes, including site work and fire separation requirements. The manufacturer does not accept any responsibility for the addition's design.
- The home with an addition must be in conformance with the HUD Manufactured Housing Code, such as exiting, light and vent, etc. The addition must be approved by the jurisdiction having authority.
- Site work shall be consistent with the objectives of site grading as described in **Prepare the Site** (p. 25).
- The manufacturer will not honor the warranty for any problem that relates to the construction of the addition (leak problems, etc).

A dormer roof can be installed on the home to match the pitch of the addition's roof. The shingles below the dormer must be removed and the dormer must be vented properly. The dormer weight, including the weight of the existing home's roof sheathing, shall be no more than 8 psf and distributed uniformly over the roof trusses of the home. Connections may be made to attach the dormer to the home, but not be used to support the roof loads of the addition. The dormer shall be shingled, flashed and sealed properly to prevent leaks (follow shingle manufacturer's instructions and ARMA guidelines for shingle valley applications).

The manufacturer is providing the above information as an accommodation only and without consideration. Accordingly, by this letter the manufacturer extends no warranties or representation either expressed or implied with regard to the recommendations herein.

► go to **Prepare Home for Occupancy** (p. 142).

# PREPARE HOME FOR OCCUPANCY

Follow these steps for final inspection and completion of the home.

**Follow the Steps below:**


▼ **Step 1.Verify Alternative Construction (A/C) Inspection** (p. 142)
▼ **Step 2. Complete Inspection Checklist** (p. 142)
▼ **Step 4. Clean the Home** (p. 142)


## Step 1. Verify Alternative Construction (A/C) Inspection

A home designated as an AC home will contain the letters “AC” in the serial number located on the data plate.Verifiying the AC inspection is the responsibility of the retailer. The installation is not complete until the alternative construction inspection has been passed and all applicable documention completed and returned to the plant.

## Step 2. Complete Inspection Checklist

After all previous steps have been accomplished, inspect the home to verify that it has been completely and properly installed using the checklist starting on p. 143. Installer must also certify that the installation is in compliance with either the manufacturer’s instructions or with an alternative design in compliance with 3285.2(c). Arrange for an inspection by the appropriate LAHJ, if required.

Correct any deficiencies found, if possible, or if not possible, inform the retailer or manufacturer immediately.

## Step 3. Clean the Home

Remove and properly dispose of all installation-generated dust, debris and packaging materials from the home and the surrounding property. Ensure that the home is in “move-in” condition.

Installation is Complete

# COMPLETE INSTALLATION CHECKLIST

## Foundation Support

- ≤ Footings properly sized and constructed for the soil conditions
- ≤ Pier spacing per data plate and applicable table and roof load zone
- ≤ Piers properly constructed and vertical
- ≤ Perimeter blocking installed (if required)
- ≤ Piers at each side of openings 48” or larger
- ≤ Piers beneath all outside exterior doors (except when located in end wall)
- ≤ Center line piers installed at columns
- ≤ Shims in place and tight

## Anchors

- ≤ Approved anchors are used
- ≤ Proper anchors installed based on soil conditions
- ≤ Anchors are installed at correct angles
- ≤ Anchor spacing and installation correct
- ≤ Longitudinal ties installed (if required)
- ≤ Anchor straps are tensioned

## Under the Home

- ≤ Moisture retarder installed
- ≤ The ground under and around the home has been properly graded to prevent water from collecting or flowing beneath the home
- ≤ HVAC ducts are supported off the ground and connected tightly to collars at all ends
- ≤ Fireplace combustion air intake free and unrestricted
- ≤ No holes or tears in bottom board
- ≤ Skirting (if used) has been installed per manufacturer’s instructions with proper venting and provision for frost heave
- ≤ Skirting has been attached in a manner that does not cause water to be trapped between the siding and trim and cannot be forced up into the wall cavity
- ≤ Dryer vent, range/cook top exhaust, water heater drain pan drain line, and AC condensate drain installed to perimeter of crawl space

## Exterior

- ≤ Shingled roofs are free of visible damage and serious defects and there are no missing or loose shingles
- ≤ Shingle close-up and ridge cap have been completed per applicable details
- ≤ All hold down straps on shingled roofs have been removed and holes have been properly sealed
- ≤ Penetrations at roof stacks, vents and chimneys have been properly sealed
- ≤ Siding and trim is free of gaps, voids, missing fasteners, damage and serious defects. All seams are sealed and hardboard edges are sealed.
- ≤ Drip edge and fascia is properly installed and free of damage and serious defects
- ≤ Gutters and downspouts are installed properly such that water is diverted away from the home
- ≤ Trees and bushes have been trimmed to prevent brushing against the home in windy conditions or under snow loads
- ≤ The HUD label is exposed, intact and legible
- ≤ The exterior of the home and immediate surroundings is clean, clear of construction materials, dust and debris

## Interior

- ≤ Ceilings, walls and floor coverings are free from damage and serious defects
- ≤ Carpeting is properly stretched and seamed
- ≤ All trim and molding is installed properly and free of damage and defects
- ≤ All cabinets, countertops, plumbing fixtures, appliances, furnishings and window coverings are free of damage or serious defects
- ≤ All cabinet doors and drawers work properly
- ≤ All interior and exterior doors and windows open, close and latch properly
- ≤ One window in each bedroom meets emergency egress requirements, has operating instruction labels on it and operates properly
- ≤ All temporary shipping hardware has been removed
- ≤ Floors are level
- ≤ The data plate is intact and legible
- ≤ Smoke alarms have been tested
- ≤ The interior of the home is clean, clear of materials, dust and debris

## Water and Drain Systems

- ≤ Crossover and service connection and splices have been properly made with correct materials
- ≤ Water and drain lines are insulated or otherwise protected from freezing
- ≤ Pipe supports are installed and properly spaced
- ≤ Proper slope has been maintained on all drain lines
- ≤ All necessary inspections and tests have been performed
- ≤ All sinks, basins, tubs and toilets operate properly
- ≤ All hot and cold water lines are properly connected to fixtures, dispense water as labeled and operate properly
- ≤ Tub, tub/shower and shower outlets have been verified to ensure the outlet water temperature does not exceed 120°F (49°C)

## Gas/Fuel Oil Systems

- ≤ The gas system pressure test has been conducted
- ≤ Connections between units are properly made with access as required
- ≤ The main fuel line has been properly connected and tested by a qualified technician
- ≤ Marriage line gasket has been installed and inspected

## Electrical Systems

- ≤ The panel amperage matches the connection to the home
- ≤ The home has been properly grounded
- ≤ All electrical crossovers have been connected
- ≤ The main power supply has been properly connected and tested by a licensed electrician
- ≤ Continuity test has been conducted
- ≤ Operational test has been conducted
- ≤ Polarity test has been conducted
- ≤ All receptacles, switches and light fixtures operate properly
- ≤ Ground fault circuit interrupters operate properly
- ≤ All exterior lights have been properly installed

## Appliance Operating and Venting

- ≤ All appliances are working properly
- ≤ Appliance venting is in accordance with the manufacturer's instructions
- ≤ Fresh air intakes are properly installed
- ≤ Whole house, kitchen and bath exhaust fan operation are correct
- ≤ Fireplace chimney stack extension and roof cap have been installed in accordance with the manufacturer's instructions
- ≤ Air conditioner/heat pump is sized properly
- ≤ Air conditioner condensate line is properly trapped and terminates outside of the skirting

## Miscellaneous

- ≤ Installation/anchoring certificates or seals have been issued and installed (if required)
- ≤ Owner's and operation manuals are available for all appliances
- ≤ This installation manual is left with home

# ZERH MH v1
# Model Manufacturer's Installation Manual

## Introduction

This is an Addendum to the Manufacturer's Installation Manual (MIM) providing instructions for the certification of Zero Energy Ready Home Manufactured Homes. This Addendum can be incorporated into the plant's MIM. Installers should be advised to read all the instructions that may apply to the specific home prior to commencing site work and home installation.

See Table 1 for requirements for different foundation types.

**Table 1 Requirements for different foundation types**

| Foundation type | Requirements |
|---|---|
| **Raised pier**[1] | 3.1, 4.1, 4.2, 4.3, 5.1 |
| **Basement**[2] | 1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 2.1, 3.1, 4.1, 4.2, 4.3, 5.1 |

**Notes:**

[1] A raised pier foundation, also called a vented crawlspace, is at or near grade level. It may or may not have a concrete slab. It is vented.

[2] A basement foundation is defined as an occupiable space below grade.

## 1 Foundation systems

### 1.1 Sealing and conditioning of basements

Insulate, seal and condition basements as follows:

a. Insulate basement perimeter walls with minimum R-5 insulation; and,

b. Seal all penetrations and joints (full length) between materials on basement perimeter walls with a moisture-proof sealant to prevent outside air infiltration; and,

c. Condition the basement in one of two methods, depending on the provisions provided by the manufacturing plant:

- Extend the supply air duct from its marked location in the floor system and into the basement. Securely fasten the duct termination so that it is suspended above head height. Seal the bottom barrier (if present) around the duct. Supply air to basement should be not less than 1 cfm per 50 sq. ft. of horizontal floor in the basement or,
- Install an exhaust fan rated for continuous operation to exhaust air from the basement at a rate of 1 cfm per 50 sq. ft. of basement floor area. Connect the fan to power receptacle provided by home manufacturer for this purpose. This option not permissible in radon zone 1.

**Exception:** Not required in Dry (B) climates (see Attachment A, Figure A.1).

### 1.2 Capillary break beneath slabs (for homes with slab, except raised pier foundations)

Install capillary break beneath slabs as follows:

a. Install a capillary break beneath slab foundations (for example, a basement slab) consisting of 4 inches of 1/2 in. diameter or greater clean aggregate stone; or, install a 4 inch uniform layer

of sand, overlain with either a layer of geotextile drainage matting throughout or strips of geotextile drainage matting along the perimeter installed according to the manufacturer's instructions; and,

b. Install a vapor barrier directly over the above consisting of either: ≥ 6 mil polyethylene sheeting lapped 6-12 in.; or, ≥ 1 in. extruded polystyrene insulation with taped joints.

**Exceptions:** Not required in Dry (B) climates (see Attachment A, Figure A.1). Not required in areas with free-draining soils – identified as Group 1 (see Attachment B, Table B.1) as verified by a certified hydrologist[1], soil scientist, or engineer through a site visit.

**1.3 Drain tiles** (for homes with basement)

Install a drain system that meets ***all*** of the following requirements:

a. Install drain pipe at basement walls, with the top of the drain pipe below the bottom of the concrete slab; and

b. Lay the pipe with enough slope to drain to a non-perforated pipe that carries the collected water to outside grade (daylight) or to a sump pump; and

c. Lay the drain pipe, perforations down, in a gravel trench with at least 6 inches of ⅛-inch to ¾-inch washed gravel or stone above the pipe and at least 2 inches below; and

d. Install landscape fabric under, around, and over the washed gravel.

**1.4 Drain or sump pump and check valve** (for homes with basement)

Install a drain or sump pump in basement, discharging to daylight at least 10 ft. outside the foundation or into an approved sewer system. For homes in EPA Radon Zone 1 (see https://www.epa.gov/radon/state-maps-radon-zones), install a check valve.

**1.5 Sump pump cover** (for homes with basement)

Select and install sump pumps that have tight-fitting lids with gaskets and mechanical fasteners.

**1.6 Damp-proofing below-grade walls** (for homes with basement

Finish the exterior surface of below-grade walls of basements and unvented crawlspaces as follows:

a. For poured concrete, masonry, and insulated concrete forms, finish with damp-proofing coating.

b. For wood framed walls finish with polyethylene and adhesive or other equivalent waterproofing.

**1.7 Vapor retarder at below grade wall** (for homes with basement)

Do not install a Class I vapor retarder on the interior side of air permeable insulation in exterior below-grade walls. A Class I vapor retarder is defined as a material or assembly with a rating of ≤ 0.1 perm. The following materials are typically ≤ 0.1 perm and shall not be used on below-grade

[1] For example, Professional Hydrologist certified by American Institute of Hydrology (AIH).

exterior walls in any climate: rubber membranes, polyethylene film, glass, aluminum foil, sheet metal, and foil-faced insulating / non-insulating sheathings. These materials can be used on the interior side of walls if air permeable insulation is not present (e.g., foil-faced rigid foam board adjacent to a below-grade concrete foundation wall is permitted).

## 2 Radon

### 2.1 Radon mitigation (for all homes in EPA Radon Zone 1 except raised pier foundation)

Check if the home is in EPA Radon Zone 1 (see https://www.epa.gov/radon/state-maps-radon-zones). If in EPA Radon Zone 1, install all the following radon-resistant features, except where noted:

a. Install capillary break under slab and at crawlspace according to 1.2 or 1.3, as applicable. Exception: In dry climates (see Attachment A, Figure A.1), a "pipe loop" in a trench of clean aggregate along the entire inside perimeter of the foundation (installed according to ANSI/AARST CCAH 403.1.1) can be used in lieu of a uniform layer of aggregate under the entire slab; and,

b. Install a 3 or 4 in. diameter gas-tight vertical vent pipe, clearly labeled as a component of a radon reduction system. Connect the vent pipe to an open T-fitting in the aggregate layer (or connect it to geotextile drainage matting according to the manufacturer's instructions) beneath the polyethylene sheeting referenced in 1.2 or 1.3, extending up through the conditioned spaces and terminating a minimum of 12 in. above the roof (see alternative in d below). Attach at least 10 ft. of horizontal perforated drain tile to the T-fitting beneath the polyethylene sheeting placed over earthen crawlspaces and below concrete slabs; and,
Note: Radon pipe suction points are not permitted on sump lids.

c. Install a radon fan (i.e., an active system) or install the above-mentioned vent pipe near the provided attic electrical receptacle (i.e., a passive system) to facilitate future fan installation if needed (see alternative d below). Provide a space surrounding the radon pipe, having a vertical height of not less than 48 inches and a diameter of not less than 21 inches, in the attic area where the radon fan can be installed, if required; or,

d. Utilize another exterior location or a garage that is not below conditioned space if the home has no accessible attic location for a fan per ANSI/AARST CCAH. Label the branch circuit supply at the electrical panel indicating its intended use; and,

e. Air seal foundation with polyurethane caulk or the equivalent at all slab openings, penetrations and control or expansion joints.

## 3 Air Sealing

### 3.1 Sealing common walls (for attached homes)

In attached homes, seal the gap between the common wall (e.g., the drywall shaft wall) and the structural framing between units at all exterior boundaries.

# 4 HVAC

## 4.1 Outdoor packaged HVAC units

Install the following for homes with outdoor packaged unit:

a. Use exterior ductwork with ≥ R-8 insulation; and,

b. Install outside duct runs less than 20 feet in length each for supply and return.

## 4.2 Garage HVAC equipment (for homes with a garage)

Do not place air-handling equipment or ductwork in a garage.

# 5 Water Management

## 5.1 Gutters and downspouts

Install gutters and downspouts as follows:

a. For homes with slab on grade, or raised pier foundation without gutters:
   - Extend the foundation walls at least 16 in. above final grade; or,
   - Provide a drip line at eaves that is horizontally 16 in. away from the edge of the foundation wall; or,
   - Use cladding materials that are decay and rot resistant and can tolerate regular wetting extending at least 16 in. above final grade and install a well-sealed, continuous drainage plane per manufacturer's instructions.

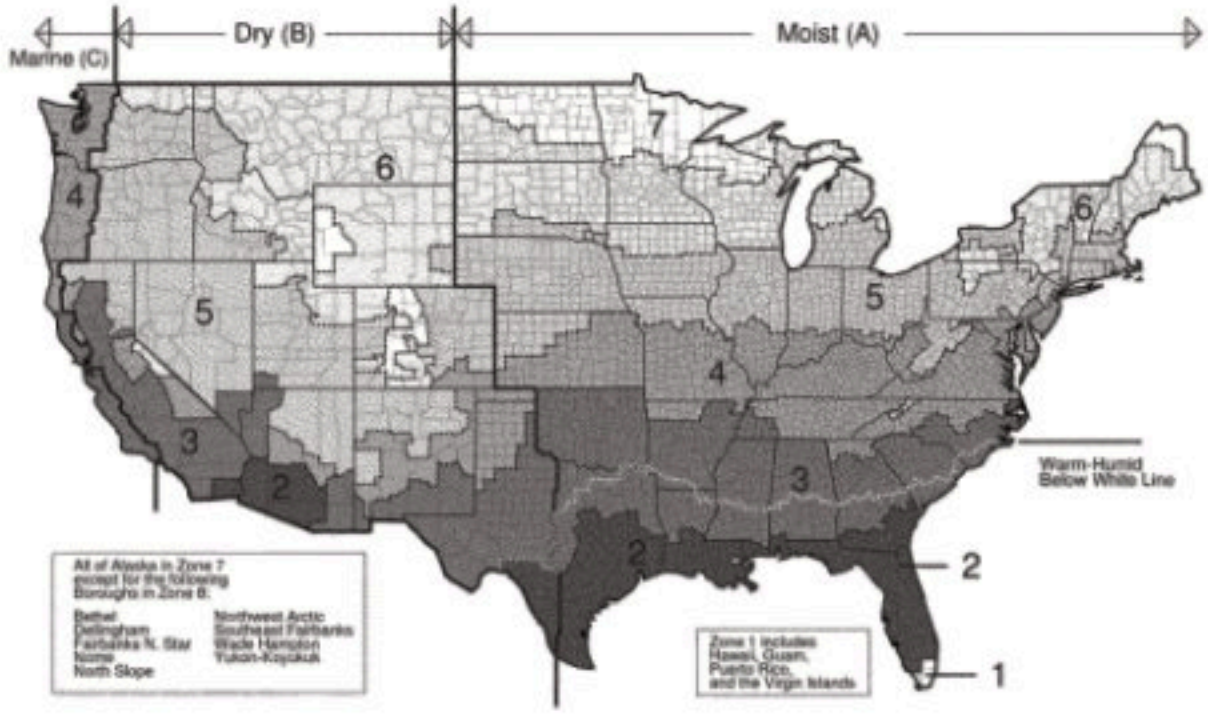
b. For homes with a basement:
   - Install adequately sized gutters and downspouts connected to piping that will carry the water to a sloping final grade ≥ 5 ft. from the foundation or to an underground catchment system ≥ 10 ft. from the foundation that is not connected to the foundation drain system.

**Exceptions:** Not required in Dry (B) climates (see Attachment A, Figure A.1). Not required in areas with free-draining soils – identified as Group 1 (see Attachment B, Table B.1) as verified by a certified hydrologist, soil scientist, or engineer through a site visit.

# ZERH MH v1
# Model Manufacturer's Installation Manual

## Attachment A

**Figure A.1 2009 IECC Climate zones. (source: 2009 IECC Figure 301.1)**




**To look up the climate zone for each County, check line here:**

https://codes.iccsafe.org/content/IECC2009/chapter-3-general-requirements

# ZERH MH v1
# Model Manufacturer's Installation Manual

## Attachment B

**Table B.1 Properties of soils classified according to the unified soil classification system (source: 2015 IRC Table R405.1)**

TABLE R405.1 PROPERTIES OF SOILS CLASSIFIED ACCORDING TO THE UNIFIED SOIL CLASSIFICATION SYSTEM

| SOIL GROUP | UNIFIED SOIL CLASSIFICATION SYSTEM SYMBOL | SOIL DESCRIPTION | DRAINAGE CHARACTERISTICS[a] | FROST HEAVE POTENTIAL | VOLUME CHANGE POTENTIAL EXPANSION[b] |
|---|---|---|---|---|---|
| Group I | GW | Well-graded gravels, gravel sand mixtures, little or no fines | Good | Low | Low |
| | GP | Poorly graded gravels or gravel sand mixtures, little or no fines | Good | Low | Low |
| | SW | Well-graded sands, gravelly sands, little or no fines | Good | Low | Low |
| | SP | Poorly graded sands or gravelly sands, little or no fines | Good | Low | Low |
| | GM | Silty gravels, gravel-sand-silt mixtures | Good | Medium | Low |
| | SM | Silty sand, sand-silt mixtures | Good | Medium | Low |
| Group II | GC | Clayey gravels, gravel-sand-clay mixtures | Medium | Medium | Low |
| | SC | Clayey sands, sand-clay mixture | Medium | Medium | Low |
| | ML | Inorganic silts and very fine sands, rock flour, silty or clayey fine sands or clayey silts with slight plasticity | Medium | High | Low |
| | CL | Inorganic clays of low to medium plasticity, gravelly clays, sandy clays, silty clays, lean clays | Medium | Medium | Medium to Low |
| Group III | CH | Inorganic clays of high plasticity, fat clays | Poor | Medium | High |
| | MH | Inorganic silts, micaceous or diatomaceous fine sandy or silty soils, elastic silts | Poor | High | High |
| Group IV | CL | Organic silts and organic silty clays of low plasticity | Poor | Medium | Medium |
| | OH | Organic clays of medium to high plasticity, organic silts | Unsatisfactory | Medium | High |
| | Pt | Peat and other highly organic soils | Unsatisfactory | Medium | High |

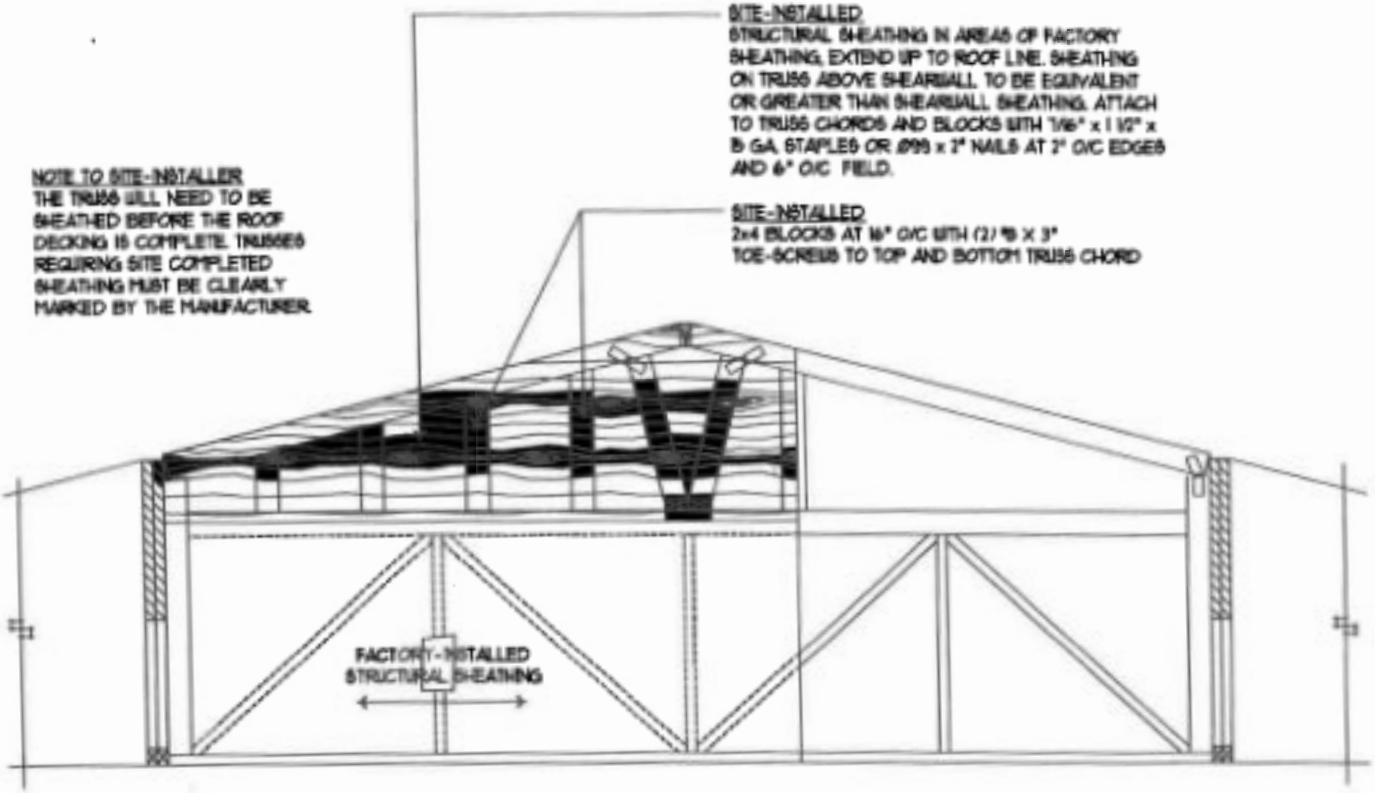
For SI: 1 inch = 25.4 mm.

a. The percolation rate for good drainage is over 4 inches per hour, medium drainage is 2 inches to 4 inches per hour, and poor is less than 2 inches per hour.

b. Soils with a low potential expansion typically have a plasticity index (PI) of 0 to 15, soils with a medium potential expansion have a PI of 16 to 35 and soils with a high potential expansion have a PI greater than 35.

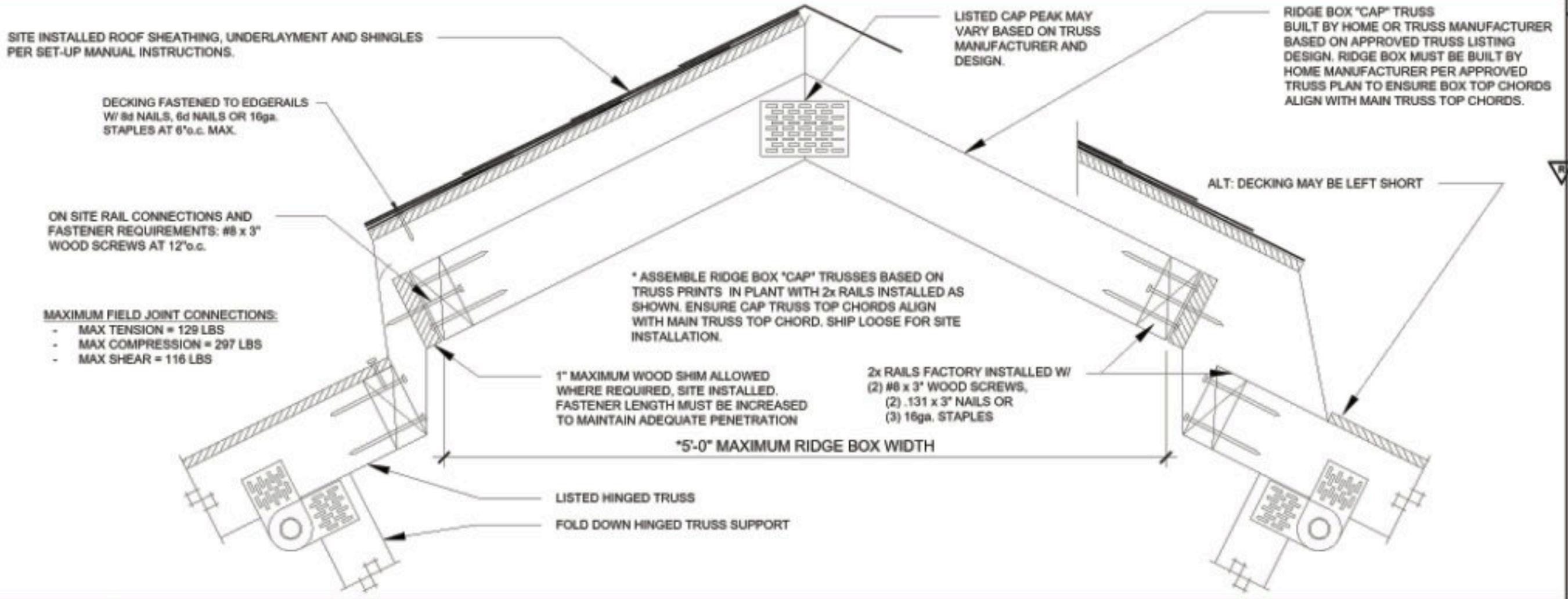
GENERAL NOTES:

1. DETAIL APPLICABLE UP TO A 445 PLF SHEARWALL MAX.
2. REFER TO OTHER DETAILS FOR ANY CONSTRUCTION DETAILS NOT EXPRESSLY MENTIONED ON THIS PAGE.

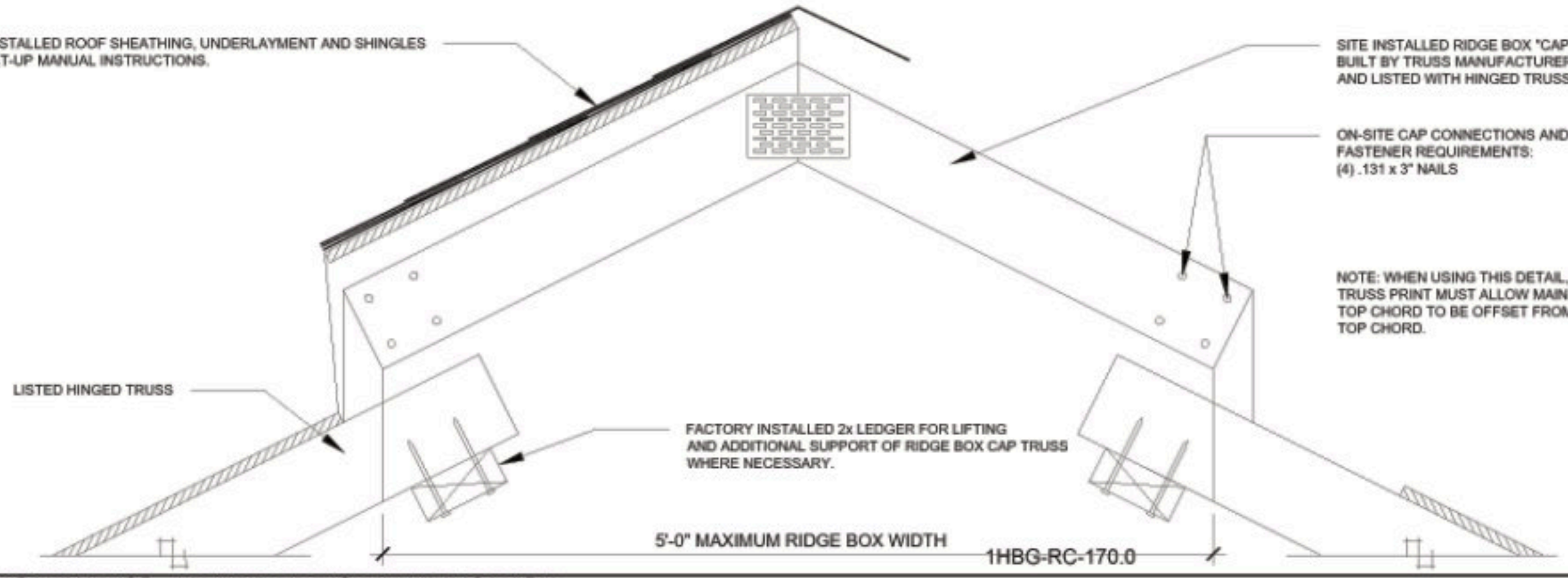



THIRD PARTY SEAL

| | | | |
|---|---|---|---|
| TITLE: HINGED TRUSS SHEATHING ABOVE SHEARWALL | | | CMH MANUFACTURING, INC. GOLDEN WEST 5960 Clayton Road, Maryville, TN 37804 Phn 865.380.3000 FAX: 865.380.3791 |
| DRAWN BY: | G. BUE | DATE: | 05/10/2019 |
| REVIEWED BY: | JRT | LAST REVISED: | 05/10/2019 |
| CHECKED BY: | RADCO | CALC REF: | SHEET: S-61.4 |




## RIDGE BOX "CAP" TRUSS - INSERT APPLICATION A




## RIDGE BOX "CAP" TRUSS - ALTERNATE INSERT APPLICATION B

## GENERAL NOTES

1. NO ROOF PENETRATIONS ALLOWED THROUGH HINGED PORTIONS OF ROOF.
2. MINIMUM ROOF PITCH = 4/12, MAXIMUM ROOF PITCH = 7/12.
3. TRUSS LISTINGS SHALL INCORPORATE THE RIDGE BOX DESIGN.
4. RIDGE BOX SIZE (OVERALL WIDTH) LIMIT = 5'-0" MAXIMUM.
5. TRUSS BOTTOM CHORDS NOT DESIGNED FOR STORAGE LOADS OR OCCUPANCY (LIVE) LOADS.
6. HOME SHALL HAVE ATTIC ACCESS PANEL IN CEILING (29" X 21" MAXIMUM) TO FACILITATE INSPECTIONS.
7. A SINGLE ROW OF SHINGLES SHALL BE LEFT OFF BY THE HOME MANUFACTURER ON EACH SIDE OF THE HINGE. SHINGLES TO BE SHIPPED LOSE WITH HOME AND INSTALLED ON SITE ONCE HINGED PORTION HAS BEEN RAISED AND FIXED IN PLACE.
8. ALL MATERIALS REQUIRED FOR COMPLETION OF HINGED ROOF SHALL BE PROVIDED BY HOME MANUFACTURER. OPTIONAL SHIMS TO BE PROVIDED BY HOME MANUFACTURER AS NEEDED.
9. REFER TO MANUFACTURER'S INSTALLATION INSTRUCTIONS FOR COMPLETE ASSEMBLY INSTRUCTIONS AND FASTENING REQUIREMENTS.

APPROVED BY NTA INC.
FEDERAL MANUFACTURED HOME CONSTRUCTION AND SAFETY STANDARDS

REVISED 5/5/2025

THIRD PARTY SEAL

| LATEST REVISION | | |
|---|---|---|
| DESCRIPTION | | REV. 1 |
| REVISED NOTE 6. | | JRT |
| | | 05/02/25 |

| | | | |
|---|---|---|---|
| DRAWN BY: | JRT | DATE: | 5/2/2025 |
| CALC REF: | - | | |

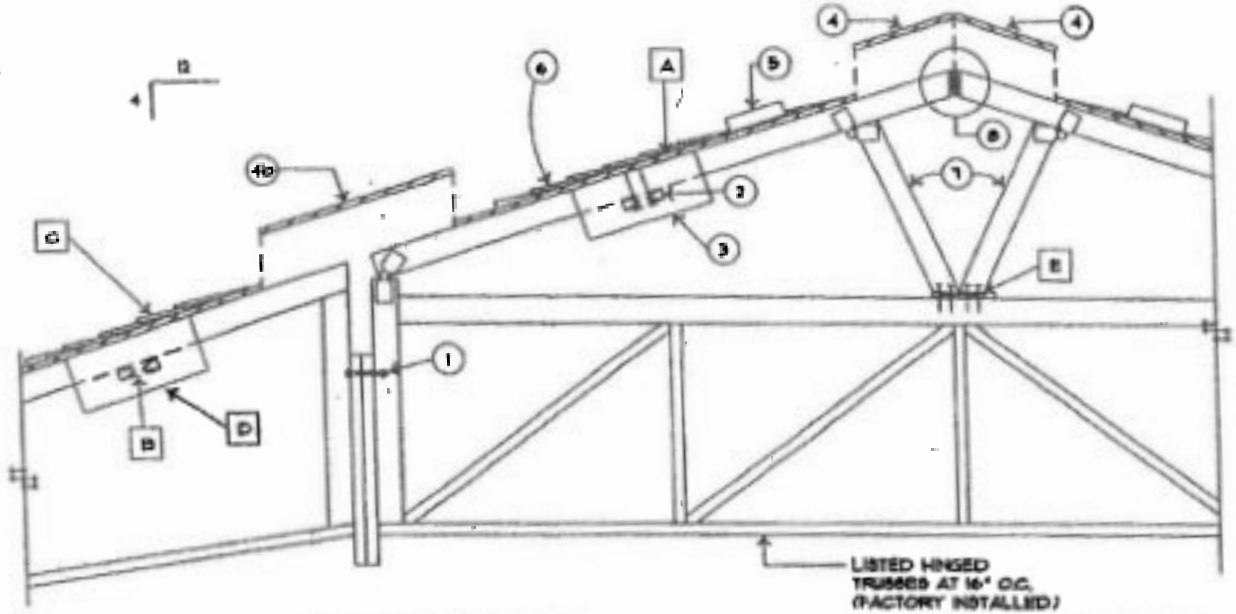
Clayton home building group

| WIND ZONE APPROVAL | ROOF LOAD PSF |
|---|---|
| NOT APPLICABLE | 20 / 30 / 40 |

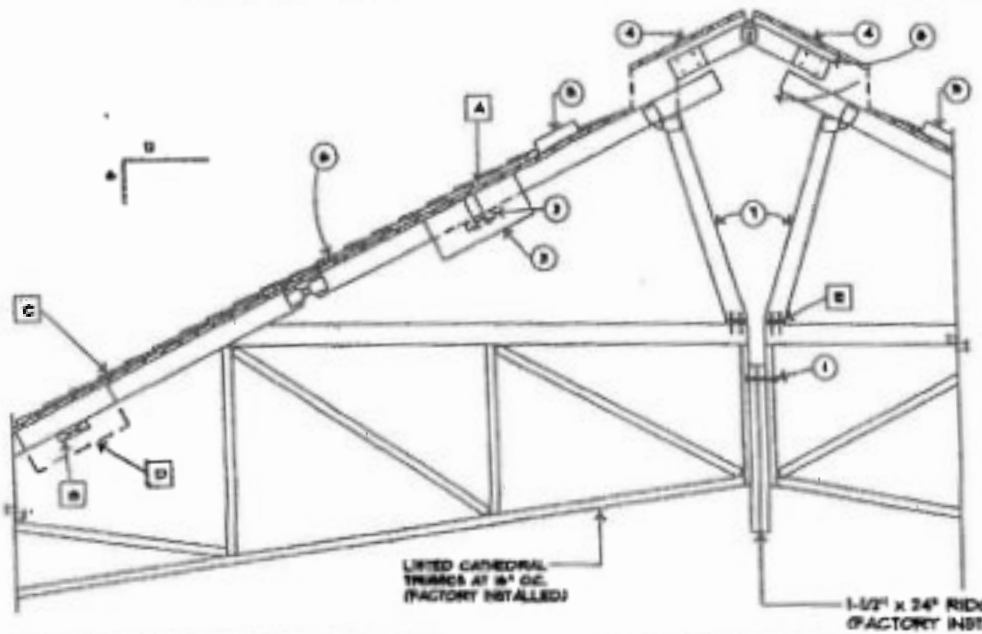
TITLE: RIDGE BOX "CAP" TRUSS TYPICAL

SHEET: 1HBG-RC-170.0

May 02, 2025 - 7:41am W:\ENG\DAPIA\DAPIA Projects\Working Projects\3120--1HBG\1HBG-RC-170.0.dwg




ROOF SECTION AT TRUSS PROFILE FOR TRIPLEWIDE




ROOF SECTION AT DOUBLEWIDE TRUSS PROFILE

REGISTERED PROFESSIONAL ENGINEER
MARK JOSEPH RIZZO
NO. C042077
Exp. 3-31-2014
CIVIL
STATE OF CALIFORNIA

LEGEND:

- SITE INSTALLED ITEM - SEE SHEET S-62 "SITE INSTRUCTIONS FOR HINGE ROOF CLOSE OFF".
- FACTORY INSTALLED ITEM - SEE SHEET S-63 "FACTORY INSTALLED NOTES".

GENERAL NOTES:

1. FOR HOMES WITHOUT "ALTERNATE CONSTRUCTION" APPROVAL, FUEL BURNING FURNACE AND WATER HEATER VENTS SHALL NOT BE POSITIONED IN HOME SUCH THAT THEY WOULD EXTEND THROUGH THE HINGED PORTION OF ROOF.
2. FACTORY SHALL PROVIDE PROTECTION FROM WEATHER DAMAGE ON UNROOFED PORTIONS OF HOME DURING TRANSIT AND STORAGE.
3. ALL MATERIALS REQUIRED FOR SITE CLOSE-OFF SUCH AS SHINGLES, SHEATHING, RIDGE BOARD, FASCIA, OUTLOOKERS, BOLTS, ATTIC VENTS, ETC., SHALL BE SUPPLIED BY GOLDEN WEST HOMES AND SHIPPED LOOSE WITH THE UNIT.

| NO. | REVISED BY | DATE |
|---|---|---|
| 1 | TSW | 5/11/2020 |

APPROVED BY
NTA INC.
9/24/2020
FEDERAL MANUFACTURED HOME CONSTRUCTION AND SAFETY STANDARDS

HINGED ROOF CLOSE-OFF

DRAWN BY: CPED
DATE: 1-23-01
ROOF LOAD: 40 PSF
S-61

# SITE INSTRUCTIONS FOR CLOSE - OFF

## DOUBLEWIDE INSTRUCTIONS

1. BOLT RIDGE BEAMS TOGETHER AT MATING LINES PER PAGES 48-51 IN THE INSTALLATION MANUAL.
2. AT EACH END OF ROOF, INSTALL 2x6 No. 2 H.F OUTLOOKERS AT 28" O/C WITH ONE OUTLOOKER DIRECTLY NEXT TO EACH SIDE OF THE PEAK. EXTEND ALL OUTLOOKERS BACK TO THIRD TRUSS FROM ENDWALL, EXTEND OUTLOOKERS OUT PAST ENDWALL TO MATCH FACTORY INSTALLED OUTLOOKER EXTENSIONS. FASTEN OUTLOOKERS TO HINGED TRUSS TOP CHORD WITH (2) No. 8x3" WOOD SCREWS AT EACH CONNECTION.
3. INSTALL CONTINUOUS 2x8 FASCIA BOARD TO ENDS OF OUTLOOKERS WITH (3) 16d NAILS AT EACH OUTLOOKER. SPLICE FASCIA WITH 1x3x7" MINIMUM SPLICE BLOCK SECURED WITH (4) 6d NAILS EACH SIDE OF SPLICE.
4. APPLY 3/8" APA RATED 24/0 ROOF SHEATHING OVER ENTIRE SURFACE OF SITE INSTALLED CAP. FASTEN SHEATHING WITH 8d NAILS AT 6" O/C EDGES AND 12" O/C FIELD.
5. INSTALL A MINIMUM OF FOUR ROOF VENTS (50 SQUARE INCH NET OPENING EACH) AT EACH HALF OF UNIT OVER CUT-OUTS IN FACTORY INSTALLED SHEATHING TO PROVIDE ATTIC VENTILATION.
   ALTERNATE: INSTALL CONTINUOUS RIDGE VENT AT A RATIO OF 1 SQ.FT. OF NET FREE AIR TO 300 SQ.FT. OF HOME AREA. HALF OF THIS AMOUNT MUST BE INSTALLED ALONG THE PEAK.
6. APPLY ROOF SHINGLES WITH UNDERLAYMENT IN ACCORDANCE WITH SHINGLE MANUFACTURERS INSTRUCTIONS. SHINGLES AND UNDERLAYMENT SHALL BE INSTALLED TO CONTINUE UPWARD FROM FACTORY APPLIED SHINGLES.
7. FASTEN EACH HINGED KING POST TO 2x4 LEDGER W/ (2) No. 8 x 3" WOOD SCREWS.
8. INSTALL CAP TRUSSES TO EACH HINGED RAFTER CHORD WITH A 6 3/4" OVERLAP EACH SIDE. FASTEN EACH SIDE WITH (4) 8d NAILS.

SU-ADD-62

## TRIPLEWIDE INSTRUCTIONS

1. BOLT RIDGE BEAMS TOGETHER AT MATING LINES PER PAGES 48-51 IN THE INSTALLATION MANUAL.
2. AT EACH END OF ROOF, INSTALL 2x6 No. 2 H.F OUTLOOKERS AT 28" O/C WITH ONE OUTLOOKER DIRECTLY NEXT TO EACH SIDE OF THE PEAK, EXTEND ALL OUTLOOKERS BACK TO THIRD TRUSS FROM ENDWALL, EXTEND OUTLOOKERS OUT PAST ENDWALL TO MATCH FACTORY INSTALLED OUTLOOKER EXTENSIONS. FASTEN OUTLOOKERS TO HINGED TRUSS TOP CHORD WITH (2) No. 8x3" WOOD SCREWS AT EACH CONNECTION.
3. INSTALL CONTINUOUS 2x8 FASCIA BOARD TO ENDS OF OUTLOOKERS WITH (3) 16d NAILS AT EACH OUTLOOKER. SPLICE FASCIA WITH 1x3x7" MINIMUM SPLICE BLOCK SECURED WITH (4) 6d NAILS EACH SIDE OF SPLICE.
4. APPLY 3/8" APA RATED 24/0 ROOF SHEATHING OVER UNSHEATHED SURFACES OF HINGED ROOF. FASTEN SHEATHING WITH 8d NAILS AT 6" O/C EDGES AND 12" O/C FIELD.

4b. APPLY 3/8" APA RATED 24/0 ROOF SHEATHING OVER THE GAP BETWEEN SECTIONS OF THE HOME. FASTEN SHEATHING WITH 8d NAILS AT 6" O/C EDGES AND 12" O/C FIELD.

5. INSTALL A MINIMUM OF SIX ROOF VENTS (50 SQUARE INCH NET OPENING EACH) AT EACH SIDE OF RIDGE OVER CUT-OUTS IN FACTORY INSTALLED SHEATHING TO PROVIDE ATTIC VENTILATION.
   ALTERNATE: INSTALL CONTINUOUS RIDGE VENT AT A RATIO OF 1 SQ.FT. OF NET FREE AIR TO 300 SQ.FT. OF HOME AREA. HALF OF THIS AMOUNT MUST BE INSTALLED ALONG THE PEAK.
6. APPLY ROOF SHINGLES WITH UNDERLAYMENT IN ACCORDANCE WITH SHINGLE MANUFACTURERS INSTRUCTIONS. SHINGLES AND UNDERLAYMENT SHALL BE INSTALLED TO CONTINUE UPWARD FROM FACTORY APPLIED SHINGLES.
7. FASTEN EACH HINGED KING POST TO 2x4 LEDGER W/ (2) No. 8 x 3" WOOD SCREWS.
8. INSTALL CONTINUOUS 1x3 MEMBERS (8' LENGTHS) TO THE TRUSSES ON EACH SIDE OF THE PEAK. FASTEN 1x TO TRUSSES WITH (2) 8d NAILS PER TRUSS. FASTEN 1x ACROSS THE PEAK WITH 8d NAILS SPACED 4" O/C.

## GENERAL NOTES

1. ALL MATERIALS REQUIRED FOR SITE CLOSE-OFF SUCH AS SHINGLES, SHEATHING, FASCIA, OUTLOOKERS, BOLTS, ATTIC VENTS, FLASHING, SIDING, ETC., SHALL BE SUPPLIED BY GOLDEN WEST HOMES AND SHIPPED LOOSE WITH THE UNIT.
2. FOR HOMES WITHOUT "ALTERNATE CONSTRUCTION" APPROVAL, FUEL BURNING FURNACE AND WATER HEATER VENTS SHALL NOT BE POSITIONED IN HOME SUCH THAT THEY WOULD EXTEND THROUGH THE HINGED PORTION OF ROOF.
3. FACTORY SHALL PROVIDE PROTECTION FROM WEATHER DAMAGE ON UNROOFED PORTIONS OF HOME DURING TRANSIT AND STORAGE.
4. ALL FASTENERS EXPOSED TO WEATHER SHALL BE CORROSION RESISTANT.
5. AT EACH END OF UNITS, FIELD INSTALL SIDING TO MATCH TYPE THAT IS FACTORY INSTALLED ON MANUFACTURED HOME. INSTALL SIDING PER MANUFACTURERS INSTRUCTIONS WITH 6d CORROSIVE RESISTANT NAILS AT 6" O.C. EDGES AND FIELD.

APPROVED BY

NTA INC. 9/24/2020

FEDERAL MANUFACTURED HOME CONSTRUCTION AND SAFETY STANDARDS

| NO. | REVISED BY | DATE |
|---|---|---|
| 1 | MSU | 5/11/2020 |
| 2 | JLH | 5/26/2021 |
| 3 | JGH | 2/18/11 |

SITE INSTRUCTIONS FOR HINGE ROOF CLOSE - OFF

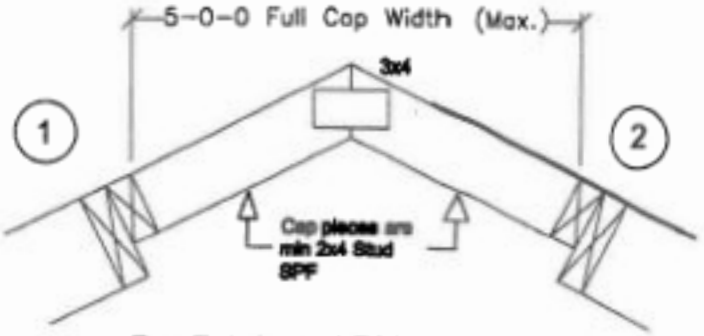
DRAWN BY CRB DATE 1-22-97

ROOF LOAD 40 PSF

S-62

Fontana Wood Products of Oregon
601 N.E. Antler Redmond, Oregon 97756
Phone: (541) 923-7747 FAX: (541) 923-7748
Copyright 2008 Fontana Wood Products of Oregon, Inc.

| Forces at Joints 1 & 2 | | | |
|---|---|---|---|
| Full Cap Width | Shear (lb) | Compression (lb) | Tension (lb) |
| 0" to 15" | 39 | 176 | 103 |
| 15.1" to 30" | 72 | 330 | 194 |
| 30.1" to 45" | 107 | 488 | 286 |
| 45.1" to 60" | 141 | 644 | 379 |




## Pre-Fabricated Ridge Cap Detail

(Connections by Others)

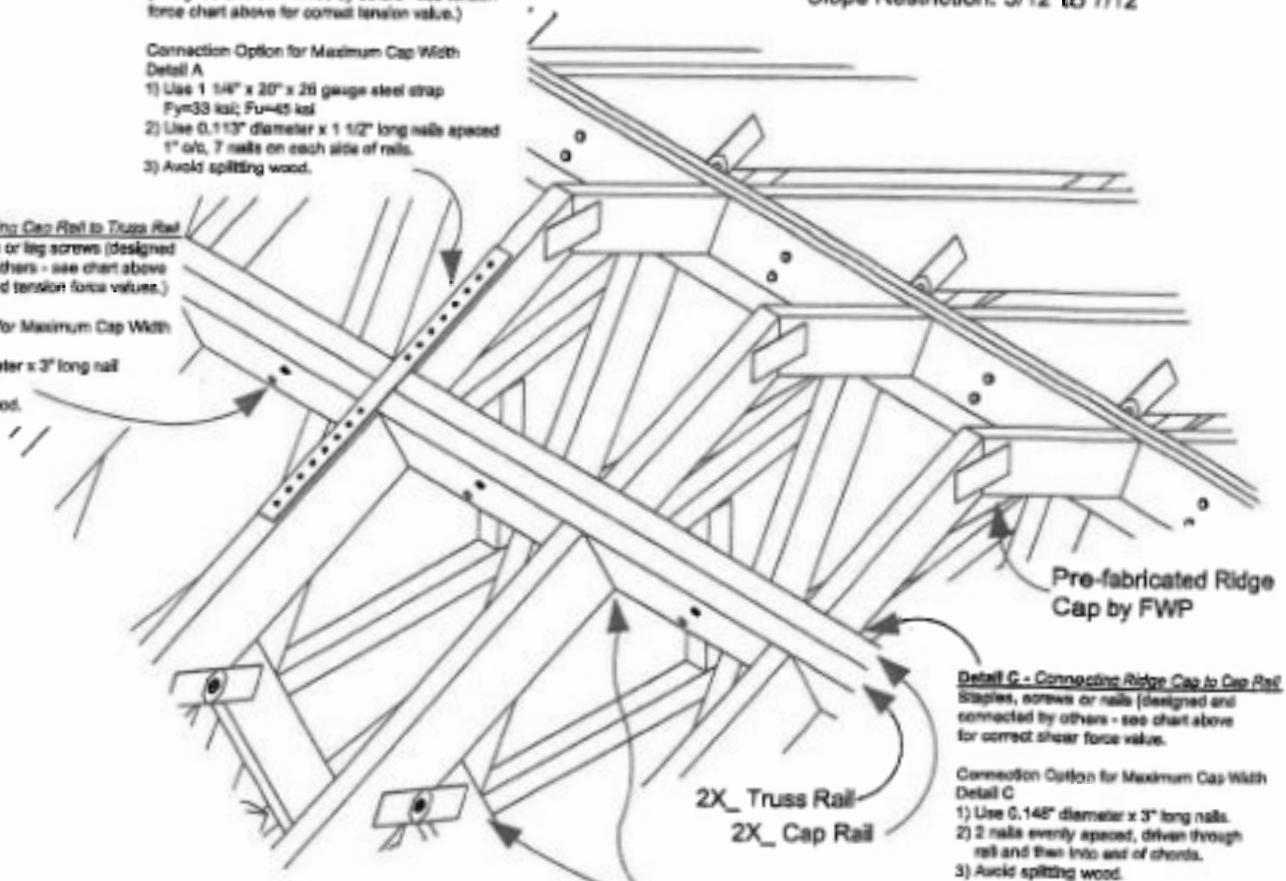
Slope Restriction: 3/12 to 7/12

**Detail A** - *Connecting for tension across Cap Rail & Truss Rail*
Steel tension strap or 7/16" span rated sheathing (designed and connected by others - see tension force chart above for correct tension value.)

Connection Option for Maximum Cap Width
Detail A
1) Use 1 1/4" x 20" x 26 gauge steel strap Fy=33 ksi; Fu=45 ksi
2) Use 0.113" diameter x 1 1/2" long nails spaced 1" o/c, 7 nails on each side of rails.
3) Avoid splitting wood.

**Detail B** - *Connecting Cap Rail to Truss Rail*
Wood screws, nails or lag screws (designed and connected by others - see chart above for correct shear and tension force values.)

Connection Option for Maximum Cap Width
Detail B
1) Use 0.148" diameter x 3" long nail
2) 2 nails per bay
3) Avoid splitting wood.




**Detail C** - *Connecting Ridge Cap to Cap Rail*
Staples, screws or nails (designed and connected by others - see chart above for correct shear force value.

Connection Option for Maximum Cap Width
Detail C
1) Use 0.148" diameter x 3" long nails.
2) 2 nails evenly spaced, driven through rail and then into end of chords.
3) Avoid splitting wood.

**Detail D** - *Connecting Truss Rail to Truss Top Chord*
Staples, screws or nails (designed and connected by others - see chart above for correct shear force value.

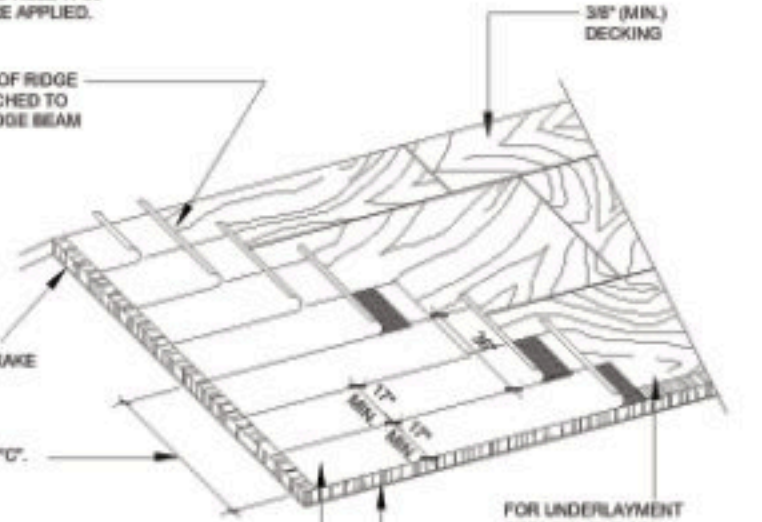
Connection Option for Maximum Cap Width
Detail D
1) Use 0.148" diameter x 3" long nails.
2) 2 nails evenly spaced, driven through rail and then into end of chords.
3) Avoid splitting wood.

| | | | |
|---|---|---|---|
| **SPACING:** 2-0-0 | **PLATES:** | WIND ZONE II @ 24 IN. O/C | WIND ZONE III @ 16 IN. O/C |
| **LOADING:** (psf) | Eagle 20ga. | -39 PSF WL | -47 PSF WL |
| **TCLL:** 45.0 | **CODE:** HUD Calc | -51 PSF WL (EAVE ONLY) | -62 PSF WL (EAVE ONLY) |
| **TCDL:** 11.0 | for Ridge Cap Only | 6 PSF DL | 6 PSF DL |

## \RNING - Verify design parameters and READ NOTES:

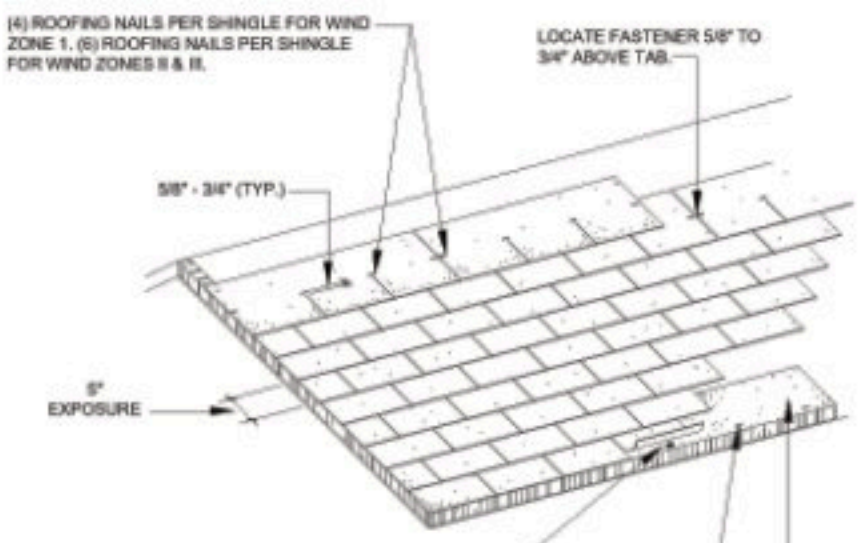
This building component has only been designed for the loads on this drawing. Construction and lifting forces have not been considered. The builder is responsible for lifting methods and system design. Builder responsibilities are defined under Section 2.3 of TPI1-2002.This design is based only upon parameters shown,and is for an individual building component to be installed and loaded vertically. Applicability of design parameters and proper incorporation of component is the responsibility of the building designer - not truss designer. Bracing shown is for lateral support of individual web members only. Additional temporary bracing to insure stability during construction is the responsibility of the erector. Additional permanent bracing of the overall structure is the responsibility of the building designer. For general guidance regarding fabrication,quality control,storage,delivery,erection and bracing,consult BCSI 1-03 from Wood Truss Council of Amerida and Truss Plate Institute recommendation availlable from Wood Truss Council of America 6300 Enterprise Ln,Madison,WI 53719

NOTE:
FASTEN UNDERLAYMENT TO DECKING WITH ONLY ENOUGH STAPLES TO HOLD IT IN PLACE UNTIL SHINGLES ARE APPLIED.




DRIP EDGE NOTES: ADJACENT PIECES OF DRIP EDGE SHALL BE OVERLAPPED A MINIMUM OF 2 INCHES. DRIP EDGES SHALL EXTEND A MINIMUM OF .25" BELOW THE ROOF SHEATHING AND EXTEND UP THE ROOF DECKING A MINIMUM OF 2". UNLESS SPECIFIED DIFFERENTLY BY THE SHINGLE MANUFACTURER, SHINGLES ARE PERMITTED TO BE FLUSH WITH THE DRIP EDGE.

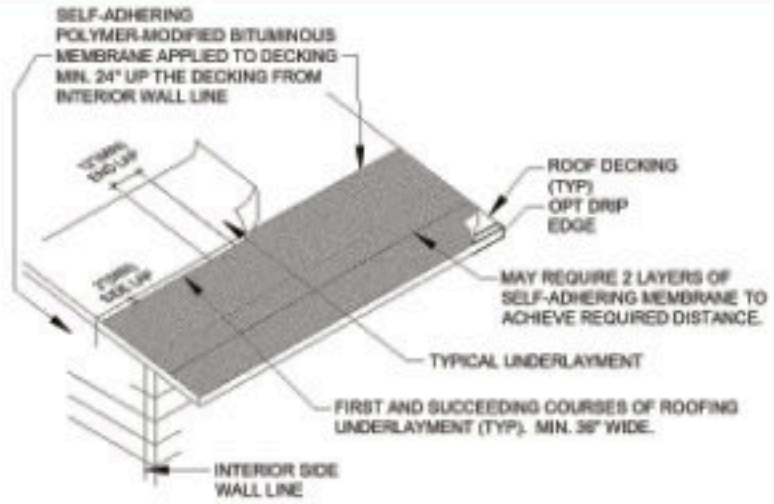
## TYPICAL UNDERLAYMENT DETAIL | A




NOTES:
1) SHINGLES WITH EXPOSED FASTENERS ONCE INSTALLED SHALL BE REMOVED AND REPLACED. COVERING EXPOSED FASTENERS WITH ROOFING TAR IS NOT AN APPROVED REPAIR METHOD.

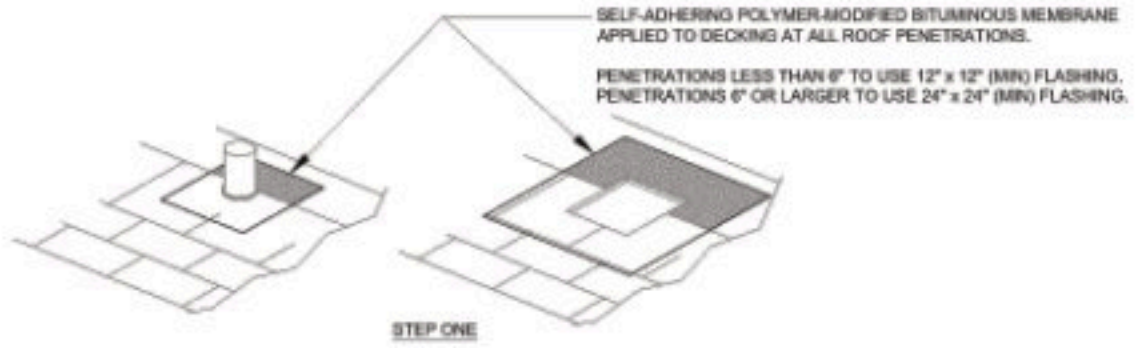
## TYPICAL SHINGLE INSTALLATION DETAIL | B

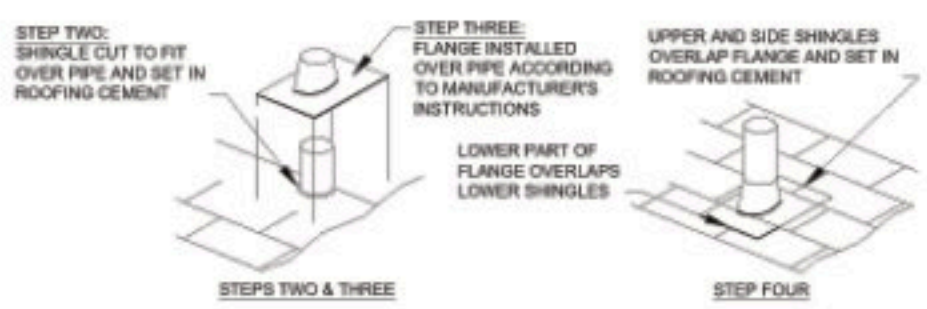
SOUTH CAROLINA LICENSED PROFESSIONAL ENGINEER
4/22/25
No. 39884
CHARLES PORTER



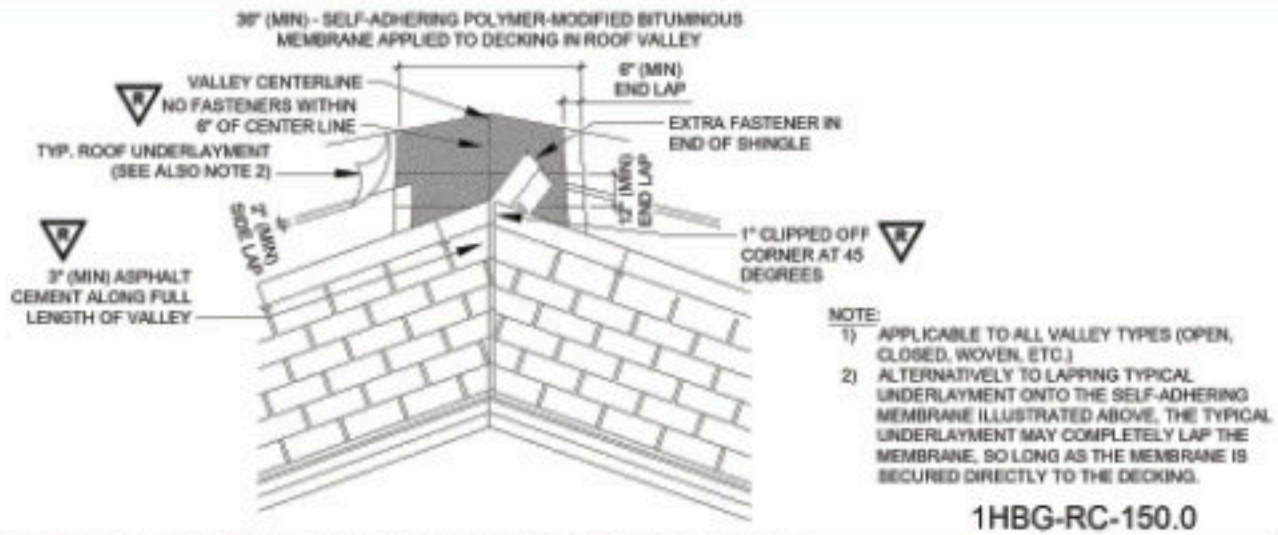

NOTES:
1) ICE DAMMING APPLICATION IS APPROVED FOR WIND ZONES I, II, AND III APPLICATIONS.
2) REFER TO OTHER DETAILS FOR INSTALLATION OF TYPICAL UNDERLAYMENTS, RAKE ENDS, PEAKS, ETC.

## TYPICAL ICE DAMMING DETAIL | C




## TYPICAL ROOF PENETRATION DETAIL | D




NOTE:
1) APPLICABLE TO ALL VALLEY TYPES (OPEN, CLOSED, WOVEN, ETC.)
2) ALTERNATIVELY TO LAPPING TYPICAL UNDERLAYMENT ONTO THE SELF-ADHERING MEMBRANE ILLUSTRATED ABOVE, THE TYPICAL UNDERLAYMENT MAY COMPLETELY LAP THE MEMBRANE, SO LONG AS THE MEMBRANE IS SECURED DIRECTLY TO THE DECKING.

1HBG-RC-150.0

## TYPICAL VALLEY UNDERLAYMENT AND FLASHING | E

## GENERAL NOTES

1. SHINGLES ARE NOT TO BE INSTALLED ON A ROOF HAVING A SLOP LESS THAN 2" IN 12" UNLESS THE SHINGLE MFG WARRANTS THE SHINGLES FOR A LESSER SLOPE.
2. SHINGLES ARE TO BE CLASS "C" WIND RESISTANT OR BETTER.
3. FOR FASTENING SHINGLES, USE 12GA. GALVANIZED ROOFING NAILS. FASTENER LENGTH SHOULD PENETRATE AT LEAST 1/8" THROUGH THE DECK.
4. AS AN ALTERNATE TO DOUBLE TYPE 15 FELT APPLICATIONS, A SINGLE LAYER OF UNDERLAYMENT LISTED BY AN APPROVED LISTING AGENCY MAY BE USED AND SHALL BE INSTALLED PER MFG'S INSTALLATION INSTRUCTIONS FOR SLOPES OVER 4" IN 12".
5. ALL ASPHALT SHALL CONFORM TO ASTM D4586 OR ASTM D2822.
6. SHINGLES SHALL EB INSTALLED TO OVERHANG THE DRIP EDGE BY 1/4" TO 3/4".
7. MANUFACTURER INSTALLATION INSTRUCTIONS WILL SUPERCEDE THIS DETAIL.
8. WATER MANAGEMENT AND ICE DAMMING REQUIRED ON ALL eBUILT HOMES AND HOMES LOCATED IN AREAS WHERE ICE DAMMING IS REQUIRED.
9. UNDERLAYMENT AND SIDE LAPS SHOWN PER ARMA GUIDELINES. ALTERNATIVELY MFG'S INSTALLATION INSTRUCTIONS MAY BE USED.
10. ALL SHINGLES PENETRATIONS TO BE SEALED IN ACCORDANCE WITH THE FLASHING MFG'S INSTALLATION INSTRUCTIONS WHEN AVAILABLE. OTHERWISE USE SHOWN DETAILS.
11. DO NOT USE PETROLEUM BASED SEALANTS ON BASE OF FLASHING WHEN USING A NO CAULK FLASHING.
12. OTHER PENETRATIONS (ATTIC VENTS, APPLIANCE VENTS, etc.) TO BE FLASHED ACCORDING TO THE VENT MFG'S INSTRUCTIONS.
13. ALL UNDERLAYMENT AND ICE DAMMING DETAILS APPLICABLE TO METAL ROOFING APPLICATIONS. METAL ROOFING TO BE INSTALLED AND FLASHED IN ACCORDANCE WITH MANUFACTURER INSTALLATION INSTRUCTIONS.

APPROVED BY NTA INC.
FEDERAL MANUFACTURED HOME CONSTRUCTION AND SAFETY STANDARDS

REVISED 5/5/2025

THIRD PARTY SEAL

LATEST REVISION

| | DESCRIPTION | REF. # |
|---|---|---|
| R | UPDATED NOTES ON DETAIL E | INITIALS 00/00/00 |

| | | | |
|---|---|---|---|
| DRAWN BY: | JRT | DATE: | 4/23/2025 |
| CALC REF: | - | | |

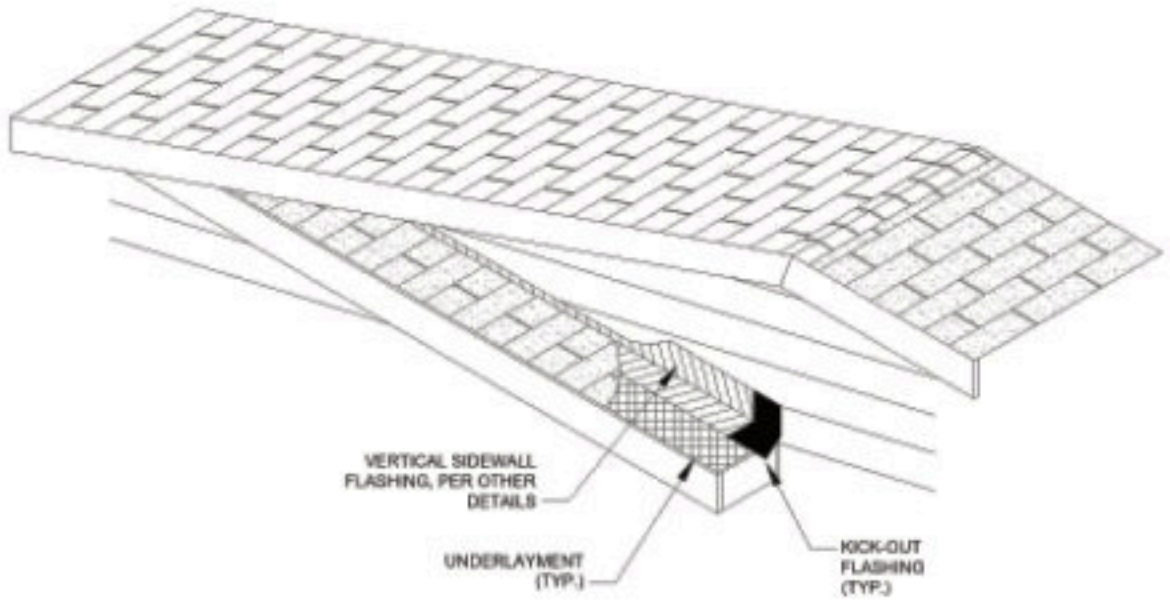
Clayton home building group

| WIND ZONE APPROVAL | ROOF LOAD PSF |
|---|---|
| NOT APPLICABLE | NOT APPLICABLE |

TITLE:
# TYPICAL SHINGLE ROOF APPLICATION DETAIL

SHEET: 1HBG-RC-150.0

Apr 23, 2025 - 3:31pm W:\ENG\DAPIA\DAPIA Projects\Working Projects\3155-R1\1HBG-RC-150.0.dwg

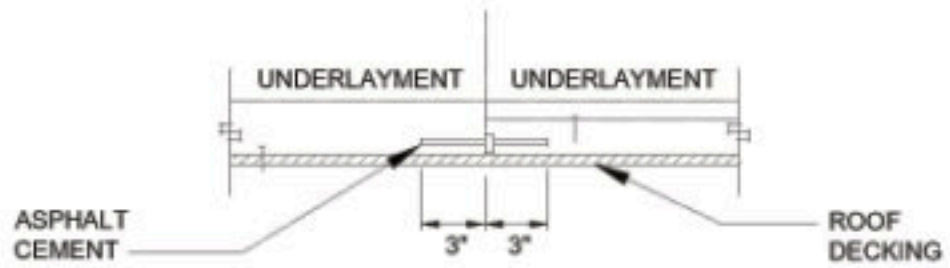



NOTE:
1) FLASHING AND KICK-OUT FLASHING (FOR ROOFS WITH ROOF-WALL INTERSECTIONS) SHALL BE INSTALLED USING THE FOLLOWING STEPS (NOTE: EQUIVALENT INNOVATIVE MATERIALS MAY BE INSTALLED PER FLASHING MATERIAL MANUFACTURER'S INSTRUCTIONS):
   a. APPLY ROOF UNDERLAYMENT OVER ROOF DECK AND UP THE SIDEWALL OVER WEATHER RESISTANT BARRIER; AND
   b. INSTALL SHINGLE STARTER STRIP THEN KICK-OUT DIVERTER; ATTACH TO ROOF DECK BUT NOT SIDEWALL; AND
   c. PLACE FIRST SHINGLE AND NEXT SECTION OF SIDEWALL FLASHING OVER UPPER EDGE OF DIVERTER; AND
   d. INSTALL REMAINING SIDEWALL FLASHING, COUNTER FLASHING, AND SHINGLES; AND
   e. APPLY SELF-ADHESIVE FLASHING OVER TOP EDGE OF THE WALL FLASHING, DIVERTER, AND WEATHER RESISTANT BARRIER; AND
   f. INSTALL THE WEATHER RESISTANT BARRIER, CUT HOUSE WRAP TO FIT OVER DIVERTER, AND TAPE TOP OF CUT.

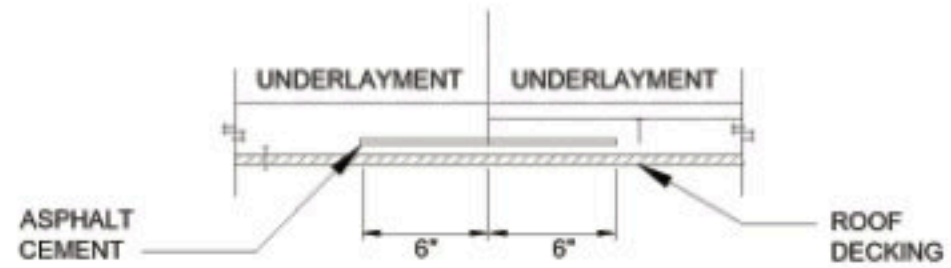
## KICK-OUT FLASHING

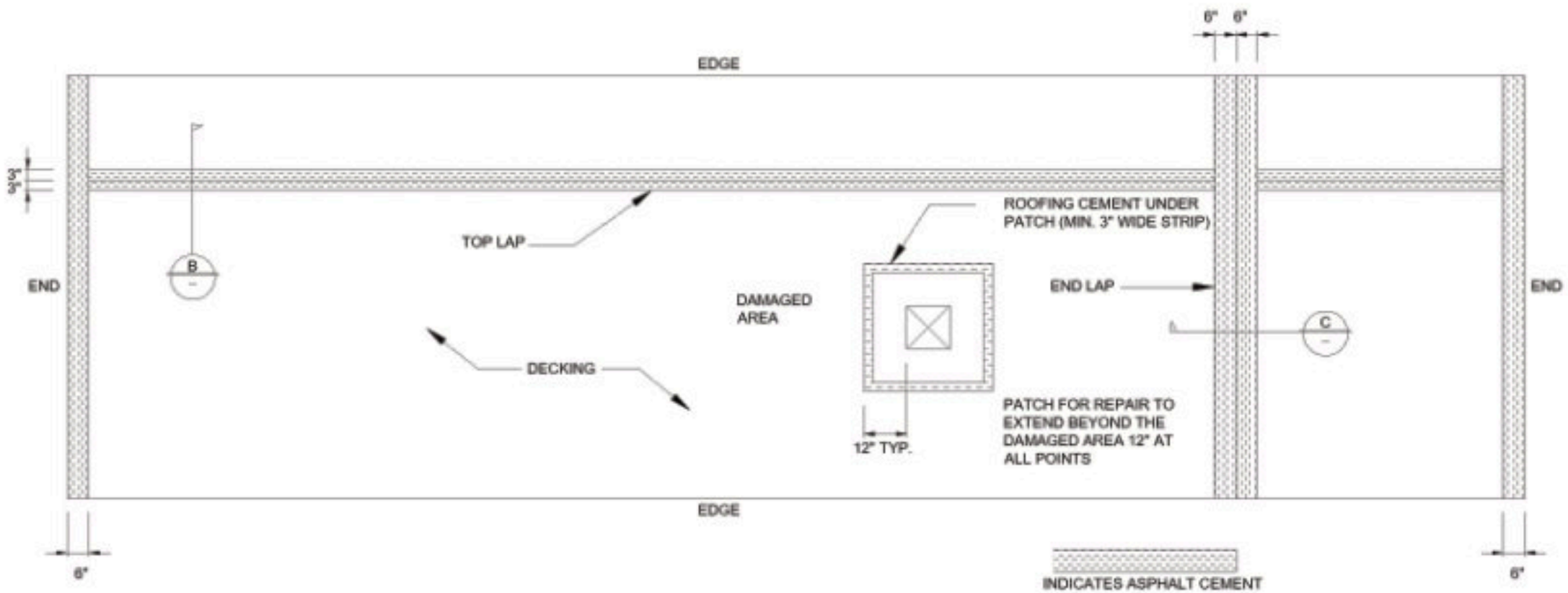
## A CEMENT AT SEAMS AND OVERLAPS (WZ 2 & 3) B

CEMENT AT OVERLAPS




CEMENT AT SEAMS




1HBG-RC-151.0

## WZ 2 & 3 UNDERLAYMENT SECUREMENT C

## GENERAL NOTES

1. WATER MANAGEMENT REQUIRED ON ALL HOMES PER ZERO ENERGY HOME PROGRAM.
2. DO NOT USE PETROLEUM BASED SEALANTS ON BASE OF FLASHING WHEN USING A NO CAULK FLASHING.
3. OTHER PENETRATIONS (ATTIC VENTS, APPLIANCE VENTS, etc.) TO BE FLASHED ACCORDING TO THE VENT MFG'S INSTRUCTIONS.
4. ANY SEAM IN THE PLY DRY MUST HAVE A 12" MIN. LAP. WITH 3/12 AND HIGHER PITCHED ROOFS 6" MIN. LAP.
5. ASPHALT CEMENT FOR WIND ZONES 2 & 3 ONLY

SOUTH CAROLINA LICENSED PROFESSIONAL ENGINEER
12/17/24
No. 39884
CHARLES PORTER

P.E. SEAL

APPROVED BY
NTA INC. 3/5/2025
FEDERAL MANUFACTURED HOME CONSTRUCTION AND SAFETY STANDARDS

THIRD PARTY SEAL

| LATEST REVISION | | |
|---|---|---|
| R | DESCRIPTION | REF. # |
| | --- | INITIALS |
| | | 00/00/00 |

DRAWN BY: JRT DATE: 12/17/2024
CALC REF: -

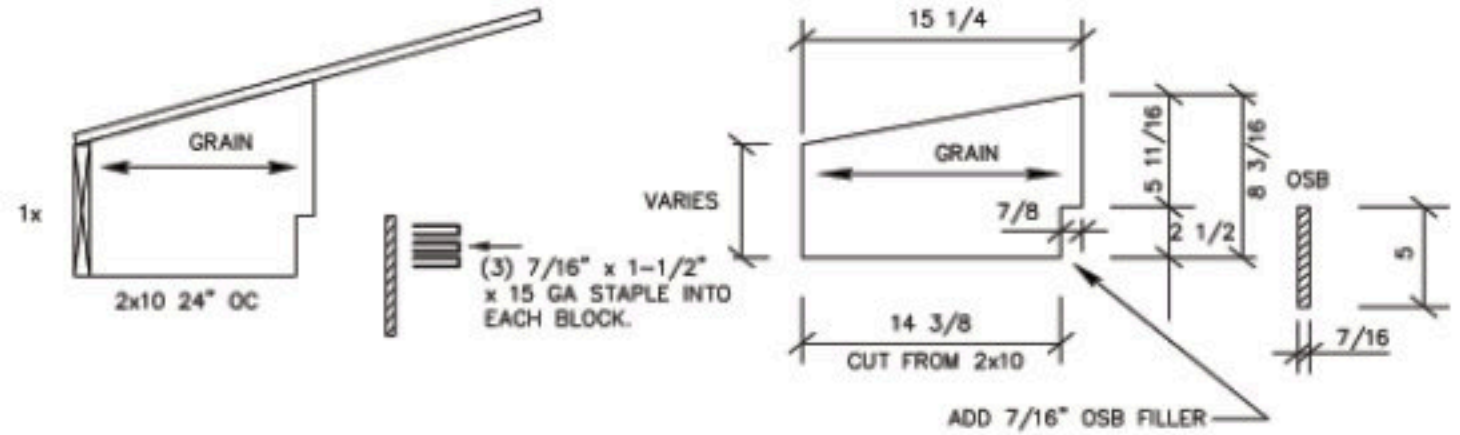
Clayton home building group

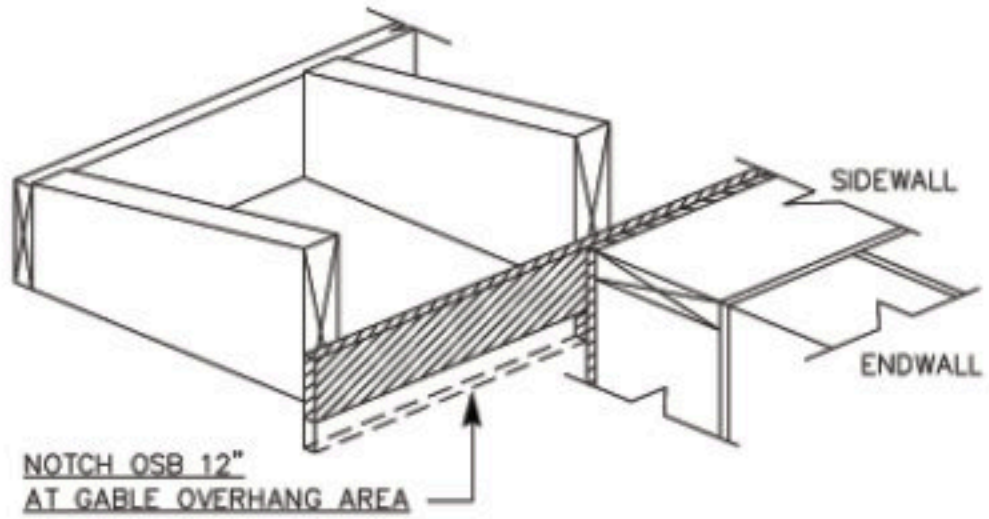
| WIND ZONE APPROVAL | ROOF LOAD PSF |
|---|---|
| NOT APPLICABLE | NOT APPLICABLE |

TITLE:
TYPICAL SHINGLE ROOF FLASHING AND UNDERLAYMENT

SHEET: 1HBG-RC-151.0

Dec 17, 2024 - 4:06pm W:\ENG\DAPIA\DAPIA Projects\Submitted Projects\2692-1HBG\1HBG-RC-151.0.dwg

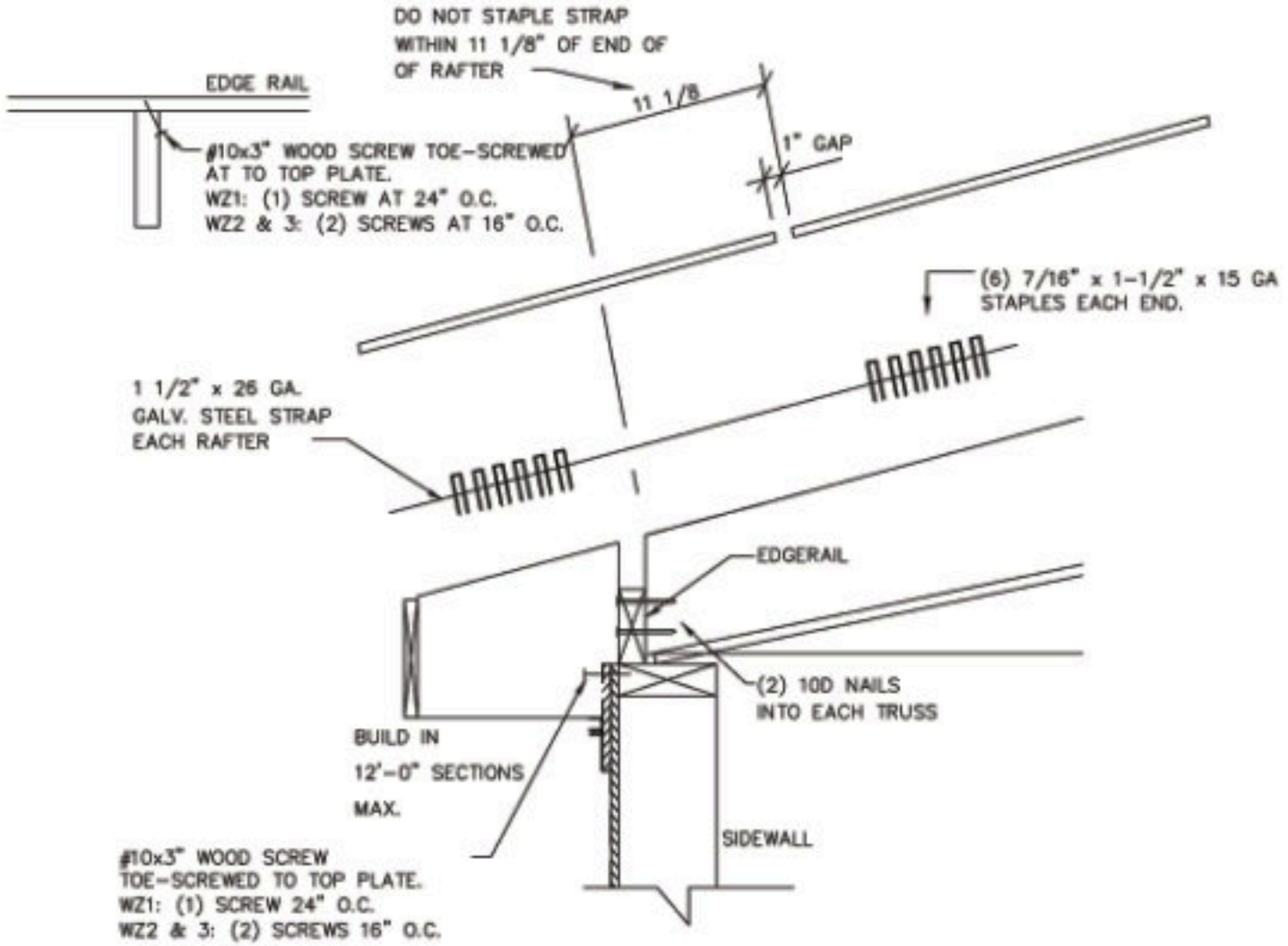


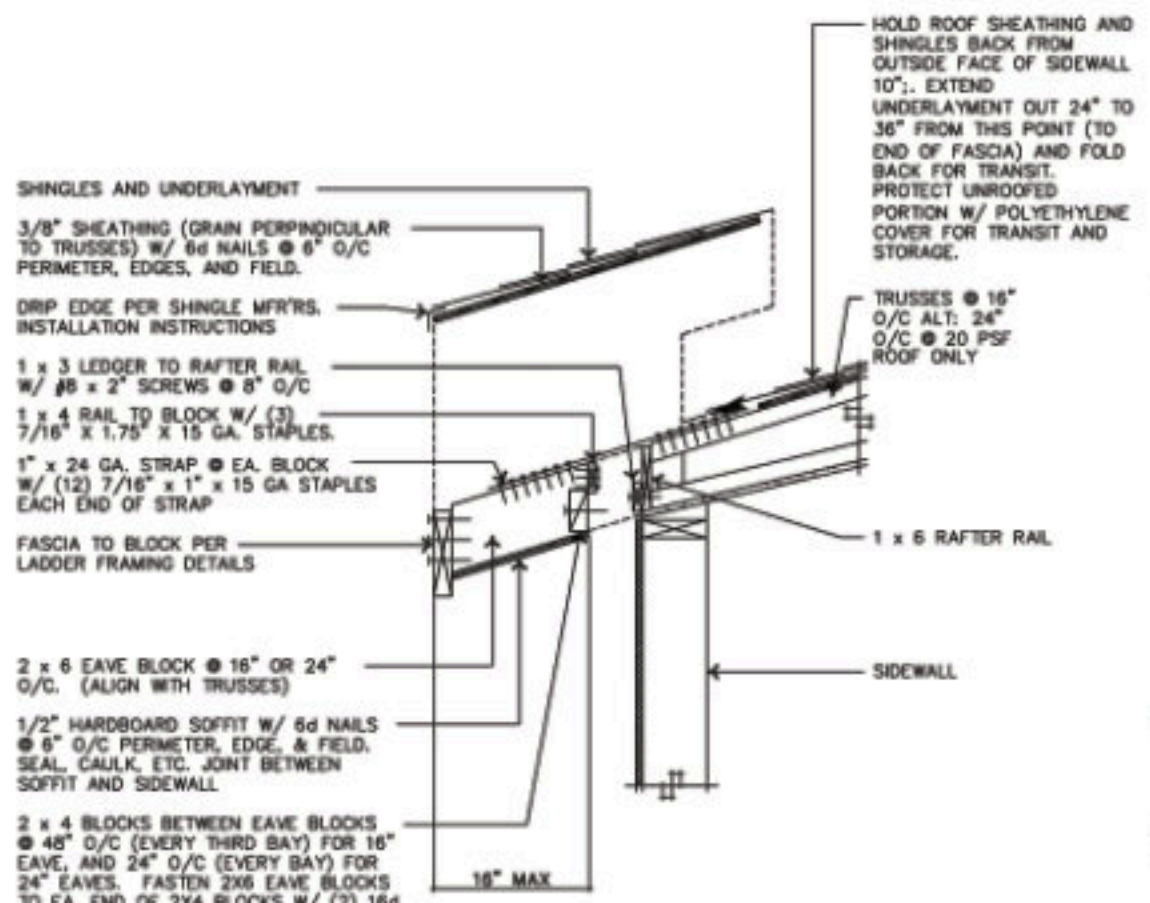

## GENERAL NOTES

1. MAX BOX WIDTH = 166"
2. MAX EAVE = 16"
3. ALL LUMBER USED MUST BE SPF #3 GRADE OR BETTER.

## END GABLE DETAIL B




## OVERHANG CONSTRUCTION DETAILS A




1HBG-RC-75.0

## ALT. ADD-ON EAVE C

SOUTH CAROLINA LICENSED PROFESSIONAL ENGINEER
12/4/24
No. 39884
CHARLES PORTER

P.E. SEAL

APPROVED BY NTA INC. 3/5/2025
FEDERAL MANUFACTURED HOME CONSTRUCTION AND SAFETY STANDARDS

THIRD PARTY SEAL

| LATEST REVISION | | |
|---|---|---|
| | DESCRIPTION | REF. # |
| R | --- | INITIALS |
| | | 00/00/00 |

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|---|---|---|---|
| DRAWN BY: | NCN | DATE: | 10/30/2024 |
| CALC REF: | C1HBG-RC-75.0-6 | | |

Clayton home building group

| WIND ZONE APPROVAL | ROOF LOAD PSF |
|---|---|
| ALL | 20 / 30 / 40 |

TITLE: 16" FLIP EAVE

SHEET: 1HBG-RC-75.0

Dec 04, 2024 - 8:50am C:\Users\NapierN\OneDrive - Clayton Homes\Documents\1. Projects\1HBG Phase 2\1HBG-RC\1HBG-RC Consolidated\1HBG-RC-75.0.dwg

## TRIPLEWIDE INSTRUCTIONS

1. BOLT RIDGE BEAMS TOGETHER AT MATING LINES PER INSTALLATION MANUAL SHEET S-11, DETAIL "A".
2. APPLY 3/8" APA RATED 24/0 ROOF SHEATHING OVER UNSHEATHED SURFACES OF ROOF. FASTEN SHEATHING WITH 6d NAILS AT 6" O/C EDGES, PERIMETER AND 12" O/C FIELD.
2b. APPLY 3/8" APA RATED 24/0 ROOF SHEATHING OVER THE GAP BETWEEN SECTIONS OF THE HOME. FASTEN SHEATHING WITH 6d NAILS AT 6" O/C EDGES, PERIMETER AND 12" O/C FIELD. SHEATHING MUST EXTEND 10" MINIMUM ONTO EACH SECTION AND THE DECKING MUST BE SECURED TO EACH TRUSS ON EACH SECTION WITH A MINIMUM OF (3) 6d NAILS.

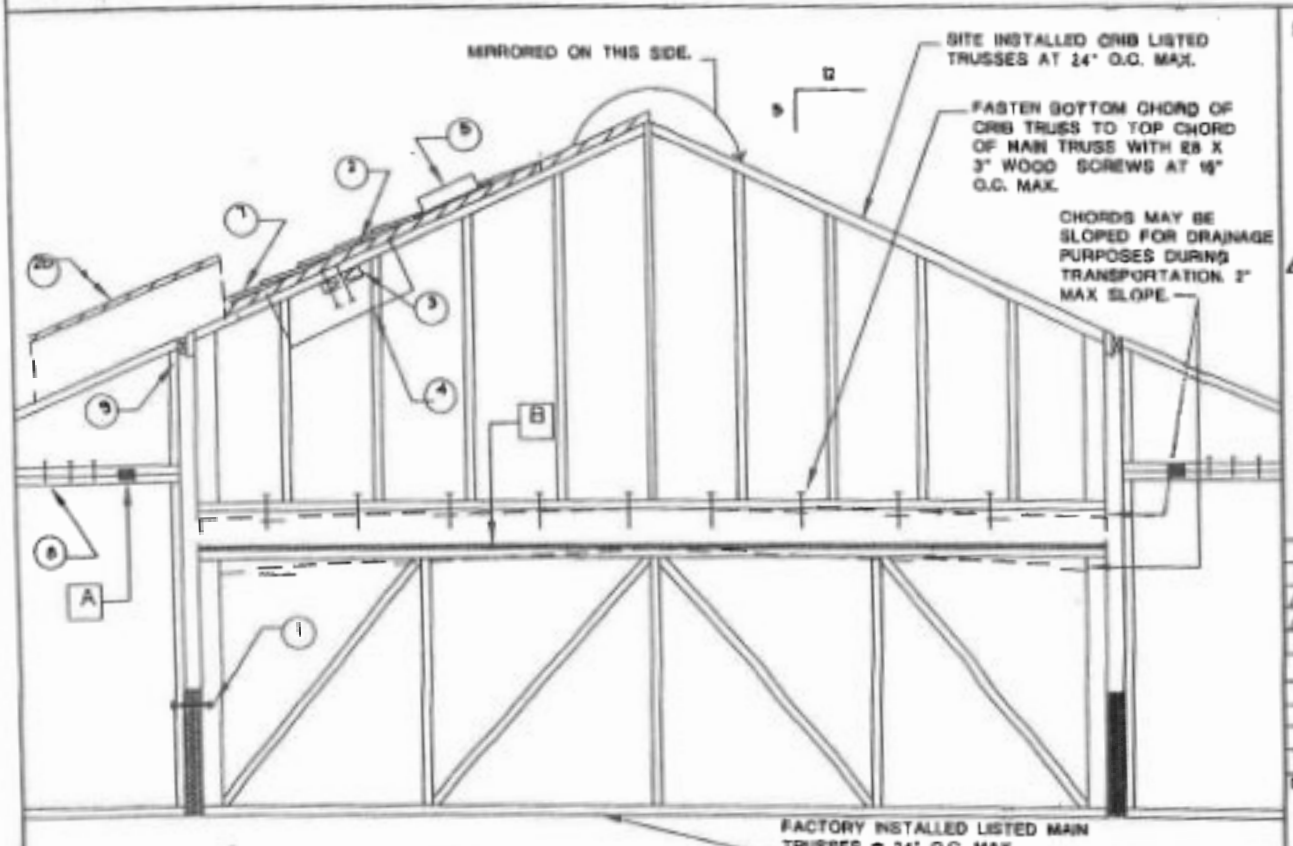
LEGEND:
(8) — SITE INSTALLED ITEM
[-] — FACTORY INSTALLED ITEM

3. AT EACH END OF ROOF, INSTALL 2x6 No. 2 H.F. OUTLOOKERS AT 28" O/C WITH ONE OUTLOOKER DIRECTLY NEXT TO EACH SIDE OF THE PEAK. EXTEND ALL OUTLOOKERS BACK TO THIRD TRUSS FROM ENDWALL. EXTEND OUTLOOKERS OUT PAST ENDWALL TO MATCH FACTORY INSTALLED OUTLOOKER EXTENSIONS. FASTEN OUTLOOKERS TO TRUSS TOP CHORD WITH (2) No. 8x3" WOOD SCREWS AT EACH CONNECTION.
4. INSTALL CONTINUOUS 2x8 FASCIA BOARD TO ENDS OF OUTLOOKERS WITH (3) 16d NAILS AT EACH OUTLOOKER. SPLICE FASCIA PER SET-UP MANUAL DRAWING "S-16".
[A] CAP TRUSS HINGED TO MAIN TRUSS WITH NYLON STRAP ENABLING THE TRUSS TO FOLD DOWN INTO ROOF CAVITY FOR TRANSPORT.
5. INSTALL A MINIMUM OF SIX ROOF VENTS (50 SQUARE INCH NET OPENING EACH) AT EACH SIDE OF RIDGE OVER CUT-OUTS IN FACTORY INSTALLED SHEATHING TO PROVIDE ATTIC VENTILATION.
6. REQUIRED VENTILATION TO BE AT LEAST 1/300 OF THE ATTIC FLOOR SPACE. THE ROOF VENTS MUST PROVIDE AT LEAST 50% OF THE REQUIRED VENTILATION AND AT LEAST 40% MUST BE PROVIDED BY EAVE, SOFFIT OR LOW GABLE VENTS. A MINIMUM HEIGHT OF 1" MUST BE PROVIDED BETWEEN THE TOP OF THE INSULATION AND THE ROOF SHEATHING.
7. APPLY ROOF SHINGLES WITH UNDERLAYMENT IN ACCORDANCE WITH SHINGLE MANUFACTURERS INSTRUCTIONS. SHINGLES AND UNDERLAYMENT SHALL BE INSTALLED TO CONTINUE UPWARD FROM FACTORY APPLIED SHINGLES.
8. SECURE CAP TRUSS TO MAIN TRUSS PER S-61.1.
9. ATTACH 2x3 EDGERAIL PER S-61.1.
[B] FACTORY INSTALLED 3/8" APA SHEATHING.
10. THIS DETAIL APPLICABLE TO WIND ZONE I.

24

## 5/12 TRIPLE WIDE CENTER SECTION SETUP INSTRUCTIONS.

A



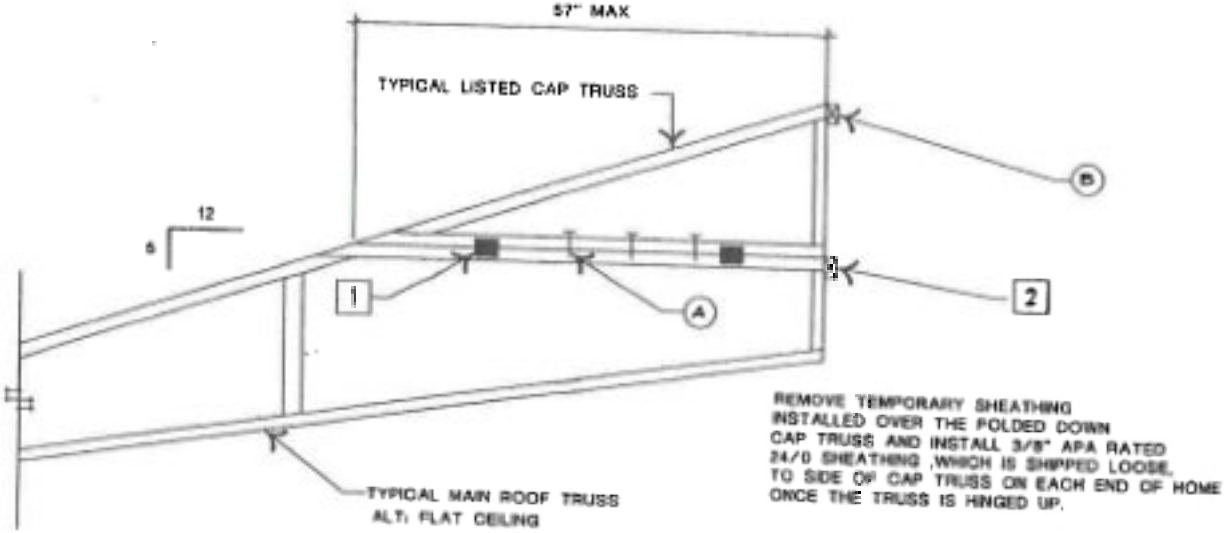

GENERAL NOTES:

1. FACTORY SHALL PROVIDE PROTECTION FROM WEATHER DAMAGE ON UNROOFED PORTIONS OF HOME DURING TRANSIT AND STORAGE.
2. ALL MATERIALS REQUIRED FOR SITE CLOSE-OFF SUCH AS TRUSSES, SHINGLES, SHEATHING, RIDGE BOARD, FASCIA, OUTLOOKERS, BOLTS, ATTIC VENTS, VENT EXTENSIONS, FLASHINGS, ETC. SHALL BE SUPPLIED BY CMH MANUFACTURING LP AND SHIPPED LOOSE WITH THE UNIT.
3. ALL FASTENERS EXPOSED TO WEATHER SHALL BE CORROSION RESISTANT.
4. AT EACH END OF UNITS, FIELD INSTALL SIDING TO MATCH TYPE THAT IS FACTORY INSTALLED ON MANUFACTURED HOME. INSTALL SIDING PER MANUFACTURERS INSTRUCTIONS WITH 6d CORROSIVE RESISTANT NAILS AT 6" O.C. EDGES AND FIELD.

| NO. | REVISED BY | DATE |
|---|---|---|
| 1 | PKT | 2-15-07 |
| 2 | PKT | 3/6/07 |

REGISTERED PROFESSIONAL ENGINEER
MARK JOSEPH OLSZO
NO. C042077
Exp. 3-31-2014
CIVIL
STATE OF CALIFORNIA

DRAWING: APPENDIX FOR SITE INSTALLED PEAK TRUSSES
ROOF LOAD: 40 PSF
DRAWN BY: TM
DATE: 4/23/04
—
S-61.2




## ROOF SECTION AT TRUSS PROFILE FOR DOUBLEWIDE

12 A

1 — CAP TRUSS HINGED TO MAIN TRUSS WITH NYLON STRAP ENABLING THE TRUSS TO FOLD DOWN INTO ROOF CAVITY FOR TRANSPORT.

2 — INSTALL TEMPORARY 2X3 STUD GRADE EDGERAIL TO EACH KINGPOST WITH (1) No. 8 X 3" WOOD SCREW

(A) — SECURE CAP TRUSS TO MAIN TRUSS W/ (3) 7/16" X 2 1/2" X 16 GA STAPLES FASTEND FROM BOTTOM CHORD OF CAP TRUSS INTO TOP CHORD OF MAIN TRUSS AS SHOWN.

(B) — REMOVE TEMPORARY EDGERAIL MENTIONED ABOVE AND RE-ATTACH TO KINGPOST OF CAP TRUSS WITH (2) 7/16" X 2-1/2" X 16 GA STAPLES.

## MISC NOTES FOR 5/12 HINGE SETUP

12 B

REGISTERED PROFESSIONAL ENGINEER
MARK JOSEPH EZZO
NO. C042077
Exp. 3-31-2014
CIVIL
STATE OF CALIFORNIA

LEGEND:

(B) — SITE INSTALLED ITEM - SEE SHEET S-62 U.O.N. "SITE INSTRUCTIONS FOR HINGE ROOF CLOSE OFF".

[ ] — FACTORY INSTALLED ITEM - SEE SHEET S-63 U.O.N. "FACTORY INSTALLED NOTES".

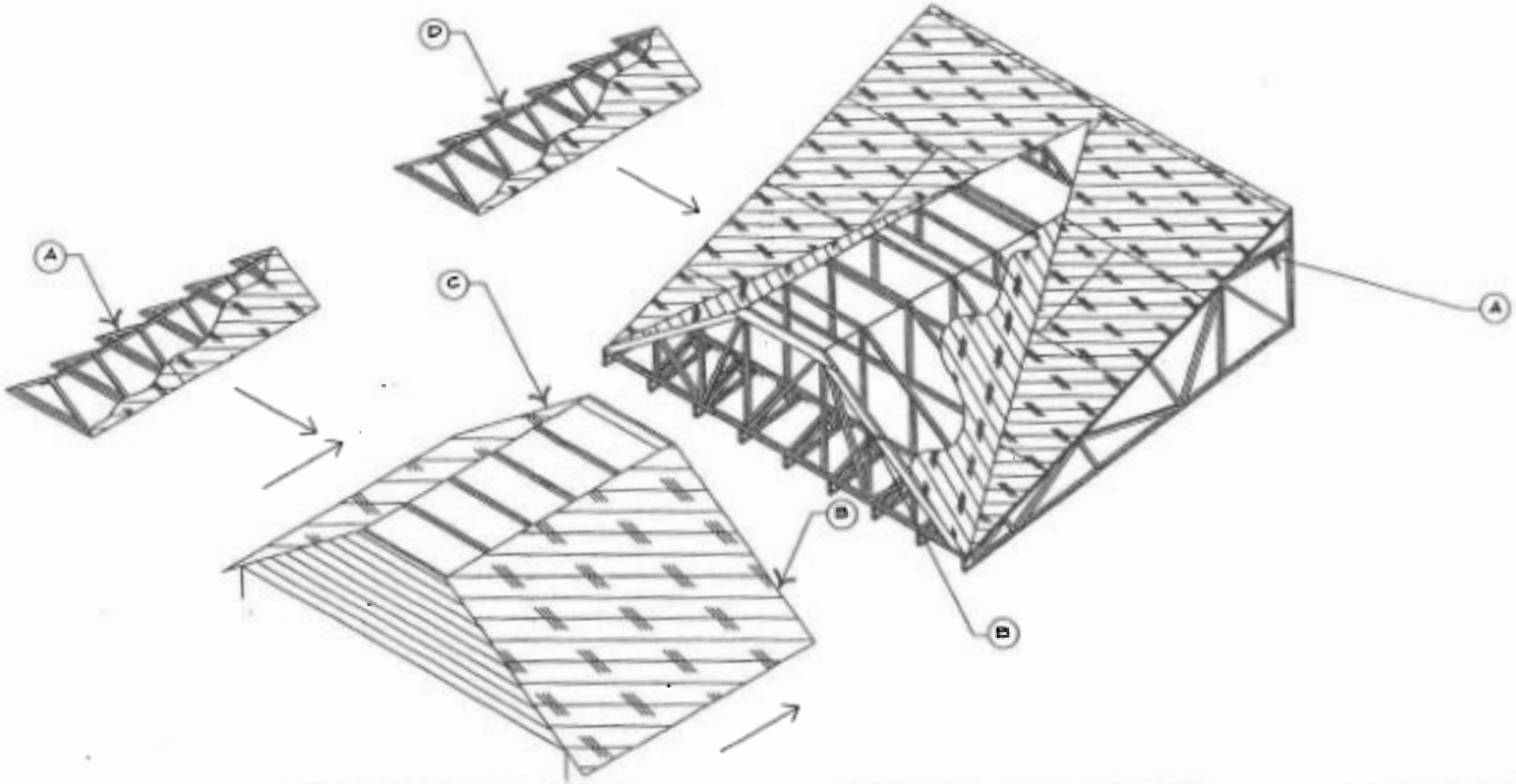
GENERAL NOTES:

1. FOR HOMES WITHOUT "ALTERNATE CONSTRUCTION" APPROVAL, FUEL BURNING FURNACE AND WATER HEATER VENTS SHALL NOT BE POSITIONED IN HOME SUCH THAT THEY WOULD EXTEND THROUGH THE HINGED PORTION OF ROOF.
2. FACTORY SHALL PROVIDE PROTECTION FROM WEATHER DAMAGE ON UNROOFED PORTIONS OF HOME DURING TRANSIT AND STORAGE.
3. ALL MATERIALS REQUIRED FOR SITE CLOSE-OFF SUCH AS SHINGLES, SHEATHING, RIDGE BOARD, FASCIA, OUTLOOKERS, BOLTS, ATTIC VENTS, ETC., SHALL BE SUPPLIED BY THE FACTORY AND SHIPPED LOOSE WITH THE UNIT.

| NO. | REVISED BY | DATE |
|---|---|---|
| 1 | TH | 1/13/04 |

HINGED ROOF CLOSE-OFF

DRAWN BY TH | DATE 7/3/02 | ROOF LOAD 40 PSF | S-61.1




ROOF SECTION AT TAG UNIT TO MAIN UNIT

LEGEND:

- (E) — SITE INSTALLED ITEM - SEE SHEET S-62 U.O.N. "SITE INSTRUCTIONS FOR HINGE ROOF CLOSE OFF".
- [-] — FACTORY INSTALLED ITEM - SEE SHEET S-63 U.O.N. "FACTORY INSTALLED NOTES".
- (A) — SECURE CAP TRUSS TO MAIN TRUSS PER DETAIL S-61.1
- (B) — TOP CHORD OF TAG TRUSS SECURED TO TOP CHORD OF MAIN UNIT TRUSS PER S-11 B OR C
- (C) — ROOF CLOSE UP PER S-31 DETAIL.
- (D) — SECURE CAP TRUSS TO MAIN TRUSS PER DETAIL S-61.2

GENERAL NOTES:

1. FOR HOMES WITHOUT "ALTERNATE CONSTRUCTION" APPROV FUEL BURNING FURNACE AND WATER HEATER VENTS SHALL NOT BE POSITIONED IN HOME SUCH THAT THEY WOULD EXTEND THROUGH THE HINGED PORTION OF ROOF.
2. FACTORY SHALL PROVIDE PROTECTION FROM WEATHER DAMAGE ON UNROOFED PORTIONS OF HOME DURING TRANSIT AND STORAGE.
3. ALL MATERIALS REQUIRED FOR SITE CLOSE-OFF SUCH AS SHINGLES, SHEATHING, RIDGE BOARD, FASCIA, OUTLOOKERS, BOLTS, ATTIC VENTS, ETC., SHALL BE SUPPLIED BY THE FACTORY AND SHIPPED LOOSE WITH THE UNIT.

MISC NOTES FOR 5/12 HINGE SETUP

| NO. | REVISED BY | DATE |
|---|---|---|
| A | - | - |

REGISTERED PROFESSIONAL ENGINEER
MARK [illegible] OLSON
NO. C042077
Exp. 3-31-2014
CIVIL
STATE OF CALIFORNIA

SITE INSTALLED PEAK TRUSSES FOR 5/12 TRANSVERSE TAG

| DRAWN BY | DATE | ROOF LOAD | |
|---|---|---|---|
| JGP1 | 8/16/2014 | 40 PSF | S-61.3 |

## FACTORY INSTALLED NOTES

### DOUBLEWIDE INSTRUCTIONS

[A] APPLY 3/8" MIN. ROOF SHEATHING TO HINGED TRUSS CHORDS PER STRUCTURAL DRAWING No. 98.00.

[B] INSTALL 2x6 FLAT OUTLOOKERS AS SHOWN PER STRUCTURAL DRAWING 96.03 AND PER STRUCTURAL DRAWING 96.04.

[C] FACTORY INSTALLED SHEATHING UP TO HINGED GANG NAIL PLATE PER STRUCTURAL DRAWING No. 98.00 AND FACTORY INSTALLED SHINGLES AND UNDERLAYMENT PER STRUCTURAL DRAWING No. 96.60

[D] FACTORY INSTALLED 2x8 FASCIA PER STRUCTURAL DRAWING 96.03 OR 96.04

[E] FACTORY INSTALLED 2x4 CONTINUOUS LEDGER FASTENED TO EACH TRUSS W/ (1) No. 8 x 3" WOOD SCREW.

### TRIPLEWIDE INSTRUCTIONS ⚠

[A] APPLY 3/8" MIN. ROOF SHEATHING TO HINGED TRUSS CHORDS PER STRUCTURAL DRAWING No. 98.00.

[B] INSTALL 2x6 FLAT OUTLOOKERS AS SHOWN PER STRUCTURAL DRAWING 96.03 AND PER STRUCTURAL DRAWING 96.04.

[C] FACTORY INSTALLED SHEATHING UP TO END OF OUTER SECTION PER STRUCTURAL DRAWING No. 98.00 AND FACTORY INSTALLED SHINGLES AND UNDERLAYMENT PER STRUCTURAL DRAWING No. 96.60

[D] FACTORY INSTALLED 2x8 FASCIA PER STRUCTURAL DRAWING 96.03 OR 96.04

[E] FACTORY INSTALLED 2x4 CONTINUOUS LEDGER FASTENED TO EACH TRUSS W/ (1) No. 8 x 3" WOOD SCREW.

## GENERAL NOTES

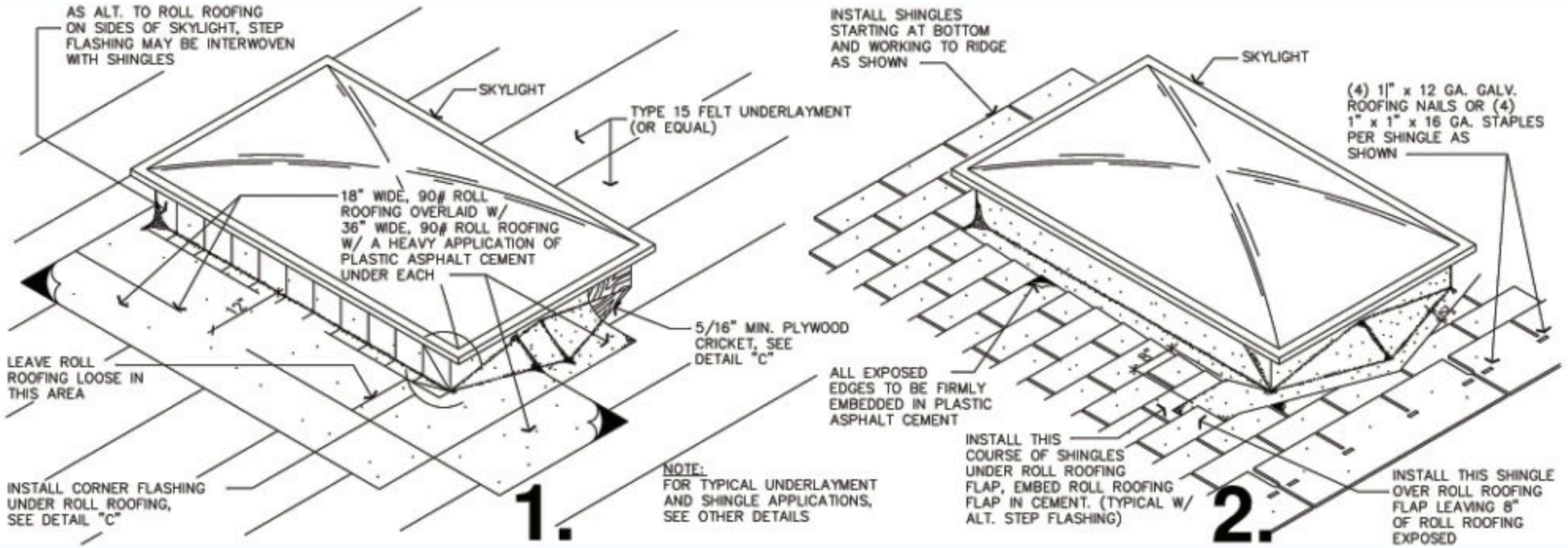
1. ALL MATERIALS REQUIRED FOR SITE CLOSE-OFF SUCH SHINGLES, SHEATHING, FASCIA, OUTLOOKERS, BOLTS, ATTIC VENTS, FLASHING, SIDING, ETC... SHALL BE SUPPLIED BY GOLDEN WEST HOMES AND SHIPPED LOOSE WITH THE UNIT.
2. FOR HOMES WITHOUT "ALTERNATE CONSTRUCTION" ⚠ APPROVAL, FUEL BURNING FURNACE AND WATER HEATER VENTS SHALL NOT BE POSITIONED IN HOME SUCH THAT THEY WOULD EXTEND THROUGH THE HINGED PORTION OF ROOF.
3. FACTORY SHALL PROVIDE PROTECTION FROM WEATHER DAMAGE ON UNROOFED PORTIONS OF HOME DURING TRANSIT AND STORAGE.
4. ALL FASTENERS EXPOSED TO WEATHER SHALL BE CORROSION RESISTANT.

| NO. | REVISED BY | DATE |
|---|---|---|
| 1 | MSB | 8/1/06 |

REGISTERED PROFESSIONAL ENGINEER
MARK J. SEBASTIAN
NO. C042077
Exp. 3-31-2014
CIVIL
STATE OF CALIFORNIA

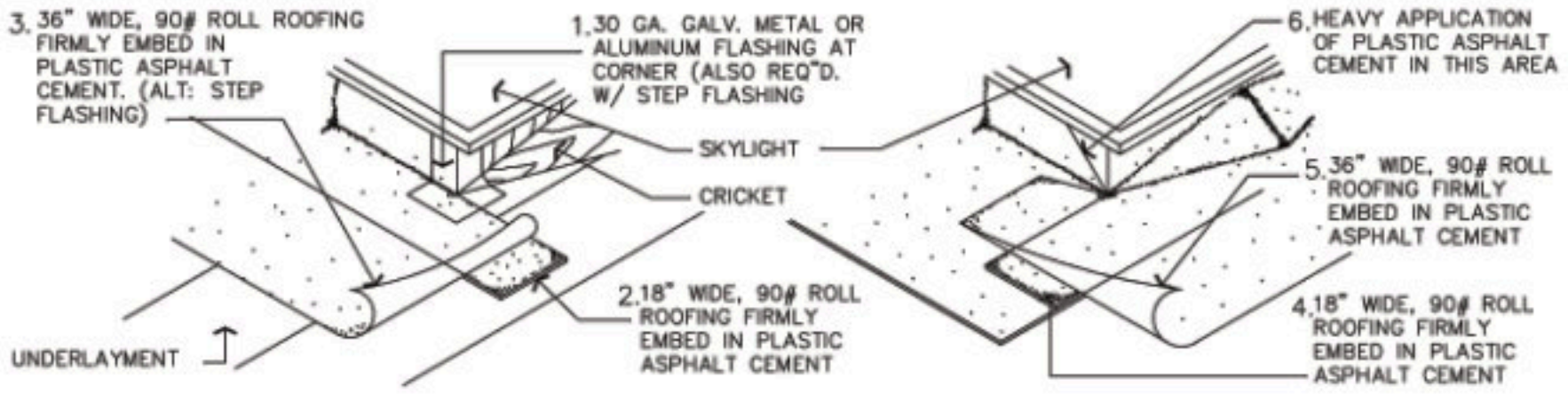
DRAWING: HINGED ROOF FACTORY INSTALLED NOTES

DRAWN BY: CRB | DATE: 1-23-91 | ROOF LOAD: 40 PSF | S-63



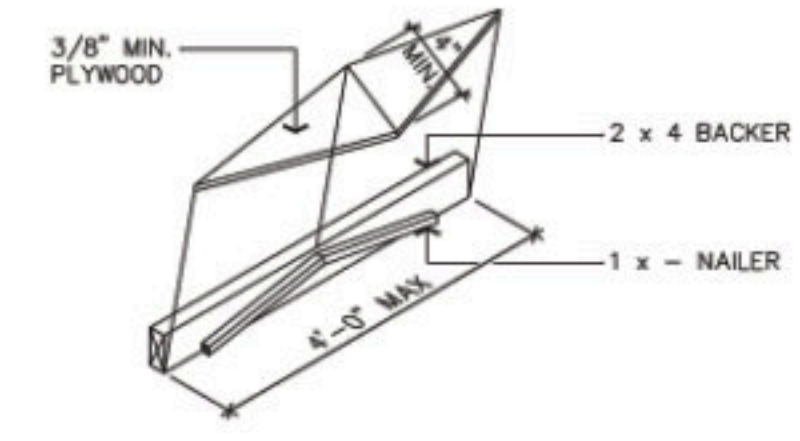

## ROOF CLOSE-UP APPLICATIONS

A




## CORNER FLASHING DETAIL

B




## CRICKET DETAIL

C

1HBG-RC-111.0

## GENERAL NOTES

1. FOR TYPICAL FELT UNDERLAYMENT AND SHINGLE APPLICATIONS SEE OTHER DETAILS

P.E. SEAL

APPROVED BY NTA INC. 3/5/2025
FEDERAL MANUFACTURED HOME CONSTRUCTION AND SAFETY STANDARDS

THIRD PARTY SEAL

| LATEST REVISION | | |
|---|---|---|
| | DESCRIPTION | REF. # |
| | --- | INITIALS |
| | | 00/00/00 |

DRAWN BY: NCN | DATE: 10/31/2024
CALC REF: -

Clayton home building group

| WIND ZONE APPROVAL | ROOF LOAD PSF |
|---|---|
| NOT APPLICABLE | 120 MAX |

TITLE: SKYLIGHT ROOFING APPLICATIONS

SHEET: 1HBG-RC-111.0

Dec 17, 2024 - 9:25am W:\ENG\DAPIA\Projects\Submitted Projects\2692-1HBG\1HBG-RC-111.0.dwg